



AURAK

الجامعة الأمريكية في رأس الخيمة
AMERICAN UNIVERSITY OF RAS AL KHAIMAH

LIFE-CHANGING EDUCATION WITH A GLOBAL REACH



AMERICAN UNIVERSITY OF RAS AL KHAIMAH
CATALOG ACADEMIC YEAR 2024-2025

Message from the President

Dear AURAK Students

Welcome to The American University of Ras Al Khaimah (AURAK).

It is a great honor to welcome you into the AURAK Community as you embark upon this incredible educational journey.

With the support of His Highness Sheikh Saud bin Saqr Al Qasimi, Ruler of Ras Al Khaimah, AURAK recognizes that it is ultimately through education that the U.A.E. and Ras Al Khaimah, will achieve the goal of sustainable prosperity and peace – requisites for a society in which each individual can contribute according to his or her unique capabilities. At AURAK, our commitment to academic excellence and an appreciation for lifelong learning defines our Mission and everything we do.

Over the past few years, the university has grown substantially with the development of new academic programs, the hiring of new faculty and staff, and a significant increase of our student population. The trajectory of this growth continues upwards and is most evident with the University's 2025 QS Ranking as one of the world's top 500 universities and number six in the UAE.

The University's Mission is to provide its students a transformational, student-centered learning experience based upon the American model. This approach prepares students to become future leaders and entrepreneurs for meaningful, fulfilling, and successful lives. AURAK has set a new standard in the Gulf Region for excellence that empowers its graduates to change the world through inspired teaching, research, creative work, and community service.

Your time at AURAK will be a time of discovery, the discovery of new ideas, friends, opportunities, and of course, your potential. That potential, at the end of your academic journey, I am confident will grow into success like it has for so many AURAK alumni before you who have journeyed down the same pathway.

I am delighted that you chose to make AURAK your educational home, and I look forward to meeting you on campus.

Sincerely,

Dr. David A. Schmidt

President, AURAK

AMERICAN UNIVERSITY OF RAS AL KHAIMAH

ACADEMIC CATALOG

2024-2025

Effective September 1, 2024

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Ms. Rania Hafiz Hussein Ali – University Registrar

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ACADEMIC CALENDAR 2024 – 2025

Fall 2024

Month	Day	Week	Activity/Event
August	1-25		Placement Tests / Continuing Student Registration
August	1-25		New Student Course Registration
August	5		Joining Date for New Faculty
August	19		Returning Date for Continuing Faculty
August	19-25		Faculty Professional Development Workshops, Including New Faculty Orientation
August	19-25		Continuing Advising & Registration for Returning Students
August	20		Deadline for Continuing Students to Complete Registration for Fall Semester without Late Registration Fee
August	22		New Student Orientation
August	23		International Student Orientation
August	26		First day of Class
August	26-Sept 1	Week 1	
September	2-8	Week 2	
September	9		AURAK Admissions Application Deadline
September	9		Last day to Add/Drop
September	9		Graduation Date for Summer 2023
September	9-15	Week 3	
September	15		Prophet's Birthday*
September	16-22	Week 4	
September	22		Census Date for Fall
September	23-29	Week 5	
September	30-Oct 6	Week 6	
October	7-13	Week 7	
October	14-20	Week 8	
October	21		Mid Term Grades Report Due
October	21-27	Week 9	
October	28		Last date to withdraw from a course with a grade of "W"
October	28-Nov 3	Week 10	
October	28		Advisement Begins For Spring 2025
November	4-10	Week 11	
November	4		Early-Registration For Spring Semester begins
November	11-17	Week 12	
November	18-24	Week 13	
November	25-Dec 1	Week 14	
November	25-Dec 11		Student Course Evaluations
December	1		UAE Commemoration Day
December	2-3		UAE National Day
December	4-11	Week 15	

December	11		Last Day of Classes for Undergraduate Programs
December	12-22	Week 16	Final Examination Period for Undergraduate Programs
December	15		Last Day of Classes for Graduate Programs
December	20-22	Week 16	Final Examination Period for Graduate Programs
December	24		Grades Available and Student Course Evaluation Results Available
December	23-Jan 12		Winter Break
January	6		Deadline for Continuing Students to Complete Registration for Spring Semester without Late Registration Fee
January	25		Graduation Date for Fall 2023
January	26		Deadline for grade appeal for courses completed in Fall Semester 2024

Spring 2025

Month	Day	Week	Activity/Event
January	6		Deadline for Continuing Students to Complete Registration for Spring Semester without Late Registration Fee
January	6-13		Continuing Advising & Registration for Returning Students
January	8		Placement Tests Begin
January	8		New Faculty Joining Date
January	8		New Student Orientation
January	9		Continuing Faculty Return Date
January	9		New Faculty Orientation
January	13		First day of Class
January	13-19	Week 1	
January	20-26	Week 2	
January	27		AURAK Admissions Application Deadline
January	27		Last day to Add/Drop
January	27-Feb 2	Week 3	
February	3-9	Week 4	
February	10		Census Date for Spring 2025
February	10-16	Week 5	
February	17-23	Week 6	
February	24-March 2	Week 7	
March	3-9	Week 8	
March	10		Mid-Term Grades Report Due
March	10-16	Week 9	
March	17		The last date to withdraw from a course with a grade of "W"
March	17-23	Week 10	
March	24-30	Week 11	

March	29-April 1		Eid Al-Fitr*
March	31-April 6	Week 12	
April	7-13		Spring Break
April	14		Classes Resume
April	14		Fall 2026 Advisement Begins
April	14-20	Week 13	
April	21		Summer 2025 Registration – Early Registration for Fall 2025 begins
April	21-27	Week 14	
April	21-May 10		Student Course Evaluations
April	28-May 4	Week 15	
May	2		Last Day of Classes for Undergraduate Programs
May	3-10		Final Examination Period Undergraduate Programs
May	4		Last Day of Classes for Graduate Programs
May	10-11	Week 16	Final Examination Period Graduate Programs
May	15		Grades and Student Course Evaluation Results Available
May	15		Graduation Date for Spring 2025
May	23		Faculty Summer Break Begins
June	5		Deadline for Grade Appeal for courses completed in Spring 2025

Summer Session 2025

Summer Session Regular Courses (6 weeks)			
May	12-19		Course Registration
May	19		Classes Begin
May	19-25	Week 1	
May	26		Last day to Add/Drop
May	26- June 1	Week 2	
May	28		Census Date for Summer 2025
June	2-8	Week 3	
June	5-8		Eid Al Adha*
June	9-15	Week 4	
June	16		The last date to withdraw from a course with a grade of "W"
June	16-22	Week 5	

June	16-29		Student Course Evaluations
June	23-29	Week 6	
June	26		Islamic New Year*
June	27		Last Day of Summer Session for Undergraduate Programs
June	29		Last Day of the Summer Session for graduate programs
July	3		Grades & Student Course Evaluation Results Available
September	9		Graduation Date for Summer 2025
September	22		Deadline for Grade Appeal for courses completed in Summer 2025
Summer Session Internship Courses (8 weeks)			
May	12-19		Internship Course Registration
May	19		Internship Classes Begin
May	19-25	Week 1	
May	26		Last day to Add/Drop
May	26- June 1	Week 2	
May	28		Census Date for Summer 2025
June	2-8	Week 3	
June	5-8		Eid Al Adha*
June	9-15	Week 4	
June	16		The last date to withdraw from a course with grade of "W"
June	16-22	Week 5	
June	16- July 11		Student Course Evaluations
June	23-29	Week 6	
June	26		Islamic New Year*
June	30-July 6	Week 7	
July	7-13	Week 8	
July	11		Internship Ends
July	15		Grades Available and Student Course Evaluation Results Available
September	9		Graduation Date for Internship Session 2025
September	22		Deadline for Grade Appeal for Internship courses completed in Summer 2025

Note: All Islamic Lunar Holidays will be fixed per announcement by the relevant Ministry.

ABOUT AMERICAN UNIVERSITY OF RAS AL KHAIMAH

Licensure and Accreditation:

The American University of Ras Al Khaimah (AURAK), located in the Emirate of Ras Al Khaimah, is officially licensed from 1 August 2009 to 15 September 2026 by the Ministry of Education of the United Arab Emirates to award degrees/qualifications in higher education.

AURAK is accredited by the Southern Association of Colleges and Schools Commission on Colleges (SACSCOC) to award baccalaureate and masters, degrees. Questions about the accreditation of AURAK may be directed in writing to the Southern Association of Colleges and Schools Commission on Colleges at 1866 Southern Lane, Decatur, GA 30033-4097, by calling (404) 679-4500, or by using information available on SACSCOC's website (www.sacscoc.org).

AURAK has earned accreditation through the Commission for Academic Accreditation of the Ministry of Education for all programs offered. In addition, the School of Engineering and Computing has the following program accreditations:

BS in Civil and Infrastructure Engineering is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org>

BS in Computer Engineering is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org>

BS in Computer Sciences is accredited by the Computing Accreditation Commission of ABET, <https://www.abet.org>

BS in Electronics and Communications Engineering is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org>

BS in Industrial Engineering is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org>

BS in Mechanical Engineering is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org>

BS in Chemical Engineering is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org>

BS in Petroleum Engineering is accredited by the Engineering Accreditation Commission of ABET, <https://www.abet.org>

The BS in Biotechnology degree in the School of Arts and Sciences is accredited by the Applied and Natural Science Accreditation Commission (ANSAC) of ABET, <https://www.abet.org/>.

AURAK's School of Business is accredited by AACSB International—The Association to Advance Collegiate Schools of Business, <https://www.aacsb.edu/accredited>

Location and Contact Information:

AURAK maintains an extensive campus located on American University of Ras Al Khaimah Road in the northern Emirate of Ras Al Khaimah. The university may be contact by mail at PO Box 10021; Ras Al Khaimah, UAE or by phone at +971 7 246 8888.

History:

With the support of His Highness Sheikh Saud bin Saqr Al Qasimi, Crown Prince of Ras Al Khaimah, the university was initially established as George Mason University, Ras Al Khaimah campus (GMU-RAK) in 2005. After George Mason University decided to discontinue its operations in Ras Al Khaimah in 2009, the Crown Prince decreed the establishment of the American University of Ras al Khaimah (AURAK) in the same location. Students who had enrolled in GMU-RAK were given the opportunity to continue at AURAK, transfer to GMU's main campus in Virginia, USA, or transfer to another institution.

Following the short-term presidencies of Professor Sharon Siverts (2009-10) and Dr. Shaukat Mirza (2010-11), AURAK's Board of Trustees appointed Professor Hassan Hamdan Al Alkim as Vice Chancellor in August 2011. Subsequent changes in the organizational structure saw Professor Al Alkim named as President in 2014. In August, 2023, Professor Al Alkim retired from AURAK, and Dr. David A. Schmidt succeeded him as President.

Since its establishment, the university has continued to develop new academic programs, attract new faculty and staff, and significantly increase its student enrollment. This rapid change has required an adaptable organizational structure to accommodate the effective and efficient delivery of programs and services. Consequently, in 2013, Vice President positions were added resulting in the organization of units in three divisions under the President, Senior Vice President Academic Affairs and Student Success and Vice President of Operational Excellence and Financial Management.

Today, AURAK's strategy focuses on cultivating excellence within the existing curricular offerings while improving policies and procedures across the institution. Student enrollment is projected to grow over the next several years. It is also expected that, with an increasingly strong global economy, AURAK will capture the attention of educators, economists, business entrepreneurs, and other professionals who want to take advantage of the programs and services the institution offers, meeting national and regional workforce needs and earning the university regional and global recognition.

Vision:

AURAK is committed to becoming an institution committed to academic excellence that embraces life-long learning, research, and sustainability and prepares graduates for careers to shape a better world.

Mission:

The Mission of the American University of Ras Al Khaimah is to prepare future leaders through a student-centered learning experience, based on the American model of higher education and informed by research, community engagement, and the pursuit of academic excellence.

The American University of Ras Al Khaimah pursues this mission as an independent, public, state-owned, non-profit, coeducational institution.

Core Values – our guiding IDEAS:

The following values guide all aspects of AURAK's operation:

Integrity:

AURAK models integrity through honesty, transparency, and accountability.

Diversity, and Inclusiveness

AURAK builds a diverse, inclusive, and culturally authentic community of students, faculty, and staff that embraces respectful dialogue and understanding.

Excellence:

AURAK strives for excellence by setting high expectations for all units and all members of the community.

Academic Freedom:

AURAK cultivates an environment of critical thought and enquiry, tolerance for views and beliefs, and an open and respectful exchange of ideas.

Sustainability:

AURAK collaborates with local and global entities to improve human lives.

AURAK Rising – Strategic Plan 2024-2029

Strategic Themes – Goals:

AURAK's strategic plan focusses on STEPS to achievement. The following goals guide all aspects of AURAK's operation.

- **S**tudent-Centered Environment: AURAK provides a student-centered experience focused on student success.
- **T**ransformational Education: AURAK employs high-impact educational practices to promote active learning, intellectual inquiry, and development of life skills.
- **E**ngagement of the Community: AURAK implements a comprehensive community engagement strategy that provides service to the community and benefits students.
- **P**ioneering Research and Scholarship: AURAK supports research and scholarship that addresses issues of significance to RAK, the region, and the world.
- **S**ustainability: AURAK will promote sustainability in all university activities including the curriculum, student life programming, administrative functions, and campus facilities management.

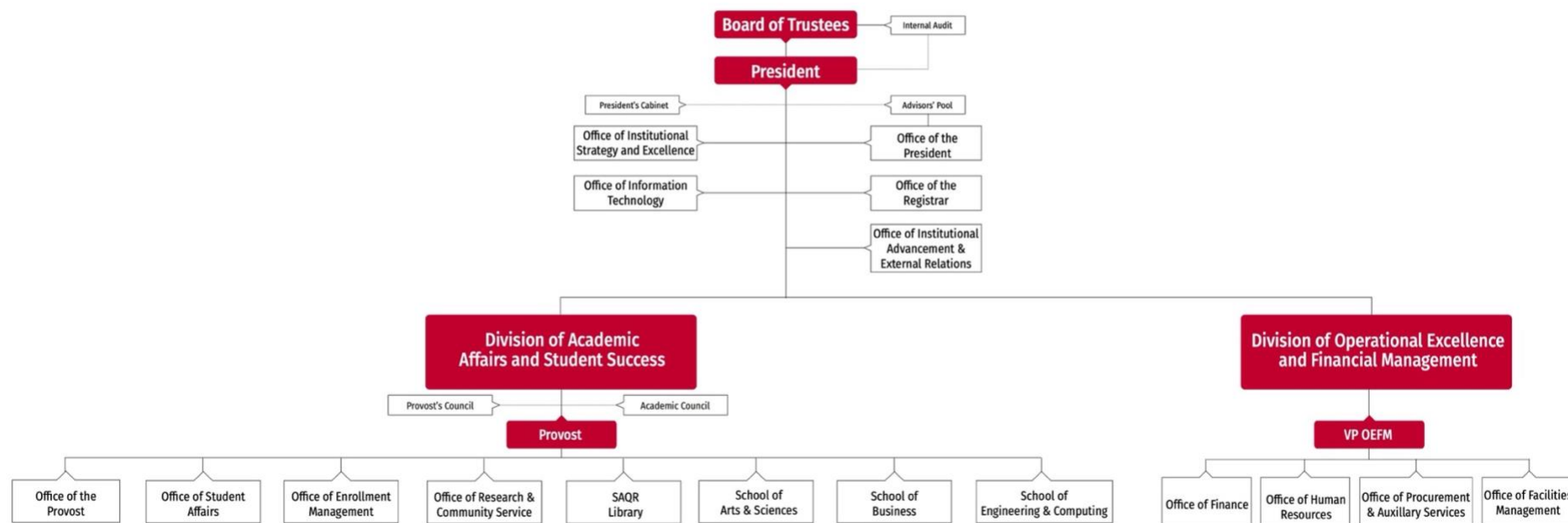
AURAK Organizational Chart

AMERICAN UNIVERSITY
OF RAS AL KHAIMAH

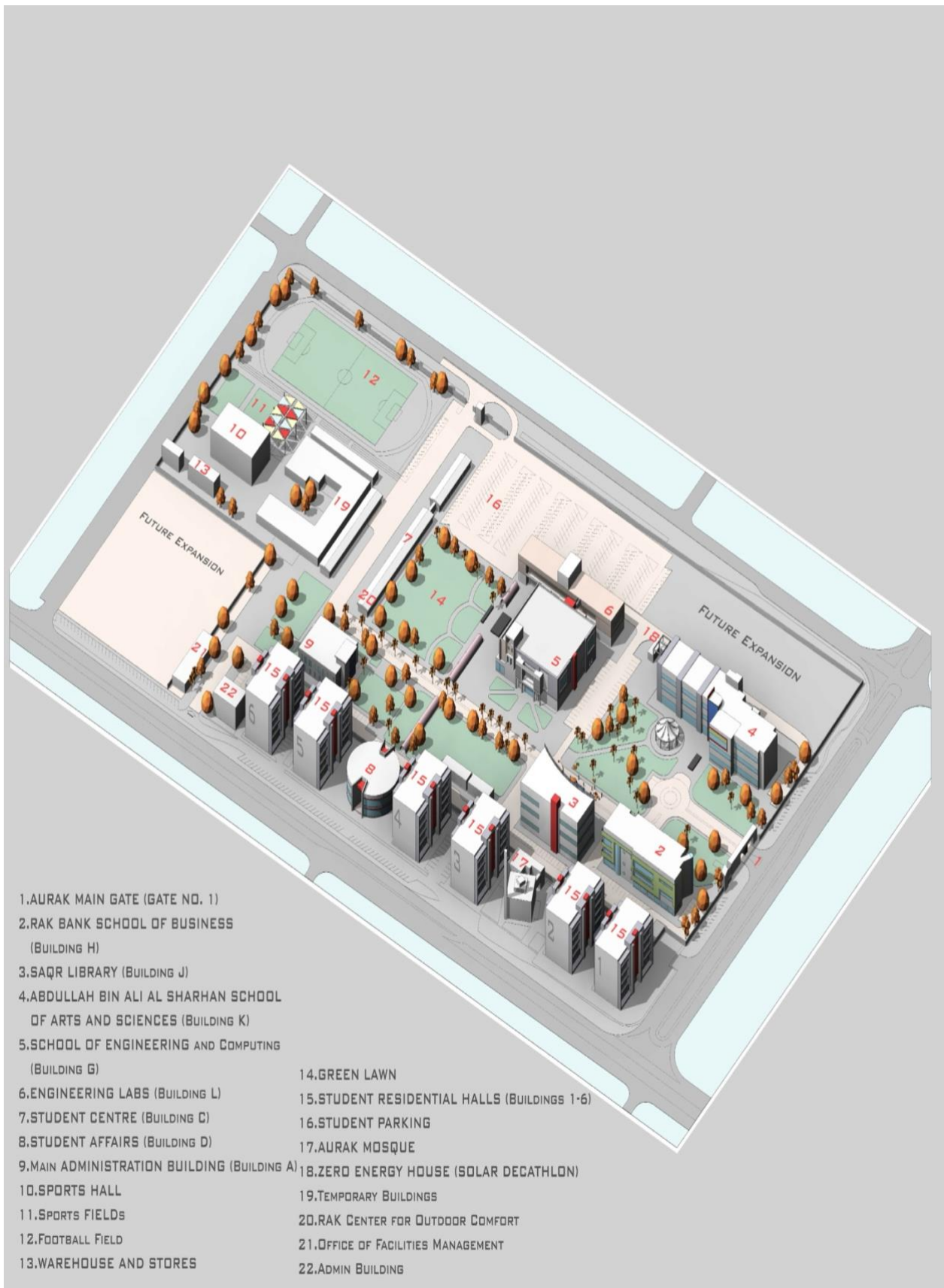


الجامعة الأمريكية
في رأس الخيمة

ORGANIZATIONAL CHART



AURAK CAMPUS MAP



ACADEMIC PARTNER INSTITUTIONS

Appalachian State University, USA

Aspen Institute (Stevens Initiatives), USA

Coastal Carolina University, USA

Hong Kong University of Science and Technology, Hongkong

Incheon National University – INU, South Korea

Majan Universty College (MUC), Oman

Narxoz University, Republic of Kazakhstan

RAK Medical & Health Sciences University, RAK, UAE

Temple University, USA

University of Texas Arlington, USA

University at Albany, State University of New York, USA

Universiti Teknologi Petronas, Malaysia

University of Wyoming, USA

Wayne State University (WSU), USA

Windesheim University of Applied Sciences, Netherlands



STUDENT RESOURCES AND SUPPORT SERVICES

STUDENT RESOURCES AND SUPPORT SERVICES

AURAK provides a range of student resources and support services. The following narratives outline seven major areas of delivery: 1) university student resources and services, 2) the Saqr Library, 3) the Office of Registrar, 4) the Office of Student Affairs, 5) the Center for Learning Support Services and e-Learning Programs, 6) the Alumni and Career Services Department, and 7) the Office of Research and Community Service.

UNIVERSITY STUDENT RESOURCES AND SERVICES

Computer Facilities and Resources

AURAK provides computing resources to support the needs of the students and faculty with their academic work. This includes Wi-Fi coverage throughout the campus, audio/video system in classrooms/laboratories equipped with overhead projectors or interactive touch panels, and a dedicated email account for each student, faculty, and staff member.

Laboratories

- ***Engineering Laboratories***

AURAK has well equipped labs aligned with international standards for all of their programs. These include: Architecture Digital Studios Labs, Surveying Laboratory, Construction Materials Laboratory, Geotechnical Engineering Laboratory, Fluid and Hydraulic Laboratory, Highway Engineering Laboratory, Computer Architecture and Microprocessors Laboratory, Computer Networking Laboratory, Computer Programming Laboratory, Embedded Systems Laboratory, Space Lab, AI Lab, Electrical/Electronics/Communications Labs, Thermal Sciences Laboratory, Manufacturing Process and Material Laboratory, Fluid & Hydraulics Laboratory, Chemical Engineering Lab that includes Reaction Lab, Unit Operation Lab, and Characterization Lab, Petroleum Engineering Lab that includes Samples Preparation Lab, Drilling Engineering Lab, Flow in Porous Media Lab, Reservoir Rock and Fluid Properties Lab, and a Computer Applications Lab. A new Engineering Laboratory Building to house all Engineering laboratories is under construction and is slated to open during the 2023-2024 academic year.

- ***Science Laboratories***

To support the laboratory-based modules of the Biotechnology Program, there are dedicated biotechnology, chemistry, and physics laboratories. In these labs, student learning is facilitated through practical exercises. The biotechnology, chemistry, and physics laboratories are fully equipped with standard and specialized instrumentation, including: microscopes, PCR and Real-Time PCR (qRT-PCR), Denovix spectrophotometer, Gel Electrophoresis and Western Blot apparatus, ELISA plate reader, High Performance Liquid Chromatography, cell culture incubator, laminar hood, freezers, liquid nitrogen tank, automatic cell counter, UV-Vis Spectrophotometers, pH meter, melting point apparatus, FTIR, Gas Chromatograph, structure modelling software for performing various experiments, PASCO software, free fall apparatus, force tables, momentum tracks, and rotational motion devices.

- ***Mass Communication Laboratory***

The Mass Communication Program has a Computer Lab with 15 Apple computers installed with software that meet the producing and editing needs for photos and videos. The software includes Full Adobe Suite and Final Cut Pro. The Mass Communication Program has also a Radio and TV studio facility including HD studio camcorders, Cinema HD Display 23-inch Widescreen, studio grade multi-

purpose tripods and studio sound monitors, and electronic news-gathering equipment including 4K Field video camera and sound recording gear.

Mosque (Masjid) and Multi-faith Room

The mosque, located on campus between the Cordoba and Granada Residence Halls (also known as Buildings 2 & 3) is open for prayers at all times. There is a separate female prayer room within the mosque. The campus also has a multi-faith room in Building H. This is an open, accepting space where all AURAK community members are invited to freely practice their various personal religions.

Parking

Parking services are offered to AURAK members on campus. Students will find information on “AURAK Parking Rules and Regulations” displayed on signs throughout the campus. Students must park only in the student designated parking, and they must display the appropriate parking permit on their vehicle. Permits can be obtained from the Office of Human Resources and Administrative Services.

Students are expected to drive carefully and follow parking rules on campus. This includes allowing pedestrians the right of way; not parking on fire lanes, curbs, special needs parking areas, or designated parking areas for university vehicles; parking within the white lines; and maintaining speeds between fifteen to twenty km/hour. Students who drive recklessly through the university parking lot and/or campus are subject to disciplinary action.

Photocopy Facilities

A coin-operated photocopier is available in the library. When using a photocopier, students are reminded that AURAK campus adheres to UAE copyright laws.

Security

The university provides twenty-four hours security services. Security guards monitor the entire campus premises including the student residence halls and all university buildings to ensure the safety and security of the campus community. CCTV cameras are installed in all campus buildings.

Student Center

The Student Center, located in Building C is the heart of the student community, serving students and the campus community by offering several ways to relax and socialize. The Student Center, open every day of the week from 8:00 am to 8:00 pm, provides wireless internet access, pool tables, table tennis, a foosball table, video games, and chess. If you would like more information, please contact the Department of Student Life located in Building D.

Food Services

Various restaurants and food serving facilities are available either on campus or immediately off campus. These services are available to ensure the health and wellbeing of AURAK’s student population.

THE SAQR LIBRARY

The spacious 3-floor new library building with 35,000 sq. ft. facility comprises the library stacks, Reading Areas, Individual and Group Study Rooms, Staff Work Areas, a Computer Laboratory for library applications, Meeting Rooms, Training Room, Conference Rooms, Auditorium, Faculty Reserved Room, and Special Collection Room. AURAK’s SAQR Library plays a significant role in supporting student academic success. The library continuously upgrades information resources and services that support the instructional and

research activity of the university. The library collection consists of many different types of library materials and information sources, either in print or electronic formats.

The SAQR Library is a key participant in the academic development of the university and actively responds to the needs of the learning community. The spacious 5,350 sq. ft. facility provides room for the library stacks, places for reading, faculty, staff, and student work areas, and a computer laboratory for library applications. Additionally, the SAQR library has sixteen individual private study carrels, a group study room, and a conference room.

The SAQR Library facilitates access to information through the sharing of resources at national, regional, and international levels, providing circulation, reference, Internet, printing, scanning services, and online access catalog services. The hours of operation, posted on the AURAK website and at the library entrance, provide students with optimal time for study and research.

SAQR Library hours of operation provides generous opportunities for students to utilize services and resources. Hours of operation vary depending on the university academic schedule and may be found on the AURAK library webpage and at the library entrance.

In addition to textbooks and reserve collections, SAQR Library houses a wide range of information sources including reference collections, periodicals, multimedia presentations, and electronic databases.

Library Circulation Services

Borrowing Service

Borrowing privileges are provided to all registered students and currently employed faculty and staff. A university ID card is required to check-out circulating items. Students, faculty, and staff have the privilege of borrowing books based on the following policy:

Loan Period

- Students – ten items for three weeks
- Faculty – twenty items for one semester
- Staff – three items for three weeks

Renewals

Circulating items may be renewed online or at the Library Circulation Desk. The borrowers may renew items twice in succession unless the item has been requested by another user.

Fines

All borrowers pay overdue fines for late return of the borrowed items. Days on which the library is closed are not counted. The current overdue fine is:

AED 10.00 per day per item (circulating and reserve materials)

Lost and/or Damaged Items

Replacement fines for lost or damaged library materials are the actual price of the material.

Reference Services

Upon request, the library provides reference services for students, faculty, staff, and library visitors to use and identify proper resources for their work. Patrons are encouraged to contact the librarian on duty for help in using reference services and the print and online resources of books and journals.

Internet Service

Workstations within the library lab have internet access, as well as access to common productivity software applications. The library is serviced by wireless internet to facilitate the use of laptops. The wireless network will allow users to access both the library catalog, as well as, the electronic databases inside and outside the university campus.

Online Public Access Catalog (OPAC)

The library catalog contains all library holdings and serves as a search guide for easy retrieval of library materials. The Online Public Access Catalog allows users to browse all existing library resources of the university online. The catalog can be accessed through the library webpage within the university homepage.

Printing and Scanning Services

Students can use the networked printer located in the library computer lab through their accounts (IDs and Password). The library also provides a scanner and a coin-operated self-service photocopier.

Library General Rules and Regulations

The Saqr Library exists to serve students. Accordingly, some rules are in place to ensure that all students are provided a positive learning environment. The library usage rules are as follows:

- Silence should be maintained in the library.
- An identification card must be presented for borrowing of library materials.
- Eating, drinking, and other activities that distract other readers are strictly prohibited.
- Online chatting and playing games are not allowed in the library computer laboratory.
- The use of mobile phones inside the library is strictly prohibited.
- Patrons should not leave valuables in the library. The library is not responsible for the loss of personal items or possessions.
- Patrons should not reshelved used library materials.
- Patrons should handle all library materials with care.
- Patrons should keep the library clean.

THE OFFICE OF THE REGISTRAR

The Office of the Registrar is committed to enhancing a culture of high academic performance and enhancing the student experience. The Registrar's team holds specialized expertise in supporting admissions and registration, maintaining student records, certifying degrees, and overseeing the university catalog. The Office of the Registrar is located in Building D.

THE OFFICE OF STUDENT AFFAIRS

The Office of Student Affairs (OSA) provides services and facilities necessary to promote the academic, professional, and social well-being of AURAK students and encourages students to become actively involved in campus life. Out-of-classroom learning experiences play a significant role in students' professional and personal development by enhancing, their social, leadership, and organizational skills. OAS offers comprehensive programs, orientation, activities, workshops, and initiatives that revolve around student needs, requests, and suggestions. OSA, in cooperation with the Student Government Association (SGA), organizes a wide range of activities for students throughout the year. These activities are carefully planned

with the purpose of exposing students to a diverse variety of cultural, social, and educational experiences. In addition to their entertainment value, these activities enhance students' creativity, collegiate, humanitarian, and social skills.

There are a number of traditional activities that OSA conducts on an annual or semester basis. These include the UAE National Day Celebration, Global Day, annual IFTAR meal, start and end of semester celebrations, and academic and student recognition award ceremonies. These activities are supplemented by cultural events – museum and theatre trips, art exhibitions, hiking trips, dhow cruises, movie nights, and beach barbecues. AURAK's very active sports program completes the comprehensive repertoire of student activities. Tournaments and leagues are organized in most sports for both men and women.

Academic Recovery, Disability Support, and Counseling Services

Academic Recovery, Disability Support, and Counseling Services assist students in both academic and personal success. Students may request the free services from the point of entry into the university through graduation. Staff are available to provide one-on-one attention to assist in meeting students' unique needs and challenges. Services include individual counseling, the loan of self-help DVD's, workshops, educational abilities testing, long-term and short-term disability services, and assisting students who are experiencing challenges both personally and academically.

AURAK provides personal and academic counseling services for individual students requiring assistance with a variety of issues. All counseling services are free and are provided to students enrolled in three or more credit hours. These services include counseling, loan of self-help DVDs, workshops, and, when necessary, referral services to qualified professionals and organizations. The service provides students the opportunity to discuss matters causing them concern or stress, with the objective of providing assistance to and support for the student so that the student can reach his or her highest potential, personally and academically. All services will be provided in a confidential, private setting.

Students with verified disabilities may request support through Disability Support Services in the Office of Student Affairs. The university will support students with special needs where resources are available. Requests are confidential and will be handled with complete discretion. The Office of Student Affairs will verify disabilities and determine the needed accommodations to provide equal access to educational facilities, programs, services, and activities. With the student's written consent, the departmental staff will communicate with appropriate faculty, staff, and other individuals regarding needed accommodations.

Office of Enrollment Management/Admissions Department

The Office of Enrollment Management/Admissions Department team leads a comprehensive, institution-wide, research-driven, systemic process committed to providing educational access for the local, regional, and international community. Admissions supports AURAK's Strategic Plan and aims to place AURAK as the first institution of choice for the pursuit of quality post-secondary education. The purpose of the Office of Enrollment Management is to achieve the university's desired enrollment objectives, engage top-quality student applicants, and guide the delivery of a superior student experience to ensure student success and satisfaction with the University. Accordingly, the admissions team works with marketing, recruitment, and retention of UAE nationals, UAE expatriates, and international students. Admissions staff also provide important information resources for students, providing them with an introduction, clarification, and guidance to the university environment, policies, and procedures. The Office of Enrollment Management is located in Building D.

Scholarships and Financial Aid Department

The Scholarships and Financial Aid Department exists to dismantle those financial obstacles standing between students and their dreams. Every student is important to us. The University actively recruits the quality of student capable of raising the institution to a higher level. To facilitate the student's entry into the university, AURAK offers a host of scholarships. Students are encouraged to explore the many options available to assist them in achieving their educational goals. For questions related to financial aid and scholarships visit the Admissions team in Building D to meet with an Admissions staff member.

Athletics

We believe that engaging in physical activity plays a significant role in promoting a healthy physical condition, self-confidence, a high quality of life, and serving as a focal point for more opportunities to sustain community interaction and building social networks.

AURAK offers athletic scholarships to qualified, outstanding student-athletes, subject to terms and conditions of the AURAK Athletic Scholarship.

Recreation and Sports Facilities

The University sports facilities include a multipurpose hall, a well-equipped gymnasium with workout machines, and several outdoor sports fields. The sports activities aim to develop and maintain physical fitness among all members of the University community by offering a variety of sports programs. Students participating in these sports activities have the opportunity to develop leadership and teamwork skills. Games, tournaments and friendly matches with other colleges/universities are regularly organized. There are also two gyms available for student use—one for men and one for women in the Sports Complex.

To know more and to participate in the sports activities contact the Student Life Department: student.life@aurak.ac.ae

Wellness Education and First Aid Station

AURAK contracts with a medical provider to deliver an on-campus Wellness Education and First Aid Station located in Building D on the first floor. It is open weekly from 9:00 am to 5:00 pm, Monday through Friday. Students requiring emergency assistance at other times should contact Campus Security.

Health Insurance

All students (undergraduate and graduate) enrolled at AURAK are required to have health insurance. International students, under sponsorship by AURAK, will automatically have the health insurance policy ordered for them and the annual expense of the insurance premium will be billed to their student account. UAE-based students who are covered by their parent's health insurance policy (or other comparable health insurance policy) are required to provide proof of valid insurance coverage by submitting details (declaration of coverage page and insurance card) in the EUMS student portal. The declaration page submitted must specify the extent of coverage and include the AED upper limit of coverage provided by the policy. This confirmation of health insurance coverage must be received by the last day of the class drop/add period each fall and spring semester. Refunds are not granted for proof of insurance that is submitted after the last scheduled day to drop/add classes each semester. The two required documents must be uploaded on the AURAK EUMS portal by accessing this [link](#).

By entering this information before the deadline date, each semester, UAE-based students will be exempted from the purchase of the University offered insurance policy. If a student does not enter proof of insurance coverage the University will automatically purchase health insurance on behalf of the student and the policy fee will be billed to the student's account. UAE nationals and citizens of other GCC countries

who are covered by the UAE Government Health Insurance policy are also exempted from the purchase of the University offered insurance policy.

For questions regarding Health Insurance, contact the Office of Student Affairs: studentaffairs@aurak.ac.ae.

Student Clubs & Student Life Activities

The Department of Student Life facilitates student growth through a variety of clubs. It supports activities to complement courses such as academic, civic, social, or recreational clubs for fun and leisure. Membership in these clubs not only helps students make new friends and meet people with common interests, but it also helps them to acquire important skills such as leadership, organization, teamwork, and event planning. All clubs work very closely with the SGA. All student clubs must be authorized and registered annually with the Department of Student Life. The university encourages all students to establish or join clubs to explore or further their interests.

Student Government Association (SGA)

The SGA represents the student body, provides an opportunity for students to demonstrate leadership, and serves as a liaison between the students and the university administration. The SGA also plays an important role in planning and sponsoring student activities and events. The representative of the SGA serves on the different committees in the university.

The SGA at AURAK follows a constitution, which defines SGA's goals and objectives and establishes the roles and responsibilities of each SGA member. The constitution also defines the policies and procedures to follow for planning and organizing university events. The SGA constitution also establishes the minimum qualifications for students who want to run for office. Further information about the SGA can be obtained by contacting the Department of Student Life.

International Learning Opportunities: International Student Exchange and Study Trips

The Office of Student Affairs also supports visiting study abroad and exchange students to our campus as well as assisting current AURAK students with plans to study abroad/exchange. The study abroad and exchange programs provide students rich experiences to encourage tolerance, acceptance, and discovery. Students participating in the study abroad or exchange programs have the opportunity to build character, adaptability, increase knowledge of international issues, and foster multicultural understanding needed to successfully navigate an ever-changing global society. The Office of Student Affairs also assists in broadening AURAK's outreach and world standing by assisting in the development of university linkages and partnerships that enhance the academic and study abroad/exchange experience for the campus community.

CENTER FOR LEARNING SUPPORT SERVICES AND e-LEARNING PROGRAMS

AURAK's Center for Learning Support Services and e-Learning Programs operates to ensure that every AURAK student is provided the opportunity to maximize his or her academic success. Specifically, the Center seeks to help students achieve their academic goals by: 1) overseeing the Math and Science Center, the Writing Center and Peer Tutoring Center, 2) supporting the first-year experience, and 3) guiding the improvement of AURAK's advising and educational experiences.

The Center for Learning Support Services and e-Learning Programs facilitates the application of the latest pedagogical theories and research to enhance knowledge instruction and acquisition at AURAK for all students, faculty, and staff. The center strives to identify and respond to the dynamic nature of student needs and learning environments. Students who request academic assistance are provided free tutoring by qualified peer tutors in various subject areas.

The Center for Learning Support Services and e-arning Programs, located in Building D, provides qualified peer tutors to aid students in a wide array of subjects including English, mathematics, biology, chemistry, physics, and a variety of computer courses. Tutoring services provided are at no cost to the students. All students are strongly encouraged to take advantage of this key academic support service. Students interested in tutoring are asked to visit the Center or contact the Director of this Center.

ALUMNI AND CAREER SERVICES DEPARTMENT (ACSD)

AURAK is committed to providing students with a wide range of career services. Students visit with a career counselor who can assist in self-awareness exploration (values, skills, abilities, and personal style), career exploration (job descriptions, educational requirements, salary range, and advancement possibilities), and employment and employability skills (writing cover letters, curriculum vitas/resume, interviewing skills, and employee decorum). Students access job listings, internship opportunities, and career fairs.

AURAK celebrates the success of its distinguished alumni and encourages alumni to stay in contact with the University. AURAK's alumni are also encouraged to utilize the University's student support services and the Saqr Library.

OFFICE OF RESEARCH AND COMMUNITY SERVICE

The Office of Research and Community Service promotes research and community engagement activities among AURAK's faculty and students. The Office also supervises AURAK's research-oriented programs and makes available small seed grants to faculty on a competitive basis. These seed grants often support students engaged in research under a research assistant contract. The Office promotes a culture of research and scholarship. Students are encouraged to attend sponsored events in order to broaden their knowledge and gain awareness of the different research activities of the AURAK faculty.

The Office of Research and Community Service manages the Ras Al Khaimah Research Innovation Center (RAKRIC), the Center of Executive and Professional Education (CEPE), the Community and Business Development Department, and the RAK Center for Outdoor Comfort. RAKRIC is a state-of-the-art research and development center specializing in sustainable system solution development related to solar technologies. RAKRIC accommodates several senior research projects of AURAK's students under the supervision of the faculty and with the assistance of the RAKRIC staff. CEPE provides professional training courses and English language training/testing including the EmSAT, TOEFL, and IELTS tests. The Community and Business Development Center provides leadership in implementing the university's Community Engagement Strategy, and the RAK Center for Outdoor Comfort focuses on improving outdoor comfort in hot and humid climates.

COST OF ATTENDANCE – TUITION & FEES

TUITION FEES for Fall 2024 - Spring 2025

NOTE: AURAK reserves the right to increase tuition fees and other charges up to a maximum of 10% per annum.

Undergraduate Program	Less than 12 Credit Hours	12-16 Credit Hours	More than 16 Credit Hours	Payment Terms
All undergraduate programs except otherwise mentioned for specific programs	2,310 Per CH	27,900	1,815 Per CH	Per Semester
Architecture Engineering	2,310 Per CH	29,100	1,815 Per CH	Per Semester
Petroleum Engineering	2,625 Per CH	32,600	2,030 Per CH	Per Semester
Chemical Engineering	2,625 Per CH	32,600	2,030 Per CH	Per Semester
Artificial Intelligence	2,425 Per CH	29,300	1,905 Per CH	Per Semester
Biotechnology	2,425 Per CH	29,300	1,905 Per CH	Per Semester
Computer Engineering	2,425 Per CH	29,300	1,905 Per CH	Per Semester
Computer Science	2,425 Per CH	29,300	1,905 Per CH	Per Semester
Electrical and Electronics Engineering	2,425 Per CH	29,300	1,905 Per CH	Per Semester

Tuition fees for Summer 2025 shall apply on per credit hour basis as per below table

Undergraduate Programs – Tuition Fee (Summer)	Fee in AED
All undergraduate programs except where mentioned otherwise for specific programs	1,815 Per CH
Architecture Engineering	1,815 Per CH
Petroleum Engineering	2,030 Per CH
Chemical Engineering	2,030 Per CH
Artificial Intelligence	1,905 Per CH
Biotechnology	1,905 Per CH
Computer Engineering	1,905 Per CH
Computer Science	1,905 Per CH
Electrical and Electronics Engineering	1,905 Per CH

Undergraduate Program – Non Tuition Fees in AED	Fall/Spring	Summer	Additional Information
Application	400	400	Application Submission, Non-refundable
Admission	750	750	Issuance of Admission Letter, Non-refundable
Technology Fees	1,000	400	Per semester
Service Fees	500	500	Per Semester
Health Insurance	Actual Price	Actual Price	Subject to Market Price
Transportation	Free	Free	N/A
Medical Services	100	100	Per Semester

Other Fees – Undergraduate & Graduate – in AED	Amount	Additional Information
Visa Fees	2,000	Non-refundable, due at the time of admission
Visa Change Status	900	Non-refundable, due at the time of admission
Visa Security Deposit	2,000	Due at the time of admission, Refundable after Visa cancellation
Security Deposit	1,500	One time – Refundable – if no damages
Graduation Fee	1,500	Non-refundable
Late Payment (if tuition and fees are not settled by due date)	500	Applicable to first payment, if not paid by due date
Reinstatement fee	1,500	To reinstate access after block due to non-payment
Cheque Returned Fee	500	Per Returned Cheque
Transcript – Official	100	Per Transcript
Transcript – True	50	Per Transcript
ID Card Replacement Fee	100	Per Instance
Official Letter Fee	50	Per letter
Gymnasium	500	Per Month – Non-refundable
Library Books overdue Fine	10	Per day, Per item
Library Books Lost/Damage Fine	Actual Cost	Per item

Tuition and Fees for graduate studies apply as per the below tables

NOTE: AURAK reserves the right to increase tuition fees and other charges up to a maximum of 10% per annum.

Graduate Programs – Tuition Fees		
Program	Amount (AED)	Payment Terms
Master of Business Administration (MBA – 36 total credit hours)	2,520	Per Credit Hour
Executive Master of Business Administration (EMBA -36 total credit hours)	3,150	Per Credit Hour
Master of Education in Educational Leadership (33 total credit hours)	2,100	Per Credit Hour
Master of Engineering Project Management (MEPM 36 total credit hours)	2,730	Per Credit Hour
Master in Sustainable and Renewable Energy (36 total credit hours)	3,000	Per Credit Hour

Graduate Program – Non Tuition Fees	Fall/Spring Fees in AED	Summer Fees in AED	Due by:
Application	500	500	Application Submission, Non-refundable
Admission	2,000	2,000	Issuance of Admission Letter, Non-refundable
Technology	750	400	Per Semester Non-Refundable
Health Insurance	Actual Cost	Actual Cost	Subject to Market Price
Transportation	Free	Free	N/A
Graduation Fee	1,500	1,500	One Time, Non-Refundable
Medical Service	100	100	Per Semester

The Housing Accommodations for both undergraduate and graduate students are outlined in the following table.

Housing and Accommodation			
Accommodation Type (Amount in AED)	Fall 2024 & Spring 2025	Summer 2025	Payment Terms
Housing (Large Room) Single Occupancy	11,000	2,400	Per Semester
Housing (Small Room) Single Occupancy	10,000	2,200	Per Semester
Housing (Large Room) Shared Occupancy	6,500	1,600	Per Semester
Housing (Small Room) Shared Occupancy	5,500	1,400	Per Semester
Security Deposit	2,000	2,000	Refundable
VAT shall apply wherever applicable as per UAE Law and Federal Tax Authority (FTA) guidelines.			

Payment Policies and Procedures

Tuition and all other fees must be paid by the first day of classes unless necessary arrangements are made with the Office of Finance. Fees can be paid by cash, check, wire transfer, or credit card.

Students who do not pay their fees by the specified deadline will be put on financial hold. Students on financial hold are not allowed to register for classes or take final exams and will not be issued official

transcripts until all fees are paid.

Students are responsible for full tuition payment for all courses in which they are registered unless their registration is canceled for nonpayment; canceled administratively due to academic suspension, dismissal, or termination; or the course is dropped before the tuition liability begins.

Refund Policies

Tuition fees are refundable subject to the following rate for the Fall and Spring semesters:

Withdraw by	% of Refund
The end of the First Week of the semester	100%
The end of the Second Week of the semester	75%
The end of the Third Week of the semester	50%
The end of the Fourth Week of the semester	25%
Beyond the end of the Fourth Week of the semester	No Refund

Please check the university Academic Calendar for dates of Drop/Add.

Tuition fees are refundable subject to the following rate for the Summer terms:

Withdraw by	% of Refund
The end of the First Week of the term	50%
Beyond the end of the First Week of the term	No Refund

Payment by Installment

If full fees cannot be paid immediately, installments can be arranged with the Office of Finance.

Deferment of tuition and fees is approved only if the following conditions are met:

- The first installment shall be paid through cash, current dated cheque, credit card or bank/wire transfer and the remaining installments through post-dated cheques (PDCs).
- The remaining installments must be paid within the same semester.
- If any installment is not paid on a due date, the student shall be placed on financial hold.
- For Fall and spring, registration shall be approved only if the first installment of 25% or 50% is paid and PDCs are received for remaining installments.
- For Summer, installments shall not be allowed, and students must settle full payment before the start of classes.

Failure to attend a course (no show) does not result in a refund of charges. Students not planning to attend must drop their registration during drop/add period. Students reported as “no show” in class will be withdrawn from the class without a refund. These students shall be placed on a drop/delete hold and will not be able to register in future semesters unless the fees are paid in full in advance.

For detailed guide on payment deadlines please refer the University website payment guide section.

Payment and Plan Violations

Where the student does not meet a scheduled payment, the University will place the student on a financial hold and block the student's access to the portal. **This may also result in the student not being able to attend the exams.**

For any clarification, please visit the Office of Finance or send an email at: student.finance@aurak.ac.ae.

Financial Responsibilities

The following outline the financial responsibilities relating to attendance at AURAK.

- Students are responsible for maintaining current addresses with AURAK and for activating and checking their personal AURAK e-mail accounts.
- Registration shall not be considered complete unless all outstanding balances from the prior term are paid in full.
- By registering for classes, students accept responsibility for charges for the entire semester.
- Failure to receive a reminder bill confirming charges does not waive the requirement for payment when due.
- Students are responsible for dropping unwanted courses by the drop dates and using the drop and withdrawal procedures published in each term's Schedule of Classes. Full or partial tuition liability may apply.
- Non-returning students are responsible for submitting a written withdrawal to the Associate Provost of Academic Affairs. Penalties may apply.
- Students may not attend classes without completing the registration process.

Residence Halls

AURAK residence halls are located on campus. These residence halls are managed by the university's Student Residential Life Department under the supervision of the Office of Student Affairs. For information about AURAK's residence halls, please contact the Office of Student Affairs, studentaffairs@aurak.ac.ae

Issues Relating to AURAK Representatives

AURAK utilizes representatives to assist and recruit applicants from countries external to the UAE. These representatives are held to the same standards as all AURAK employees. Applicants should never pay money to any representative. All funds are to be paid to the Office of Finance for which official university receipts will be issued. **Should an issue ever arise in dealing with any AURAK representative, concern should be addressed with the Office of Student Affairs for investigation and resolution as appropriate.**

SCHOLARSHIPS AND FINANCIAL AID

AURAK is committed to providing incentives in the form of scholarships and financial grants to attract and retain high academic achievers. In addition to meeting the expectations set for each scholarship/grant, students must meet other conditions set including, at a minimum, meeting all requirements for admission to the program of interest and being enrolled as a full-time student. To this end, AURAK will establish a range of scholarships and financial grants for new and continuing students. The eligibility criteria for such awards will be benchmarked against other institutions to ensure that the scholarships and financial grants will be useful in attracting and retaining students who might otherwise choose to enroll at other institutions. These criteria will also be crafted to promote the recruitment and retention of students who

will contribute positively to the academic reputation of the university and who will bring talents, skills, and diversity to the campus that contribute to achievement of the university's strategic objectives.

Categories of Scholarships and Financial Aid

The following institutional scholarships will be in effect for students joining AURAK beginning in AY 2023-2024 and onward. The awards will be determined by the following standards. All of the scholarships and grants described below will be awarded to eligible admitted students upon their signing a formal Agreement of Undertaking that will be subject to renewal for up to four years (five years if admitted to the Bachelor of Architecture), contingent on the students meeting the performance expectations for continuation of the scholarship or discount.

New Undergraduate Student Scholarships

New Undergraduate Student Scholarships will be awarded to full-time students beginning their undergraduate studies at AURAK in the following categories:

- **Scholastic achievement scholarships:**

Scholarship	High School Percentage	Scholarship Amount	Basis for Award	CGPA Requirement for Continuation of Scholarship in Subsequent Years
Presidential Scholarship	99% or Higher	70%	Competitive	3.75
Provost's Scholarship	97% to 98.9%	40%	Competitive	3.60
Dean's Scholarship	94% to 96.9%	30%	Competitive	3.40
Faculty Scholarship	90% or Higher	20%	Competitive	3.25

For all scholarships based on high school scholastic achievement, the high school average to qualify will be adapted to be equivalent for different curricula with different grading system.

Subject to the CGPA requirements shown in the table above, the Scholastic Achievement Scholarships will be renewed annually up to the number of years specified for the completion of a program of study as long as recipients abide by the Academic Integrity Policy and the Student Code of Conduct.

- **First-year recruitment grant:**

- AURAK Recruitment Grant – For first-year students not eligible for a scholarship, a 15% reduction in tuition and associated instruction-related fees may be available for the first semester of study. To retain the grant for the second semester, students must have a CGPA of 2.75 or higher at the end of the first semester.

- **Other scholarships:**

- Sibling Scholarships – for students admitted who have a sibling currently enrolled at AURAK – 20% scholarship.
- Alumni Scholarships – for AURAK alumni admitted to pursue an additional graduate degree – 20% scholarship.
- Partnership Scholarships – for students admitted who are employees of entities that have been designated as "AURAK Scholarship Partners" – 20% scholarship.

- Scholarships for employees and their relatives in accordance with the relevant remuneration and benefits policy.

For continuation of these “other” scholarships in subsequent years, scholarship recipients must be in good academic standing and must abide by the Academic Integrity Policy and the Student Code of Conduct.

- **Need-Based Financial Grants:**

- For students who can demonstrate that their family’s financial resources are not sufficient to cover tuition and fees at AURAK. Recipients of Need-Based Financial Grants who remain unable to pay the full cost of tuition and fees at the time of registration for subsequent academic years and who are not eligible for one of the other scholarships available must re-apply for a Need-based Financial Grant. The availability of Need-based Financial Grants cannot be guaranteed beyond the initial award of a Need-based Financial Grant. A Need-Based Financial Grant can be combined with a scholastic achievement scholarship or a talent/skill-based scholarship if the scholarship recipient demonstrates that the family is unable to pay the tuition and fees not covered by the scholarship.

Undergraduate Continuing Student Scholarships

Students completing their first or second year of full-time study at AURAK who were not previously awarded one of the following scholarships may apply for the following scholarships or a Needs-based Financial Grant subject to meeting the eligibility criteria specified:

- Presidential Scholarship – the student has a CGPA of 3.75 or higher and who has appeared at least once on the President’s List for Academic Excellence.
- Provosts’ Scholarship – the student has a CGPA of 3.60 to 3.7449 and who has appeared at least once on the Dean’s List for Academic Excellence.
- Dean’s Scholarship – the student has a CGPA of 3.40 to 3.599 and appeared at least once on the Dean’s List for Academic Excellence.
- Faculty Scholarship – the student has a CGPA of 3.25 to 3.399.
- Need-based Financial Grants – the student is in good academic standing and can demonstrate that the family’s financial resources are not sufficient to cover tuition and fees at AURAK. A Need-Based Financial Grant can be combined with a scholastic achievement scholarship if the scholarship recipient demonstrates that the family is unable to pay the tuition and fees not covered by the scholarship.

In order to retain the scholarship the following criteria will be applied:

- Presidential Scholarship recipients must have a CGPA of 3.75 or higher at the end of each year of study following award of the scholarship.
- Provost’s Scholarship recipients must have a CGPA of 3.60 to 3.7449 at the end of each year of study following award of the scholarship.
- Dean’s Scholarship recipients must have a CGPA of 3.40 to 3.599 at the end of each year of study following award of the scholarship.
- Faculty Scholarship recipients must have CGPA of 3.25 to 3.399 at the end of each year of study following award of the scholarship.

- Recipients of Need-Based Financial Grants who remain unable to pay the full cost of tuition and fees at the time of registration for subsequent academic years must re-apply for a Need-based Financial Grant. The availability of Need-based Financial Grants cannot be guaranteed beyond the initial award of a Need-based Financial Grant.

New Graduate Student Scholarships

Scholarships will be awarded to students beginning their graduate studies at AURAK in the following categories:

- Scholastic Achievement Scholarship
 - Graduate Study Scholarship (20%) – for students with an undergraduate CGPA of 3.25 or above.
- Other scholarships:
 - Alumni Scholarship (20%) – for AURAK alumni admitted to pursue an additional graduate degree.

In order to retain a Graduate Study Scholarship or Alumni Scholarship recipients must have a CGPA of 3.50 or higher at the end of the year of study following the award of the scholarship.

External Scholarships

External scholarships are available for UAE Nationals from various entities such as the Sheikh Saqr Program for Government Excellence (SSPGE) and the Telecommunications Regulatory Authority (TRA). Other external scholarships may be available for students of all nationalities. All external scholarships are subject to the selection criteria of the indicated entities.

Obtaining Additional Information Concerning Scholarships and Financial Aid

Information relative to the process for applying for Scholarships and Financial Aid can be obtained by contacting the Office of Enrollment Management.

STUDENT RIGHTS AND RESPONSIBILITIES

AURAK adheres to the following values that underlie the philosophy and goals of the university:

- The importance of personal integrity.
- The right of every individual to be treated with respect and dignity.
- The freedom of intellectual inquiry in the pursuit of truth.
- The freedom of speech and the open exchange of ideas.
- The acceptance and appreciation of diversity with regards to race, gender, religion, age, disability, and ethnicity.
- The recognition that community service is an important component of the intellectual development of students.

STUDENT RIGHTS

AURAK students have the right to pursue their education in a free and fair manner so long as they maintain their eligibility to remain as a member of the student community by meeting its academic and behavioral standards. It is the responsibility of students to observe the regulations imposed by the university for the governance of the academic community.

- Students have a right to expect AURAK to maintain a climate conducive to thinking and learning. University teaching should reflect consideration for the dignity of students and their rights as persons. Students have a right to be treated with courtesy and respect.
- Students have a right to reasonable notice of the general content of the course, what will be required of them, and the criteria upon which their performance will be evaluated. Students have a right to have their performance evaluated promptly, conscientiously, without prejudice or favoritism, and consistently by the criteria stated at the beginning of the course.
- Students have a right to perform student evaluations considered in the retention and promotion reviews of faculty members.
- Students have a right to be free from discrimination and sexual harassment. Campus policy prohibits discrimination and harassment of a student because of his/her race, color, religion, national origin, sex, age, or status as an individual with a disability.
- Students have a right to privacy and confidentiality subject to reasonable campus rules and regulations, right to protect against unauthorized disclosures of confidential information contained in their educational records. Students have a right to examine the information contained in their educational records.
- Students have a right to form student organizations for any lawful purpose consistent with the goals and mission of AURAK.
- Students have the right to excellent faculty, academic technology, classrooms, libraries, presentations and other resources necessary for the learning process.

Student Responsibilities

AURAK students are required to abide by the expectations within the Code of Student Conduct, maintain current contact information with the University, including permanent and local addresses, telephone numbers, student ID number, and legal name (as written in their passport). Each student must also maintain the University e-mail account assigned at the time of admission. Students are responsible for official communications directed to AURAK e-mail accounts. Students are responsible for providing a copy of their University ID when requested by AURAK administration, faculty staff members and/or security guards and to comply with directives of a university official.

Student Academic Grievance Policy and Procedures

Academic grievance include grievance students may have against a faculty member, procedures implemented in class, or related academic issues. All students have the right to express their dissatisfaction. If the student has a disagreement with the faculty, grades, or the classroom the grievance is a Student Academic Grievance. To appeal a grade, the Grade Appeal Form, obtained from the Office of the Registrar and filled out by the student is the most expeditious and efficient way to address a grade appeal with a professor. The following is the process for a student academic grievance.

If any student wishes to file a formal grievance, these steps apply:

1. The grievant must file a formal written complaint with the Associate Provost for Student Affairs.
2. The Associate Provost for Student Affairs will ascertain if the complaint falls within the scope of an action appropriate for a grievance and will attempt to resolve the complaint informally before instituting the official grievance process.
3. If such action is deemed appropriate, the Associate Provost for Student Affairs will convene a

committee of a minimum of three people including faculty, administration, and student representative(s) within three working days.

4. The committee will review the complaint, interview the complainant and any witnesses cited in the complaint.
5. The Committee will recommend a cause of action to the Associate Provost for Student Affairs within seven working days.
6. The Associate Provost for Student Affairs will forward to the student the recommendation of the Committee or a modification of its recommendation deemed appropriate after consultation with the committee within three working days.
7. If the student does not agree with the decision, the student has the right to file an appeal with the Provost. The Provost will review the decision and render a final decision in the matter within three working days. The Provost is the final authority to whom appeal may be made.
8. If a grievance is with the Provost, then the student may submit the grievance in writing to the President to initiate the process.

Student Non-Academic Grievance Policy and Procedures

All students have the right to express their dissatisfaction and misconduct issues are handled under the non-academic grievance policy and procedures. Incidents that occur outside the classroom or disagreements with the staff fall under the non-academic grievance policy. If a student wishes to file a grievance, these steps apply.

1. The grievant must file a formal written complaint with the Associate Provost for Student Affairs.
2. The Associate Provost for Student Affairs will ascertain if the complaint falls within the scope of an action appropriate for a grievance and will attempt to resolve the complaint informally before instituting the official grievance process.
3. If such action is deemed appropriate, the Associate Provost will convene a committee of a minimum of three people including faculty, administration, and student representative(s) within three working days.
4. The committee will review the complaint, interview the complainant and any witnesses cited in the complaint.
5. The committee will recommend a cause of action to the Associate Provost for Student Affairs within seven working days.
6. The Associate Provost for Student Provost will forward to the student the recommendation of the committee or a modification of its recommendation deemed appropriate after consultation with the committee within three working days.
7. If the student does not agree with the decision, the student has the right to file an appeal with the Provost. The Provost will review the decision and render a final decision in the matter within three working days. The Provost is the final authority to whom appeal may be made.
8. If a student's grievance is with the Associate Provost for Student Affairs, it may be submitted to the Provost to begin the process.
9. If a grievance is with the VPAASS or other Vice-President, the student may submit the grievance in writing to the President to begin the process.

Obtaining Additional Information

The Office of Student Affairs is available to answer any questions that may arise concerning AURAK's grievance policies and procedures. Please feel free to visit should you require additional information.

STUDENT RESPONSIBILITIES

AURAK students are expected to conduct themselves according to AURAK policy. The following policies are of particular interest in maintaining these expectations.

AURAK Communication Policy

AURAK students are required to maintain current contact information with the university, including permanent and local addresses, telephone numbers, student ID number, and legal name (as written in their passport). Each student must also maintain the university e-mail account assigned at the time of admission. Students are responsible for official communications directed to AURAK e-mail accounts.

Student Publication Policy

Any student publication at AURAK will be required to conform to the norms of responsible journalism and avoid libelous, indecent, or harassing material. The same publication policies will apply when deciding what material can be included on the website, or any other social media, developed and managed by the students. All the publications should particularly keep in mind the cultural norms of Ras al Khaimah and the UAE.

Students should get all articles, stories, or any other related materials reviewed by the editor to ensure the correctness of the English language. Students should take prior approval from the Associate Provost of Student Affairs before distributing, posting, or publishing any printed/online material in the name of AURAK.

Guidelines for Using AURAK Name, Logo, and Mark

Students may use the AURAK name in association with their university-sanctioned activities. Recognized student groups producing merchandise for sale that incorporates AURAK's name, marks, or logos must comply and obtain permission from the Associate Provost of Student Affairs. Students may use the name of a school, department, or other AURAK programs outside the university (other than on a resume) only with approval of the appropriate authorized personnel.

STUDENT CONDUCT

The American University of Ras Al Khaimah adopts the following Student Code of Conduct to ensure that students conduct themselves appropriately at all times as members of the University community. Students will:

- Attend classes regularly and punctually and complete assignments on time.
- Participate actively in class discussions and other course-related classroom activities.
- Be knowledgeable of and plan appropriately to meet the requirements for course and program completion.
- Adhere to the published test or examination rules and regulations.
- Demonstrate sensitivity to the cultural and religious norms and customs of the UAE and ensure that their conduct is in line with those cultural expectations.
- Display principles of integrity at all times, including those associated with academic integrity, ethics, and honesty.

- Treat others as they would themselves wish to be treated.
- Show proper respect to fellow students as well as to AURAK employees and representatives. This includes security personnel. Respectful behavior includes refraining from disruptive behavior such as:
 - Talking during speeches or classes when the speaker or instructor is talking
 - Arriving late for classes or leaving early
 - Receiving or initiating telephone calls on mobile phones during classes
 - Sending text messages during class
 - Making negative or rude comments during class about other students, faculty members, or the opinions of other students
- Have knowledge of and observe the regulations of AURAK.
- Use University property/facilities responsibly.
- Provide security guards with personal identification (ID card) and appropriate documentation when requested.
- Seek permission for using someone else's property, and refrain from using another person's name and/or ID number for any reason.
- Plan and organize all extra-curricular activities on the AURAK campus in collaboration with the Department of Student Life, obtaining all necessary permissions.
- Obtain permission from the Department of Student Life to organize events in the name of AURAK.
- Obtain permission from the Department of Student Life to distribute leaflets or journals, post notices, or collect signatures on the premises of AURAK.
- Use the internet responsibly, observing all applicable IT policies.
- Refrain from the possession, use, or sale of illegal substances or materials.
- Refrain from exchanging any gifts of significant monetary value with members of the faculty, staff, or administration.
- Refrain from carrying mobile phones or any other electronic devices inside classrooms in which examinations are being administered.

Student Behavioral Misconduct

The Associate Provost for Student Affairs is responsible for the administration of disciplinary procedures related to student behavioral misconduct. Reports of alleged student violations of university regulations may be submitted by any individual to the Associate Provost of Student Affairs. Allegations of violations of the AURAK Code of Conduct are accepted for investigation only when the apparent infractions are observed on AURAK property, at AURAK student events and activities off campus or at other locations at which AURAK provides services. Infractions of UAE or Ras Al Khaimah laws occurring off campus shall be the sole concern of the civil authorities, except when such actions: 1) directly affect the health, safety, or security of AURAK community, 2) affect AURAK's pursuit of its educational goals, or 3) occur as a direct result of an AURAK-related activity.

All student behavioral misconduct will, if possible, be resolved by the Associate Provost for Student Affairs through mediation or through verbal and written warning. For examples of behavioral misconduct, please see the *AURAK Student Handbook*.

The resolution of cases of behavioral misconduct by the Associate Provost may involve issuance of a verbal or written warning. However, the Associate Provost may judge that the conduct violation requires the student's probation, suspension or dismissal. In this situation, the case will be referred to the Disciplinary Committee, following any immediate action warranted by the Associate Provost. The following summarizes possible violations and disciplinary action to be taken by the Disciplinary Committee.

Student Behavioral Misconduct	Possible Disciplinary Action to be taken by the Disciplinary Committee
Inappropriate Student Conduct Level 1	Disciplinary Written Warning Disciplinary Probation
Inappropriate Student Conduct Level 2	Disciplinary Probation Dismissal
Damage to University Property	Orders by committee to pay actual repair or replacement costs – Failure to pay may lead to additional disciplinary actions and/or Registration Hold Transcript Hold Disciplinary Probation Dismissal

For all levels of behavioral misconduct the Disciplinary Committee may remove the privilege of participating in or representing the university in extracurricular activities, including athletics, as well as the privilege of running for or holding office in any student organization that is allowed to use university facilities or receives university funds, or other restrictions as determined by the Disciplinary Committee.

Academic Integrity Violations

Behavior that violates the Student Academic Integrity Policy is addressed through specific procedures associated with that policy.

Release of Student Course Evaluation Results

Each semester AURAK conducts course evaluations. Students are assured of anonymity under Board Policy, which does not allow release of the results until all grades are posted. Should any student feel that anonymity of the course evaluation has been violated, a grievance may be filed with the Associate Provost of Student Affairs.

POLICIES RELATING TO STUDENT CONDUCT AND SAFEGUARDS

Drug and Alcohol Policy

The AURAK campus adheres to the laws of the UAE concerning alcohol and drug use. Any student found using drugs will face automatic dismissal. Any student found drinking alcohol or having it in their possession on campus will be automatically dismissed.

Individuals with Disabilities Policy

The campus is committed to providing equal access to employment and educational opportunities for people of determination. AURAK recognizes that people of determination may need reasonable accommodations to have equally effective opportunities to participate in or benefit from campus

educational programs, services, and activities and to have equal employment opportunities. The campus will adhere to all applicable UAE federal and state laws, regulations, and guidelines with respect to providing reasonable accommodations as necessary to afford equal employment opportunity and equal access to programs for qualified people of determination. Students who seek reasonable accommodations for a disability should contact the Office of Student Affairs.

Non-discrimination Policy

AURAK's non-discrimination policy prohibits discrimination based on race, color, religion, national origin, disability, sexual orientation, gender, or age. AURAK is committed to providing faculty, staff, and students with an environment in which they may pursue their careers or studies free from discrimination.

Statement Relating to Sexual Assault

The following policy applies equally to all members of the AURAK community: students, faculty, administrators, staff, contract employees, and visitors.

AURAK and the AURAK campus is committed to providing an institutional environment where all people may pursue their studies, careers, duties, and activities in an atmosphere free of threat of unwelcome and unwanted sexual actions. It strongly condemns sexual offenses, will not tolerate sexual offenders, and supports those who have been victimized.

The campus will respond promptly, fairly, and decisively to all reports of sexual assault. Members of the campus community accused of sexual assault will be subject to campus disciplinary procedures when the alleged incident has occurred, on or off-campus, and affects the learning environment or operations of the campus. For those seeking assistance, please contact the Office of Student Affairs.

Statement Relating to Sexual Harassment and Associated Grievances

The university is committed to providing an institutional environment in which all persons may pursue their studies, careers, duties, and activities in an atmosphere free of threat of unwelcome and unwanted sexual advances. The university strongly condemns sexual offenses, will not tolerate offenders, and supports those who have been harassed. Students who believe they have been the victim of sexual harassment should report the incident to the Office of Student Affairs. This policy applies equally to all members of university community: students, faculty, administrators, staff, contract employees, and visitors.

Unwelcome sexual advances, requests for sexual favors, and other verbal or physical conduct of a sexual nature constitute harassment when:

- The submission to or rejection of such conduct is made either explicitly or implicitly a term or condition of an individual's academic performance or employment.
- The submission to or rejection of such conduct by an individual is used as the basis for decisions about academic evaluation, employment, promotion, transfer, selection for training, performance evaluation, or selection for academic awards or benefits.
- The conduct has the purpose or effect of creating an intimidating, hostile, or offensive educational or work environment or substantially interferes with a student's academic or an employee's work performance.

Students who believe they have been the victim of sexual harassment should immediately report the incident to the Associate Provost for Student Affairs. Every effort will be taken to provide protection should any student feel threatened in any way.

Statement Relating to Stalking

Stalking is prohibited and will not be tolerated. Stalking is defined as behaviors or activities, which, when taken as a whole, give the victim reason to believe that his or her safety is at risk or affects the learning experience or participation at AURAK. This policy applies to students, faculty, staff, contractors, and visitors, and applies to behaviors and activities occurring both on and off campus. Victims of stalking are encouraged to report incidents to the Office of Student Affairs.

Pets on Campus

No pets, except those assisting people of determination, are permitted on campus at any time. Pets on campus grounds must be on a leash and under supervision at all times.

Facilities Management & Maintenance Policy - Smoke-free Environment

Smoking is not permitted in any building on the AURAK campus including the student residences. There are designated areas for smoking outside the buildings.

STUDENT ATTENDANCE POLICY

AURAK'S Student Code of Conduct requires that students attend classes regularly and punctually, complete assignments on time, and participate actively in class discussions and other course-related classroom activities. This section of the catalog outlines the University's expectations of regular student attendance in class meeting in all courses offered by the institution.

AURAK students are expected to:

- Attend all learning and teaching sessions associated with the courses in which they are enrolled.
- Notify their course instructors in advance that they will be absent from scheduled class sessions.
- Obtain prior permission from their instructors for planned absences of two or more consecutive class sessions during the semester.
- Provide a medical certificate or other corroborating evidence to explain an absence based on a health or medical condition.

Unsatisfactory student attendance includes failure to regularly attend class meetings without providing a satisfactory reason to instructors for absence and failure to observe protocols for online course attendance (e.g., having camera turned on). Unsatisfactory student attendance also includes persistent late arrival at, or early departure from, class meetings. At their discretion, instructors may count students absent if they arrive late for or depart early from scheduled class meetings as long as the information is included in the syllabus.

Where students fail to attend 10% of the scheduled classes for the semester without the University's permission, students will receive a "non-attendance warning," (NAW) and will be required to provide satisfactory explanation for their non-attendance. Where students fail to attend 15% of the scheduled classes for the semester without the University's permission, a second NAW will be issued. With each NAW issued, a formal report on the student's non-attendance is made to the student's academic advisor and to the course instructor.

Where students fail to attend 20% of the scheduled classes for the semester, students will receive a notice that they have been withdrawn from the course with a grade of W if the 20% non-attendance is recorded prior to the deadline for withdrawing with a grade of W. If the 20% non-attendance is recorded after the deadline for withdrawing with a grade of W, students will receive a notice that they have failed the course and will receive an "F (Fail)" or "U (Unsatisfactory)" grade, as appropriate.

All students must adhere to AURAK expectations regarding class attendance.

ACADEMIC INTEGRITY

AURAK HONOR CODE

To promote a stronger sense of mutual responsibility, respect, trust, and fairness among all members of the AURAK community and with the desire for greater academic and personal achievement, we, the student members of the university community, have set forth this honor code:

“As a member of the AURAK family, I pledge to be reliable, responsible, honorable, and embrace only the success and recognition I earn and deserve. I will act with integrity and honesty and will strive daily to become the best version of myself – a decent human being, a conscientious student, a good citizen, and a successful professional.”

The role of the Honor Code and associated Academic Integrity Policy is to protect the academic integrity of the university, encourage consistent ethical behavior among students, and foster a climate of honorable academic achievement. The Honor Code is an integral part of university life and students are responsible, therefore, for understanding and abiding by the code’s provisions. While a student's commitment to honesty and personal integrity is assumed and expected, this Code and associated policy and procedures provides clarity of expectations. To maintain an academic community according to these standards, students and faculty follow the reporting policy and procedures herein.

Expectations

AURAK expects students and all other members of the AURAK community to be honest in academic endeavors and enforces academic integrity within the university.

At the beginning of each semester, faculty members have the responsibility to explain the Honor Code and their expectations for students in each class with special focus on plagiarism as well as specifics related to anticipated coursework and assignments. They must also explain the extent to which aid from others, if any, is permitted on academic work. Students should request an explanation of any aspect of an instructor’s expectations regarding the Honor Code that they do not fully understand. Students have an obligation not only to follow the code themselves, but to encourage respect for, and compliance with, provisions of the code by other students. This includes the student obligation to report potential violations by fellow students and to understand that not doing so may also violate the Honor Code.

Faculty members are responsible for maintaining the integrity of the learning and evaluation process. Faculty members are expected to actively proctor exams, and should enforce the prohibition on possession of any unauthorized electronic devices and/or materials during exams. Faculty may also take any additional action they believe is warranted to assure academic integrity.

While the internet and easy access to information and knowledge is commonplace, the University is committed to students understanding and demonstrating the ethical foundational values of scholarship. For all academic assignments, project work, and presentations, students need to ensure that due acknowledgement is given to the source of any information which they incorporate in their work.

Violations of Academic Integrity

AURAK regards violations of academic integrity as academic fraud and a serious breach of university policy. Academic fraud consists of any action that serves to undermine the integrity of the academic process, including assignments, assessments, and research, or that gives the student an unfair advantage, including but not limited to:

- Inspecting, duplicating or distributing test materials without authorization.
- Cheating, attempting to cheat, or assisting others to cheat – relevant here is the prohibition on being in possession of a mobile telephone or similar electronic device during a test or examination. In case such devices are found with a student, the student will be deemed to have attempted to cheat and will be subject, as with all instances of cheating, to disciplinary action under the Student Academic Integrity Policy.
- Altering work after it has been submitted for a grade.
- Plagiarism.
- Using or attempting to use anything that constitutes unauthorized assistance on an assignment or assessment, including any form of collaboration unless such collaboration is explicitly identified as being acceptable on the assignment or assessment.
- Fabricating, falsifying, distorting, or inventing any information, documentation, or citation.

Faculty members may prohibit the use of generative AI, including though not limited to, AI writers such as Open AI ChatGPT, in completing assignments. When such prohibitions have been communicated by the faculty member, incorporating information from such sources into an assignment submission will be treated as a serious violation of academic integrity expectations.

Programs may have on record specific acts particular to a discipline that constitutes academic fraud and these will be conveyed in writing to students by the faculty.

Statement on Plagiarism

AURAK recognizes that one of the most common violations of academic integrity is plagiarism which can be intentional or unintentional. AURAK therefore ensures that students are informed about plagiarism in all its forms and therefore unintentional plagiarism is as unacceptable as intentional plagiarism and the commission of it brings the same penalties as intentional plagiarism.

AURAK regards plagiarism as theft - students are required to give credit through specific citations where information is used from sources whether quoted, paraphrased, or summarized. Merely rearranging a sentence or changing a few words is not acceptable. The citation style must be appropriate for the discipline and should clearly indicate the beginning and end of the referenced material. All sources used in the preparation of an academic paper must also be listed with full bibliographic details at the end of the paper, as appropriate in the discipline. While plagiarism detection software can assist identifying plagiarism, there is no “percentage of matching content” threshold for determining that content in a written assignment has been plagiarized. Indeed, the presentation of a single striking phrase originally written by another without attribution to the original source can constitute plagiarism, even though the percentage of matching content found by plagiarism-checking software might be very small.

Faculty and Student Responsibilities

- Every student, faculty member, and administrator is responsible for upholding the highest standards of academic integrity. Every member of the AURAK community shall honor the spirit of this policy by refusing to tolerate academic fraud.
- When expectations for a course are not addressed in this policy, it is the responsibility of the instructor to provide students with additional guidelines for what constitutes “authorized” and “unauthorized” assistance.
- It is the responsibility of every student to seek clarification if in doubt about what constitutes

“authorized” and “unauthorized” assistance. In cases involving collaborative work, all students within the collaborative group may be held responsible for violating the code if any member of the group receives, accepts, or utilizes “unauthorized” assistance.

- Students are required to obtain permission prior to submitting work, any part of which was previously or will be submitted in another course. The instructor has the option of accepting, rejecting, or requiring modification of the content of previously or simultaneously submitted work. A student who suspects that a violation of academic integrity has occurred should report that violation to the dean or to the Office of the Provost. In this report, the student should describe any action taken, such as talking with the person involved or with a faculty or staff member. Every effort will be made to preserve the anonymity of the student reporting the incident; however, confidentiality cannot be guaranteed.

Resolution at the Faculty/Student Level

Where a faculty member suspects a student of a possible violation of academic integrity, the faculty member writes formally to the student, using the Student Academic Integrity Violation Submission Form, with a copy of the communication sent to the Academic Integrity Officer. The communication provides the student with five working days to respond, prior to the faculty member making a final determination regarding an academic integrity violation.

Taking into account any response provided by the student, the faculty member decides, following the five-day period for the student’s response, if the student has breached academic integrity and whether a penalty is warranted. The faculty member then prepares a report detailing the alleged academic fraud, with a copy sent to the:

- Dean of the School in which the affected course is located.
- Dean of the School in which the student is enrolled.
- Student
- Academic Integrity Office (as Office of Record).

The suggested penalty for academic fraud in any course is failure in the course. However, faculty members may impose lesser penalties. Penalties available to faculty members include:

- Formal warning – appropriate only for early assignments in first-year undergraduate courses.
- Reduction in grade for the assignment or assessment (test) up to and including a grade of zero for the assignment or assessment – a reduction in grade less severe than a grade of zero is appropriate only for early assignments/assessments in first-year undergraduate courses.
- Reduction in the grade for the course up to and including a grade of F for the course.
A reduction in grade for an assignment or assessment is likely to result in a reduction in the grade for the course. However, in instances in which the assignment or assessment is weighted such that a grade of zero does not result in a grade of F for the course, the faculty member may impose the penalty of a grade of F in the course. A faculty member can also choose to impose this penalty for an egregious act of academic dishonesty, such as attempting to steal an examination or submitting the work of another person as one’s own.

Review by the Academic Review Board

All reports of violations of academic integrity are reviewed by the Academic Integrity Officer to determine whether there is a record of previous academic integrity violations and if the violation warrants a penalty different from that imposed by the faculty member. Based on the Academic Integrity Officer’s review, possible actions by the Provost include:

- Dismissal of the student from the university if the student has received a failing grade (F) in a second course because of an academic integrity violation.
- Dismissal of the student from the university if the student has stolen or attempted to steal an examination.
- Referral of the case to the Academic Review Board if the student has multiple academic integrity violations but did not receive a failing course grade for both offenses. Penalties available to the Academic Review Board include:
 - **An F grade in the course.** An F grade assigned for a course as a result of academic fraud will replace any other grade previously posted for the course.
 - **Academic Dishonesty Probation.** When a student is found to have a second academic integrity violation, but the offenses are not sufficiently egregious to warrant dismissal, a student will be placed on Academic Dishonesty Probation for a designated period of time, e.g., until graduation.
 - **Academic Dishonesty Suspension.** Any academic integrity violation by a student while on Academic Dishonesty Probation will result in suspension from the university for at least one semester and possibly up to one academic year. Students suspended for academic dishonesty cannot transfer into AURAK any credits earned at another institution during the suspension. To qualify for readmission to AURAK following suspension, a suspended student will be asked to complete online training related to academic integrity. Readmission is not assured but will be based on approval by the Academic Integrity Officer in consultation with the Dean of the School in which the student is enrolled and the Provost.
 - **Academic Dishonesty Dismissal.** Any additional instance of academic dishonesty following Academic Dishonesty Suspension will result in Academic Dishonesty Dismissal. A student dismissed for academic dishonesty will not be readmitted to the university.

Appeal to the Academic Review Board

Students may appeal a decision on academic integrity made by either a faculty member or the Provost to the Academic Review Board within ten (10) working days of notice of the decision. When a case is referred to the Academic Review Board, the Academic Integrity Officer notifies the student in writing of the time and location of the Academic Review Board hearing. For decisions made by the Academic Review Board, the student may request reconsideration of the decision by the Academic Review Board within ten (10) working days of notice of the decision. Grounds for appeal are limited to:

- failure to follow university procedure for cases of alleged academic integrity violations;
- presentation of evidence not considered when the original decision was reached.

At an appeal before the Academic Review Board, the student has the opportunity to be heard and the right to produce witnesses or introduce evidence subject to the reasonable discretion of the Academic Review Board. The student may also be accompanied by a representative of his or her choosing who may not participate in the appeal.

The results of all appeals to the Academic Review Board, permitted in the Academic Integrity Policy, shall be final. The Academic Review Board will communicate its final decision to the student, the dean of the school in which the student is pursuing a program of study, the Provost, and the Department of Registrar.

Any matter submitted to a faculty member or the university for decision or review under this policy shall be decided in a timely manner. The parties shall use their best efforts to conclude the proceedings within the semester in which the alleged offense has occurred or the appeal is received. However, any delay in the proceedings shall in no way operate as a waiver of the university's right to assess any or all of the sanctions permitted under the Academic Integrity Policy.

STUDENT RECORDS

CUSTODY OF RECORDS

All transcripts and other documents submitted from other institutions at the time of admission, or later, are the property of AURAK, and are part of the student record that is under the custody of the AURAK Registrar. The university is not required to provide copies of these documents. Transcripts submitted to AURAK for admission or credit transfer cannot be returned to the student or forwarded to other institutions.

Students Privacy Rights

Students have the right to inspect and review information contained in their educational records. The university is not required to provide copies of these documents. The university allows, nevertheless, copies to be made of specific documents included in a student's record. Copies of such documents will only be provided upon submission of a signed request from the student concerned.

Academic Transcripts

A permanent record reflecting the academic achievements of each student who registers at the university (i.e., transcript) is maintained by the Office of the Registrar. Official transcripts are available on request.

Records on Academic Integrity Code Violations

The retention of records on academic integrity code violations is governed by the following:

All records pertaining to the infringement of the code are maintained by the student's school. If the student does not graduate from AURAK, the records are retained for five years after the student's last registration. If the student graduates from AURAK, these records are destroyed by the school upon the student's graduation.

Official Verifications and Certifications

All certificates related to a student's status at AURAK must be requested from the Office of the Registrar; a nominal fee applies.

RELEASE OF INFORMATION

Access to a student's education records is available to authorized officials of AURAK for academic purposes, such as recording grades, attendance, advising, or review of academic progress.

The confidentiality of student information is maintained, and periodic reviews are made in order to check the efficacy of the system. Student records are not disclosed to any third party unless a student gives permission for this in writing. The only exceptions regarding release of student information include the release of information to official accreditation bodies conducting a review or to law enforcement officials conducting pertinent individual case reviews.



UNDERGRADUATE PROGRAMS

UNDERGRADUATE ADMISSIONS

AURAK seeks to admit qualified students who wish to become part of an innovative global academic institution based on the American model. It is the goal of AURAK to admit those students most likely to benefit from its academic programs and diverse student body. Applications are welcome from all students without regard to race, color, religion, national origin, disability, sex, or age. To be considered for admission, each applicant must present sufficient evidence of appropriate qualifications as outlined by AURAK Board Policy and AURAK Presidential Procedure. The factors considered include, but are not limited to, scores on school leaving certificates and on standardized exams, English proficiency, and level of difficulty and appropriateness of coursework, essays, and recommendations. An offer of admission is valid only for the semester for which the student applies but may be extended at the discretion of the university. ***As AURAK must comply with the expectations of licensure and accreditation, the university reserves the right to modify admissions criteria at any time by action of the Board of Trustees or the President.***

English Language Proficiency

The language of instruction at AURAK is English, therefore prospective undergraduate students must demonstrate a reasonable level of proficiency when admitted to a degree program. AURAK accepts the following assessments at the indicated levels as evidence of English proficiency.

Type of English Proficiency Exam	Minimum Score
EmSAT	1100
TOEFL Paper-based	500
TOEFL iBT	61
TOEFL CBT	173
IELTS Academic	5.0

Institutional TOEFL scores from other institutions are not valid except if the TOEFL was completed in AURAK or AMIDEAST offices in the UAE. International TOEFL scores should be sent directly to AURAK's Department of Admissions or ordered online through the official ETS website at www.ets.org. AURAK's TOEFL school code is 4574.

IELTS (Academic) scores are valid for admission to the university, but IELTS (General) scores are not. The exam can be conducted in any of the test centers listed on www.ielts.org. Once the IELTS result is received, a copy of the results must be provided to the Department of Admissions by the student.

Upon submission of their application to the Office of Admissions, applicants who have not submitted the required evidence of proficiency in English will register for AURAK's English Placement Test. AURAK uses the Oxford Online Placement Test (OOPT), which is designed to measure test takers' ability to communicate at different levels of English language proficiency in accordance with the Common European Framework of Reference (CEFR). Students who demonstrate that they meet the English language proficiency requirements for entry to AURAK (EmSAT 1100, IELTS Academic 5.0) are enrolled in ENGL 101. Students who do not meet the English language proficiency requirement are placed in the English Language Program (ELP), ENGL 095 (non-degree credit course), or ENGL 099 (non-degree credit course) depending on their demonstrated level of English.

The results of the OOPT are used to determine which English course(s) students will be required to take:

- Students scoring 30 or below are placed in the ELP
- Students scoring 31-46 (A1, A2 beginner's level) are placed in ENGL 095

- Students scoring 47-59 (B1, lower intermediate) are placed in ENGL 099
- Students scoring 60 and above (high intermediate up to proficient) are placed in ENGL 101

Students who are placed in ENGL 095 must successfully complete the course and the following courses in sequence: ENGL 099, ENGL 101. Students who are placed in ENGL 099 must successfully complete the course and the following in sequence: ENGL 101. Non-degree credits for foundation courses ENGL 095, ENGL 099 will not count toward the undergraduate degree or the student's CGPA.

English Language Program (ELP)

The ELP prepares students with the general English language skills necessary to succeed in an English-medium institution. The program includes four courses (15 credit hours of instruction per week) that aim to develop key skills in listening, speaking, reading, and writing. Upon completion of the program, and based on their OOPT results, students will ideally progress to ENGL 095 to begin academic English instruction, although some may place into a higher-level course if their OOPT results allow. Students who place in the ELP must enroll in and successfully complete the following courses:

- ENGL 091 – ELP 1: Intensive Listening and Speaking (6 credit hours)
- ENGL 092 – ELP 2: Intensive Reading (3 credit hours)
- ENGL 093 – ELP 3: Intensive Writing (3 credit hours)
- ENGL 094 – ELP 4: Communication Skills Integration (3 credit hours)

Admissions Criteria

When reviewing applications, the Department of Admissions considers the academic rigor of the stream of study.

AURAK Academic Program	Admission Criteria Based on High School Performance and EmSAT Test Scores			
BA in Mass Communication	UAE Curriculum	Advanced (Scientific)	70%	
		General (Literary)	70%	
	Non UAE Curriculum	The University recognizes all the other certificates and converts their grades to the equivalent grade.		
Bachelor of Architecture	UAE Curriculum	Advanced (Scientific)	70%	
		General (Literary)	70%	
	Non UAE Curriculum	The University recognizes all the other certificates and converts their grades to the equivalent grade.		

AURAK Academic Program	Admission Criteria Based on High School Performance and EmSAT Test Scores			
	➤ Achieving an average of 70% or higher in the secondary school certificate. ➤ Once accepted into the Architecture program, students will be placed into an appropriate initial course in Mathematics based either on their Mathematics high school score or their score on the Accuplacer Advanced Algebra and Functions Placement Test. The Accuplacer Advanced Algebra and Functions Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education. ➤ Once accepted into the program in Architecture, students will be placed into an appropriate initial course in Physics based either on their score on the EmSAT in Physics or their score on the AURAK Physics Placement Test. The AURAK Physics Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education.			
BS in Biotechnology	UAE Curriculum	Advanced (Scientific)	70%	
		General (Literary)	70%	Successful completion of BIOL 095, MATH 095, and CHEM 095 at AURAK.
	Non UAE Curriculum	The University recognizes all the other certificates and converts their grades to the equivalent grade.		
	➤ Passing EmSAT in Mathematics with a score of 700, or achieving on the Accuplacer Advanced Algebra and Functions Placement Test a score corresponding to placement in AURAK’s Mathematics course MATH 111 – Calculus for the Life Sciences. The Accuplacer Advanced Algebra and Functions Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed. ➤ Passing EmSAT in Chemistry or EmSAT in Biology with a score of 700 or equivalent, or achieving a score on the AURAK Chemistry Placement Test corresponding to placement in AURAK’s Chemistry Course CHEM 111 – Principles of General Chemistry or CHEM 120 – Analytical Chemistry. The AURAK Chemistry Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed.			
BS in Artificial Intelligence BS in Computer Science	UAE Curriculum	Advanced (Scientific)	70%	
		General (Literary)	70%	
	Non UAE Curriculum	The University recognizes all the other certificates and converts their grades to the equivalent grade.		
	➤ Passing EmSAT in Mathematics with a score of 700, or achieving on the Accuplacer Advanced Algebra and Functions Placement Test a score corresponding to placement in AURAK’s Mathematics course MATH 113 – Calculus 1. The Accuplacer Advanced Algebra and Functions Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed. ➤ Passing EmSAT in Physics with a score of 700 or equivalent, or achieving a score on the AURAK Physics Placement Test corresponding to placement in AURAK’s Physics Course PHYS 110 – University Physics 1. The AURAK Physics Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed.			

AURAK Academic Program	Admission Criteria Based on High School Performance and EmSAT Test Scores			
All Business Programs	UAE Curriculum	Advanced and Elite (Scientific)	70%	
		General (Literary)	75%	
	Non UAE Curriculum	The University recognizes all the other certificates and converts their grades to the equivalent grade.		
	➤ Passing EmSAT in Mathematics with a score of 600, or achieving on the Accuplacer Quantitative Reasoning, Algebra and Statistics Test a score corresponding to placement in AURAK’s Mathematics course STAT 100 – Introductory Probability & Statistics. The Accuplacer Quantitative Reasoning, Algebra and Statistics Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed. ➤ Personal interview is also required.			
All Engineering Programs	UAE Curriculum	Elite Track	75%	
		Advanced Track	80%	
		Or Equivalent in Standardized International Systems		
	Non UAE Curriculum	The University Recognizes all the other certificates and converts their grades to the equivalent grade.		
	➤ Passing EmSAT in Mathematics with a score of 900, or achieving on the Accuplacer Advanced Algebra and Functions Placement Test a score corresponding to placement in AURAK’s Mathematics course MATH 113 – Calculus 1. The Accuplacer Advanced Algebra and Functions Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed. ➤ Passing EmSAT in Physics with a score of 800 or equivalent, or achieving a score on the AURAK Physics Placement Test corresponding to placement in AURAK’s Physics Course PHYS 110 – University Physics 1. The AURAK Physics Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed. ➤ Passing EmSAT in Chemistry with a score of 800 or equivalent, or achieving a score on the AURAK Chemistry Placement Test corresponding to placement in AURAK’s Chemistry Course CHEM 211 – General Chemistry 1. The AURAK Chemistry Placement Test will be administered on the AURAK campus through the Center for Executive and Professional Education prior to applicants having their admission status confirmed.			

Minimum EmSAT Scores in Subjects Other than English

For students from school systems that use standardized and/or centralized examinations (such as the British, the International Baccalaureate, or the Emirati School system), achieving 80% or higher (or the equivalent based on the Scholaro GPA Calculator) in required subjects can substitute for the required EmSAT score for all subjects except English. Placement of students in CHEM 111, CHEM 120, CHEM 211, MATH 111, MATH 113, and PHYS 110 depends on their proficiency level in each respective subject, as evidenced by high school transcripts, EmSAT scores, or placement test results.

Secondary School Completion

For admission, applicants must provide a Secondary School Certificate confirming completion of 12 years of secondary school at a Ministry of Education School, or an equivalence certificate issued by the Ministry of Education for private schools.

Transfer Admission Requirement

AURAK will award credit transfer where a student is transferring from:

- a UAE institution recorded in the National Register of Licensed Higher Education Institutions (HEIs) (or approved registry), or
- other organizations in the UAE approved by the Commission for Academic Accreditation, or
- recognized institutions of higher learning located outside the UAE, which are eligible for transfer admission.

Students will only be eligible for the above credit transfer arrangements, where they:

- Provide valid evidence that they meet the English language competency scores for full admission.
- Are in good academic standing and have a CGPA of 2.0 on a 4.0 scale, or equivalent.

Transfer students not in good academic standing at their previous institution, but who meet the English language competency scores for full admission, may receive credit transfer to a program that is not related to the program that they previously studied.

In all cases, students seeking credit transfer may be admitted as per the Ministry of Education requirements, and credit transfer will be restricted to:

- Related courses where students have earned a grade of C (i.e., 2.0 on a 4.0 scale (or equivalent)).
- Courses that deliver similar outcomes.
- A maximum of fifty-percent (50%) of the credit for the receiving program.
- Courses other than graduation projects or thesis.

Where credit transfer is awarded for courses, it will only be awarded once for a given course and there will be no duplication of credit for a course. In addition, credit awarded by credit transfer will not contribute to the calculation of the CGPA.

AURAK will not award credit for experiential or other non-classroom based learning unless the student has undertaken a challenge examination and the process for awarding such credit has been pre-approved by the Commission for Academic Accreditation.

Other Transfer Requirements

The Department Chair/Program Coordinator, with the assistance of faculty specializing in the field covered by courses, will review the application and supporting documentation to identify whether credit transfer can be applied for the courses. The following will be applied:

- The courses for which credit transfer is being sought must have the same number of credits and equivalent learning outcomes as the AURAK courses.
- The courses for which credit transfer is being sought must be at the same level (upper or lower).

- Credit cannot be given for two courses which are substantially the same course.
- For Undergraduate credit transfer will only be permitted for up to 50% of the program credits and only courses graded as “C” or above are considered for transfer credit.
- Graduation projects and thesis are not eligible for transfer credit.

The Department Chair/Program Coordinator will complete an Evaluation Form setting out which courses and credits are recommended for transfer credit, and will forward this to the Dean of School who will validate the appropriateness of the transfer credit recommended by the Department Chair/Program Coordinator.

Where there is disagreement the Dean of School will discuss with the Department Chair/Program Coordinator but the Dean of School decision is final. The Dean of School will sign the evaluation form with the final decision and notify the Department of Admissions and the Registrar.

The Registrar will finalize the number of credits to be transferred and will ensure that the:

- Transferred courses are listed on the student’s transcript courses.
- Transferred credits and grades are not used in calculating the cumulative grade point average (CGPA).

The Admissions Department will notify the transfer applicant of the final decision.

The Registrar will place the documentation related to the transfer credit on the student’s file.

Application Procedures (For Both Freshman and Transfer Applicants)

Prospective students can apply online or in person. The online application is available on <http://apply.aurak.ac.ae>

The following documents are required with every application:

- Passport copy
- Copy of health card
- Copy of valid Emirates ID (UAE residents only)
- Health History Form (Completed and signed by a physician)
- Four (4) recent passport-size photographs
- Exemption letter from the National and Reserve Service Authority (UAE male applicants between the ages of 18 and 30 only)
- Birth Certificate
- Family Book (UAE nationals only)
- Army Exemption/Completion Letter (UAE nationals only)

Attestation of Documents

Admitted students are required to submit officially attested education certificates by the deadline mentioned in their admission letter. Students with certificates obtained outside the UAE **may** be given an extension until the end of the first semester of their studies subject to the approval of the Manager of

Admissions. Subsequent extensions are subject to approval of the Associate Provost for Academic Affairs. Failure to provide appropriate attested education certificates will lead to expulsion from the institution. AURAK transcripts are not provided until all admission requirements are satisfied. For more information regarding attestations, contact the Enrollment Management Department: admissions@aurak.ac.ae.

Admission Interview

The Enrollment Management Department reserves the right to require an applicant to attend an admissions interview as part of the application process when required by the prospective program or there is a need for additional information. Interviews may take place on campus, in person, by digital means, or by phone where appropriate.

Admission Categories

Full Admission

Applicants must meet all outlined requirements to obtain full admission status. To qualify for full admission a new undergraduate student must hold a UAE valid secondary stage completion certificate (SSCC) or attested equivalent along with a detailed statement of the grades for each subject. Applicants may be granted full admission on the basis of unofficial copies of such documents but must provide the official copies of required documents within two semesters following initial enrollment or face withdrawal from the program of study.

Conditional Admission

Applicants considered marginal may be conditionally admitted subject to a recommendation from the program coordinator requesting exceptional consideration.

Applicants lacking the following may also be conditionally admitted:

- English language proficiency.
- Required EmSat score in Mathematics, Physics, Biology, or Chemistry for the desired program of study.

During the initial semester of study as conditionally admitted students, students may take up to 12 semester credits of General Education in addition to possible foundational coursework corresponding to the missing EmSAT score(s). The foundational coursework will not count toward the undergraduate degree or the students' CGPA.

Admission of Visiting Students

A visiting student is one who is not formally admitted to AURAK and who is not participating in a formal exchange agreement between AURAK and a partner university. A visiting student is typically enrolled in a degree program at another university. This student may take courses at AURAK for transfer to the student's home institution. Visiting students are advised to check with their home institutions to confirm the transferability of AURAK credits.

Applicants seeking visiting student status must submit to AURAK's Department of Admissions the following documents:

- Copy of high school certificate
- An official letter from the university allowing the student to enroll for the course(s) at AURAK
- English proficiency scores (EmSAT, TOEFL, IELTS or Equivalent).
- Passport copy, Emirates ID, and residence visa (If Applicable)

- Four (4) passport-size photographs
- Proof of acceptable health insurance or purchase AURAK health insurance

Admission Requirements Subject to Program Admission Specifications as Outlined by the Ministry of Education

From time to time, the Ministry of Education establishes expectations for admission to specific programs. These expectations are communicated by MOE Letters of Reform. In all cases, AURAK conforms to the program admission standards outlined in MOE Letters of Reform.

Application Deadlines

The application deadline for fall admission is notified through newspapers or through university notice boards for freshmen and transfer applicants. Admission decisions for first year students are usually made after receipt of the first-semester grades of the senior year of secondary school and all appropriate test results. Admission is contingent upon satisfactory completion of in-progress course work and graduation from the secondary school. Transfer decisions are made as files become complete.

AURAK Admissions Application Deadline for the Fall 2024 Semester is **9 September 2024**.

AURAK Admissions Final Application Deadline for the Spring 2025 Semester is **20 January 2025**.

Admission deadlines are subject to extension by the Department of Admissions where circumstances warrant.

Note: International students must apply two weeks in advance of the stated deadline to meet visa timeline requirements.

Acceptance of Admission Offer

Students will be required to confirm their acceptance of an admission offer by completing the enrollment confirmation form in the admission packet and submitting it to AURAK with the enrollment deposit, admission fee, visa and health insurance fee (if applicable), and residence hall fee (if applicable).

Application for a Second Bachelor's Degree

Existing and former AURAK students may apply to undertake a second degree at AURAK provided such degree will enable the completion of at least thirty (30) credit hours beyond the first degree undertaken at the institution. For existing students, this will only be permitted where they have met all the requirements of their first degree.

New Student Placement Exams

An English language placement exam is administered to entering students. The local English language placement exam is utilized for placement in the appropriate level English course. AURAK uses a math placement exam to determine which mathematics course new students may take. Students who were awarded transfer credit for a math course taken at another institution are not required to take the math placement exam.

Scores from the EmSAT can be utilized for placement into mathematics, physics, and English courses.

Student Residence Visa

The Human Resources Department arranges student visas under the sponsorship of AURAK. Visas must be renewed every year. Students on an AURAK visa who discontinue their studies or transfer to another university must submit their passport for visa cancellation prior to their departure.

The visa process takes approximately two weeks during which students cannot leave the country as they must hand over their passports temporarily. The visa application requires:

- Eight (8) passport photos
- Passport copy
- Copy of the secondary school certificate
- Visa Request Form
- Payment of visa fee and security deposit for visa

All student applicants for a residence visa must undergo medical tests conducted by the Preventive Health Department of the RAK Government and fingerprinting by the Police Department. The medical tests include a blood test and an X-ray. Failure to pass either one of the medical tests leads to deportation to country of origin.

Right to Withdraw Offer

AURAK may on the approval of the Registrar withdraw offers of admission if an applicant fails to satisfy all requirements, or if it is determined that admission was obtained through the use of falsified, altered, or embellished information (or similar). In the case of withdrawal of admission from a matriculated student, credit earned may be withheld.

Readmission

Students who have missed two or more consecutive semesters (excluding Summer Semester) of enrollment at AURAK will be required to apply for re-admission on the following basis:

- A student returning after any absence during which he or she studied at another institution without prior written permission of the Vice President of Academic Affairs and Student Success (VPAASS) or designee will be treated as a **transfer student**.
- A student who failed to demonstrate good academic standing during their previous studies at AURAK, will be required to reapply for **full admission** in order to be considered for admission.
- A student who was in good academic standing during the previous studies at AURAK, may be readmitted on completion of a Re-Enrollment Form available through the Office of the Registrar.

UNDERGRADUATE ACADEMIC POLICIES

Each student is responsible for knowing AURAK's rules, regulations, requirements, and academic policies. The Student Handbook, the University Catalog, and the institutional website are repositories of policy statements. Corrections, changes, or interpretations may be communicated by other means, including electronically. Any student in doubt about an academic matter should consult the Office of the Registrar or their assigned faculty advisor. Students are subject to the university's stated policies regarding patents and copyrights.

Academic Advising

AURAK is committed to helping students achieve success in their course of study. Ensuring that students receive appropriate academic advising is part of that commitment.

New students are first advised during a required orientation program in which they learn about university policies and procedures and receive a current university catalog and a student handbook. At the initial meeting with their faculty advisor during orientation, each student is informed of program procedures, course of studies, and the importance of understanding the information in the university catalog. Advisors also answer specific questions concerning the academic program.

Each student is responsible for his or her progress toward degree and for meeting all requirements for graduation. The advisement process is available to assist the student in completing his or her studies, but responsibility for doing so lies ultimately with each student.

Faculty advisors are required to post office hours for advising that coincide with students' availability. The faculty advisor is to maintain advising files for all advisees that contain a log of meetings with a brief description of major issues discussed or advice given. Any information contained in these files shall remain confidential.

Students on probation are required to meet with their advisor on a regular basis to assess their progress and to report their use of AURAK's various resources for academic support. Students violating AURAK's attendance policy or doing poorly at mid-term are also required to meet with their advisors.

If students have concerns, comments, or recommendations about their educational experiences at AURAK, they should contact the dean of the school in which their degree program is located.

Courses and Course Schedules

Each course offered by the university has a designated course prefix (or code) and number. The course prefix represents the discipline or field of study; the number indicates the level of the course content. MATH 108 thus indicates that the course is Mathematics and is appropriate for students in their first year; CHEN 472 indicates that the course is in Chemical Engineering and is intended for fourth year students.

Courses are offered at the discretion of the Senior Vice President for Academic Affairs and Student Success and the respective deans and department chairs. Not every course is offered every semester but are offered at a frequency that enables students to make satisfactory progress toward their degrees.

Course Credit

Each course has a credit value. A credit represents the in-class instruction and out-of-class study per week during the sixteen-week semester. Normally, one credit represents fifty minutes of class instruction per week per semester or 150 minutes of laboratory experience per week per semester.

Class Periods

Classes normally meet two times a week in one hour fifteen-minute periods or two. In some cases, usually advanced courses, classes may meet once per week for 150 minutes. The university traditionally holds classes four days a week from Monday through Thursday. Graduate courses are usually held on Friday, Saturday or Sunday. Though university administrative offices are usually closed on Saturday and Sunday, appointments may be arranged to meet students during the weekends.

All classes are expected to meet on the days and times published in the course schedule. Changes may only be made with the approval of the Senior Vice President for Academic Affairs and Student Success.

Independent study or research, internships, other programs for experimental learning, and other study opportunities may follow a different, approved time frame and schedule.

Course Prerequisites and Corequisites

Course prerequisites or corequisites state requirements for student entry into courses and reflect necessary preparation for enrolling in a course. It is the student's responsibility to be aware of these requirements as stated in the university catalog and to register for courses only if stated prerequisites have been completed. Similarly, if a course has a corequisite, a student can only register for the course if the corequisite is also selected for registration in the same semester. The Student Information System has been designed to prevent students from registering for courses for which prerequisites have not been completed or for which corequisites have not been selected for simultaneous registration. However, if a student succeeds in enrolling in a course without having completed pre-requisites or corequisites for the course, the student will be dropped from the courses as soon as the registration error is detected. No waivers of prerequisites or corequisites will be granted.

Course Selection and Registration

Typically, by mid-semester, the schedule of classes for the following semester is available through the Office of the Registrar and on the website. Students may select courses in consultation with their faculty/academic advisor, and they can then register online or submit a completed "Course Request Form" to the Office of the Registrar by the published deadlines.

Registration Criteria

Only registered students may participate in AURAK classes. Full-time status requires that an undergraduate student enroll in 12 or more semester credit hours in Fall and Spring semesters. Undergraduate students enrolling in fewer than 12 semester credit hours are considered to be enrolled on a part-time basis, but no undergraduate student may enroll for fewer than 6 semester credit hours, except in cases in which students need fewer than 6 credit hours to complete the degree. Undergraduate students whose semester credit hour loads are limited due to probation requirements are considered to hold full-time status. All undergraduate students are accepted to the university on the assumption that they will be studying full time, and the timeframe for completing the program of study described in this catalog is based on full-time study. Registering as a part-time student results in a significantly higher per credit tuition fee, makes students ineligible for scholarships, and lengthens the program of study.

Independent Study

Occasionally, students may have the opportunity to benefit from an independent study experience that will significantly add to their preparation for a career or ensure timely completion of their program of study. Only one course (maximum of 4 credits) may be completed as an independent study in a student's program of study. The independent study must be approved by the instructor(s) involved. Students must have completed a minimum of sixty credits, must have a CGPA of 2.0 or higher, and seek approval through completion of the appropriate "Permission for Independent Study" Form. The instructor must hold a regular rank faculty appointment at AURAK within the department or program requesting the independent study and is responsible for submitting the final grade, and ensuring adherence to academic standards, policies, and procedures. In addition to the individual effort of the student, a student enrolled for independent study must meet weekly with the instructor for guidance in research, analysis, and composition of the project report or presentation. Independent study courses may not duplicate available course offerings in the semester in which the independent study is being taken.

The independent study must provide a rigorous academic experience equivalent to that of any other undergraduate course at the same level at AURAK. Grading of an independent study adheres to the established grading system.

Concurrent Enrollment at Other Institutions

Students who apply for admission to AURAK usually do not seek simultaneous enrollment with another institution. Should students seek concurrent enrollment, they must obtain written approval in advance from the VPAASS/Provost. Such approval enables a student to enroll elsewhere in a course unavailable at AURAK.

Catalog numbers and descriptions of courses to be taken elsewhere must be submitted with the request for approval. As for any transferred course, students must submit an official transcript to the Office of the Registrar to earn credit, and grades are not included in the GPA. Students who enroll elsewhere without advance written permission while enrolled at AURAK may not receive transfer credit for the course work taken.

Summer Term

During the summer term, students may register for one or two courses. All attendance, grading, and other academic policies apply to the summer term as well as to the two academic semesters.

Student Coursework Assessment Policy

Academic assessment of students must be robust, reliable, fair, accurate, aligned with learning outcomes and program goals, and undertaken at an appropriate level. As such, undergraduate students will be required to undergo a minimum of three (3) assessments distributed evenly throughout the semester. Acceptable assessment tools include:

- (a) Exam
- (b) Project
- (c) Paper
- (d) Presentation
- (e) Homework assignment
- (f) In class quiz
- (g) Case study/Analysis

'In class participation' will also be acceptable as an assessment tool, but will only be used to calculate the final grade where it does not exceed ten percent (10%) of the final grade.

Irrespective of the assessment tool used, it must be appropriate for the learning outcome(s) under assessment. Assessments may be adjusted to accommodate people of determination, however, such special adjustments must not invalidate the assessment.

Marking of Assessments and Moderation

AURAK will implement methods for the moderation and assessment of student work in which more than one individual independently marks, or moderates an assessment, or evaluates student performance.

Weighting of Assessment

- No single assessment tool can count for more than forty percent (40%) of the total grade.

- “In class participation” may be used in calculating final grade provided it does not exceed ten percent (10%) of the total grade.
- Attendance has no weight in the final grade as attendance requirements are explained under “attendance policy.”

Midterm Exams

Midterm progress is reported for all semester-long courses so that students will have a clear idea of their standing in each course midway through the semester. The reporting period extends from the fifth through the ninth week of the semester, allowing flexibility as to when individual faculty provide reports for their classes. Midterm grades are reported as S (satisfactory - C+ and above) or U (unsatisfactory - C and below). Students should check with their instructors as to when reports will be complete and available for viewing. These progress reports, which appear as ‘Midterm Grades,’ are not part of the student's official record. They are not calculated in the GPA, nor do they appear on any official or unofficial transcript. The midterm progress report is distributed to the student, his or her faculty advisor, the Office of Student Affairs.

Final Exams

Final exams are usually given at the end of all undergraduate courses. Except for project submissions, laboratory courses, and studio courses, no exam or assessment tools of any kind may be given during the last week of classes. Exams may not exceed the scheduled length of two (2) hours for most courses. The days following the last day of classes each semester that comprise the final exam period are identified in the Academic Calendar found on the university website. Students must plan to be available for final exams during the entire final exam period. The schedule showing the specific days and times for final exams for each course is posted each semester on the university website six weeks prior to the last day of classes.

An instructor who is assigning a take-home exam or significant end-of-semester paper or project should so inform the students on the syllabus at the beginning of the semester. Take home exams should be distributed by the beginning of the last week of classes so that students can coordinate them with preparation for other exams. Students must not be required to submit exams before the date of the regularly scheduled exam for a course. Accommodations will be made for students with more than one final exam scheduled in the same time period. Retaking (or re-sitting) a final exam is not permitted. Students are not expected to take more than two final exams in one day.

Absence from Final Exams

Absences from final exams are not to be excused except for sickness on the day of the exam or for other causes approved by the Office of Student Affairs. The effect of an unexcused absence from an undergraduate final exam shall be determined by the weighted value of the exam as stated in the course syllabus provided by the instructor. If an absence from a final exam is unexcused, a grade of zero will be entered for the final exam.

Final Semester Grades

Final grades are available through the student portal on the university website and/or collected from the Office of the Registrar. Official transcripts and official semester grade reports for tuition reimbursement are obtained through the Office of the Registrar.

Students may access and print an unofficial record of their semester grades and other academic information through the student portal on the university website by logging onto their Student Information System (SIS) account. Official transcripts are available upon request.

Change of Grade/Grade Appeal

Once a final grade has been recorded by the faculty member, it can be changed only through the Grade Appeal process. Where students believe their final course grades to be in error, students must request a meeting with the member of faculty to ask that the faculty member reconsider the grade and ensure that the appeal is submitted within the required timeline. Completed Grade Appeal Forms must be submitted within the following timeframe:

- For Fall Semester courses, completed Grade Appeal Forms must be submitted within the first three weeks of the following Spring Semester.
- For Spring Semester courses, completed Grade Appeal Forms must be submitted within three weeks of the “Grades Available” date listed in the Academic Calendar.
- For Summer Session courses, completed Grade Appeal Forms must be submitted within the first three weeks of the following Fall Semester.

The grade appeal will only be considered where:

- There was an error or unfairness in the assessment of the exam grade.
- There was an error or unfairness in the evaluation process for determining the course grade.

The faculty member will meet formally with the student and a record of the meeting will be recorded on the Grade Appeal Review Form. In the meeting with the student, the faculty member must explain how the grade in question was arrived at. If the faculty member decides, based on the discussion with the student, that a change of grade is warranted, a Change of Grade Form will be completed by the faculty member, explaining the basis for the grade change and this will be forwarded to the Department Chair.

If the faculty member decides a change of grade is not warranted, the faculty member will complete the Grade Appeal Review Form, explaining why the original grade should not be changed and this will be forwarded to the Department Chair. The Change of Grade Form/Grade Appeal Review Form (as applicable) will then be reviewed in succession by the Department Chair, the Dean, and the Associate Provost for Academic Affairs, and a decision will be made.

The Associate Provost for Academic Affairs will render a decision only after the faculty member has responded to any requests for further consideration made by the Department Chair and the Dean. The decision by the Associate Provost for Academic Affairs will be final and not subject to further appeal. The Registrar will not change a student's final grade in a course for any other reason or through any other procedure.

Where an appeal is successful, the Associate Provost for Academic Affairs will forward the Change of Grade Form to the Registrar's Department so that the student's grade is changed on the Student Information System (SIS). A copy of the Grade Change Form will be filed in the student's record.

Pending Grade Appeal

A student may request the VPAASS/Provost to delay imposing an academic suspension because of a pending grade appeal that could change the student's status. An approved delay allows the student to register. If the grade appeal is successful, the official transcript is corrected, and the student continues in classes as a student in good academic standing. If the grade appeal is not successful, the student is required to stop attending all classes and settle the case with VPAASS/Provost appropriately.

Grading System

University course work is measured in terms of quantity and quality. A credit normally represents one hour per week of lecture or recitation or not fewer than two hours per week of independent or laboratory work

throughout a semester. The number of credits is a measure of quantity. The grade is a measure of quality. The university system for undergraduate grading is as follows:

Grade	Quality Points	Percentage Scores
A	4.0	95-100
A-	3.7	90-94
B+	3.3	86-89
B	3.0	83-85
B-	2.7	80-82
C+	2.3	76-79
C	2.0	73-75
C-	1.7	70-72
D+	1.3	66-69
D	1.0	60-65
F	0	0-59

No credit toward graduation is received from a failing grade or additional credit gained from a grade that is replaced by a repeated course.

Grade Point Average (GPA)

Grade or quality point values are assigned to letter grades as indicated in the grading system table. A quality or grade point score is computed by multiplying the value of a letter grade by the number of credits for the course. For example, a student receiving an A in a three-credit course earns twelve quality points. The GPA is computed by dividing the quality points earned by the number of credits graded "A" through "F" (GPA hours).

For undergraduates, the GPA computed for the current semester gives the current GPA, which is the measure of academic performance in one semester and affects eligibility for the honor's list. The GPA computed for all institutional credit gives the CGPA, which is the basis for the university's retention policies, including good standing, warning, probation, suspension, and dismissal. The CGPA also determines students' eligibility to graduate and to have university honors posted to their records upon graduation.

Additional Grade Notations

Satisfactory/Unsatisfactory (S/U):

An "S" grade reflects satisfactory, or passing, work in a course; a "U" grade reflects unsatisfactory, or failing, work in a course. S and U will have no effect on GPA.

Incomplete (I):

This grade is given to students who are passing a course but are unable to complete all the course work or the final exam for a verified reason beyond their control or the instructor agrees to allow the student additional time to complete substantial aspects of the course's required work. The student must complete the final exam or the required work for the course by the end of the subsequent registered semester, not including summer terms, and the instructor must turn in the final grade by the end of that semester's grading deadline. Unless an explicit written extension is filed, the grade of "I" is changed to an "F" if the course requirements have not been fulfilled. Students who have filed their intention to graduate have six weeks from the date of degree conferral to resolve any incomplete grades. An "I" is not calculated in a student's GPA.

In Progress (IP):

The grade of “IP” indicates that the student is making progress in a course, writing a thesis or dissertation, or participating in an internship that extends beyond one semester or summer term. The grade of “IP” has no effect on the student’s GPA.

Withdrawal (W):

The grade of “W” on a course indicates that the student has dropped the course. While it has no effect on the GPA, dropped courses are part of attempted course credits that serve as the basis for the student’s credit level. A “W” for all courses in a given semester and the transcript notation “withdrawn” indicates that the student withdrew from AURAK.

Academic Status

Each semester, a student’s academic status based on the CGPA is recorded on the transcript of both undergraduate and graduate students using the following CGPA ranges:

Academic Status Range	
Excellent	3.60 to 4.00
Very Good	3.00 to 3.59
Good	2.50 to 2.99
Pass	2.00 to 2.49

Latin Honors at Graduation

The university recognizes undergraduate students who have excelled in their studies with three distinct Latin Honor designations at the time of graduation. The honors are given to undergraduates based on the CGPA at the time of graduation.

Latin Honors	
Designation	Required CGPA
Summa Cum Laude	3.90 and above
Magna Cum Laude	3.80 to 3.89
Cum Laude	3.65 to 3.79

Repeating a Course

Undergraduate students may repeat courses for which they seek a higher grade. A grade received in a repeated course replaces the previous grade in the same course in the calculation of the GPA, even if the more recent grade is lower. Every registration for a course and its grade remains as part of a student’s transcript. Duplicate credit is not given. No adjustment to the CGPA will be made if the grade in the repeated course is a W. No more than three courses can be repeated at AURAK over the course of a given degree program.

A grade in an AURAK course will not be excluded from the CGPA based on the subsequent taking of an equivalent course at a transfer institution. Note that individual programs may disallow students from retaking certain high-demand courses simply for the purpose of improving their grade.

Transfer Credits

Courses taken at other institutions of higher education may be counted toward a degree but are not computed in the AURAK GPA.

Academic Progress

Full-time/Part-time Status

Only registered students may participate in AURAK classes. Fulltime status requires that an undergraduate student enroll in 12 or more semester credit hours in Fall and Spring semesters. Undergraduate students enrolling in less than 12 semester credit hours are considered to be enrolled on a part-time basis. Undergraduate students whose semester credit hour loads are limited due to probation requirements are considered to hold fulltime status. All undergraduate students are expected to enroll on a fulltime basis unless granted permission by the dean of the respective school to enroll on a part-time basis.

All students are required to demonstrate sufficient academic performance to maintain enrollment with the university. Accordingly, AURAK establishes the following to recognize high academic performance and establish parameters for improving academic performance falling below minimal expectations.

Academic Excellence

An undergraduate student will be named to the Dean's List for a fall or spring semester if during that semester the student has:

- Completed twelve or more semester credit hours in courses counted in the CGPA;
- Received no grade below B;
- Received no grade of I (Incomplete) in any course; and
- Attained a semester's GPA of 3.60 or above at the end of the given semester

The dean of the school will notify students who have been named to the Dean's List.

An undergraduate student who qualifies for the Dean's List in a given semester with a GPA of 3.90 or above will also be named to the President's List. Students on the President's List are not included on the Dean's List, as the President's List represents a higher standard.

The aforementioned lists are only compiled for fall and spring semesters.

Satisfactory Academic Progress

In order to be considered in good standings, an undergraduate student must maintain a CGPA of 2.0 or higher. To be considered in good standing, a graduate student must maintain a CGPA of 3.0 or higher. A student, undergraduate or graduate, must be in good standing to be eligible for graduation.

Every student, undergraduate or graduate, must evidence sufficient academic progress towards the completion of his or her program requirements. Academic progress is evaluated based on the student's maintenance of a satisfactory CGPA and satisfactory completion of courses at a pace consistent with completing the program within the maximum timeframe for degree completion.

Academic Warning, Probation, Suspension, and Dismissal

The first time the CGPA of an undergraduate student falls below 2.0, the student will be given an Academic Warning. If a student has completed fewer than 60 credits at the time of receiving an Academic Warning, the student will be required to complete successfully the university's study skills course in the subsequent semester. If the student's CGPA remains below 2.0 at the end of the subsequent semester or the student

fails to complete successfully the study skills course, the student will be placed on Academic Probation 1. All students on academic probation will be required to sign the AURAK Probation Contract before being allowed to register for any coursework.

A student on Academic Probation 1 will be allowed to take a maximum of 13 semester credit hours. Students earning a CGPA of 2.0 or higher in the subsequent semester will be removed from Academic Probation 1 and returned to good academic standing. An Academic Warning is only given once. Therefore, a student whose CGPA falls below 2.0 a second time will be placed on Academic Probation 1.

Students on Academic Probation 1 earning a CGPA of 1.5 to 1.99 will be placed on Academic Probation 2. Students on Academic Probation 1 earning a CGPA below 1.5 will be academically suspended. A student on Academic Probation 2 is allowed a maximum of 10 semester credit hours. Students earning a CGPA of 2.0 or higher in the subsequent semester will be removed from Academic Probation 2 and returned to good academic standing. Students completing a semester of Academic Probation 2 with a CGPA less than 2.0 will be academically suspended.

Suspended students will not be allowed to enroll in AURAK courses for the subsequent semester unless they change their program of study to a distinctly different academic discipline, utilizing the New Beginnings procedure. Suspended students who elect not to change their program of study to a different academic discipline can only request re-admission once they have been out for one full academic fall or spring semester and have completed during that time not-for-credit remedial coursework. Students readmitted following suspension will be dismissed from the university if they are suspended for a second time unless they change their program of study to a different academic discipline.

New Beginnings for Suspended Students

A student who has been academically suspended from the university may apply for re-admission without being separated from the university for one regular semester as part of the New Beginnings procedure. The New Beginnings procedure will only enable the suspended student to undertake an alternative program that is distinctly different from the program from which the student was suspended. Students suspended from a math/science intensive program such as an Engineering program or Biotechnology cannot undertake an alternative program in another math/science intensive program. A student who has been academically suspended from the university for the first time may apply for re-admission by submitting a suspension appeal that includes a request to change program to the Associate Provost for Academic Affairs immediately following suspension from the original program of study. The New Beginnings procedure cannot be enacted more than once.

A suspended student pursuing the New Beginnings option must seek the approval of the admissions manager, current academic advisor, new department chair, new academic advisor, and the associate provost for academic affairs.

Then, the Department of the Registrar will verify that the student has not previously been re-admitted under the New Beginnings procedure.

Once a student is re-admitted to a different program, the Registrar will remove from the CGPA calculation the grades for courses that do not meet the requirements of the new degree program, as selected by the student and the new academic advisor.

Courses in which a student was guilty of academic dishonesty (e.g., cheating, plagiarism) and general education courses that continue to meet requirements in the new degree program will not be removed from the calculation of the CGPA.

The Registrar will recalculate both the semester GPA and CGPA based on the above and the adjusted courses will remain on the transcript and will receive an “E” notation, meaning the grades and the credit hours will be “Excluded from GPA calculations.”

New Beginnings for Students Who Have Not Been Suspended

A student who has not been academically suspended from the university may apply once during enrollment at the university to transfer to another program utilizing the New Beginnings option with the same conditions described above for suspended students. The student does so by submitting a Change of Major form to the Associate Provost for Academic Affairs.

Intellectual Property (Student Work)

Intellectual property (IP) is property (an idea, invention, or process) that derives from the work of the mind or intellect. IP is also an application, right or registration of an idea, invention, or prose. Copyrightable works, including publications and patentable works developed in connection with course work by students who are not AURAK employees, are deemed to be intellectual property that belongs to the student. However, AURAK may claim copyright ownership of a work or ownership of a patentable invention when extraordinary use of university facilities, personnel, or resources is made in the development of the materials or invention, especially when unrelated to course work. Students are subject to the university’s stated policies regarding patents and copyrights.

University Withdrawal

Official withdrawal removes students from any academic program and cancels student status at AURAK. In order to return to the university, students need to apply for readmission through the Department of Admissions. Students who wish to withdraw have to complete a withdrawal form available at the Office of the Registrar and get clearance from all university departments. If the withdrawal process is completed satisfactorily and all financial obligations to the university are cleared, the effective date of withdrawal is noted on the student’s permanent academic record. The effective date is the date used for calculating billing or refunds. No grades other than a W are recorded for the current semester.

Leave of Absence

Occasionally, students are forced to leave the university for a semester or two because of circumstances beyond their control. Others find they simply need a break from studying. In such circumstances, taking a leave of absence might be wise. Students who have an approved leave of absence for a semester or a year may register for the semester in which they plan to return without applying for readmission. Unless there are extenuating circumstances such as illness, a leave of absence is not normally given to a student who leaves the university during a term.

Fields of Study - Degree Offerings

American University of Ras Al Khaimah has three schools that offer both undergraduate and graduate degree programs. Undergraduate programs are listed below:

- Bachelor of Architecture
- Bachelor of Arts in Mass Communication
- Bachelor of Science in Artificial Intelligence
- Bachelor of Science in Biotechnology

- Bachelor of Science in Business Administration
- Bachelor of Science in Chemical Engineering
- Bachelor of Science in Civil and Infrastructure Engineering
- Bachelor of Science in Computer Engineering
- Bachelor of Science in Computer Science
- Bachelor of Science in Electrical and Electronics Engineering
- Bachelor of Science in Mechanical Engineering

Concentrations

Undergraduate and graduate degree programs may offer students one or more concentrations within the major from which to choose. A concentration is a set of related courses that focus on a specific sub-field within the general discipline. A concentration must consist of no fewer than fifteen (15) credit hours of study.

The concentration is specified on the student's diploma and transcript. To be recognized as completing an official concentration, a student must complete and submit the Request for Concentration Form before having completed 105 credit hours towards the degree.

Minor

A minor is a secondary area of specialization in a field of study outside a student's major or concentration, requiring less coursework than the major. AURAK has recently introduced a minor in Artificial Intelligence (AI) for eligible students in the following undergraduate programs: Electrical and Electronics Engineering, Civil and Infrastructure Engineering, Mechanical Engineering, Chemical Engineering, and Architecture. For more details on the AI minor requirements, please refer to the School of Engineering and Computing section.

Double Major

An undergraduate student may elect to complete a double major, which is a single degree with two majors.

A double major may be undertaken under the following conditions:

- A student must have a CGPA of 2.5 or above at the time of electing the second major.
- A student must have completed no more than 105 credit hours when electing to pursue a second major.
- A student must have completed two full years of study toward the first major (sixty credit hours) before applying for a double major.
- The second major may not be in the same academic department as the first.

To complete a double major, the following conditions must be met:

- A student must meet all the degree requirements for both majors.
- A student must complete at least twelve credits in addition to those required by the first major.
- No course substitutions will be available to a student in order to complete the requirements for the second major.

A student who completes a double major will receive one degree that indicates both majors on the transcript.

Change of Major

A student may change his or her major at any point during matriculation at AURAK. At the time of the major change, the student must indicate which major and the General Education requirements that will be met: those in the catalog stipulated at the time the student entered AURAK as a matriculating student or those stipulated at the time of the change of major. The student must choose one set of General Education requirements or the other; they may not split the General Education requirements listed in either catalog.

Unless the student invokes the GPA Adjustment Policy, all previous courses taken remain on the student's transcript and grades received are calculated in the student's CGPA.

To take effect, the student must complete and submit the Change of Major form with all signatures.

Course Substitution

All degree programs, concentrations, and minors have published requirements for graduation, including GPA requirements applicable to all university students. All requirements are met by successful completion of specific courses.

The need occasionally arises for a course substitution in a student's program, which AURAK permits under certain circumstances:

- If a required course has not been offered in at least two different semesters during the student's period of full-time matriculation

OR

- Under exceptional circumstances (e.g., certain disabilities make it impossible for a student to enroll in a required course)

In such cases, a course substitution is possible only if:

- The substitute course covers material similar in content and skills to the initial requirement.
- The substitute course is at the same level as, or more advanced than, the course for which it is being substituted.
- The replacement is approved by the student's academic advisor, a content specialist, the chair of the department of the program for which the substitution is being made, and the dean of the appropriate school, or, in the case of General Education, by the chair of the General Education Committee.

A course being transferred from another university may be used to substitute for an AURAK requirement if all other conditions are met.

Permission for a course substitution is granted only rarely, and a student may receive no more than two course substitutions in any program, including General Education. For a double major, no course substitutions are permissible in the second major.

Among others, the following will not be considered appropriate justifications for applying for a course substitution:

- A desire to complete the program of study quickly
- Failure to enroll and complete a required course that is offered regularly
- Dislike of a professor teaching a required course
- Substitution for a required course which the student has failed previously

- An independent choice to diversify courses taken for personal preference

Undergraduate Enrollment in Graduate Courses

AURAK recognizes that there are occasions where it is academically valuable and appropriate for an undergraduate student to enroll in and receive credit for a graduate level course. To be eligible to undertake such courses the undergraduate student must gain the approval of their Academic Advisor and Department Chair and must have:

- senior status
- a CGPA of no less than 3.0

Students granted permission to undertake graduate level courses must complete the same assignments as the graduate students enrolled on the course(s) and will only be permitted to enroll in **two** graduate level courses.

The credit achieved may count toward the student's Bachelor degree as determined by the Program's Specification and contribute to the relevant graduate program should the student later matriculate into the graduate program.

This policy is subject to approval by the Commission for Academic Accreditation (CAA).

Internship Policy

AURAK considers Internships to be an important education-related experience in the workplace and an opportunity for learning that integrates theory learned in the classroom with practical application. The Internship is also seen as an opportunity for skills development on the job that contribute to the development of personal and professional maturity as well as ethical decision-making.

Undergraduate students will only be permitted to register for the Internship if they:

- (a) have completed a minimum of two years of coursework (or higher if determined by the applicable School). *In exceptional circumstances a student who has not met this condition, may be permitted to undertake the Internship on the recommendation of the Dean of School and with approval of the Provost.*
- (b) are in good standing (at least 2.0 CGPA). *In exceptional circumstances a student who has not met this condition, may be permitted to undertake the Internship on the recommendation of the Dean of School and with the approval of the Provost.*
- (c) have the written approval of their Academic Advisor and the School's Internship Coordinator.

Internship Arrangements

In compliance with accrediting body requirements, AURAK will ensure that the Office/Department responsible for Internships:

- (a) Has an effective method for matching students to potential placement employers.
- (b) Prepares an Internship Manual which covers the following:
 - i. Policies and procedures related to Internships and other experiential learning opportunities.
 - ii. A delineation of the responsibilities of the faculty or academic supervisor, the provider and the student.
 - iii. A delineation of the placement of the internship in the curriculum or program of the student.

- (c) Prepares official agreements that clearly specify the responsibilities of AURAK, its student and the employer during the Internship, obtains the signature of all parties to the agreement in advance of the placement of students within an organization and retains a copy on file.
- (d) Organizes an Internship for each undergraduate student of a duration that is compliant with the relevant accrediting body(ies) for the program in which the student is enrolled.
- (e) Provides an orientation for students as to the skills, behaviors and knowledge which they need to succeed in the work environment.
- (f) Has an effective process in place to monitor the employment site and gather feedback from students and the provider.

Student Petitions and Appeals

Students' Responsibility

All official university communications are distributed through the AURAK-issued email address. These are considered official notifications. Students are responsible for checking their AURAK email accounts and for acting upon messages accordingly.

Students should keep their own records of all transactions with the university (e.g., registration schedules and forms, grade reports, payment records, etc.). It is also advisable to keep copies of all tests, digital files, papers, and so forth submitted in fulfillment of course work. Students should keep copies of all course syllabi.

Appeal of Academic-Related Issues

Academic grievances include grievances students may have against a faculty member, procedures implemented in class, or related academic issues. Note that grade appeals follow a different procedure and are described elsewhere in the Student Handbook and the AURAK catalog. For information regarding grievances please contact the Associate Provost of Student Affairs.

- The grievant must file a formal, written complaint with the VPAASS/Provost.
- The VPAASS/Provost will ascertain if the complaint falls within the scope of an action appropriate for a grievance and will attempt to resolve the complaint informally before instituting the official grievance process.
- If such action is deemed appropriate, the VPAASS/Provost will convene a committee of a minimum of three people including faculty, administration, and student representative(s) within three working days.
- The committee will review the complaint, interview the complainant and any witnesses cited in the complaint.
- The committee will recommend a cause of action to the VPAASS/Provost within seven working days.
- The VPAASS/Provost will forward to the student the recommendation of the Committee or a modification of its recommendation deemed appropriate after consultation with the committee within three working days.
- If the student does not agree with the decision, the student has the right to file an appeal with the President. The President will review the decision and render a final decision in the matter within three working days. The President is the final authority to whom appeal may be made.

- If a grievance is with the VPAASS/Provost, then the student may submit the grievance in writing to the President to initiate the process.

Graduation

Undergraduate Graduation Requirements

The graduation requirements for any individual student are normally determined by the catalog that was effective when the student began matriculated studies in the degree program. Every individual student is personally responsible for meeting all graduation requirements as detailed in his or her catalog year.

If a required course within an undergraduate program changes its number of credits, then the number of credits required by the program for graduation may, at the discretion of the school, change by the same amount provided the minimum total number of credits for graduation is 120 and the CGPA is at least 2.00. In case of substantial changes, in course offerings, equivalent graduation requirements are determined by the dean of the student's school.

Important: Course information, content, and prerequisites may be subject to change as a result of the university's commitment to a process of continual improvement in academic programs. Students must comply with the most up-to-date course requirements.

Academic Standing Requirement

A student must be in good academic standing to be eligible for graduation.

Application for Graduation

Candidates for undergraduate degrees must file an Application for Graduation form with the Office of the Registrar during the registration period of the last expected term of study. Only after an Application for Graduation form has been filed can the Office of the Registrar begin processing the necessary information for final certification for graduation.

Students who fail to complete all degree requirements by the end of the term for which they apply to graduate need not reapply for graduation. Their previous application will be automatically moved to the following semester.

Conferral of Degrees

Only students who have successfully completed degree requirements by the end of the term for which they have applied to graduate qualify for conferral of a degree. Degrees are conferred at the end of the semester in which requirements have been met. Conferral of the degree is noted on the academic transcript of the graduate with the date of graduation.

Names on Degrees

The names of AURAK students will be spelled in English exactly as they appear on the student's passport or identity card when printed on degrees. If a name on a passport or an identity card does not appear in English, then the spelling of the name will be printed according to the personal preference of the student.

Attestation of Degrees and Transcripts

The Office of the Registrar provides information relevant to the attestation of degrees and transcripts to the UAE Ministry of Education.

ACADEMIC PROGRAMS

A degree program—the major or field—is a program of study that normally requires at least thirty credits of coursework in a specific field. The degree and major, or field, appear on the diploma. The Bachelor of Science degree programs at AURAK are professionally oriented, and all require the completion of a minimum of 120 credits, approximately forty courses, including General Education courses. Some programs require additional coursework, and all require an internship.

Degrees normally take four years to complete with a minimum of three years, assuming summer enrollment. If a degree is not completed within six years, all coursework in the major is re-evaluated for its current relevance. The Bachelor of Architecture degree is a five-year program.

AURAK currently offers the following eleven undergraduate majors:

- Bachelor of Architecture
- Bachelor of Arts in Mass Communication
- Bachelor of Science in Artificial Intelligence
- Bachelor of Science in Biotechnology
- Bachelor of Science in Business Administration
 - Major in Business Administration
 - Major in Accounting
 - Major in Business Analytics
 - Major Finance
 - Major in Hospitality and Tourism Management
 - Major in Human Resource Management
 - Major in Marketing
- Bachelor of Science in Chemical Engineering
- Bachelor of Science in Civil and Infrastructure Engineering
- Bachelor of Science in Computer Engineering
- Bachelor of Science in Computer Science
- Bachelor of Science in Electrical and Electronics Engineering

- Bachelor of Science in Mechanical Engineering

AURAK Allocates Academic Credit According to the Following Criteria

AURAK follows the standard US calibration of academic credit, such that 1 semester credit hour implies 15 hours of class time per semester. Two (2) semester credit hours of credit implies 30 hours of class time. Three (3) semester credit hours implies 45 hours of class time. Four (4) semester credit hours implies 60 hours of class time. The class meetings are typically distributed evenly across a 15-week semester in fall and spring, excluding vacation days. Summer sessions and accelerated (short) sessions/courses vary in length.

Each course has a credit value. A credit represents the in-class instruction and out-of-class study per week during the 15-week semester. Normally one credit represents 50 minutes of class instruction per week per semester.

- Lectures carry one academic credit for 50 minutes of class instruction per week for 15 weeks.
- Laboratories are usually 2.5 contact hours per week and carry one credit.
- Courses in drawing and design, currently taught in the Bachelor of Architecture, carry one academic credit for three contact hours in class.
- Independent Study courses carry 1-4 credits, depending on the amount of work required for the specific project as determined by the supervisor of the project.
- Internships carry 3 credits for a full-time experience.

Degree Completion Requirements

- Successful completion of all credit hours as specified by the program curriculum with a minimum CGPA of 2.0
- Completion of the minimum enrollment period and not exceeding the maximum enrollment period
- Transfer or concurrently enrolled students are required to complete a minimum of fifty percent of the required credit hours at AURAK including the majority of the final thirty credit hours (please refer to this catalog for additional information on transfer credit policies)

Enrollment Periods

Degrees normally take four years to complete with a minimum of three years, assuming summer enrollment. At AURAK the degree must be completed within a specified period in order to remain current and the maxima for programs is as follows:

- Programs requiring up to 140 credits must be completed within seven (7) years from initial registration (inclusive of periods of non-enrollment).
- Programs requiring more than 140 credits must be completed within eight (8) years from initial registration (inclusive of periods of non-enrollment).

AURAK will permit an appeal for extenuating circumstances to extend the period by up to two additional semesters.

GENERAL EDUCATION COMPONENT

Introduction

The General Education courses required of each baccalaureate degree at AURAK are based on a coherent rationale. Courses offered to meet the General Education Component requirements are designed to provide students with the foundational knowledge necessary to engage in higher order thinking, analysis of information, and problem-solving. These courses are intended to cultivate an informed, capable, and progressive citizenry. All courses included in the General Education Component are university-level courses and emphasize experiential and activity-based learning. Specifically, the General Education Component examines the application of knowledge in concrete situations to foster critical thinking skills, communication, teamwork, empirical and quantitative skills, and personal and social responsibility. While the General Education Component is not a degree program, all undergraduate degree students must complete the University's General Education Component requirements as a part of their academic program of study.

Mission

The General Education Component is the fundamental constituent of all degree-granting programs. It provides students with fundamental knowledge that broadens their acquaintance with disciplines. As the component providing transferable skills of cortical thinking, numeracy, and synthesis, the General Education Component supports the University mission by preparing students to face the challenges of a globalized, multicultural, and rapidly changing world. The General Education Component complements the students' major program of study by examining achievements in areas of human inquiry and creative endeavor to cultivate those abilities so essential for the acquisition of knowledge.

Purpose

The purpose of the General Education Component is to explore broad disciplinary areas in order to expand students' perspectives and improve their skills in analytical and critical thinking. The General Education Component is intended to facilitate a well-rounded and informed educational experience established on a holistic awareness of a broad spectrum of knowledge areas to ensure that students are exposed to a wide range of disciplines and topics. The General Education Component is designed to provide a relevant and challenging educational experience.

Completion of General Education Requirements

AURAK's General Education Component embodies the vision, mission, and purpose of the University. The courses are required of all graduates as the intellectual foundation upon which AURAK's educational framework is built. They are an essential element in the University's educational process and lay the foundation for professional success, personal growth, and responsible citizenship. Students have the opportunity to acquire knowledge, learn communication skills, and examine their goals, values, and potentialities. Students are encouraged to complete their General Education Component requirements by the end of their second year of study.

Opportunities for Students to Develop and Demonstrate Competencies in Four Areas

The General Education Component provides opportunities for students to develop and demonstrate competencies in the following four areas:

Critical Thinking Skills: including reasoning, inquiry, reflection, creativity, and innovation, and the ability to analyze and synthesize information.

Communication and Teamwork: including the development, interpretation, and expression of ideas, the ability to consider and integrate multiple points of views and to work with others on shared objectives.

Empirical and Quantitative Skills: including the ability to derive knowledge from observable facts, the manipulation and analysis of data, resulting in informed, verifiable solutions and conclusions. Effective understanding of technological concepts and the appropriate use of technology as a problem-solving tool.

Personal and Social Responsibility: including the ability to consider ethical and moral aspects of actions and consequences, intercultural awareness, and knowledge of regional, national, and global responsibility.

General Education Component Learning Outcomes

Upon completion of the General Education Component requirements, completers will be able to:

Learning Outcome One: Solve abstract, familiar, and non-routine problems without assistance using appropriate information drawn from relevant fields of work or disciplines.

Learning Outcome Two: Present collaborative group analyses of topics that include complex ideas and integrate multiple point of view.

Learning Outcome Three: Analyze empirical data using appropriate information retrieval and analysis methods, tools and techniques to arrive at informed, verifiable solutions to abstract, familiar, and non-routine problems.

Learning Outcome Four: Articulate the ethical and moral aspects of possible courses of action in particular situations and explain the consequences that might follow with reference to intercultural issues at regional, national, and global levels.

Required Pathway to Completion of General Education Component Learning Outcomes:

AURAK ensures that graduates acquire a holistic awareness of a broad range of knowledge areas. The General Education Component addresses these needs in a diverse, connect, globalized environment through courses that are not only broad and deep in content but use up-to-date pedagogical methods. The diverse range of topics in the General Education Component, the interdisciplinary and globalized focus of the content, the pedagogical shift to experiential and activity-based learning, the focus on research, innovation and enhanced aesthetic sensitivity all combine to aid in the development of responsible, self-directed individuals who will engage in life-long learning.

The General Education Component at AURAK fulfills the requirements of a liberal arts and sciences education by providing a comprehensive foundation in the competencies that enable students to grow academically as well as professionally. To achieve the General Education Component learning outcomes, students must complete 31-33 semester credit hours consisting of:

- Orientation Courses (O) – 13-14 semester credit hours
- Humanities/Fine Arts (H) – minimum of 6 semester credit hours
- Social and Behavioral Sciences (S) – minimum of 6 semester credit hours
- Natural Sciences/Mathematics (N) – minimum of 6 semester credit hours

AURAK distinguishes between categories of **Orientation Courses** (O) that provide a coherent foundation and academic orientation, and **Knowledge Domains** that offer students opportunities to develop and pursue a variety of academic interests. The Knowledge Domains are delineated into **Humanities/Fine Arts** (H), **Social and Behavioral Sciences** (S), and **Natural Sciences/Mathematics** (N). In addition, the General Education Component requires that of the courses taken to fulfil General Education Component requirements, five (5)

courses must qualify as **Writing Intensive** courses. The codes “O”, “H”, “S”, and “N” identify the category to which a course belongs.

All AURAK undergraduate students must complete all General Education Component requirements to graduate. Completion of these requirements will amount to a total of 31-33 credit hours. All students must complete the Orientation Course requirements (amounting to 13-14 credits), and a minimum of 18 credits from the three remaining Knowledge Domains (amounting to a 18-19 credits). The courses for each of these are as follows.

ORIENTATION COURSES (13-14 semester credit hours required)

Students must complete from 13 to 14 semester credit hours from the following list of courses.

Orientation Courses		
Required Courses	Title	Semester Credit Hours
ENGL 101 (O)	Composition (Writing Intensive)	3
UNIV 100 (O)	University First-Year Transition	1
UNIV 200 (O)	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
ITEC 103 (O) or CSCI 112/113 (O) or CSCI 114/115 (O)	Fundamentals of Information Technology Introduction to Computer Programming Applied Computational Thinking/ Applied Computational Tools	3-4
ARAB 101 (O) or ARAB 110 (O)	Arabic Language and Culture for Non-Native Learners I or Arabic Language and Culture for Native Arabic Speakers I	3
Subtotal =		13 to 14 Credit Hours

KNOWLEDGE DOMAIN (18-19 semester credit hours required)

Students must complete a minimum of 18 semester credit hours in this area. The requirements are divided into the following three categories: 1) Humanities/Fine Arts (H), 2) Social and Behavioral Sciences (S), and 3) Natural Sciences/Mathematics (N). Expectations are outlined below.

Humanities/Fine Arts		
Required Courses	Title	Semester Credit Hours
PHIL 100 (H) OR ENGL 200 (H)	Critical Thinking and Reasoning (Writing Intensive) Advanced Composition (Writing Intensive)	3
MEST 100 (H)	Introduction to Islam in World Culture (Writing Intensive)	3
Subtotal =		6 Credit Hours
Social and Behavioral Sciences		

Required Courses	Title	Semester Credit Hours
UAES 200 (S)	Survey of United Arab Emirates Studies (Writing Intensive)	3
PSYC 100 (S) ECON 103 (S) POLI 100 (S) POLI 101 (S) COMM 101 (S)	Introduction to Psychology Principles of Microeconomics Contemporary Global Issues Politics of Scarcity Interpersonal Communication and Group Interaction	3
Subtotal =		6 Credit Hours
Natural Sciences/Mathematics		
Required Courses	Title	Credit Hours
MATH 101 (N) STAT 100 (N) MATH 108 (N) MATH 111 (N) MATH 113 (N)	Numbers and Data Interpretation Statistics Calculus with Business Applications Calculus with Life Sciences Applications Calculus I	3
BIOL 100 (N) CHEM 100/101 (N) CHEM 211/212 (N) ENVS 102 (N)	Humankind in a Biological World Chemistry in Everyday Life General Chemistry I Sustainability and Human-Environment Relations	3-4
Subtotal =		6 to 7 Credit Hours
TOTAL = 31 to 33 Credit Hours		

Academic programs are encouraged to include additional courses focused on promoting creative and aesthetic understanding from *COMM 102 Reading Image and Film* and *ENGL 102 Public Speaking*. Academic programs are also encouraged to include additional courses focused on promoting cross-cultural and ethical understanding from *HIST 100 Contemporary Middle Eastern History*, *HIST 101 Ancient History of the Arabian Peninsula*, *PHIL 101 Ethics in Today's World*, and *PHIL 102 World Philosophies*.

Writing Intensive Courses

To promote students' achievement of the Written Communication competency, the University's General Education Component also requires that students complete five writing-intensive courses. One of these writing-intensive courses is ENGL 101. Of the remaining four required writing-intensive courses, three must be other writing-intensive General Education courses, and the fifth writing-intensive course must be a course in the major program of study. Programs are encouraged to require students to take more than the university minimum of five writing-intensive courses.

The criteria for writing-intensive courses are provided below:

- The course includes a sustained focus on individual student writing, as demonstrated in the syllabus through multiple writing assignments.
- Students receive actionable feedback on their writing from the course instructor.

- Students substantially revise and resubmit written drafts, incorporating feedback from the course instructor and using their own critical thought to improve their writing.
- The final versions of written assignments total to at least 14 double-spaced, printed pages.

To be designated as “Writing-Intensive,” courses must be approved by the General Education Committee as satisfying the criteria listed above. Courses that have been approved as Writing-Intensive are clearly designated as such in the catalog, on transcripts, and in study plans for all programs.

The General Education Component Requirements and Course Substitutions

The degree requirements for the individual undergraduate programs outline the course options from which a student may select courses to meet the General Education requirements. Other General Education courses that fulfill the requirements listed in the General Education Component may be substituted to complete respective General Education requirements in accordance with the established Course Substitution Policy and Procedure. Although students are encouraged to identify and select courses that meet their needs, faculty advisors must approve these course selections to ensure that they satisfy General Education Component requirements.



SCHOOL OF ARTS & SCIENCES

SCHOOL OF ARTS AND SCIENCES

Bachelor of Arts in Mass Communication

The Bachelor of Arts degree in Mass Communication provides students with a high-quality education in line with emerging market trends in the media industry within the UAE, the Middle East, and the world. Rapid advancements in the field of communication and new media have created demand for qualified professionals and leaders who possess the knowledge to address global issues and, by so doing, shape the future of the media industry.

The Bachelor of Arts in Mass Communication is overseen by the School of Arts and Sciences and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Arts in Mass Communication with concentrations in Digital Media or Public Relations will produce competent graduates with the ability to apply both knowledge and skills of their area of concentration to engage in research to benefit the community and the nation, to ensure quality in the learners' professional lives.

Program Educational Objectives

The goals and objectives of the Bachelor of Arts in Mass Communication are simultaneously aligned with the mission statement of the university and seek to achieve the objectives embedded in the university mission.

The Goals of the Mass Communication Program are to:

1. Provide the student with a broad and deep knowledge about the study of communication, the similarities and differences between communication delivery modes, including their structure, processes, an ethical practice thereof, and the significance in social and professional uses;
2. Foster an understanding of the interdisciplinary nature and the cultural, ethical, and historical contexts of human communication;
3. Enhance the student's proficiency in research skills, the use of technology and media, critical thinking, creativity, and analytical abilities to pursue knowledge independently in the pursuit of advanced higher education;
4. Produce graduates who possess advanced practical and interpersonal skills that are grounded in truth and ethically practiced who demonstrate the capacity to establish and evaluate communication relationships in order to lead and serve the community; and
5. Produce responsible and competent professionals with practical skills, social commitment, and ethical standards that satisfy the demands of employment in the UAE, the region, and the world.

Program Learning Outcomes

AURAK Mass Communication graduates should demonstrate the ability to:

1. Explain communication theory and demonstrate applications of relevant theoretical concepts for different mass communication fields;
2. Recognize key skills and employ contemporary tools for communicating effectively in realistic media environments;
3. Utilize appropriate messages and techniques for communicating effectively;
4. Apply research methodologies and models and implement these in communication work;

5. Identify factors characterizing the global nature of modern media systems;
6. Distinguish cultural issues in planning various forms of communication; and
7. Analyze principles of ethics, fairness, and regulation as they relate to media practices.

Goals for the Digital Media Concentration

The goals of the Digital Media Concentration are to:

1. Provide a strong foundation in the theoretical principles, approaches, techniques, and practices of communication through digital media;
2. Produce graduates that have proficiency in communication and communication delivery modes that is applicable in diverse contexts;
3. Enhance the student's proficiency in research skills by developing their critical thinking skills, creativity, analytical abilities, and their ability to work with communication technology and communication media; and
4. Produce responsible professionals competent in communication in digital media who satisfy the demands of employment and the needs of the community in the UAE and the wider region.

Program Learning Outcomes for the Digital Media Concentration

Upon completing the Bachelor of Arts in Mass Communication with a concentration in Digital Media, graduates will demonstrate the ability to:

1. Employ convergent technologies for production and expression of communication pieces for news, documentaries, entertainment, and persuasive communication; and
2. Apply latest trends in multimedia practices and uses for professional digital communication to a local or international context.

Goals for the Public Relations Concentration

The goals of the Public Relations Concentration are to:

1. Foster an understanding of the interdisciplinary nature and the historical and cultural contexts of human communication in English and in another modern language in all public relations roles;
2. Enhance the learner's proficiency in research skills, the use of technology and the media, critical thinking, creativity, and analytical abilities to pursue public relations communication knowledge independently in advanced higher education;
3. Demonstrate how contemporary public relation approaches and strategies may be used to improve communication with the audience in both public and private spheres;
4. Listen to and to speak, and to read and write public relations texts thoughtfully and to demonstrate an ability to support any communication with truthful and honest responses that increase the trust of the audience; and
5. Write effective public relations texts and to create public relations in media.

Program Learning Outcomes for the Public Relations Concentration

Upon completing the program in Mass Communication with a concentration in Public Relations, graduates will demonstrate the ability to:

1. Discuss the characteristics of unique publics and design strategies to best establish symmetrical relationships with them; and
2. Summarize modern principles of public relations practices and analyze how these can be used in the UAE and globally.

Degree Requirements

To earn a BA degree in Mass Communication, students must satisfactorily complete at least 120 approved credits, fulfill all the requirements for the BA in Mass Communication degree, and achieve a GPA of 2.00 or higher.

Thirty-one credit hours from approved courses are required to fulfill the general education requirements. Thirty-five credit hours from the program's core courses are similarly required. The BA in Mass Communication offers concentrations in Digital Media and Public Relations; these courses are offered in the second, third, and fourth years of study. Students complete eighteen credits in their concentration, an additional fifteen credits through the completion of concentration electives, and then fifteen credits in free electives. After completing 90 credits, students are required to complete an internship (3 credit hours) which is required for their concentration (Digital Media or Public Relations).

Students also take an Arabic for Media course. This three credit-hour course is offered in two versions: one for native-Arabic Speakers and a one for non-native Arabic speakers. No native Arabic speaker is permitted to take Arabic courses that are designed for non-native learners.

The BA degree in Mass Communication is designed to be completed in four years, assuming students do not interrupt their study. Students who withdraw or take a leave of absence from the program must meet requirements for returning that are outlined in the AURAK Catalog. Students must meet specific standards to progress, as well as the maximum time allowed to complete the program, which are also detailed in the catalog. If a degree is not completed within a period of six years, all coursework in the major will be re-evaluated for its current relevance.

BA MASS COMMUNICATION DEGREE COMPLETION REQUIREMENTS

The Bachelor of Arts in Mass Communications degree requires the completion of the following requirements:

Requirement	Credits
I. University General Education Requirements	31
II. Program Core Course Requirements	35
III. Concentration Course Requirements: Digital Media or Public Relations	18
IV. Mass Communication Electives	15
V. Free Electives	15
VI. Arabic for Media (native speaking or non-native speaking)	3
VII. Internship	3
Total	120

University General Education Requirements

31 Credits

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education

Component.” You must speak with your advisor to ensure that the General Education Component requirements are satisfied. ***The fifth writing intensive course for the BA in Mass Communication is COMM 450 Selected Topics in Communication.***

Course Code	Title	Credits
ARAB 101 OR ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
ITEC 103	Fundamentals of Information Technology	3
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive Course)	3
PHIL 100 OR ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
Gen Ed Course	Social and Behavioral Science Course	3
Gen Ed Course	Mathematics Course	3
Gen Ed Course	Natural Science Course	3

Program Core Course Requirements

35 Credits

Course No.	Title	No. of Credits
COMM 111	Introduction to Mass Communication	3
COMM 112	Introduction to Public Relations	3
COMM 113	Introduction to Digital Media	3
COMM 212	Media Writing	3
COMM 215	Feature Writing	3
COMM 222	Intercultural Mass Communication	3
COMM 223	Globalization and Media Culture	3
COMM 311	Media Law and Ethics	3
COMM 321	Theories of Media Communication	3
COMM 391	Communication Research Methodology	2
COMM 492	Mass Media and Public Opinion	3
COMM 450	Selected Topics in Communication (Writing Intensive)	3

Concentration Course Requirements

Concentration in Digital Media

18 Credits

Course No.	Title	No. of Credits
COMM 214	New Media and Digital Culture	3
COMM 224	Visual Storytelling	3
COMM 323	News Reporting	3
COMM 334	Broadcast Journalism	3
COMM 423	Interactive Multimedia	3

COMM 424	Capstone: Multimedia Storytelling	3
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Concentration in Public Relations

18 Credits

Course No.	Title	No. of Credits
COMM 213	Public Relations Writing	3
COMM 334	Broadcast Journalism	3
COMM 337	Public Relations Cases	3
COMM 344	Public Relations & Event Management	3
COMM 421	Social Media: Audiences and Messages	3
COMM 427	Capstone: Public Relations Campaigns	3

Mass Communications Electives

15 Credits

Course No.	Title	No. of Credits
COMM 221	Communication Analysis and Criticism	3
COMM 225	Visual Communication and Digital Photography	3
COMM 226	Graphic Communication Design and Print Production	3
COMM 227	Film and Video Editing	3
COMM 322	Digital Resources and Content	3
COMM 422	Streaming Media, Podcasting and Radio Production	3
COMM 425	Media Management and Entrepreneurship	3

* Note: Students may choose one course from the other concentration as an elective course.

Modern Language Competency Requirement

3 Credits

Arabic for Non-Native Arabic Learners

Course No.	Title	Prerequisites and Corequisites	No. of Credits
ARAB 211	Arabic for Media: Non-Native Speakers	ARAB 101	3

Arabic for Native Arabic Learners

Course No.	Title	Prerequisites and Corequisites	No. of Credits
ARAB 210	Arabic for Media: Native Speakers	ARAB 110	3

Free Electives

15 Credits

Internship

3 Credits

Course No.	Title	No. of Credits
COMM 390	Internship	3

TOTAL PROGRAM COURSE REQUIREMENTS = 120 CREDIT HOURS

B.A. Degree in Mass Communication with Concentration in Digital Media (Four-Year Schedule)

Year 1 Semester 1			Year 1 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 111	Introduction to Mass Communication	3	COMM 112	Introduction to Public Relations	3
ENGL 101	Composition	3	COMM 113	Introduction to Digital Media	3
ITEC 103	Fundamentals of Information Technology	3	MEST 100	Introduction to Islam in World Culture	3
UNIV 100	University First-Year Transition	1	UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3	PHIL 100 or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
Gen Ed	Social and Behavioral Science	3			
Subtotal =		16	Subtotal =		15
Year 2 Semester 1			Year 2 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COM 212	Media Writing	3	COMM 215	Feature Writing	3
COMM 222	Intercultural Mass Communication	3	COMM 223	Globalization and Media Culture	3
COMM 214	New Media and Digital Culture	3	COMM 224	Visual Storytelling	3
UAES 200	Survey of United Arab Emirates Studies	3		Mass Communication Elective	3
Gen Ed	Mathematics Course	3	ARAB 210 OR ARAB 211	Arabic for Media: Native Speakers OR Arabic for Media: Non-Native Speakers	3
Subtotal =		15	Subtotal =		15
Year 3 Semester 1			Year 3 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 311	Media Law and Ethics	3	COMM 391	Communication Research Methodology	2
COMM 321	Theories of Media Communication	3	COMM 334	Broadcast Journalism	3
COMM 323	News Reporting	3		Mass Communication Elective	3
Gen Ed	Natural Science Course	3		Free Elective	3
	Mass Communication Elective	3		Free Elective	3
Subtotal =		15	Subtotal =		14
Year Three, Summer Semester					
Course No.	Description	CR			
COMM 390	Internship for Mass Communication	3			
Year 4 Semester 1			Year 4 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 492	Mass Media and Public Opinion	3	COMM 424	Capstone: Multimedia Storytelling	3
COMM 423	Interactive Multimedia	3		Free Elective	3
COMM 450	Selected Topics in Communication (Writing Intensive)	3		Free Elective	3
	Mass Communication Elective	3		Mass Communication Elective	3
	Free Elective	3			
Subtotal =		15	Subtotal =		12
TOTAL CREDIT HOURS 120					

**B.A. Degree in Mass Communication with Concentration in Public Relations
(Four-Year Schedule)**

Year 1 Semester 1			Year 1 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 111	Introduction to Mass Communication	3	COMM 112	Introduction to Public Relations	3
ENGL 101	Composition	3	COMM 113	Introduction to Digital Media	3
ITEC 103	Fundamentals of Information Technology	3	MEST 100	Introduction to Islam in World Culture	3
UNIV 100	University First-Year Transition	1	UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
ARAB 101 OR ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3	PHIL 100 OR ENGL 200	Critical Thinking and Reasoning OR Advanced Composition	3
Gen Ed	Social and Behavioral Science	3			
Subtotal =		16	Subtotal =		15
Year 2 Semester 1			Year 2 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 212	Media Writing	3	COMM 215	Feature Writing	3
COMM 222	Intercultural Mass Communication	3	COMM 223	Globalization and Media Culture	3
	Mass Communication Elective	3	COMM 213	Public Relations Writing	3
UAES 200	Survey of United Arab Emirates Studies	3		Mass Communication Elective	3
Gen Ed	Mathematics Course	3	ARAB 210 OR ARAB 211	Arabic for Media: Native Speakers OR Arabic for Media: Non-Native speakers	3
Subtotal =		15	Subtotal =		15
Year 3 Semester 1			Year 3 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 311	Media Law and Ethics	3	COMM 391	Communication Research Methodology	2
COMM 321	Theories of Media Communication	3	COMM 334	Broadcast Journalism	3
COMM 337	Public Relations Cases	3	COMM 344	Public Relations & Event Management	3
Gen Ed	Natural Science Course	3		Free Elective	3
	Mass Communication Elective	3		Free Elective	3
Subtotal =		15	Subtotal =		14
Year Three, Summer Semester					
Course No.	Description	CR			
COMM 390	Internship for Mass Communication	3			
Year 4 Semester 1			Year 4 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 492	Mass Media and Public Opinion	3	COMM 427	Capstone: Public Relations Campaigns	3
COMM 421	Social Media: Audiences and Message	3		Free Elective	3
COMM 450	Selected Topics in Communication (Writing Intensive)	3		Free Elective	3
	Mass Communication Elective	3		Mass Communication Elective	3
	Free Elective	3			
Subtotal =		15	Subtotal =		12
TOTAL CREDIT HOURS 120					

Bachelor of Science in Biotechnology

Biotechnology is the modern face of the biological sciences. It applies technological advances, particularly those involving the sciences of molecular genetics, recombinant DNA techniques, and molecular diagnostics to solve biological problems through the production of materials, processes, and services to benefit our environment.

In medicine, Biotechnology aids in the manufacture of therapeutics, enzymes, antibiotics, vitamins, and vaccines. In agriculture, Biotechnology helps improve food quality, quantity, and processing. Additionally, Biotechnology has applications in green manufacturing technologies. It minimizes contamination and pollution and locates alternative energy sources. In the past decade, industry has increasingly turned to biotechnology to improve product quality and to make the production process easier, cleaner, and more cost-effective.

Biotechnology graduates work in fields such as molecular biology, genetics, forensics, plant and animal sciences, food production, therapeutics, agriculture, and environmental sciences.

The Bachelor of Science in Biotechnology is overseen by the School of Arts and Sciences and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Science in Biotechnology Program, with separate concentrations in Cell and Molecular Biotechnology, Medical Biotechnology, and Forensic Biology, aims to produce skilled and knowledgeable biotechnologists who are capable of applying their acquired knowledge to scientific research and development in medicine, science, agriculture, forensics and the environment for the benefit of their societies, while maintaining the highest standards of professional ethics in their careers.

Program Educational Objectives

The goals of the Biotechnology Program are to:

1. Supply society with biologically literate citizens who are capable of advising and making decisions;
2. Produce ambitious, creative graduates who are interested in continuing their education in the biosciences;
3. Practice scientific inquiry and appreciate its role in the development of research, science, technology, and society;
4. Produce responsible biotechnology professionals to fulfill the employment and research needs in the biotechnology industry in the UAE and the region;
5. Contribute to the advancement of agriculture, medicine, and environmental sciences through the application of biotechnology theory and recombinant DNA technology; and
6. Enhance the students' ability to integrate their acquired math, computer, and bioscience knowledge and skills to investigate and solve biological problems.

Program Learning Outcomes

On completion of the program graduates will be able to:

1. **PLO1:** Identify, formulate, and solve broadly defined technical or scientific problems by applying knowledge of mathematics and science and/or technical topics to areas relevant to the discipline.
2. **PLO2:** Formulate or design a system, process, procedure or program to meet desired needs.

3. **PLO3:** Develop and conduct experiments or test hypotheses, analyze and interpret data and use scientific judgment to draw conclusions.
4. **PLO4:** Communicate effectively with a range of audiences.
5. **PLO5:** Discuss ethical and professional responsibilities and the impact of technical and/or scientific solutions in global, economic, environmental, and societal contexts.
6. **PLO6:** Function effectively on teams that establish goals, plan tasks, meet deadlines, and analyze risk and uncertainty.

Program Concentration Learning Outcomes – Cell and Molecular Biotechnology

7. **PLO7:** Lead and implement appropriate research strategies to produce verifiable data in Cell and Molecular Biotechnology.
8. **PLO8:** Evaluate and manage complex and unpredictable work procedures to be able to function successfully within the Cell and Molecular Biotechnology graduate programs and employment sectors.

Program Concentration Learning Outcomes – Medical Biotechnology

7. **PLO7:** Lead and implement appropriate research strategies to produce verifiable data in Medical Biotechnology.
8. **PLO8:** Evaluate and manage complex and unpredictable work procedures to be able to function successfully within the Medical Biotechnology graduate programs and employment sectors.

Program Concentration Learning Outcomes – Forensic Biology

7. **PLO7:** Lead and implement appropriate research strategies to produce verifiable data in Forensic Biology.
8. **PLO8:** Evaluate and manage complex and unpredictable work procedures to be able to function successfully within the Forensic Biology graduate programs and employment sectors.

Degree Requirements

The Bachelor of Science in Biotechnology degree is a four-year degree program with concentration in Cell and Molecular Biotechnology, Medical Biotechnology, and Forensic Biology. For the first three years of the Bachelor of Science in Biotechnology program, students follow a common sequence of courses, with courses specific to the respective concentration taken in the fourth year of the program. To earn a bachelor's degree, students must satisfactorily complete at least 128 approved credits, fulfill all the requirements for the Bachelor of Science in Biotechnology degree, and achieve a GPA of 2.00 or higher. Students must apply for at least forty-eight credits of upper-level courses (designated 300- level or higher) toward graduation requirements. Thirty-one to Thirty-four credit hours from approved courses are required to fulfill the General Education Program requirements. The degree is designed to be completed in four years, assuming students do not interrupt their study. Students who withdraw or take a leave of absence from the program must meet requirements for returning that are outlined in AURAK catalog. Students are required to meet specific standards to progress, as well as the maximum time allowed to complete the program, which are also detailed in the catalog. If a degree is not completed within a period of six years, all coursework in the major will be re-evaluated for its current relevance.

The Bachelor of Science in Biotechnology degree necessitates the completion of the following requirements:

Requirements	Credits
University General Education Requirements	31
Biotechnology Program Core Courses	63
Biotechnology Program Elective courses	7
Concentration (Cell and Molecular Biotechnology, Medical Biotechnology, or Forensic Biology) Course Requirements	21
Free Electives	6
TOTAL	128

University General Education Requirements

31 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, “General Education Component.” You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The 5th writing intensive course for the BS in Biotechnology is BIOL 450 Bioethics.**

Course Code	Title	Credits
ENGL 101	Composition (Writing Intensive)	3
UNIV 100	University First-Year Transition	1
ITEC 103	Fundamentals of Information Technology	3
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 111	Calculus for Life Sciences	3
Gen Ed	General Education Social Sciences Course	3
BIOL 100	Humankind in a Biological World	3

Biotechnology Program Core Courses

63 Credit Hours

Course Code	Title	No. of Credits
BIOL 112	University Biology I	3
BIOL 113	University Biology I Lab	1
BIOL 114	University Biology II	3
BIOL 115	University Biology II Lab	1
BIOL 230	General Microbiology	3
BIOL 231	General Microbiology Lab	1

BIOL 270	General Genetics	3
BIOL 271	General Genetics Lab	1
BIOL 330	Applied and Industrial Microbiology	3
BIOL 331	Techniques in Applied and Industrial Microbiology	1
BIOL 250	Biochemistry I	3
BIOL 251	Biochemistry I Lab	1
BIOL 350	Biochemistry II	3
BIOL 356	Virology	3
BIOL 380	Biotechnology and Genetic Engineering	3
BIOL 381	Biotechnology Lab Methods and Techniques	1
BIOL 450	Bioethics (<i>Writing Intensive</i>)	3
BIOL 491	Senior Seminar	2
BIOL 490	Research Methods in Biology	3
CHEM 111	Principles of General Chemistry	3
CHEM 112	Principles of General Chemistry Lab	1
CHEM 120	Analytical Chemistry	3
CHEM 121	Analytical Chemistry Laboratory	1
CHEM 215	Organic Chemistry I	3
CHEM 216	General Organic Chemistry Lab I	1
ENVS 102	Sustainability and Human-Environment Relations	3
STAT 100	Introductory Probability and Statistics	3
BIOL 390	Internship	3

Biotechnology Program Elective Courses

7 Credit Hours

Course Code	Title	No. of Credits
BIOL 322	Microbial Genetics	3
BIOL 357	Artificial Intelligence in Biotechnology	3
BIOL 480	Food Biotechnology	3
BIOL 481	Bioprocessing Technology in the Pharmaceutical Industry	3
	Students may choose one course and its corresponding lab from the other concentration as an elective.	4

Concentration Course Requirements**21 Credit Hours****Concentration in Cell and Molecular Biotechnology****21 Credit Hours**

Course Code	Course Title	No. of Credits
BIOL 420	Molecular Biology	3
BIOL 421	Molecular Biology and Recombinant DNA Lab	1
BIOL 425	Advanced Genetics	3
BIOL 430	Cell Culture Theory and Technology	3
BIOL 431	Cell Culture Techniques Lab	1
BIOL 434	Cell Communication and Signal Transduction	3
BIOL 473	Bioinformatics and Computational Biology	3
BIOL 494	Senior Project in Cell and Molecular Biotechnology	4

Concentration in Medical Biotechnology**21 Credit Hours**

Course Code	Course Title	No. of Credits
BIOL 411	Drug Discovery, Design and Development	3
BIOL 435	Stem Cell Biology and Regenerative Medicine	3
BIOL 436	Immunology	3
BIOL 440	Medical Microbiology	3
BIOL 441	Medical Microbiology Lab	1
BIOL 442	Molecular Diagnostics	3
BIOL 443	Molecular Diagnostics Lab	1
BIOL 495	Senior Project in Medical Biotechnology	4

Concentration in Forensic Biology**21 Credit Hours**

Course Code	Course Title	No. of Credits
CHEM 420	Forensic Chemistry	3
CHEM 421	Forensic Chemistry Lab	1
BIOL 423	Advanced Molecular Biology and Forensic Sciences	3
BIOL 424	DNA Fingerprinting and Serology Lab	1
BIOL 462	Crime Scene Investigation	3
BIOL 464	Criminal Justice	3
BIOL 473	Bioinformatics and Computational Biology	3
BIOL 496	Senior Project in Forensic Biology	4

Program Free Electives**6 Credits Hours**

Course Code	Course Title	No. of Credits
	Free Electives	6

B.S. Degree in Biotechnology with Concentrations
Proposed 4 Year Schedule

Year 1 Semester 1			Year 1 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
BIOL 112	University Biology I	3	BIOL 114	University Biology II	3
BIOL 113	University Biology I Lab	1	BIOL 115	University Biology II Lab	1
MATH 111	Calculus for the Life Sciences	3	MEST 100	Introduction to Islam in World Culture	3
ENGL 101	Composition	3	STAT 100	Introductory Probability and Statistics	3
CHEM 111	Principles of General Chemistry	3	CHEM 120	Analytical Chemistry	3
CHEM 112	Principles of General Chemistry Lab	1	CHEM 121	Analytical Chemistry Lab	1
UNIV 100	University First-Year Transition	1	Gen Ed	Social Sciences Course	3
Subtotal =		15	Subtotal =		17
Year 2 Semester 1			Year 2 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
BIOL 230	General Microbiology	3	BIOL 270	General Genetics	3
BIOL 231	General Microbiology Lab	1	BIOL 271	General Genetics Lab	1
UNIV 200	Innovation, Entrepreneurship, & Sustainability	3	BIOL 250	Biochemistry I	3
CHEM 215	Organic Chemistry I	3	BIOL 251	Biochemistry I Lab	1
CHEM 216	General Organic Chemistry I Lab	1	ENVS 102	Sustainability and Human-Environment Relations	3
ITEC 103	Fundamentals of Information Technology	3	PHIL 100 or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
Subtotal =		14	Subtotal =		14
Year 3 Semester 1			Year 3 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
BIOL 350	Biochemistry II	3	UAES 200	Survey of United Arab Emirates Studies	3
BIOL 380	Biotechnology and Genetic Engineering	3	BIOL 330	Applied and Industrial Microbiology	3
BIOL 381	Biotechnology Lab Methods and Techniques	1	BIOL 331	Techniques in Applied and Industrial Microbiology	1
BIOL 356	Virology	3	Program Electives	Biotechnology Elective	4
Elective	Free Elective	3	BIOL 100	Humankind in a Biological World	3
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR	3	Elective	Free Elective	3

	Arabic Language and Culture for Native Arabic Speakers I				
Subtotal =		16	Subtotal =		17
Year 3, Summer Semester					
Course No.	Description	CR			
BIOL 390	Internship	3			
Year 4 with Concentration in Medical Biotechnology					
Year 4 Semester 1			Year 4, Semester 2		
Course No.	Description	CR	Course No.	Description	CR
BIOL 435	Stem Cell Biology and Regenerative Medicine	3	BIOL 440	Medical Microbiology	3
BIOL 442	Molecular Diagnostics	3	BIOL 441	Medical Microbiology Lab	1
BIOL 443	Molecular Diagnostics Lab	1	BIOL 436	Immunology	3
BIOL 490	Research Methods in Biology	3	BIOL 411	Drug Discovery, Design and Development	3
BIOL 450	Bioethics	3	BIOL 495	Senior Project Medical Biotechnology	4
BIOL 491	Senior Seminar	2			
Program Elective	Biotechnology Elective	3			
Subtotal =		18	Subtotal =		14
Year 4 with Concentration in Cell and Molecular Biotechnology					
Year 4 Semester 1			Year 4, Semester 2		
BIOL 425	Advanced Genetics	3	BIOL 420	Molecular Biology	3
BIOL 430	Cell Culture Theory and Technology	3	BIOL 421	Molecular Biology Recombinant DNA Lab	1
BIOL 431	Cell Culture Theory and Technology Lab	1	BIOL 434	Cell Communication and Signal Transduction	3
BIOL 490	Research Methods in Biology	3	BIOL 473	Bioinformatics and Computational Biology	3
BIOL 450	Bioethics	3	BIOL 494	Senior Project Cell and Molecular Biotechnology	4
BIOL 491	Senior Seminar	2			
Program Elective	Biotechnology Elective	3			
Subtotal =		18	Subtotal =		14

Year 4 with Concentration in Forensic Biology					
Year 4 Semester 1			Year 4, Semester 2		

BIOL 423	Advanced Molecular Biology and Forensic Sciences	3	CHEM 420	Forensic Chemistry	3
BIOL 424	DNA Fingerprinting and Serology Lab	1	CHEM 421	Forensic Chemistry Lab	1
BIOL 462	Crime Scene Investigation	3	BIOL 464	Criminal Justice	3
BIOL 490	Research Methods in Biology	3	BIOL 473	Bioinformatics and Computational Biology	3
BIOL 450	Bioethics	3	BIOL 496	Senior Project Forensic Biology	4
BIOL 491	Senior Seminar	2			
Program Elective	Biotechnology Elective	3			
Subtotal =		17	Subtotal =		14
TOTAL CREDIT HOURS 128					



SCHOOL OF BUSINESS

SCHOOL OF BUSINESS

Bachelor of Science in Business Administration

Program Description

AURAK's Bachelor of Science in Business Administration offers a broad foundation in various disciplines, including accounting, finance, marketing, human resources, and operations management.

It provides graduates with essential critical thinking, problem-solving, communication, teamwork, and leadership skills. It is ideal for students interested in pursuing a career in business and who want to develop the skills needed to succeed in today's ever-changing business landscape.

Graduates of the Bachelor of Science in Business Administration program have a wide range of career options, including management, marketing, finance, accounting, human resources, and entrepreneurship positions.

BSBA majors may choose to specialize in accounting, finance, human resource management, marketing, business analytics or hospitality and tourism management by taking seven major required courses and three major electives in one of these areas or choose to be a generalist, by taking a combination of courses from all the six specialist areas.

The Bachelor of Science in Business Administration is overseen by the School of Business and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The mission of the Business Administration Program is to prepare graduates who are knowledgeable, skilled, globally oriented, and ethical for entrance into business, government, other professions, and/or graduate programs.

Program Educational Objectives

The overall aims of the program are to:

1. Develop a breadth of knowledge across business functions
2. Create an understanding of ethical and social responsibility
3. Demonstrate clarity of written and oral communication using appropriate technology
4. Develop teamwork and leadership skills
5. Enable Professional reflection and self-development
6. Develop critical thinking, analytical and problem-solving skills
7. Develop proficiency in a chosen business discipline

Program Learning Outcomes

On completion of the program graduates will be able to demonstrate

PLO1: (CORE) Breadth of Business Knowledge

Critically expand on the basic principles of economics, accounting, finance, management, information systems, marketing and operations in the context of a global economy.

PLO2: (CORE) Ethical Reasoning

Apply concepts and theories of ethics and social responsibility to practical business dilemmas, recognizing the implications of management decisions for the interests of key internal and external stakeholders.

PLO3: (CORE) Communication Skills

Demonstrate competency in presentation and writing skills using the latest business communication tools.

PLO4: (CORE) Teamwork

Effectively work in teams and take the lead in team initiatives.

PLO5: (CORE) Professional Reflection

Engage in critical self-analysis and professional reflection about potential roles in a business context.

PLO6: (CORE) Critical Thinking and Problem Solving

Evaluate business situations and critique managerial decisions, using financial statements, statistical tools, and other appropriate methods to organize, analyze and present data.

Major Learning Outcomes (MLOs): Disciplinary Proficiency

Achievement of the learning outcomes specified for one of the BSBA majors.

Degree Requirements

The Bachelor of Science in Business Administration (BSBA) program includes forty-eight (48) credits of core courses in business administration covering a broad knowledge of business areas and functions, emphasizing global business environment and practices.

BSBA majors may choose to specialize in accounting, finance, human resource management, marketing, business analytics or hospitality and tourism management by taking 21 credits of major required courses and nine credits of major electives in one of these areas.

Additionally, students must fulfill the General Education requirements with thirty-one credits and complete 9 credits of general business electives and a three-credit general electives to complete the 121-credit program.

The Bachelor of Science in Business Administration degree requires the completion of the following degree requirements:

Program Requirements	BSBA Major in Accounting/ Business Analytics/ Finance/ HRM/ Marketing/ Hospitality & Tourism Management	BSBA – Generalist
	No. of Credits	No. of Credits
University General Education Requirements	31	31
School Requirements	48	48
Major Requirements	21	-
Major Elective Requirements	9	-
Business Electives	9	39
Free Electives	3	3
Total	121	121

University General Education Component Requirements**31 Credits**

The General Education Component requires undergraduate students to take 31-34 credits in designated categories. To review the General Education Component requirements, please refer to the section of the catalog entitled, "General Education Component." ***The fifth writing intensive course for the BS in Business Administration is BUSN 301 Business Communication.***

Note that those seeking the BS of Business Administration are required to take ECON 103 Principles of Economics to satisfy the Social and Behavioral Sciences requirements and STAT 100 Statistics to satisfy the mathematics requirement for the Natural Sciences. For further guidance in selecting courses to satisfy the General Education Component requirements, please consult your academic advisor.

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
ITEC 103	Fundamentals of Information Technology	3
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
ECON 103	Principles of Microeconomics	3
STAT 100	Introduction to Probability & Statistics	3
Gen Ed Course	Natural Science Course	3

School of Business Core Course Requirements**48 Credits**

Course Code	Title	Prerequisites	Credits
ACCT 204	Principles of Financial Accounting	ECON 103	3
ACCT 205	Principles of Managerial Accounting	ECON 103	3
BUAN 301	Analytics for Decision Making	OPMT 311	3
BUSN 301	Business Communication (A) (Writing Intensive)	ENGL 101	3
BUSN 304	Business Law	Completion of 58 Cr.	3
BUSN 306	International Business (A)	ECON 104	3
BUSN 307	Research Methodology (A)	Completion of 58 Cr.	3
BUSN 312	Public Policy and Business Ethics (A)	ECON 103	3
BUSN 498	Strategic Management (A)	Completion of 90 Cr.	3
ECON 104	Macroeconomic Policy	ECON 103	3
FNAN 300	Principles of Business Finance	ACCT 204	3
MGMT 301	Principles of Management	Sophomore Standing	3
MGMT 401	Organizational Behavior	Completion of 90 Cr.	3
MIST 301	Introduction to Business Information Systems	Sophomore Standing	3
MKTG 301	Principles of Marketing	ECON 103	3
OPMT 311	Methods and Models of Management Sciences	STAT 100	3

(A) – Applied Course

Major Requirements

21 credits

BSBA Major Specialization

To complete a major, graduates must complete thirty (30) credits in the chosen discipline, twenty-one (21) credits from the major required courses, and nine (9) credits from major elective courses

BSBA (Major in Accounting)

Bachelor of Science in Business Administration (BSBA), major in Accounting, offers students a strong foundation in the theory and practices of the accounting field. This bachelor's degree equips students with the necessary accounting business knowledge and skills to pursue a wide range of careers in the accounting field of their choice. The program is designed to deliver a comprehensive education covering all the essential accounting areas, including financial accounting, financial analysis, managerial accounting, taxation, auditing, and information systems.

Graduates of BSBA (Major in Accounting) will be well-prepared to enter the workforce in various roles in profit-making and non-profit organizations, such as public accounting, corporate accounting, government accounting, and non-profit accounting.

The BSBA Major in Accounting program is accredited by an international professional accountancy body;- the Association of Chartered Certified Accountants (ACCA) in the UK, to ensure that the AURAK graduates are internationally recognized in the accounting profession. This recognition helps them work in different countries in the accounting profession. AURAK BSBA (Major in Accounting) graduates are granted course exemptions while enrolling for ACCA qualification.

Accounting Major Program Learning Outcomes (AMLO)

On completion of a BSBA (Major in Accounting) students will be able to:

AMLO 1: Evaluate, aggregate, and synthesize knowledge relating to the accounting function from diverse sources for planning, controlling and performance evaluation;

AMLO 2: Assess how auditing and fiscal control measures can be implemented in manual and electronic accounting information systems: and

AMLO 3: Prepare, analyze, and interpret financial statements using International Financial Reporting Standards on accounting practice for business organizations and other corporate entities.

BSBA (Major in Accounting) Required Courses

21 Credits

Course Code	Course Title	Credits	Prerequisites
ACCT 311	Managerial and Cost Accounting (A)	3	ACCT 205
ACCT 331	Intermediate Accounting I (A)	3	ACCT 204
ACCT 332	Intermediate Accounting II (A)	3	ACCT 331
ACCT 361	Accounting Information Systems (A)	3	ACCT 331
ACCT 390	Internship in Accounting (A)	3	Completion of 90 Cr.
ACCT 412	Auditing (A)	3	ACCT 331
ACCT 444	Accounting Project Based Internship (A)	3	ACCT 331

(A) – Applied Course

BSBA (Major in Accounting) Elective courses (Any 3 courses to be taken)**9 Credits**

Course Code	Course Title	Credits	Prerequisites
ACCT 321	Governmental and Not-for-Profit Accounting	3	ACCT 204
ACCT 352	Taxation and Managerial Decisions (A)	3	ACCT 331
ACCT 472	International Accounting (A)	3	ACCT 331
ACCT 491	Special Topics in Accounting (A)	3	ACCT 331
FNAN 303	Financial Analysis, Forecasting, and Valuation (A)	3	ACCT 204

(A) – Applied Course

BSBA (Major in Finance)

Bachelor of Science in Business Administration (BSBA), major in Finance, provides students with a broad range of knowledge and skills in corporate investment decisions, corporate financing decisions, corporate dividend decisions, financial statements analysis, risk management that are applied in a variety of roles and settings.

The program also offers a broad foundation in various disciplines, including accounting, finance, marketing, human resources, and operations management. It provides graduates with essential critical thinking, problem-solving, communication, teamwork, and leadership skills. Graduates of the BSBA majoring in Finance can look forward to various career opportunities in government and non-government organizations, such as banking, investment management, financial planning, corporate finance, and risk management, to name a few.

The BSBA Major in Finance program is accredited by an international professional accountancy body, - the Association of Chartered Certified Accountants (ACCA), in the UK to ensure that the AURAK graduates are internationally recognized in the accounting profession. This recognition helps the graduates find and work in different countries in the accounting profession. AURAK BSBA (Major in Finance) graduates are granted exemptions while registering for ACCA qualification.

The BSBA major in Finance is affiliated with the Certified Financial Association (CFA) Institute, and the course contents align with the CFA level 1 curriculum. As a CFA-affiliated major, students attending BSBA major in finance will be eligible to apply for the CFA student scholarships to help them sit for the CFA level 1 exam.

Finance Major Program Learning Outcomes (FMLO)

On completion of a BSBA (Major in Finance) students will be able to:

FMLO 1: Assess the salience to specific decision scenarios of the structures and regulatory frameworks encompassing financial institutions and markets

FMLO 2: Evaluate, aggregate and synthesize knowledge relating to financial operations from diverse sources for the benefit of the organization; and

FMLO 3: Formulate and craft effective solutions to finance-related business problems, based on sound application of underlying financial principles.

BSBA (Major in Finance) Required Courses**21 Credits**

Course Code	Title	Credits	Prerequisite
FNAN 303	Financial Analysis, Forecasting, and Valuation (A)	3	ACCT 204
FNAN 304	Risk Management and Insurance (A)	3	FNAN 303
FNAN 321	Financial Institutions (A)	3	FNAN 300
FNAN 390	Internship in Finance (A)	3	Completion of 90 Cr.

FNAN 401	Advanced Financial Management (A)	3	FNAN 303
FNAN 411	Investment Analysis and Portfolio Management (A)	3	STAT 100, FNAN 303
FNAN 444	Finance Project Based Internship (A)	3	FNAN 303

(A) – Applied Course

BSBA (Major in Finance) Elective Courses (Any 3 courses to be taken)

9 Credits

Course	Title	Credits	Prerequisites
ACCT 331	Intermediate Accounting I (A)	3	ACCT 204
FNAN 331	Fundamentals of Sustainable Finance	3	FNAN 300
FNAN 412	Futures and Options Markets (A)	3	FNAN 303, FNAN 411, OPMT 311
FNAN 421	Money and Capital Markets (A)	3	FNAN 303
FNAN 431	International Financial Management (A)	3	FNAN 303

(A) – Applied Course

BSBA (Major in Human Resource Management)

Bachelor of Science in Business Administration, major in Human Resource Management, provides students with a well-rounded education in business and will develop their skills and knowledge in various areas such as employee recruitment, selection and retention, training and development, compensation and benefits, labor relations, and employee safety and health. In addition, students will have the opportunity to gain practical experience through internships, hands-on projects, and simulations, which will allow them to apply the knowledge and skills learned in the classroom to real-world situations.

Graduates of the bachelor's degree in Human Resource Management program can look forward to various career roles, including human resource managers, talent acquisition specialists, employee relations managers, and compensation and benefits specialists.

Human Resource Management Major Program Learning Outcomes (HMLO)

On completion of a BSBA (Major in Human Resource Management) students will be able to

HMLO 1: Assess sustainable paths to develop the human resource within an organization;

HMLO 2: Evaluate, aggregate, and synthesize knowledge relating to people management functions from diverse sources for the benefit of the organization; and

HMLO 3: Formulate and craft viable solutions to people management problems at both higher strategic and operational levels.

BSBA (Major in Human Resource Management) Required Courses

21 Credits

Course Code	Course Title	Credits	Prerequisites
MGHR 301	Human Resource Management	3	MGMT 301
MGHR 302	Compensation Management	3	MGHR 301
MGHR 304	Human Resource Recruitment & Selection (A)	3	MGHR 301
MGHR 390	Internship in Human Resource Management (A)	3	Completion of 90 Cr.
MGHR 403	Cross Cultural and International Human Resource Management (A)	3	MGHR 301
MGHR 431	Employee Relations	3	MGHR 301
MGHR 444	HRM Project Based Internship (A)	3	MGHR 301

(A) – Applied Course

BSBA (Major in Human Resource Management) Elective Courses**9 Credits**

Course Code	Course Title	Credits	Prerequisites
MGHR 461	Diversity in organization	3	MGMT 301
MGHR 462	Strategic Human Resource Management (A)	3	MGHR 301
MGHR 463	Negotiations in Organizations (A)	3	MGHR 301
MGHR 464	Training and Development (A)	3	MGMT 301

(A) – Applied Course

BSBA (Major in Marketing)

Bachelor of Science in Business Administration, major in Marketing, provides students with a well-rounded education in business and will develop their skills and knowledge necessary to effectively promote products and services to consumers. In addition, students will have the opportunity to gain practical experience through internships, hands-on projects, and simulations, which will allow them to apply the knowledge and skills learned in the classroom to real-world situations.

Graduates of the bachelor's degree in Marketing program can look forward to various career roles, including marketing managers, digital marketers, brand managers, advertising executives, and public relations specialists.

Marketing Major Program Learning Outcomes (MMLO)

On completion of a BSBA (Major in Marketing) students will be able to

MMLO 1: Apply salient marketing techniques to product/service positioning within market segments;

MMLO 2: Evaluate, aggregate, and synthesize knowledge relating to the marketing function from diverse sources for the benefit of the organization; and

MMLO 3: Formulate and craft integrated market plans and marketing strategies for various business domains.

BSBA (Major in Marketing) Required Courses**21 Credits**

Course Code	Course Title	Credits	Prerequisites
MKTG 312	Consumer behavior	3	MKTG 301
MKTG 313	Integrated Marketing Communications (A)	3	MKTG 301
MKTG 351	Marketing Research Techniques & Applications (A)	3	MKTG 301, OPMT 311
MKTG 390	Internship in Marketing (A)	3	90 Cr. Hrs.
MKTG 412	Marketing of Services (A)	3	MKTG 301
MKTG 471	Marketing Strategy	3	90 Cr. Hrs.
MKTG 444	Marketing Project Based Internship (A)	3	MKTG 301

(A) – Applied Course

BSBA (Major in Marketing) Elective Courses**9 Credits**

Course Code	Course Title	Credits	Prerequisites
MKTG 311	Sales Management	3	MKTG 301
MKTG 315	Social media and Digital Marketing (A)	3	MKTG 301
MKTG 316	Luxury Brand Management (A)	3	MKTG 301
MKTG 332	Retailing and E-Commerce (A)	3	MKTG 301
OPMT 405	Supply Chain Management	3	OPMT 311

MKTG 407	International Marketing (A)	3	MKTG 301
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(A) – Applied Course

BSBA (Major in Business Analytics)

Bachelor of Science in Business Administration, major in Business Analytics, provides students with the skills and knowledge needed to make sense of vast amounts of data and to use that data to inform business decisions.

Students will learn to apply statistical models, data visualization techniques, and machine learning algorithms to identify trends and insights that can improve business performance. It is the ideal program for students who enjoy working with data and are interested in business.

Graduates of the Bachelor's degree in Business Analytics program have a wide range of career opportunities, including positions as data analysts, business intelligence analysts, data scientists, and operations research analysts within industries such as finance, healthcare, technology, and marketing.

Business Analytics Major Program Learning Outcomes (BMLO)

On completion of a BSBA (Major in Business Analytics) students will be able to

BMLO 1: Integrate information technologies with quantitative analysis and data science methods to extract business value from data sets, and to enable data-driven business decisions.

BMLO 2: Identify opportunities, needs and constraints for data analytics within specific organizational contexts.

BMLO 3: Demonstrate proficiency and creativity with the integrated use of analytics tools and techniques to formulate useful data analytics solutions.

BSBA (Major in Business Analytics) Required Courses

21 Credits

Course	Course Title	Credits	Prerequisites
BUAN 311	Business Data Mining (A)	3	BUAN 301
BUAN 312	Visual Analytics (A)	3	BUAN 301
BUAN 390	Internship in Business Analytics (A)	3	Completion of 90 Cr.
BUAN 401	Business Data Engineering (A)	3	MIST 301, BUAN 311
BUAN 411	Business Intelligence & Big Data Analytics	3	MIST 301, BUAN 311
MKTG 351	Marketing Research Techniques and Applications (A)	3	MKTG 301, OPMT 311
BUAN 444	Business Analytics Project Based Internship (A)	3	BUAN 301

(A) – Applied Course

BSBA (Major in Business Analytics) Elective Courses

9 Credits

Course Code	Course Title	Credits	Prerequisites
ACCT 361	Accounting Information Systems (A)	3	ACCT 331
FNAN 303	Financial Analysis, Forecasting, and Valuation (A)	3	ACCT 204
OPMT 405	Supply Chain Management	3	OPMT 311
OPMT 301	Operations Management	3	Sophomore Standing

(A) – Applied Course

BSBA (Major in Hospitality and Tourism Management)

Bachelor of Science in Business Administration, major in Hospitality and Tourism Management, offers a comprehensive curriculum that covers a wide range of topics, including hospitality operations, food and beverage management, event planning, and tourism development. In addition, students will have the opportunity to gain practical experience through internships, hands-on projects, and simulations, which will allow them to apply the knowledge and skills learned in the classroom to real-world situations.

The BSBA (Major in Hospitality and Tourism Management) is offered in collaboration with Appalachian State University. Students will undertake one semester in a partner organization in the United States of America (USA). Graduates of the Bachelor's degree in Hospitality & Tourism Management program can look forward to various career opportunities, including positions in hotels, restaurants, cruise lines, theme parks, convention centers, travel agencies, and event planning.

Hospitality and Tourism Management Major Program Learning Outcomes (HTMLO)

On completion of a BSBA (Major in Hospitality and Tourism Management) students will be able to

HTMLO 1: Apply core hospitality and tourism management principles to an evaluation of hospitality and tourism systems, structures and processes;

HTMLO 2: Apply practical considerations to an evaluation of hospitality and tourism systems, structures and processes; and

HTMLO 3: Integrate an analysis of the external environment into an evaluation of management decisions in the hospitality and tourism sector.

BSBA (Major in Hospitality and Tourism Management) Required Courses

21 Credits

Course Code	Course Title	Credits	Prerequisites
HTMT 311	Survey of Hospitality and Tourism	3	None
HTMT 321	Hospitality Operations I – F&B	3	HTMT 311
HTMT 322	Hotel Operations (A)	3	HTMT 311
HTMT 390	Internship in Hospitality & Tourism Management (A)	3	Completion of 90 Cr.
HTMT 411	Destination Management (A)	3	HTMT 311
HTMT 412	Meetings and Conventions (A) (C)	3	Completion of 84 Credits
HTMT 444	Hospitality and Tourism Management Project Based Internship (A)	3	HTMT 311

(A) – Applied Course, (C) Compulsory Course, (O) Optional Course

BSBA (Major in Hospitality and Tourism Management) Elective Courses

9 Credits

Course Code	Course Title	Credits	Prerequisites
HTMT 312	Sustainable Tourism	3	HTMT 311
BUSN 391	International Internship (A)	3	Completion of 90 Credits
MKTG 313	Integrated Marketing	3	MKTG 301
MKTG 412	Marketing of Services (A)	3	MKTG 301
MGHR 301	Human Resource Management (O)	3	MKTG 301

(A) – Applied Course, (C) Compulsory Course, (O) Optional Course

Business Elective Courses**9 Credits**

Any three courses from the major requirements and major electives listed above other than the declared major required and major elective courses taken to complete a major.

Free Elective Courses**3 Credits**

Any Course from other Programs and not already taken.

Other Paths to complete the 121 credit hours requirement for a BSBA degree**The Generalist Option**

Instead of completing thirty (30) credits (21 + 9) of major elective courses in a specific major, students can complete thirteen (13) courses, 39 credit hours of business electives (including 2 internship courses in any of the majors) provided these comprise major required and major elective courses in any of the 6 majors.

TOTAL DEGREE REQUIREMENT**121 Credit Hours**

Bachelor of Business Administration Four-Year Sample Schedule of Delivery (Major in Accounting/Business Analytics/Finance/HRM/Marketing)

Year One, First Semester			Year One, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
ENGL 101	Composition	3	ECON 104	Principles of Macroeconomics	3
ECON 103	Principles of Microeconomics	3	UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
ITEC 103	Introduction to Computing	3	ARAB 101 or ARAB110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
STAT 100	Introductory Probability & Statistics	3	PHIL 100	Critical Thinking & Reasoning	3
UNIV 100	University First-Year Transition	1	UAES 200	Survey of United Arab Emirates Studies	3
MEST 100	Introduction to Islam in World Culture	3			
Subtotal =		16	Subtotal =		15
Year Two, First Semester			Year Two, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
GEN ED	Any Natural Science Course	3	ACCT 205	Principles of Managerial Accounting	3
ACCT 204	Principles of Financial Accounting	3	MGMT 301	Principles of Management	3
MKTG 301	Principles of Marketing	3	BUAN 301	Analytics for Decision Making	3
BUSN 301	Business Communication	3	FNAN 300	Principles of Business Finance	3
OPMT 311	Methods and Models of Management Science	3	MIST 301	Introduction to Business Information Systems	3
Subtotal =		15	Subtotal =		15
Year Three, First Semester			Year Three, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
BUSN 304	Business Law	3	BUSN 307	Research Methodology	3
BUSN 312	Public Policy and Ethics	3	MAJOR	PROGRAM MAJOR 3	3
BUSN 306	International Business	3	MAJOR	PROGRAM MAJOR 4	3
MAJOR	PROGRAM MAJOR 1	3	Major Elective	MAJOR ELECTIVE 1	3
MAJOR	PROGRAM MAJOR 2	3	Elective	BUSINESS ELECTIVE 1	3
Subtotal =		15	Subtotal =		15
Year Three, Summer Session					
Major Internship 390	Major Specific Internship	3			
Year Four, First Semester			Year Four, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
BUSN 498	Strategic Management	3	Major Elective	MAJOR ELECTIVE 3	3
MGMT 401	Organizational Behavior	3	Elective	BUSINESS ELECTIVE 2	3
MAJOR	PROGRAM MAJOR 5	3	Elective	BUSINESS ELECTIVE 3	3
Major Elective	MAJOR ELECTIVE 2	3	Elective	FREE ELECTIVE	3
Major Specific Internship 444	Major Specific Project Based Internship	3			3
Subtotal =		15	Subtotal =		12
121 Semester Credit Hours					

Bachelor of Business Administration Four-Year Sample Schedule of Delivery (Major in Hospitality and Tourism Management)

Year One, First Semester			Year One, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
ENGL 101	Composition	3	ECON 104	Principles of Macroeconomics	3
ECON 103	Principles of Microeconomics	3	UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
ITEC 103	Introduction to Computing	3	ARAB 101 or ARAB110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
STAT 100	Introductory Probability & Statistics	3	PHIL 100	Critical Thinking & Reasoning	3
UNIV 100	University First-Year Transition	1	UAES 200	Survey of United Arab Emirates Studies	3
MEST 100	Introduction to Islam in World Culture	3			
Subtotal =		16	Subtotal =		15
Year Two, First Semester			Year Two, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
GEN ED	Any Natural Science Course	3	ACCT 205	Principles of Managerial Accounting	3
ACCT 204	Principles of Financial Accounting	3	MGMT 301	Principles of Management	3
MKTG 301	Principles of Marketing	3	BUAN 301	Analytics for Decision Making	3
BUSN 301	Business Communication	3	FNAN 300	Principles of Business Finance	3
OPMT 311	Methods and Models of Management Science	3	MIST 301	Introduction to Business Information Systems	3
Subtotal =		15	Subtotal =		15
Year Three, First Semester			Year Three, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
BUSN 304	Business Law	3	BUSN 307	Research Methodology	3
BUSN 312	Public Policy and Ethics	3	BUSN 498	Strategic Management	3
BUSN 306	International Business	3	MGMT 401	Organizational Behavior	3
HTMT 311	Survey of the Hospitality & Tourism Industry	3	Elective	BUSINESS ELECTIVE 2	3
Elective	BUSINESS ELECTIVE 1	3	Elective	BUSINESS ELECTIVE 3	3
Subtotal =		15	Subtotal =		15
Year Three, Summer Session					
HTMT 390	Internship in Hospitality & Tourism Management	3			
Major Elective	MAJOR ELECTIVE 1	3	HTMT 412	Meetings and Conventions	3
Major Elective	MAJOR ELECTIVE 2	3	HTMT 411	Destination Management	3
HTMT 322	Hotel Operations	3	Major Elective	MAJOR ELECTIVE 3	3
HTMT 321	Hospitality Operations I – F&B Management	3			
HTMT 444	Hospitality & Tourism Management Project Based Internship	3	Elective	FREE ELECTIVE	3
Subtotal =		15	Subtotal =		12
121 Semester Credit Hours					

Bachelor of Business Administration Four-Year Sample Schedule of Delivery (Generalist)

Year One, First Semester			Year One, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
ENGL 101	Composition	3	ECON 104	Principles of Macroeconomics	3
ECON 103	Principles of Microeconomics	3	UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
ITEC 103	Introduction to Computing	3	ARAB 101 or ARAB110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
STAT 100	Introductory Probability & Statistics	3	PHIL 100	Critical Thinking & Reasoning	3
UNIV 100	University First-Year Transition	1	UAES 200	Survey of United Arab Emirates Studies	3
MEST 100	Introduction to Islam in World Culture	3			
Subtotal =		16	Subtotal =		15
Year Two, First Semester			Year Two, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
GEN ED	Any Natural Science Course	3	ACCT 205	Principles of Managerial Accounting	3
ACCT 204	Principles of Financial Accounting	3	MGMT 301	Principles of Management	3
MKTG 301	Principles of Marketing	3	BUAN 301	Analytics for Decision Making	3
BUSN 301	Business Communication	3	FNAN 300	Principles of Business Finance	3
OPMT 311	Methods and Models of Management Science	3	MIST 301	Introduction to Business Information Systems	3
Subtotal =		15	Subtotal =		15
Year Three, First Semester			Year Three, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
BUSN 304	Business Law	3	BUSN 307	Research Methodology	3
BUSN 312	Public Policy and Ethics	3	Elective 3	Business Elective 3	3
BUSN 306	International Business	3	Elective 4	Business Elective 4	3
Elective 1	Business Elective 1	3	Elective 5	Business Elective 5	3
Elective 2	Business Elective 2	3	Elective 6	Business Elective 6	3
Subtotal =		15	Subtotal =		15
Year Three, Summer Session					
Major Internship 390 (Elective 7)	Major Specific Internship	3			
Year Four, First Semester			Year Four, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
BUSN 498	Strategic Management	3	Elective 11	Business Elective 11	3
MGMT 401	Organizational Behavior	3	Elective 12	Business Elective 12	3
Elective 8	Business Elective 8	3	Elective 13	Business Elective 13	3
Elective 9	Business Elective 9	3	Free Elective	Free Elective	3
Major Specific 444 (Elective 10)	Major Specific Project Based Internship	3			3
Subtotal =		15	Subtotal =		12
121 Semester Credit Hours					



SCHOOL OF ENGINEERING

SCHOOL OF ENGINEERING AND COMPUTING

Bachelor of Architecture

Program Description

Bachelor of Architecture program is designed to prepare well-educated, critical-thinkers and problem-solvers who possess and can effectively apply highly refined aesthetic, technical knowledge, are capable of facing professional and global challenges. Focused on sustainability challenges, which are as broadly cultural as they are technical, the program graduates are prepared to become credible stewards of the built and surrounding environment. The program leads students towards the application of principles of architecture as a science to analyze physical environments, components, and processes to identify problems and propose solutions through design and innovative products. It navigates graduates to integrate into multidisciplinary teams as architects, to respect professional ethics, and through highly professional attitude contribute to the local, regional, and global sustainability and community development.

The Bachelor of Architecture is overseen by the School of Engineering and Computing and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

Bachelor of Architecture program is focused on sustainability challenges and prepares graduates that are critical-thinkers, problem-solvers and capable of facing professional and global challenges. Equipped with highly refined aesthetic and technical knowledge, capable of respecting professional ethics and work in multidisciplinary teams, graduates can easily integrate into architectural practices for built environment problem solving.

Program Educational Objectives

The Bachelor of Architecture Program at AURAK produces graduates who are armed with knowledge and skills as articulated in its student outcomes to become:

1. Industrious architects who create healthy and safe spaces of different scale (buildings, cities) that are responsive to the environment and users.
2. Practitioners who value professional ethics and architecture business as influencing the built environment and human wellness.
3. Professionals capable of continuous learning to meet the requirements of altering built environment and comply with the regional/ national laws and regulations.
4. Designers experienced in evaluating solutions as it relates to building systems design, materials, technologies against project objectives, cost and building performance.
5. Team leaders capable of making decisions within architectural projects for universal design with minimal impact on environment.
6. Active contributors to the local community and the world through an integrative design process that takes cultural values into consideration with a multidisciplinary team.

Program Learning Outcomes

Graduates of the Bachelor of Architecture program will have the ability to:

1. Employ critical thinking that takes into consideration alternative perspectives by analyzing, evaluating, and synthesizing ideas and information gathered through research grounded in acquired information.

2. Implement complex two and three-dimensional graphic representation techniques using a wide variety of manual and digital media as well as oral and written forms to reflect on and convey the architectural design process to a wide range of stakeholders.
3. Demonstrate a solid base of knowledge in history and theory grounded in the intellectual understanding of social, cultural, and economic aspects and through applied research by raising clear and precise questions to clarify and express information to reach logical conclusions and evaluate options against relevant design criteria, building standards, and program requirements.
4. Utilize and integrate technical skills and professional knowledge in comprehensive application of life safety, accessibility, and sustainability issues in integrating the principles of environmental stewardship and making appropriate design decisions of varying scales and levels of complexity through a technical design process based on appropriate documentation in a manner that is client-centered, sustainable, aesthetic, cost effective, and socially responsible.
5. Develop the capability to synthesize variable components into an integrated design solution through the employment of appropriate building materials, building systems, and construction practices which are environmentally friendly and based on research and design decisions for a variety of systems and levels of complexity.
6. Show the ability to lead teams of stakeholders collaboratively within the process of understanding, developing, and applying solutions to problems in the built as well as the natural environments, utilizing knowledge and parameters of professional practice associated with ethical, legal, financial and social responsibilities.
7. Apply reasoning skills including math, physics, and logic to investigate problems related to gravity and other forces in structural systems, thermal heat gain and loss in buildings, material quantity estimates, budget management, and life-cycle cost analysis.

Degree Requirements

Students complete the Bachelor of Architecture by completing twelve (12) courses (32 credit hours) of University General Education, five (5) course (13 credits) of Faculty Requirements, three (3) course (9 credits) of Department of Architecture requirements, thirty-one (31) courses (101 credits) of Architecture Major requirements and two (2) courses (6 credits) of Architecture Major Electives. Total number of credits – 161.

The required courses are:

The University General Education Component Requirements

32 Credit Hours

Course Code	Title	Credit
MATH 113	Calculus I	3
ENGL 101	Composition (Writing Intensive)	3
UNIV 100	University First-Year Transition	1
CSCI 114	Applied Computational Thinking	3
CSCI 115	Applied Computational Tools	1
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
Required Electives – The field of Art and Humanities (6 credits)		

MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
Electives – The field of Social and Behavioral Sciences (6 credits)		
UAES 200	Survey of United Arab Emirates Studies (Required)	3
PSYC 100	Introduction to Psychology (Or another course from the same list)	3
Required Elective – The field of natural Sciences (3 credits)		
ENVS 102	Sustainability and Human Environment Relations	3

The fifth writing intensive course for the Bachelor of Architecture is ARCH 591 Graduation Project Thesis Research.

School of Engineering and Computing Requirements

13 Credit Hours

Course Code	Title	Credit
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab.	1
MENG 323	Engineering Economy	3
ARCH 390	Internship I	3
ARCH 391	Internship II	3

Department Requirements

9 Credit Hours

Course	Title	Credit
INDS 112	Design Communication 1: Sketching and Drawing for Interiors	3
INDS 121	Interior Design Process, Human Factors and Ergonomic	3
INDS 122	Design Communication 2: Digital Media for Interiors	3

Major Compulsory Requirements

101 Credit Hours

Course	Title	Credit
ARCH 121	Basic Design 1	4
ARCH 122	Basic Design 2	4
ARCH 223	Computer Architectural Drawing	3
ARCH 221	Architectural Design Studio 1	4
ARCH 222	Architectural Design Studio 2	4
ARCH 231	Building Construction 1	3
ARCH 232	Building Construction 2	3
ARCH 241	History and Theory of Architecture 1	3
ARCH 242	History and Theory of Architecture 2	3
ARCH 321	Architectural Design Studio 3	4
ARCH 322	Architectural Design Studio 4	4
ARCH 331	Building Construction 3	3
ARCH 336	Construction Drawing	3
ARCH 341	History and Theory of Architecture 3	3
ARCH 342	History and Theory of Contemporary Architecture	3
ARCH 372	Sustainable Architecture	3

ARCH 421	Architectural Design Studio 5	4
ARCH 422	Architectural Design Studio 6	4
ARCH 454	Urban Planning	3
ARCH 457	Urban Design	3
ARCH 463	Landscape Architecture	3
ARCH 466	Building Utilities II: Illumination, Acoustics, and Electrical Building Services	3
ARCH 485	Professional Practice	3
ARCH 487	Project Management	3
ARCH 521	Architectural Design Studio 7	4
ARCH 591	Graduation Project Thesis Research (Writing Intensive)	2
ARCH 592	Graduation Project Design	5
CIEN 216	Structures for Architecture 1	3
CIEN 261	Surveying	1
CIEN 316	Structures for Architecture 2	3
MENG 468	Building Utilities: HVAC and Mechanical Building Services	3

Major Electives

6 Credit Hours

Course Code	Title	Credits
ARCH 449	Technology of the Built Environment	3
ARCH 456	Sustainable Housing	3
ARCH 459	Conservation of Historic Environment	3
ARCH 473	Environmental Control	3
INDS 351	Parametric Design	3
INDS 353	Landscape and Garden Design	3
INDS 451	Portfolio Design	3
INDS 453	Advanced BIM: Scheduling and coordination	3

Bachelor of Architecture Five-Year Sample Schedule

Year 1 Semester 1			Year 1 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
ARCH 121	Basic Design 1	4	ARCH 122	Basic Design 2	4
MATH 113	Calculus I	3	INDS 121	Interior Design Process, Human Factors and Ergonomics	3
INDS 112	Design Communication 1: Sketching and Drawing for Interiors	3	INDS 122	Design Communication 2: Digital Media for Interiors	3
ENGL 101	Composition	3	PHYS 110	University Physics I	3
UNIV 100	University First-Year transition	1	PHYS 111	University Physics I Lab.	1
			CSCI 114	Applied Computational Thinking	3
			CSCI 115	Applied Computational Tools	1
Subtotal =		14	Subtotal =		18
Year 2 Semester 1			Year 2 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
ARCH 221	Architectural Design Studio 1	4	ARCH 222	Architectural Design Studio 2	4
ARCH 231	Building Construction 1	3	ARCH 232	Building Construction 2	3
ARCH 241	History & Theory of Architecture 1	3	ARCH 242	History & Theory of Architecture 2	3
CIEN 216	Structures for Architecture 1	3	CIEN 261	Surveying	1
ARCH 223	Computer Architectural Drawing	3	ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
			PSYC 100	Introduction to Psychology (or other from the same list)	3
Subtotal =		16	Subtotal =		17
Year 3 Semester 1			Year 3 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
ARCH 321	Architectural Design Studio 3	4	ARCH 322	Architectural Design Studio 4	4
ARCH 331	Building Construction 3	3	ARCH 336	Construction Drawings	3
ARCH 341	History and Theory of Architecture 3	3	ARCH 342	History and Theory of Contemporary Architecture	3
CIEN 316	Structures for Architecture 2	3	ARCH 372	Sustainable Architecture	3
ENVS 102	Sustainability and Human-Environment Relations	3	ARCH	Technical Elective 1	3
Subtotal =		16	Subtotal =		16
Summer Semester					
Code	Course Title	Credits			
ARCH 390	Internship I	3			
Year 4 Semester 1			Year 4 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
ARCH 421	Architectural Design Studio 5	4	ARCH 422	Architectural Design Studio 6	4
ARCH 463	Landscape Architecture	3	ARCH 466	Building Utilities II: Illumination, Acoustics, and Electrical Building Services	3
MENG 468	Building Utilities I: HVAC and Mechanical Building Services	3	ARCH 454	Urban Planning	3
MEST 100	Introduction to Islam in World Culture	3	UAES 200	Survey of United Arab Studies	3
ARCH 457	Urban Design	3	ENGL 200 or PHIL 100	Advanced Composition or Critical Thinking and Reasoning	3
Subtotal =		16	Subtotal =		16
Summer Semester					
Course No.	Description	CR			
ARCH 391	Internship II	3			
Year 5 Semester 1			Year 4 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
ARCH 521	Architectural Design Studio 7	4	ARCH 592	Graduation Project Design	5
ARCH 591	Graduation Project Thesis Research	2	MENG 323	Engineering Economy	3
ARCH	Technical Elective 2	3	ARCH 485	Professional Practice	3
ARCH 487	Project Management	3	UNIV 200	Innovation, Entrepreneurship, & Sustainability	3
Subtotal =		12	Subtotal =		14
TOTAL CREDIT HOURS 161					

Bachelor of Science in Artificial Intelligence

Program Description

The Bachelor of Science in Artificial Intelligence (BSAI) degree program at AURAK is designed to equip learners with an in-depth knowledge of computer science theory and a thorough understanding of artificial intelligence (AI) and data science technologies. Students are exposed to state-of-the-art AI-related methods relevant to knowledge representation and reasoning. Such techniques will enhance human capabilities through automated analysis of complex inputs and facilitate automated decision making. The program includes techniques and topics such as machine learning, computer vision, robotics, data analysis and visualization, natural language processing, and data mining.

The global demand for AI and data science professionals is on the rise due to the integration of the field into most industries. This need is further exacerbated by a worldwide shortage of AI talent.

The Bachelor of Science in Artificial Intelligence is overseen by the School of Engineering and Computing and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The mission of the Bachelor of Science in Artificial Intelligence program is to graduate students with the knowledge and skills to enable them to design and develop computer systems and data models using the latest advances in the field and hence become effective leaders capable of utilizing artificial intelligence and data science locally and globally across various domains.

Program Educational Objectives

A few years after earning their degree, our graduates will be:

1. Successful professionals and innovators in theoretical and practical areas of computer science, artificial intelligence, and data science;
2. Engaged in creating a positive technological impact with a sufficient awareness of the ethical, legal, and security issues related to computing, artificial intelligence, and data science; and
3. Equipped with the skills required for professional practice including functioning in teams and communicating effectively.

Program Learning Outcomes

Graduates of the program will have the ability to:

1. Analyze a complex computing problem and to apply principles of computing, artificial intelligence, statistics, and other relevant disciplines to identify solutions.
2. Design, implement, and evaluate computer science or artificial intelligence solutions to meet a given set of computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. Apply computer science and artificial intelligence theory and software development fundamentals to produce computing-based solutions.
7. Build, apply, and evaluate data science models to solve problems using theoretical fundamentals.

Degree Requirements

The BS in Artificial Intelligence requires the completion of 128 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering and Computing Requirements	26
Artificial Intelligence Program Requirements	70 (64 compulsory & 6 technical electives)

General Education Requirements

32 Credits

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied.

The fifth writing intensive course for the BS in Artificial Intelligence is CSAI 451 Machine Learning Lab.

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
Gen Ed Course	Social and Behavioral Sciences Course	3
ENVS 102	Sustainability and Human-Environment Relations	3

School of Engineering and Computing Requirements

26 Credits

Course Code	Title	Credits
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab	1
MATH 114	Calculus II	3
MATH 213	Calculus III	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab	1
MATH 203	Linear Algebra	3
ENGR 200	Engineering Statistics	3
CSAI 390	Internship I	3
CSAI 391	Internship II	3

Artificial Intelligence Program Requirements**70 Credits****Core Course****64 Credit Hours**

Course Code	Title	Credits
MATH 225	Discrete Mathematics	3
CSCI 104	Introduction to Computing	3
CSCI 211	Object Oriented Programming	3
CSCI 215	Data Structures and Algorithms	3
CSCI 232	Computer Organization	3
CSAI 350	Introduction to Artificial Intelligence	3
CSCI 315	Design and Analysis of Algorithms	3
CSCI 326	Database Systems	3
CSCI 312	Operating System Fundamentals	3
CSAI 351	Data Science	3
EEEN 331	Digital Systems Design	3
EEEN 332	Digital Systems Design Lab	1
CENG 411	Software Engineering	3
CSCI 415	Introduction to Parallel Programming	3
CSAI 484	Internet of Things System	3
CSAI 450	Machine Learning	3
CSAI 451	Machine Learning Lab (Writing Intensive)	1
CSCI 462	Data Communications and Computer Networks	3
CSAI 452	Natural Language Processing	3
CSAI 453	Data Visualization	3
CSAI 490	Professional Software Practice	2
CSCI 492	Senior Design Project I	2
CSCI 493	Senior Design Project II	4

Technical Electives**6 Credit Hours**

Course Code	Title	Credits
CSCI 450	Information Security and Privacy	3
CSAI 480	Big Data	3
CSAI 481	Computer Vision	3
CSAI 482	Data Mining	3
CSAI 483	Information Retrieval	3
CSAI 485	Introduction to Deep Learning	3
CSCI 416	Human Computer Interaction	3
CENG 431	Embedded Systems	3
CENG 432	Embedded Systems Lab	1
CSAI 486	Special Topics in Artificial Intelligence	3
CSAI 487	Introduction to Robotics	3
ENGR 399	Undergraduate Research Project	3

Bachelor of Science in Artificial Intelligence Four-Year Study Plan

Year 1 Semester 1			Year 1 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
CSCI 104	Introduction to Computing	3	ARAB 101 Or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition	3	CSCI 112	Introduction to Computer Programming	3
MATH 113	Calculus I	3	CSCI 113	Introduction to Computer Programming Lab	1
PHYS 110	University Physics I	3	MATH 114	Calculus II	3
PHYS 111	University Physics I Lab	1	PHYS 220	University Physics II	3
UNIV 100	University First-Year Transition	1	PHYS 221	University Physics II Lab	1
			MEST 100	Introduction to Islam in World Culture	3
Subtotal =		14	Subtotal =		17
Year 2 Semester 1			Year 2 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
CSCI 211	Object Oriented Programming	3	CSCI 215	Data Structures and Algorithms	3
MATH 213	Calculus III	3	EEEN 331	Digital Systems Design	3
MATH 225	Discrete Mathematics	3	EEEN 332	Digital Systems Design Lab	1
CSCI 232	Computer Organization	3	ENGR 200	Engineering Statistics	3
GEN ED	Social and Behavioral Sciences	3	PHIL 100 Or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
			MATH 203	Linear Algebra	3
Subtotal=		15	Subtotal=		16
Year 2 Summer Session					
Course No.	Description	CR			
CSAI 390	Internship I	3			
Year 3 Semester 1			Year 3 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
CSAI 350	Introduction to Artificial Intelligence	3	CSCI 315	Design and Analysis of Algorithms	3
ENVS 102	Sustainability and Human-Environment Relations	3	CSAI 351	Data Science	3
CSCI 312	Operating System Fundamentals	3	CSCI 462	Data Comm. and Computer Networks	3
CSCI 326	Database Systems	3	CSCI 415	Introduction to Parallel Programming	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability	3	UAES 200	Survey of United Arab Emirates Studies	3
Subtotal =		15	Subtotal =		15
Year 3 Summer Session					
Course No.	Description	CR			
CSAI 391	Internship II	3			
Year 4 Semester 1			Year 4 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
CSAI 450	Machine Learning	3	CSCI 493	Senior Design Project II	4
CSAI 451	Machine Learning Lab	1	CSAI 452	Natural Language Processing	3
CENG 411	Software Engineering	3	CSAI 453	Data Visualization	3
CSCI 492	Senior Design Project I	2	Elective	Technical Elective I	3
CSAI 490	Professional Software Practice	2	Elective	Technical Elective II	3
CSAI 484	Internet of Things System	3			
Subtotal =		14	Subtotal =		16
TOTAL CREDIT HOURS 128					

Bachelor of Science in Chemical Engineering

Program Description

The Bachelor of Science in Chemical Engineering (CHEN) program was established in 2014 and started accepting students in Fall 2014. In Fall 2024, AURAK added a concentration to the program which provides a pathway for those students who are interested in petroleum engineering.

The CHEN program is a unique four-year undergraduate program. The program covers areas of sustainable engineering, global issues, and multidisciplinary problem solving and system analysis components across its curriculum to solve society and industrial most challenging problems. In addition, the program closely matches local professional opportunities and the direction of professions in chemical engineering and through the petroleum engineering concentration, to employment in that industry.

The CHEN program has 136 credit hours over the duration of eight semesters including six credit hours of internship program. The curriculum of the program covers reaction engineering, process design and control, fluid and material transportation, renewable energy, mass and heat transfer and water and environmental engineering. The program was prepared taking into consideration criteria set by the Accreditation Board for Engineering and Technology (ABET) and the Commission for Academic Accreditation (CAA) at the Ministry of Education, United Arab of Emirates. As mentioned earlier there is also a concentration which allows students to opt to take 24 Credits specifically for petroleum engineering.

The faculty in the Department of Chemical and Petroleum Engineering have graduated from well-known universities in the world, and their expertise covers a broad range of technical areas, including reaction engineering, process design, renewable energy and management, environmental engineering, and wastewater treatment.

The Bachelor of Science in Chemical Engineering is overseen by the School of Engineering and Computing and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Science in Chemical Engineering program at AURAK educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of Chemical Engineering and, for those studying the petroleum engineering, in the field of Petroleum Engineering. The graduates of the program will be well versed in technology and in social and environmental issues.

To fulfill this mission, the general Chemical Engineering program provides students with a thorough foundation in the basic tenets of Chemical Engineering and technologies and a broad introduction into chemistry, engineering materials, thermal sciences, reaction and separation sciences, energy and environment protection, chemical process design, integration and optimization and data science in chemical engineering. For those studying the petroleum concentration, students cover reservoir rock and fluid properties, drilling, petroleum reservoir engineering, well logging, production engineering and reservoir simulation.

The program provides a strong background for graduate study in the diverse areas branching out of the Chemical and Petroleum Engineering fields. The technical focus is complemented with topics in general education leading to a well-rounded member of the global society.

Program Educational Objectives

The Program Educational Objectives (PEOs) describe the expected accomplishments of the graduates of the Bachelor of Science in Chemical Engineering Engineering general pathway and concentration in Petroleum Engineering. The PEOs have been derived from and support the mission statement of AURAK. The chemical

engineering program has a strong focus on the integration of engineering, science, and process design with complementary areas of study in environmental engineering, fuels and energy, pharmaceutical, petrochemicals, materials, and renewable energy.

The graduates of the Chemical Engineering Program are expected to:

1. Pursue a successful professional career in engineering, scientific, or complementary disciplines through technical competence, effective communication, teamwork, and leadership;
2. Engaged in creating a positive impact with a sufficient awareness of the ethical, cultural, legal, and environmental issues; and
3. Maintain a lifelong interest in learning for personal and professional development.

In addition to the above, Graduates from the Petroleum Engineering concentration are expected to apply broad knowledge of upstream and downstream processes and emerging technology for maximum economic recovery and processing of oil, gas and related resources.

Program Learning Outcomes

AURAK has adopted ABET's student outcomes for the Chemical Engineering Program. The outcomes for the program are for the student to acquire:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Degree Requirements

The Bachelor of Science in Chemical Engineering program requires the completion of 136 credits in the following areas:

Area	Credit
University General Education Requirements	33
School of Engineering and Computing Requirements	33
Chemical Engineering Program Requirements (Core Courses)	46
Chemical Engineering (General) or Petroleum Engineering (Concentration)	

Applied Courses	18
Technical Electives	6

University General Education Requirements

33 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS Chemical Engineering and Petroleum Engineering Concentration is CHEN 471 Unit Operation Lab.**

Course Code	Course Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 114	Applied Computational Thinking	3
CSCI 115	Applied Computational Tools	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
ECON 103	Principles of Microeconomics	3
CHEM 211	General Chemistry I	3
CHEM 212	General Chemistry I Lab	1

School of Engineering and Computing Requirements

33 Credit Hours

Course Code	Course Title	Credits
ENGR 107	Introduction to Engineering	2
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab.	1
MATH 114	Calculus II	3
MATH 203	Linear Algebra	3
MATH 214	Elementary Differential Equations	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab.	1
ENGR 200	Engineering Statistics	3
ENGR 210	Sustainability in Engineering	2

ENGR 390	Internship I	3
ENGR 391	Internship II	3
MENG 323	Engineering Economy	3

Chemical Engineering Program Requirements

70 Credit Hours

Program Requirements are 70 Credit Hours as follow:

Core Courses:

46 Credit Hours

Courses include the following:

Course Code	Course Title	Credits
CHEM 213	General Chemistry II	3
CHEM 214	General Chemistry II Lab	1
MENG 241	Engineering Materials	3
CHEM 215	Organic Chemistry I	3
CHEM 315	Physical Chemistry	3
CHEN 202	Principles of Chemical and Petroleum Engineering	3
CIEN 251	Fluid Mechanics	3
CHEN 301	Computational Tools in Chemical and Petroleum Engineering	1
CHEN 312	Chemical Engineering Thermodynamic	3
CHEN 351	Chemical Reaction Engineering	3
CHEN 370	Heat and Mass Transport	3
CHEN 471	Unit Operation Lab (Writing Intensive)	1
CHEN 472	Separation Processes	3
CHEN 481	Process & Plant Design	3
CHEN 482	Chemical Process Dynamics & Control	3
CHEN 483	Advance Simulation Lab	1
CHEN 491	Senior Design Project I	3
CHEN 492	Senior Design Project II	3

Chemical Engineering (General)

Applied Core Courses:

18 Credit Hours

Course Code	Course Title	Credits
BIOL 112	University Biology I	3
MENG 242	Materials Lab	1
CHEM 216	Organic Chemistry Lab	1
CHEM 316	Physical Chemistry Lab	1
CHEN 352	Chemical Reaction Lab	1
CHEN 372	Thermal and Fluid Sciences Lab	1
CHEN 404	Safety and Environmental Impact	3
CHEN 484	Process Integration and Optimization	3
CHEN 486	Data Science in Chemical Engineering	3

ENGR 450	Engineering Seminar	1
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Technical Electives:

6 Credit Hours

To be selected from the following list of courses:

Course Code	Course Title	Credits
CHEN 403	Fundamentals of Biochemical Engineering	3
CHEN 452	Introduction to Hydrogen Technologies	3
CHEN 462	Petroleum Refining Engineering	3
CHEN 463	Natural Gas Engineering	3
CHEN 464	Industrial Catalysis	3
CHEN 473	Water Desalination	3
CHEN 474	Industrial and Wastewater Treatment	3
CHEN 476	Fundamentals of Nanotechnology	3
CHEN 485	Carbon Capture, Utilization and Storage	3
CHEN 493	Special Topics in Chemical Engineering	3
ENGR 399	Undergraduate Research Project	3

Chemical Engineering (Petroleum Engineering Concentration)

Applied Core Courses

18 credit hours

Course Code	Course Title	Credits
PENG 351	Reservoir Rock and Fluid Properties	3
PENG 362	Drilling Engineering	3
PENG 371	Petroleum Reservoir Engineering	3
PENG 381	Well Logging	3
PENG 384	Petroleum Production Engineering	3
PENG 473	Reservoir Simulation	3

Technical Electives:

6 Credit Hours

To be selected from the following list of courses:

Course Code	Course Title	Credits
CHEN 404	Safety and Environmental Impact	3
CHEN 452	Introduction to Hydrogen Technologies	3
CHEN 463	Natural Gas Engineering	3
CHEN 485	Carbon Capture, Utilization and Storage	3
PENG 404	Petroleum Economics	3
PENG 464	Data Science in Petroleum Engineering	3
PENG 474	Applied Reservoir Simulation	3
PENG 483	Well Testing	3
PENG 486	Enhanced Oil Recovery	3

PENG 493	Special Topics in Petroleum Engineering	3
ENGR 399	Undergraduate Research Project	3

Chemical Engineering Four-Year Sample Schedule

Freshman Year, First Semester			Freshman Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PHYS 110	University Physics I	3	ARAB 101 or ARAB 110	Beginner Level Arabic and Culture I Arabic Language and Culture for Native Arabic Speakers I	3
PHYS 111	University Physics I Lab.	1	MATH 114	Calculus II	3
ENGR 107	Introduction to Engineering	2	MEST 100	Introduction to Islam in World Culture	3
ENGL 101	Composition	3	PHYS 220	University Physics II	3
MATH 113	Calculus I	3	PHYS 221	University Physics II Lab.	1
UNIV 100	University Freshman Transition	1	CSCI 114	Applied Computational Thinking	3
CHEM 211	General Chemistry I	3	CSCI 115	Applied Computational Tools	1
CHEM 212	General Chemistry I Lab. I	1			
Total		17	Total		17
Sophomore Year, First Semester			Sophomore Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CHEM 213	General Chemistry II	3	MENG 241	Engineering Materials	3
CHEM 214	General Chemistry II Lab.	1	MENG 242	Materials Lab.	1
ENGR 210	Sustainability in Engineering	2	CHEM 215	Organic Chemistry I	3
MATH 203	Linear Algebra	3	CHEM 216	Organic Chemistry Lab. I	1
UNIV 200	Innovation and Entrepreneurship	3	MATH 214	Elementary Differential Equations	3
BIOL 112	University Biology I	3	CHEN 202	Principles of Chemical and Petroleum Engineering	3
ECON 103	Principles of Micromechanics	3	CIEN 251	Fluid Mechanics	3
Total		18	Total		17
Sophomore Year, Summer Semester					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 390	Internship I	3			
Junior Year, First Semester			Junior Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CHEN 301	Computational Tools in Chemical and Petroleum Engineering	1	PHIL 100 or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
ENGR 200	Engineering Statistics	3	CHEN 312	Chemical Engineering Thermodynamics	3
CHEM 315	Physical Chemistry	3	CHEN 351	Chemical Reaction Engineering	3
CHEM 316	Physical Chemistry Lab.	1	CHEN 352	Chemical Reaction Lab	1
MENG 323	Engineering Economy	3	CHEN 370	Heat and Mass Transport	3
UAES 200	Survey of United Arab Emirates Studies	3	CHEN 372	Thermal and Fluid Sciences Lab	1
Total		14	Total		14
Junior Year, Summer Semester					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 391	Internship II	3			
Senior Year, First Semester			Senior Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CHEN 404	Safety and Environment Impact	3	CHEN 482	Chemical Process Dynamics and Control	3
CHEN 471	Unit Operation Lab	1	CHEN 484	Process Integration and Optimization	3
CHEN 472	Separation Processes	3	CHEN 486	Data Science in Chemical Engineering	3
CHEN 481	Process & Plant Design	3	CHEN 492	Senior Design Project II	3
	CHEN Technical Elective 1	3		CHEN Technical Elective II	3
CHEN 483	Advance Simulation Lab	1	ENGR 450	Engineering Seminar	1
CHEN 491	Senior Design Project I	3			
Total		17	Total		16
Total 136 Credits					

Chemical Engineering (Petroleum Engineering Concentration) Four-Year Sample Schedule

Freshman Year, First Semester			Freshman Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PHYS 110	University Physics I	3	ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I or Arabic Language and Culture for Native Arabic Speakers I	3
PHYS 111	University Physics I Lab.	1	MATH 114	Calculus II	3
ENGR 107	Introduction to Engineering	2	MEST 100	Introduction to Islam in World Culture	3
ENGL 101	Composition	3	PHYS 220	University Physics II	3
MATH 113	Calculus I	3	PHYS 221	University Physics II Lab.	1
UNIV 100	University First-Year Transition	1	CSCI 114	Applied Computational Thinking	3
CHEM 211	General Chemistry I	3	CSCI 115	Applied Computational Tools	1
CHEM 212	General Chemistry I Lab. I	1			
Total		17	Total		17
Sophomore Year, First Semester			Sophomore Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CHEM 213	General Chemistry II	3	MENG 241	Engineering Materials	3
CHEM 214	General Chemistry II Lab.	1	CHEM 215	Organic Chemistry I	3
ENGR 210	Sustainability in Engineering	2	MATH 214	Elementary Differential Equations	3
MATH 203	Linear Algebra	3	CHEN 202	Principles of Chemical and Petroleum Engineering	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability	3	CIEN 251	Fluid Mechanics	3
ECON 103	Principles of Microeconomics	3	PENG 351	Reservoir Rock and Fluid Properties	3
Total		15	Total		18
Sophomore Year, Summer Semester					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 390	Internship I	3			
Junior Year, First Semester			Junior Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PHIL 100 or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3	CHEN 312	Chemical Engineering Thermodynamics	3
ENGR 200	Engineering Statistics	3	CHEN 351	Chemical Reaction Engineering	3
CHEM 315	Physical Chemistry	3	CHEN 370	Heat and Mass Transport	3
MENG 323	Engineering Economy	3	PENG 362	Drilling Engineering	3
UAES 200	Survey of United Arab Emirates Studies	3	PENG 371	Petroleum Reservoir Engineering	3
CHEN 301	Computational Tools in Chemical and Petroleum Engineering	1	PENG 384	Petroleum Production Engineering	3
Total		16	Total		18
Junior Year, Summer Semester					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 391	Internship II	3			
Senior Year, First Semester			Senior Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CHEN 471	Unit Operation Lab	1	CHEN 482	Chemical Process Dynamics and Control	3
CHEN 472	Separation Processes	3	CHEN 492	Senior Design Project II	3
CHEN 481	Process & Plant Design	3	PENG 381	Well Logging	3
CHEN 483	Advance Simulation Lab	1	PENG 473	Reservoir Simulation	3
	Technical Elective 1	3		Technical Elective II	3
CHEN 491	Senior Design Project I	3			
Total		14	Total		15
Total 136 Credits					

Bachelor of Science in Civil and Infrastructure Engineering

Program Description

The Civil and Infrastructure Engineering (CIEN) program is a unique undergraduate program in the United Arab Emirates and in the region as it concentrates on infrastructure engineering in addition to traditional civil engineering areas. The CIEN program prepares students to excel in several sectors including building design and construction, highways and transportation, planning and management, and consulting.

The program has 134 credit hours over the duration of eight semesters including hours at internship programs. The curriculum of the program covers construction engineering and management, structural and materials engineering, highway and transportation engineering, geotechnical engineering, and water and environmental engineering. The program was prepared according to the International Standards of Accreditation Board for Engineering and Technology (ABET) and has passed all the conditions and criteria set by the Commission for Academic Accreditation (CAA) at the Ministry of Higher Education and Scientific Research, United Arab of Emirates.

The faculty members of the Civil and Infrastructure Engineering program have graduated from top universities in the world and their expertise covers a broad range of technical areas, including structural engineering, geotechnical engineering, construction engineering and management, environmental systems, hydraulics and water resources, and transportation systems engineering.

The Bachelor of Science in Civil and Infrastructure Engineering is overseen by the School of Engineering and Computing and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Science in Civil and Infrastructure Engineering program at AURAK educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of Civil and Infrastructure Engineering. The graduates of the program will be well versed in technology and social and environmental issues.

To fulfill this mission, the program provides the undergraduate student with a thorough foundation in the basic tenets of civil and infrastructure technologies and a broad introduction into structures, engineering materials, transport systems, soil engineering, and environment protection. The program provides a strong background for graduate study in the diverse areas branching out of the civil and infrastructure field. The technical focus is complemented with topics in general education leading to a well-rounded member of a global society.

Program Educational Objectives

The Bachelor of Science in Civil and infrastructure Engineering program at AURAK, based on student outcomes, produces graduates, who are armed with knowledge and skills to engage in lifelong learning and application of sustainable technologies to become:

1. Leaders in their profession who serve society in the construction industry, the transportation sector, project management, and in shaping public environmental and infrastructure policies;
2. Competent civil engineers who are engaged in the design of civil engineering and infrastructure systems; and
3. Contributors to the advancement of the civil engineering discipline at the national and global levels.

Program Learning Outcomes

AURAK has adopted ABET's student outcomes for the Civil and Infrastructure Engineering program. The outcomes for the program are for the student to acquire:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Degree Requirements

The Bachelor of Science in Civil and Infrastructure Engineering requires the completion of 134 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering and Computing Requirements	30
Civil and Infrastructure Engineering Department	72

University General Education Requirements

32 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS in Civil and Infrastructure Engineering is CIEN 334 Civil Engineering Testing and Materials.**

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 114	Applied Computational Thinking	3
CSCI 115	Applied Computational Tools	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
ECON 103	Principles of Microeconomics	3
CHEM 211	General Chemistry I	3

School of Engineering and Computing Requirements

30 Credit Hours

Course Code	Course Title	Credits
ENGR 107	Introduction to Engineering	2
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab.	1
MATH 114	Calculus II	3
MATH 213	Calculus III	3
MATH 214	Elementary Differential Equations	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab.	1
ENGR 200	Engineering Statistics	3
ENGR 390	Internship I	3
ENGR 391	Internship II	3
ENGR 450	Engineering Seminar	1
CHEM 212	General Chemistry I Lab	1

Civil and Infrastructure Engineering Department Requirements**72 Credit Hours****Core Courses****63 Credit Hours**

Course Code	Course Title	Credits
CIEN 201	Computer Aided Drawings	3
CIEN 211	Statics	3
CIEN 212	Mechanics of Materials	3
CIEN 241	Infrastructure Management	3
ENGR 210	Sustainability in Engineering	2
CIEN 251	Fluid Mechanics	3
CIEN 261	Surveying	1
CIEN 265	Geographical Information System	2
CIEN 301	Numerical Analysis	3
CIEN 311	Structural Analysis	3
CIEN 321	Reinforced Concrete Design	3
CIEN 331	Construction Materials	3
CIEN 332	Construction Materials Lab	1
CIEN 333	Geotechnical Engineering	3
CIEN 334	Civil Engineering Testing and Materials (Writing Intensive Course)	1
CIEN 351	Environmental Engineering	2
CIEN 361	Highway Engineering and Design	3
CIEN 362	Transportation Engineering	3
CIEN 421	Structural Steel Design	3
CIEN 440	Infrastructure Financing	3
CIEN 441	Construction Management	3
CIEN 451	Infrastructure Systems	3
CIEN 491	Senior Design Project (1)	2
CIEN 492	Senior Design Project (2)	4

Basic Science Electives**3 Credit Hours**

Course Code	Course Title	Credits
PENG 101	Physical Geology	3
BIOL 112	University Biology	3

Technical Electives**6 Credit Hours**

Course Code	Course Title	Credits
CIEN 422	Advanced Reinforced Concrete Design	3
CIEN 424	Bridge Design	3
CIEN 431	Foundation Engineering	3
CIEN 442	Construction Planning and Scheduling	3
CIEN 444	Construction Cost Analysis and Estimating	3
CIEN 457	Water Resources Sustainability	3
ENGR 410	Climate Change	3
CIEN 456	Sustainable Urban Building Sites	3
CIEN 462	Advanced Pavement Design	3
CIEN 464	Airport Planning and Design	3
CIEN 493	Special Topics in Civil Engineering	3
ENGR 399	Undergraduate Research Project	3

Civil and Infrastructure Engineering Four-Year Sample Schedule

First Year, First Semester			First Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGL 101	Composition	3	ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGR 107	Introduction to Engineering	2	MATH 114	Calculus II	3
MATH 113	Calculus I	3	MEST 100	Introduction to Islam in World Culture	3
UNIV 100	University First-Year Transition	1	CSCI 114	Applied Computational Thinking	3
PHYS 110	University Physics I	3	CSCI 115	Applied Computational Tools	1
PHYS 111	University Physics I Lab.	1	PHYS 220	University Physics II	3
CHEM 211	General Chemistry I	3	PHYS 221	University Physics II Lab.	1
CHEM 212	General Chemistry I Lab.	1			
Subtotal =		17	Subtotal =		17
Second Year, First Semester			Second Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CIEI 201	Computer Aided Drawings	3	CIEI 212	Mechanics of Materials	3
CIEI 211	Statics	3	CIEI 251	Fluid Mechanics	3
CIEI 241	Infrastructure Management	3	CIEI 261	Surveying	1
ENGR 210	Sustainability in Engineering	2	CIEI 265	Geographical Information System	2
MATH 213	Calculus III	3	MATH 214	Elementary Differential Equations	3
ECON 103	Principles of Microeconomics	3	PENG 101 or BIOL 112	Physical Geology or University Biology I	3
Subtotal =		17	Subtotal =		15
Second Year, Summer Session					
Course Code	Course Title	Credits			
ENGR 390	Internship I	3			
Third Year, First Semester			Third Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CIEI 301	Numerical Analysis	3	CIEI 321	Reinforced Concrete Design	3
CIEI 311	Structural Analysis	3	CIEI 334	Civil Engineering Testing and Materials	1
CIEI 331	Construction Materials	3	CIEI 351	Environmental Engineering	2
CIEI 332	Construction Materials Lab.	1	CIEI 361	Highway Engineering and Design	3
CIEI 333	Geotechnical Engineering	3	CIEI 362	Transportation Engineering	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability	3	ENGR 200	Engineering Statistics	3
Subtotal =		16	Subtotal =		15
Third Year, Summer Semester					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 391	Internship II	3			
Fourth Year, First Semester			Fourth Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CIEI 421	Structural Steel Design	3	CIEI ---	Technical Elective II	3
CIEI 440	Infrastructure Financing	3	CIEI 492	Senior Design Project (2)	4
CIEI 441	Construction Management	3	PHIL 100 or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
ENGR 450	Engineering Seminar	1	UAES 200	Survey of United Arab Emirates Studies	3
CIEI 491	Senior Design Project (1)	2			
CIEI 451	Infrastructure Systems	3			
CIEI ---	Technical Elective I	3			
Subtotal =		18	Subtotal =		13
Total 134 Semester Credit Hours					

Bachelor of Science in Computer Engineering

Program Description

The Computer Engineering program integrates both electronics engineering and computer science in the development of computer systems. Computer Engineers are involved in many hardware and software aspects of computing including design of microcontrollers, computers, and circuit design. These engineers focus not just on how computers work but on how to integrate them into larger systems.

Students learn the fundamentals of circuits, systems, and software associated with the design of programmable systems as used for general purpose computing, communications, control, or signal processing. Because of their broad skills in both hardware and software, students in computer engineering are in high demand for employment after graduation. Computer engineering students are also positioned to pursue graduate programs in either computer science or electronics engineering.

The Bachelor of Science in Computer Engineering is overseen by the School of Engineering and Computing and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The mission of the Computer Engineering program is to produce graduates who have the necessary theoretical background and the technical know-how and skills to create successful innovations and to solve problems in the field and to understand the social and environmental issues that may be involved. Graduates will have a thorough foundation in the basic principles of electrical engineering and information technology and a broad introduction to electronics, the design of digital computer systems, software development, data communications, and networking. Students will be prepared for employment in diverse areas of computer engineering or for further study in a graduate program.

Program Educational Objectives

A few years after earning their degree, our graduates will be:

1. Successful professionals in the theoretical and practical areas of Computer Engineering;
2. Ethical and responsible engineers with sufficient awareness of the impact of engineering solutions in various contexts; and
3. Life-long learners equipped with the skills required for professional practice including functioning in teams and communicating effectively.

Program Learning Outcomes

Graduates of the program will have the ability to:

1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
3. Communicate effectively with a range of audiences;

4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts;
5. Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions; and
7. Acquire and apply new knowledge as needed, using appropriate learning strategies.

Degree Requirements

The BS in Computer Engineering (CE) requires the completion of 132 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering and Computing Requirements	32
CE Program Requirements	68

University General Education Requirements

32 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS Computer Engineering is CSCI 463 Data Communication and Computer Networks Laboratory.**

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
Gen Ed	Social and Behavioral Sciences Course	3
ENVS 102	Sustainability and Human-Environment Relations	3

School of Engineering and Computing Requirements**32 Credit Hours**

Course Code	Course Title	Credits
ENGR 107	Introduction to Engineering	2
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab	1
MATH 114	Calculus II	3
MATH 213	Calculus III	3
MATH 214	Elementary Differential Equations	3
MATH 203	Linear Algebra	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab	1
ENGR 200	Engineering Statistics	3
ENGR 450	Engineering Seminar	1
ENGR 390	Internship I	3
ENGR 391	Internship II	3

CE Program Requirements**68 Credit Hours****Core Courses****62 Credit Hours**

Course Code	Course Title	Credits
CSCI 211	Object Oriented Programming	3
MATH 225	Discrete Mathematics	3
CSCI 215	Data Structures and Algorithms	3
CSCI 312	Operating System Fundamentals	3
CENG 315	Microprocessors	3
CENG 316	Microprocessors Lab	1
CENG 335	Computer Architecture	3
CENG 336	Computer Architecture Lab	1
CENG 411	Software Engineering	3
CENG 431	Embedded Systems Design	3
CENG 432	Embedded Systems Design Lab	1
CSAI 350	Introduction to Artificial Intelligence	3
CSAI 484	Internet of Things System	3
EEEN 220	Signals & Systems	3
EEEN 221	Signals & Systems Lab	1
EEEN 280	Electric Circuit Analysis I	3
EEEN 281	Electric Circuit Analysis Lab	1
EEEN 331	Digital System Design	3

Course Code	Course Title	Credits
EEEN 332	Digital System Design Lab	1
EEEN 333	Linear Electronics I	3
EEEN 334	Linear Electronics I Lab	1
CSCI 326	Database Systems	3
CSCI 462	Data Communications and Computer Networks	3
CSCI 463	Data Communications and Computer Networks Lab (Writing Intensive)	1
CENG 492	Senior Design Project I	2
CENG 493	Senior Design Project II	4

Technical Electives

6 Credit Hours

Course Code	Course Title	Credits
CSCI 415	Introduction to Parallel Programming	3
CSCI 450	Information Security and Privacy	3
CENG 435	Parallel Computer Architectures	3
CENG 437	Introduction to Robotics	3
CENG 461	Network Security	3
CENG 401	Network Servers & Architecture	3
CSCI 499	Special Topics in Computing	3
EEEN 431	Digital Circuit Design	3
EEEN 481	Concepts of Multimedia Processing and Transmission	3
ENGR 399	Undergraduate Research Project	3

BS in Computer Engineering Four-Year Sample Schedule

First Year, First Semester			First Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PHYS 110	University Physics I	3	ARAB 101 Or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
PHYS 111	University Physics I Lab	1	CSCI 112	Introduction to Computer Programming	3
ENGL 101	Composition	3	CSCI 113	Intro. to Computer Programming Lab	1
ENGR 107	Introduction to Engineering	2	MATH 114	Calculus II	3
MATH 113	Calculus I	3	PHYS 220	University Physics II	3
ENVS 102	Sustainability and Human-Environment Relations	3	PHYS 221	University Physics II Lab	1
UNIV 100	University First-Year Transition	1	MEST 100	Introduction to Islam in World Culture	3
Subtotal =		16	Subtotal =		17
Second Year, First Semester			Second Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MATH 225	Discrete Mathematics	3	CSCI 215	Data Structures and Algorithms	3
CSCI 211	Object-Oriented Programming	3	EEEN 220	Signals & Systems	3
EEEN 280	Electric Circuit Analysis I	3	EEEN 221	Signals & Systems Lab	1
EEEN 281	Electric Circuit Analysis Lab	1	EEEN 331	Digital System Design	3
MATH 213	Calculus III	3	EEEN 332	Digital System Design Lab	1
ENGR 200	Engineering Statistics	3	MATH 203	Linear Algebra	3
			PHIL 100 Or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
Subtotal =		16	Subtotal =		17
Second Year, Summer Semester					
Course Code	Course Title	Credits			
ENGR 390	Internship I	3			
Third Year, First Semester			Third Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CENG 335	Computer Architecture	3	CENG 315	Microprocessor	3
CENG 336	Computer Architecture Lab	1	CENG 316	Microprocessor Lab	1
CSCI 326	Database Systems	3	MATH 214	Elementary Differential Equations	3
EEEN 333	Linear Electronics I	3	CSCI 462	Data Comm. and Computer Networks	3
EEEN 334	Linear Electronics Lab I	1	CSCI 463	Data Comm. and Computer Networks Lab	1
CSCI 312	Operating System Fundamentals	3	UAES 200	Survey of United Arab Emirates Studies	3
			UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
Subtotal =		14	Subtotal =		17
Third Year, Summer Semester					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 391	Internship II	3			
Fourth Year, First Semester			Fourth Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 450	Engineering Seminar	1	CENG 493	Senior Design Project II	4
CSAI 484	Internet of Things System	3	CENG 431	Embedded Systems Design	3
Gen Ed	Social and Behavioral Sciences	3	CENG 432	Embedded Systems Design Lab	1
CENG 411	Software Engineering	3		Technical Elective I	3
CENG 492	Senior Design Project I	2		Technical Elective II	3
CSAI 350	Introduction to Artificial Intelligence	3			
Subtotal =		15	Subtotal =		14
Total 132 Credits					

Bachelor of Science in Computer Science

Program Description

The Bachelor of Science in Computer Science (CS) degree program provides a solid foundation in computing and technology related sciences, along with exploration of the various disciplines of liberal arts and humanities. By examining the latest industry tools and techniques and developing valuable experience and professional skills in areas such as object-oriented programming, database systems, and network security, students are equipped with the in-depth understanding of their chosen professions.

The CS program aims to develop and prepare individuals for rewarding, cutting- edge careers in software engineering, system administration, and management. Furthermore, the CS graduates will be well trained to handle the problems faced in industry, academia, and daily life.

The Bachelor of Science in Computer Science is overseen by the School of Engineering and Computing and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Science in Computer Science aims to provide students with high-quality education covering a broad and well-integrated knowledge in the concepts and methodologies underlying the analysis, design, and utilization of computer software, algorithms, and systems. The graduates of the program will be well versed in computer technology and its impact on the global society.

Program Educational Objectives

A few years after graduation, our alumni will be:

1. Innovators in applied areas of computer science, effectively accomplishing software-related projects to meet market needs, and actively engaging in research.
2. Professionals making informed judgments and creating a positive technological impact with awareness of the ethical and legal issues of computer science.
3. Leaders in the computing field capable of working in teams, communicating effectively, and engaging in life-long learning and community service.

Program Learning Outcomes

Graduates of the program will have the ability to:

1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions;
2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline;
3. Communicate effectively in a variety of professional contexts;
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles;
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline; and

6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

Degree Requirements

The BS in Computer Science (CS) requires the completion of 127 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering and Computing Requirements	26
CS Program Requirements	66
Free Electives	3

University General Education Requirements

32 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS Computer Engineering is CSCI 463 Data Communication and Computer Networks Laboratory.**

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
Gen Ed	Social and Behavioral Sciences Course	3
ENVS 102	Sustainability and Human-Environment Relations	3

School of Engineering and Computing Requirements

26 Credit Hours

Course Code	Course Title	Credits
PHYS 110	University Physics I	3

Course Code	Course Title	Credits
PHYS 111	University Physics I Lab	1
MATH 114	Calculus II	3
MATH 213	Calculus III	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab	1
MATH 203	Linear Algebra	3
ENGR 200	Engineering Statistics	3
CSCI 390	Internship I	3
CSCI 391	Internship II	3

CS Program Requirements

66 Credit Hours

Core Courses

60 Credit Hours

Course Code	Course Title	Credits
CSCI 104	Introduction to Computing	3
CSCI 211	Object Oriented Programming	3
MATH 225	Discrete Mathematics	3
CSCI 215	Data Structures and Algorithms	3
CSCI 232	Computer Organization	3
CSAI 350	Introduction to Artificial Intelligence	3
CSAI 351	Data Science	3
CSCI 315	Design and Analysis of Algorithms	3
CSCI 326	Database Systems	3
CSCI 312	Operating System Fundamentals	3
CSCI 372	Compiler Design	3
CSCI 388	Programming Languages	3
CENG 335	Computer Architecture	3
CENG 336	Computer Architecture Lab	1
EEEN 331	Digital Systems Design	3
EEEN 332	Digital Systems Design Lab	1
CENG 411	Software Engineering	3
CSCI 462	Data Communications and Computer	3
CSCI 463	Data Communications and Computer Networks Lab (Writing Intensive)	1
CENG 461	Network Security	3
CSCI 492	Senior Design Project I	2
CSCI 493	Senior Design Project II	4

Technical Electives**6 Credit Hours**

Course Code	Course Title	Credits
CENG 437	Introduction to Robotics	3
CSCI 411	Computer Graphics	3
CSCI 412	Computer Graphics Lab	1
CSCI 415	Introduction to Parallel	3
CSCI 416	Human Computer Interaction	3
CENG 435	Parallel Computer Architectures	3
EEEN 481	Concepts of Multimedia Processing and Transmission	3
CENG 401	Network Servers & Architecture	3
CSCI 450	Information Security and Privacy	3
CSCI 499	Special Topics in Computing	3
ENGR 399	Undergraduate Research Project	3

Free Electives**3 Credit Hours**

Students must complete one course (3 credits) of free electives.

BS in Computer Science Four-Year Sample Schedule

First Year, First Semester			First Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CSCI 104	Introduction to Computing	3	ARAB 101 Or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition	3	CSCI 112	Introduction to Computer Programming	3
MATH 113	Calculus I	3	CSCI 113	Intro. to Computer Programming Lab	1
PHYS 110	University Physics I	3	MATH 114	Calculus II	3
PHYS 111	University Physics I Lab	1	PHYS 220	University Physics II	3
UNIV 100	University First-Year Transition	1	PHYS 221	University Physics II Lab	1
			MEST 100	Introduction to Islam in World Culture	3
Subtotal =		14	Subtotal =		17
Second Year, First Semester			Second Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CSCI 211	Object Oriented Programming	3	CSCI 215	Data Structures and Algorithms	3
MATH 213	Calculus III	3	EEEN 331	Digital Systems Design	3
MATH 225	Discrete Mathematics	3	EEEN 332	Digital Systems Design Lab	1
CSCI 232	Computer Organization	3	MATH 203	Linear Algebra	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability	3	ENGR 200	Engineering Statistics	3
			PHIL 100 Or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
Subtotal =		15	Subtotal =		16
Second Year, Summer Session					
Course Code	Course Title	Credits			
CSCI 390	Internship I	3			
Subtotal =		3			
Third Year, First Semester			Third Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CENG 335	Computer Architecture	3	Gen Ed	Social and Behavioral Sciences	3
CENG 336	Computer Architecture Lab	1	CSCI 372	Compiler Design	3
ENVS 102	Sustainability and Human-Environment Relations	3	CSCI 462	Data Comm. and Computer Networks	3
CSCI 326	Database Systems	3	CSCI 463	Data Comm. and Computer Networks Lab	1
UAES 200	Survey of United Arab Emirates Studies	3	CSCI 388	Programming Languages	3
CSCI 312	Operating System Fundamentals	3	CSCI 315	Design and Analysis of Algorithms	3
Subtotal =		16	Subtotal =		16
Third Year, Summer Session					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CSCI 391	Internship II	3			
Subtotal =		3			
Fourth Year, First Semester			Fourth Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CSAI 350	Introduction to Artificial Intelligence	3	CSCI 493	Senior Design Project II	4
CENG 411	Software Engineering	3	CENG 461	Network Security	3
CSCI 492	Senior Design Project I	2	CSAI 351	Data Science	3
	Free Elective	3		Technical Elective II	3
	Technical Elective I	3			
Subtotal =		14	Subtotal =		13
Total 127 Credits					

Bachelor of Science in Electrical and Electronics Engineering

Program Description

The Bachelor of Science in Electrical and Electronics Engineering provides graduates with a sound understanding of fundamental engineering principles and applications in the field.

Graduates are prepared for post-graduate study or employment and are able to demonstrate their ability to apply the knowledge and methodologies of electrical and electronics engineering essential for a successful career. The American University of Ras Al Khaimah's (AURAK) Bachelor of Electrical and Electronics Engineering aims to produce effective, well educated, and globally competitive electrical and electronics engineers for the region and the wider world. The program aims to be amongst the top in the GCC region in the fields of electrical and electronics engineering with highly qualified and dedicated faculty and staff. It also aims to have a strong link with the local and regional government, public-sector organizations, and private sector enterprises. Moreover, the program aims to provide service to the community by sharing the program's technical expertise and human resources.

The Bachelor of Science in Electrical and Electronics Engineering is overseen by the School of Engineering and Computing and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The mission of the Electrical and Electronics Engineering program is to provide graduates with a sound understanding of fundamental engineering principles and applications in the field. Graduates will have a comprehensive understanding of subjects like signals and systems, digital logic, circuits and electronics, and the theory and applications of digital communications, particularly networks and wireless communications, in addition to power systems, control systems, and electric machines. Graduates will be prepared for both graduate study and employment. They will be able to demonstrate their ability to apply the knowledge and methodologies in power systems, electronics, and communications engineering essential for a successful career.

Program Educational Objectives

The graduates of Electrical and Electronics Engineering Program, within a few years will be:

1. Become successful practitioners and innovators in the region and worldwide;
2. Continue their professional development throughout their careers; and
3. Work productively and ethically in teams, and function effectively in diverse cultural environments.

Program Learning Outcomes

AURAK Electrical and Electronics Engineering graduates should have:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
3. An ability to communicate effectively with a range of audiences;

4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts;
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions; and
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Degree Requirements

The BS in Electrical and Electronics Engineering (EEE) requires the completion of 135 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering and Computing Requirements	34
EEE Department Requirements	69 (63 CH compulsory and 6 CH technical electives)

University General Education Requirements

32 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS in Electrical and Electronics Engineering is EEEN 434 Linear Electronics II Laboratory.**

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
Gen Ed	Social and Behavioral Science Course – ECON 103 is recommended.	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
CHEM 211	General Chemistry I	3

School of Engineering and Computing Requirements**34 Credit Hours**

Course Code	Course Title	Credits
ENGR 107	Introduction to Engineering	2
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab	1
MATH 114	Calculus II	3
MATH 203	Linear Algebra	3
MATH 213	Calculus III	3
MATH 214	Elementary Differential Equations	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab	1
ENGR 200	Engineering Statistics	3
ENGR 210	Sustainability in Engineering	2
ENGR 390	Internship I	3
ENGR 391	Internship II	3
ENGR 450	Engineering Seminar	1

EEE Department Requirements**69 Credit Hours****Core Courses****63 Credit Hours**

Course Code	Course Title	Credits
EEEN 220	Signals & Systems	3
EEEN 221	Signals & Systems Lab	1
EEEN 280	Electric Circuit Analysis I	3
EEEN 281	Electric Circuit Analysis Lab	1
EEEN 282	Electric Circuit Analysis II	3
EEEN 305	Electromagnetic Theory	3
EEEN 331	Digital System Design	3
EEEN 332	Digital System Design Lab	1
EEEN 333	Linear Electronics I	3
EEEN 334	Linear Electronics I Lab	1
EEEN 350	Electric Machines	3
EEEN 351	Electric Machines Lab	1
EEEN 412	Power Systems	3
EEEN 413	Microcontrollers and Embedded Systems	3
EEEN 414	Microcontrollers and Embedded Systems Lab	1
EEEN 431	Digital Circuit Design	3
EEEN 433	Linear Electronics II	3
EEEN 434	Linear Electronics II Lab	1
EEEN 437	Power Electronics	3
EEEN 451	Control Theory	3
EEEN 452	Control Lab	1
EEEN 460	Communication Systems	3
EEEN 461	Communication Engineering Lab	1
EEEN 464	Digital Communication Systems	3
EEEN 466	Digital Signal Processing	3

Course Code	Course Title	Credits
EEEN 492	Senior Design Project I	2
EEEN 493	Senior Design Project II	4

Technical Electives

6 Credit Hours

Course Code	Course Title	Credits
CENG 315	Microprocessor	3
EEEN 455	Digital Image Processing	3
EEEN 462	Data and Computer Communication	3
EEEN 467	Mobile and Wireless Communications	3
EEEN 472	Antenna Theory and Design	3
EEEN 474	Advanced Information Theory and Coding	3
EEEN 487	Robotics Engineering	3
EEEN 481	Concepts of Multimedia Processing & Transmission	3
EEEN 499	Special Topics in EEEN	3
EEEN 422	High Voltage Engineering	3
EEEN 423	Electrical Energy Systems and Fault Analysis	3
EEEN 425	Smart Power Grid Systems Theory and Implementation	3
EEEN 426	Renewable Energy Systems	3
ENGR 399	Undergraduate Research Project	3

Electrical and Electronics Engineering Program EEE 4-Year Study Plan

First Year, First Semester			First Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PHYS 110	Physics I	3	ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
PHYS 111	Physics I Lab	1			
ENGL 101	Composition	3	CSCI 112	Intro. to Computer Programming	3
ENGR 107	Introduction to Engineering	2	CSCI 113	Intro. to Computer Programming Lab	1
MATH 113	Calculus I	3	MATH 114	Calculus II	3
UNIV 100	University First-Year Transition	1	PHYS 220	Physics II	3
CHEM 211	General Chemistry I	3	PHYS 221	Physics II Lab	1
			MEST 100	Introduction to Islam in World Culture	3
Subtotal =		16	Subtotal =		17
Second Year, First Semester			Second Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
EEEN 280	Electric Circuit Analysis I	3	EEEN 220	Signals & Systems	3
EEEN 281	Electric Circuit Analysis I Lab	1	EEEN 221	Signals & Systems Lab	1
UAES 200	Survey of United Arab Emirates Studies	3	EEEN 282	Electric Circuit Analysis II	3
MATH 203	Linear Algebra	3	ENGR 200	Engineering Statistics	3
MATH 213	Calculus III	3	MATH 214	Elementary Differential Equations	3
EEEN 331	Digital System Design	3	ENGR 210	Sustainability in Engineering	2
EEEN 332	Digital System Design Lab	1			
Subtotal =		17	Subtotal =		15
Second Year, Summer Session					
Course Code	Course Title	Credits			
ENGR 390	Internship I	3			
Third Year, First Semester			Third Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credit
EEEN 305	Electromagnetic Theory	3	EEEN 412	Power Systems	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability	3	EEEN 433	Linear Electronics II	3
EEEN 350	Electric Machines	3	EEEN 434	Linear Electronics II Lab	1
EEEN 351	Electric Machines Lab	1	EEEN 431	Digital Circuit Design	3
EEEN 333	Linear Electronics I	3	EEEN 451	Control Theory	3
EEEN 334	Linear Electronics I Lab	1	EEEN 452	Control Lab	1
EEEN 413	Microcontrollers and Embedded Systems	3	EEEN 460	Communication Systems	3
EEEN 414	Microcontrollers and Embedded Systems Lab	1			
Subtotal =		18	Subtotal =		17
Third Year, Summer Semester					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 391	Internship II	3			
Fourth Year, First Semester			Fourth Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PHIL 100 or ENGL 200	Critical Thinking and Reasoning OR Advanced Composition	3	EEEN 464	Digital Communication Systems	3
EEEN 437	Power Electronics	3	ENGR 450	Engineering Seminar	1
EEEN 492	Senior Design Project I	2	EEEN 493	Senior Design Project II	4
EEEN 466	Digital Signal Processing	3	EEEN 461	Communication Engineering Lab	1
	Technical Elective I	3		Technical Elective II	3
			ECON 103	Principles of Microeconomics	3
Subtotal =		14	Subtotal =		15
Total 135 Credits					

Bachelor of Science in Mechanical Engineering

Program Description

The Bachelor of Science degree in mechanical engineering requires a minimum of 134 credit hours of coursework. Mechanical engineering is one of the largest, broadest, and oldest engineering disciplines. Mechanical engineers use the principles of energy, materials, and mechanics to design and manufacture machines and devices of all types. They create the processes and systems that drive technology and industry.

The key characteristics of the profession are its breadth, flexibility, and individuality. The career paths of mechanical engineers are largely determined by individual choices, a decided advantage in a changing world.

Mechanics, energy and heat, mathematics, engineering sciences, design and manufacturing form the foundation of mechanical engineering. Mechanics includes fluids, ranging from still water to hypersonic gases flowing around a space vehicle; it involves the motion of anything from a particle to a machine or complex structure.

The Bachelor of Science in Mechanical Engineering is overseen by the School of Engineering and Computing and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Science in the mechanical engineering program at the American University of Ras Al Khaimah (AURAK) educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of mechanical engineering. The graduates of the program will be well versed in technology and in social and environmental issues.

To fulfill this mission, the program provides the undergraduate student with a thorough foundation in the basic tenets of mechanical engineering and technologies, and a broad introduction into machine design, engineering materials, thermal sciences, energy and environment protection and mechanical systems. The program provides a strong background for graduate study in the diverse areas branching out of the mechanical engineering field. The technical focus is complemented with topics in general education leading to a well-rounded member of the global society.

Program Educational Objectives

A few years after earning their degree, our graduates will:

1. Take leadership positions in energy and industrial sectors in response to regional and global market needs.
2. Pursue higher education, research and development, and/or other creative and innovative efforts in science, engineering and technology.
3. Create start-up companies that provide creative products and services for industry and society.

Program Learning Outcomes

Graduates of the program will have the ability to:

- 1 Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

- 2 Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
- 3 Communicate effectively with a range of audiences
- 4 Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
- 5 Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
- 6 Develop and conduct appropriate experimentation, analyze and interpretation of data, and use engineering judgment to draw conclusions
- 7 Acquire and apply new knowledge as needed, using appropriate learning strategies

Degree Requirements

The BS in Mechanical Engineering (MENG) requires the completion of 134 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering and Computing Requirements	32
Mechanical Engineering Department Requirements	64
Technical Electives	6

University General Education Requirements

32 credit hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS in Mechanical Engineering is MENG 313 Manufacturing Processes Lab.**

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 114	Applied Computational Thinking	3
CSCI 115	Applied Computational Tools	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing)	3
ECON 103	Principles of Microeconomics	3

UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
CHEM 211	General Chemistry I	3

School of Engineering and Computing Requirements

32 Credit Hours

Course Code	Course Title	Credits
ENGR 107	Introduction to Engineering	2
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab	1
MATH 114	Calculus II	3
MATH 213	Calculus III	3
MATH 214	Elementary Differential Equations	3
MATH 203	Linear Algebra	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab	1
ENGR 210	Sustainability in Engineering	2
ENGR 450	Engineering Seminar	1
ENGR 390	Internship I	3
ENGR 391	Internship II	3
CHEM 212	General Chemistry I Lab	1

MENG Program Requirements

70 Credit Hours

Core Courses

64 Credit Hours

Course Code	Course Title	Credits
CIEN 211	Statics	3
CIEN 212	Mechanics of Materials	3
CIEN 251	Fluid Mechanics	3
CIEN 301	Numerical Analysis	3
EEEN 280	Electric Circuits Analysis I	3
EEEN 281	Electric Circuits Analysis Lab.	1
MENG 241	Engineering Materials	3
MENG 242	Engineering Materials Lab.	1
ENGR 200	Engineering Statistics	3
MENG 312	Manufacturing Processes	3
MENG 313	Manufacturing Processes Lab.	1
MENG 201	Mechanical Engineering Drawing	3
MENG 211	Thermodynamics I	3
MENG 212	Thermodynamics II	3
MENG 221	Dynamics	3
MENG 231	Engineering Measurements	2
MENG 321	Vibration and Control	3
MENG 342	Fluid Mechanics Lab.	1
MENG 361	Heat Transfer	3
MENG 362	Thermal Sciences Lab.	1
MENG 421	Theory of Machines	3

Course Code	Course Title	Credits
MENG 451	Mechanical Design I	3
MENG 452	Mechanical Design II	3
MENG 491	Senior Design Project I	2
MENG 492	Senior Design Project II	4

Technical Electives

6 Credit Hours

Course Code	Course Title	Credits
MENG 311	Internal Combustion Engines	3
MENG 323	Engineering Economy	3
MENG 441	Turbo Machinery	3
MENG 453	Computer Aided Design	3
MENG 455	Finite Elements in Machine Design	3
MENG 461	HVAC and Refrigeration Systems	3
MENG 462	Design of Thermal System	3
MENG 463	Energy Conversion and Management	3
MENG 471	Sustainable Product Design	3
MENG 493	Special Topics in Mechanical Engineering	3
ENGR 399	Undergraduate Research Project	3
ENGR 420	Life Cycle Assessment	3
EEEN 426	Renewable Energy Systems	3
EEEN 451	Control Theory	3

BS in Mechanical Engineering Four-Year Sample Schedule

First Year, First Semester			First Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PHYS 110	University Physics I	3	ARAB 101 Or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
PHYS 111	University Physics I Lab	1	CSCI 114	Applied Computational Thinking	3
ENGL 101	Composition	3	CSCI 115	Applied Computational Tools	1
ENGR 107	Introduction to Engineering	2	MATH 114	Calculus II	3
MATH 113	Calculus I	3	PHYS 220	University Physics II	3
CHEM 211	General Chemistry I	3	PHYS 221	University Physics II Lab	1
CHEM 212	General Chemistry I Lab	1	MEST 100	Introduction to Islam in World Culture	3
UNIV 100	University First-Year Transition	1			
Subtotal =		17	Subtotal =		17
Second Year, First Semester			Second Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MENG 201	Mechanical Engineering Drawing	3	CIEN 212	Mechanics of Materials	3
CIEN 211	Statics	3	MENG 221	Dynamics	3
ENGR 210	Sustainability in Engineering	2	CIEN 251	Fluid Mechanics	3
MATH 213	Calculus III	3	MATH 214	Elementary Differential Equations	3
MENG 211	Thermodynamics I	3	MENG 212	Thermodynamics II	3
MENG 231	Engineering Measurements	2	PHIL 100 Or ENGL 200	Critical Thinking and Reasoning Or Advanced Composition	3
Subtotal =		16	Subtotal =		18
Second Year, Summer Session					
Course Code	Course Title	Credit			
ENGR 390	Internship I	3			
Third Year, First Semester			Third Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MENG 241	Engineering Materials	3	MENG 361	Heat Transfer	3
MENG 242	Engineering Materials Lab.	1	MENG 362	Thermal Sciences Lab	1
MATH 203	Linear Algebra	3	MENG 321	Vibration and Control	3
ECON 103	Principles of Microeconomics	3	ENGR 200	Engineering Statistics	3
MENG 342	Fluid Mechanics Lab	1	MENG 312	Manufacturing Processes	3
UAES 200	Survey of United Arab Emirates Studies	3	MENG 313	Manufacturing Processes Lab.	1
			EEEN 280	Electric Circuit Analysis	3
			EEEN 281	Electric Circuit Analysis Lab	1
Subtotal =		14	Subtotal =		18
Third Year, Summer Session					
Course Code	Course Title	Credits			
ENGR 391	Internship II	3			
Fouthr Year, First Semester			Fourth Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MENG 451	Mechanical Design I	3	MENG 452	Mechanical Design II	3
MENG 421	Theory of Machines	3		ME Technical Elective 2	3
	ME Technical Elective I	3	ENGR 450	Engineering Seminar	1
CIEN 301	Numerical Analysis	3	UNIV 200	Innovation, Entrepreneurship & Sustainability	3
MENG 491	Senior Design Project I	2	MENG 492	Senior Design Project II	4
Subtotal =		14	Subtotal =		14
Total 134 Credits					

Minor in Artificial Intelligence

In addition to the six undergraduate programs offered by the School of Engineering and Computing, the Department of Computer Science and Engineering offers a minor in Artificial Intelligence to students enrolled in the programs below:

- Bachelor of Science in Electrical and Electronics Engineering
- Bachelor of Science in Civil and Infrastructure Engineering
- Bachelor of Architecture
- Bachelor of Science in Mechanical Engineering
- Bachelor of Science in Chemical Engineering

AURAK maintains the existing accredited programs as a pathway to completion of each degree but will offer students the choice of taking additional courses to make up a minor.

Program Learning Outcomes

Upon successful completion of this AI minor, students will be able to

1. Design, implement, and evaluate AI-driven solutions tailored to fulfill specific engineering and computing needs.
2. Use techniques, skills, and tools necessary for AI-based solutions
3. Work effectively in teams to achieve a common objective

Minor Requirements

Students with a CGPA of 2.5 or above and who have completed 60 credit hours or more, wishing to pursue the minor will be required to complete at least one additional semester in order to fulfill all the minor course requirements and have the minor recorded on their transcripts. The Minor in AI consists of 18 credit hours distributed as follows:

Background Courses

Course Code	Title	Credits
CSCI 211	Object Oriented Programming	3
CSCI 215	Data Structures and Algorithms	3

AI Core Courses

Course Code	Title	Credits
CSAI 350	Introduction to Artificial Intelligence	3
CSAI 351	Data Science	3
CSAI 450	Machine Learning	3
CSAI 453	Data Visualization	3





GRADUATE PROGRAMS



ADMISSION TO GRADUATE STUDIES

AURAK offers five Master Programs in response to the needs of Ras Al Khaimah, the region, and the international community. AURAK's programs meet the highest international standards, and our students will be well prepared for the professional challenges they will face in their careers. The programs engage relevant contemporary issues as well as crucial intellectual contexts. Students can expect to engage with one another and their professors in stimulating discussions and challenging assignments. They will study with experienced faculty from many of the best university in North America, Europe, Asia, and the Middle East.

Degree Offerings

AURAK currently offers five (5) programs of graduate study leading to a master's degree. These are:

- Master of Education: Educational Leadership
- Executive Master of Business Administration
- Master of Business Administration
- Master of Science in Engineering Project Management
- Master of Science in Sustainable and Renewable Energy

Application Process

Matriculating Students

Applications for all AURAK graduate programs are processed through the Department of Admissions under the guidance of the appropriate school and program of study.

Prospective students can apply online or in person. The online application is available on <https://apply.aurak.ac.ae>.

The following documents would be required to submit:

- Attested transcripts for bachelor's degree(s) and certificate(s);
- English proficiency scores (EmSAT/TOEFL/IELTS or equivalent);
- Two (2) recommendation letters;
- Updated resume;
- Valid passport copy, Emirates ID, and residence visa (If Applicable);
- Health insurance card;
- Health history form; and
- Two (2) passport photos.

Incomplete applications are not processed.

Upon receiving a complete application, the Department of Admissions coordinates with the appropriate school and program to determine if the applicant meets the requirements for admission to the program. The relevant Graduate Program Coordinator makes all final admissions decisions. The Department of Admissions will notify the applicant of the university's final decision and provide all relevant information.

Application Deadlines

AURAK Admissions Application Deadline for Fall 2024 Semester
9th of September 2024

AURAK Admissions Final Application Deadline for Spring 2025 Semester
20th of January 2025

Note: International students must apply two weeks in advance of the stated deadline to meet visa timeline requirements.

International applicants (i.e., graduates of universities located outside the UAE) are required to submit completed application forms and all supporting documents to the Department of Admissions.

Graduates of universities outside the UAE are required to present a Recognition of University Certificates from the UAE Ministry of Education. The procedure for obtaining recognition can be obtained online through the following link: <https://www.moe.gov.ae/En/EServices/ServiceCard/pages/universiycertificateequilization.aspx#>

Admitted international students who need visas for the UAE should submit the visa application form, which is available from the Department of Admissions, at least three weeks prior to the first day of class.

Admission is valid only for the semester immediately following the admission decision. If an admitted student would like to have the admission date delayed, he or she must submit a formal request in writing to the Department of Admissions. After consultation with the Graduate Program Coordinator and the Dean of the school in which the admitted student's degree program is located, the student will be informed of the university's decision concerning delay of admission.

University Requirements for Admission to Graduate Programs

To be considered for admission, all applicants must meet the general university requirements for graduate admission. There are also program-specific requirements, information about which applicants will find in the section of the catalog in which the specific program is described.

For admission to a graduate degree program at AURAK, an applicant must:

- Hold a four-year bachelor's degree from an independently accredited university recognized by the UAE Ministry of Education (applicants with a bachelor's degree obtained outside the UAE must submit a Recognition for University Certificates of their degree from the UAE Ministry of Education); and
- Have attained the minimum score of one of the specified exams as indicated in the Graduate English Language Proficiency requirement table below. Note that the EmSAT is the preferred assessment of English proficiency.

Type of English Proficiency Exam	Minimum Score
EmSAT	1400
IELTS	6.0
TOEFL	550
TOEFL (Internet-based)	79

Table: Graduate English Proficiency Requirement

Program-Specific Requirements

- Master of Education in Educational Leadership:
 - A minimum of two (2) years of experience of teaching or equivalent work experience is required.
 - The minimum entry requirement for the Master of Education is a CGPA of 3.0 on a 4.0 scale.
- Executive Master of Business Administration:
 - A baccalaureate degree in Business Administration, with a minimum cumulative GPA of 3.0 on a 4.0 scale, or its established equivalent in the applicant's undergraduate degree program;
 - A baccalaureate degree, or its equivalent, with a major or minor in Business Studies, Marketing, Finance, Accounting, Economics, or Management/Business Information Systems with a minimum CGPA of 3.0 on a 4.0 scale or its recognized equivalent in the applicant's undergraduate degree program, or
 - A baccalaureate degree, or its equivalent, in a non-business field with a minimum of CGPA of 3.0 on a 4.0 scale, followed by a postgraduate certificate or diploma in a business-related field.
 - The students must also submit an experience certificate, which proves at least five years of work experience in a managerial position to be eligible.
 - Successful Personal Interview with School of Business Faculty.
- Master of Business Administration:
 - A baccalaureate degree in Business Administration, with a minimum cumulative GPA of 3.0 on a 4.0 scale, or its established equivalent in the applicant's undergraduate degree program;
 - A baccalaureate degree, or its equivalent, with a major or minor in Business Studies, Marketing, Finance, Accounting, Economics, or Management/Business Information Systems with a minimum CGPA of 3.00 on a 4.0 scale or its recognized equivalent in the applicant's undergraduate degree program, or
 - A baccalaureate degree, or its equivalent, in a non-business field with a minimum of CGPA of 3.00 on a 4.0 scale, followed by a postgraduate certificate or diploma in a business-related field.
 - This MBA program is geared toward applicants with or without work experience. Leveling courses may be required by the School of Business.
 - Successful Personal Interview with School of Business Faculty .
- Master of Science in Engineering Project Management:
 - A baccalaureate degree in Engineering, Computer Science, or Information Technology, with a minimum CGPA of 3.0 on a 4.0 scale.
 - Qualified applicants requiring prerequisite courses may be required to take such courses in addition to their regular graduate program courses.
 - Students with no prior background in probability and statistics will be required to take ENGR

- 200 Engineering Statistics.
- Students with no prior background in engineering economy will be required to take MENG 323 (Engineering Economy).
- Master of Science in Sustainable and Renewable Energy
 - A baccalaureate degree in engineering or a science discipline.
 - Completion of specific courses in the areas of thermodynamics, fluid mechanics and heat transfer.

Conditional Admission

English Language: AURAK may conditionally admit students with a recognized Bachelor's degree and an EmSAT score of 1250 or its equivalent on another standardized test approved by the CAA. Such a student must meet the following requirements during the period of conditional admission or be subject to dismissal:

- must achieve an EmSAT score of 1400 or equivalent, by the end of the student's first semester of study.
- may take a maximum of six credit hours in the first semester of study, not including intensive English courses.
- must achieve a minimum CGPA of 3.0 on a 4.0 scale, or its established equivalent, in the first six credit hours of credit-bearing courses studied for the graduate program.

CGPA: AURAK may conditionally admit students with a recognized Bachelor degree and a minimum cumulative grade point average (CGPA) of 2.5 on a 4.0 scale or its established equivalent. Such a student must take a maximum of nine credit hours of courses studied for the graduate program during the period of conditional admission and must achieve a minimum CGPA of 3.0 on a 4.0 scale, or its established equivalent, in these nine credits of courses studied for the graduate program or be subject to dismissal.

Prospective students with a CGPA of 2.0–2.49 will be admitted to the Graduate Remedial Pathway Program, designed to prepare them for graduate-level coursework. The program consists of three compulsory courses, detailed in the table below.

Compulsory Courses	MBFN 500 Business Graduate Foundation (3 CH) EDUC 500 Foundations of Education (3 CH) ENGR 500 Engineering Foundation for Graduate Students (3 CH)
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The student must achieve a minimum CGPA of 3.0 on a 4.0 scale, or its established equivalent, in these nine credits of remedial courses in order to progress to the graduate program or be subject to dismissal.

Only after passing the three remedial courses within one semesters will students be allowed to register for the graduate program for which they meet all other requirements. An unsuccessful student who has not completed the courses within one semesters will not be able to continue at the institution.

There may also be program-specific prerequisite courses for conditional admission students. Credits from these courses do not satisfy credit requirements for the degree and are not used to calculate the graduate CGPA.

Non-Degree Seeking Graduate Student Admission

A non-degree seeking graduate student is one who would like to take AURAK courses for academic credit

but who does not seek a master's degree. Students are admitted to AURAK with non-degree status if they meet the normal requirements for graduate admission or receive permission from the Dean of the appropriate school of study. Complete applications should be submitted to the Department of Admissions.

A non-degree seeking graduate student may take a maximum of six credit hours at the graduate level.

Visiting Students Admission

A visiting student is one who is not formally admitted to AURAK, and who is not participating in a formal exchange agreement between AURAK and a partner university but is typically enrolled in a degree program at another university. Such a student may take courses at AURAK for transfer to the student's home institution. It is wise for visiting students to check with their home institutions. Checking for the transferability of credit hours from a course from AURAK to another institution is the responsibility of the student.

To be admitted as a visiting graduate student, a student must be enrolled in a graduate program at an accredited institution and be in good academic standing in his/her current institution. Students must also have attained the same minimum score on the English Proficiency Exam as students gaining admission to the Graduate Degree Programs at AURAK. Applicants seeking visiting graduate student status must submit to AURAK's Department of Admissions a complete application form and evidence of current enrollment in another university's graduate program, typically an official transcript. The application will be considered for admission. Visiting students will be admitted as space allows. Registration in courses is subject to approval by the respective graduate program chair/coordinator.

Applicants seeking visiting student status must submit to AURAK's Department of Admissions the following documents:

- Transcripts for bachelor's degree and certificate;
- Evidence of enrollment in another university;
- An official letter from the university allowing the student to enroll for the course(s) at AURAK;
- English proficiency scores (TOEFL/IELTS or equivalent) – expired will also be accepted;
- Two (2) recommendation letters;
- Updated resume;
- Valid passport copy, Emirates ID, and residence visa (If Applicable); and
- Two (2) passport photos.

A complete application form is required with evidence of current enrollment in another university's graduate program, typically an official transcript. The application will be considered and visiting students will be admitted as space allows. Registration in courses is subject to approval by the relevant graduate program director.

Change of Status

A student may apply at any time for a change of status from non-degree to degree status, or from visiting to degree status. To change status, a student should submit a complete application through the Department of Admissions. All admissions and program requirements in place at the time of the change of status request must be met to be admitted and to graduate. Grades earned in courses taken as a visiting student in AURAK will be counted towards the CGPA (CGPA).

Courses already taken at previous institution may be accepted with the approval of the **relevant school and program**. University rules and regulations governing transfer courses and credits will apply.

Transfer Credit Policy for Graduate Studies

Applicants may be allowed to transfer up to 25% of the total program credit hours subject to review and recommendation of the program faculty. Partial transfer credit will not be awarded. Transfer credit may not be awarded for graduation projects or theses courses.

Grades earned in transferred courses do not count in the student's CGPA, though transferred credits count toward the cumulative earned hours and may apply towards meeting graduation requirements. Applicants must request that credit transfers be reviewed at the time of application.

GRADUATE ACADEMIC POLICIES

Each student is responsible for knowing AURAK's rules, regulations, requirements, and academic policies. Both the graduate information in the catalog and the institutional website are repositories of policy statements. Corrections, changes, or interpretations may be communicated by other means including electronically. Any student in doubt about an academic matter should consult the Associate Provost for Academic Affairs, the Associate Provost for Research and Community Service, Department of the Registrar, Dean of the appropriate school, or their assigned faculty advisor. Students are subject to the university's stated policies regarding patents and copyrights.

Academic Advising

Academic Advisement is a critical dimension of the AURAK student experience, one that involves both students and faculty. For students it helps them to make acceptable progress toward their degree, and for faculty it is an important aspect of their teaching responsibilities.

Each student is responsible for his or her progress towards his or her degree completion and for meeting all requirements for graduation. The advisement process is available to assist the student in completing his or her studies, but responsibility for doing so lies ultimately with each student.

Academic advising is the responsibility of full-time faculty members. A full-time faculty member will be assigned to serve as a student's academic advisor from each department where a degree or minor program is housed. Typically, advisees are distributed equally among the full-time faculty in the relevant academic departments. The advisor's responsibilities include:

- Meeting with each advisee in the two-week period before early registration each semester to guide the advisee in course selection;
- Following up on the student's course requests during early and regular registration to facilitate the student's successful registration;
- Meeting with each advisee early in the semester to ensure that the student has registered for the appropriate courses;
- Meeting with each advisee roughly at mid-semester to gauge the student's progress and advise accordingly; and
- Serving as a mentor to advisees to facilitate their academic success and to provide guidance when appropriate on their future studies and careers.

Academic advisors are expected:

- To be familiar with all academic requirements for graduation; and
- To collaborate with university colleagues to provide the strongest academic advisement possible.

Attendance Policy

Please see the Student Attendance Policy section of this catalog.

Courses and Course Schedules

Each course offered by the university has a designated course prefix (or code) and number. The course prefix represents the discipline or field of study; the number indicates the level of the course content.

Courses are offered at the discretion of the VPAASS/Provost, the Associate Provost of Academic Affairs, the Registrar, the respective deans, and department chairs. Not every course is offered every semester. The Registrar has primary responsibility for approving the course schedule.

Course Credit

Each course has a credit value. A credit represents the in-class instruction and out-of-class study per week during the sixteen-week semester. Normally one credit represents fifty minutes of class instruction per week per semester.

Course Prerequisites and Corequisites

Course prerequisites or corequisites state requirements for student entry into courses and reflect necessary preparation for attempting a course. It is the student's responsibility to be aware of these requirements as stated in the catalog, and to have taken prerequisites recently enough to be of value. The instructor of the course may drop students who have enrolled in a course for which they have not met the prerequisites.

Course Selection and Registration

By mid-semester, the class schedule for the following semester is available through the Department of Registration and on the website. Students may select courses in consultation with their faculty/academic advisor, and they can then register online or submit a completed "Course Request Form" to the Department of the Registrar by the published deadlines.

Registration Criteria

Only registered students may participate in AURAK classes. Full-time status requires that a graduate student enroll in 9 semester credit hours in Fall and Spring semesters, EXCEPT for Master of Education students in their first semester of study. In their first semester of study, the full-time course load for Master of Education students is 6 semester credit hours. Apart from the first-semester exception for Education students, graduate students enrolling in fewer than 9 semester credit hours in Fall and Spring semesters are considered to be enrolled on a part-time basis. However, please note the following:

- A student with a CGPA of less than 3.0 is limited to registering for six credit hours the following semester subject to the approval of the academic advisor.
- No more than six (6) credit hours are allowed during summer terms.
- A student with a CGPA of 3.50 or higher can enroll in 12 semester credit in the subsequent semester unless the student's program restricts registration to no more than nine credit hours in a semester.

Independent Study

Independent study courses are not allowed for graduate programs.

Summer Courses Outside AURAK

Requirements

An enrolled student is eligible to apply to take courses at another college/university during the summer with the aim of transferring credits to AURAK provided the following conditions are met:

- The student must be in good academic standing at AURAK.
- The summer courses at the host university must not be taken as attempts to repeat AURAK courses in which F grades were previously earned.
- The host university must be recognized by the UAE Ministry of Education.

- The host university must provide learning experiences similar to those offered by AURAK.
- Contact/credit hours for the course at the host university must be equivalent to, or greater than, the contact/credit hours required for its equivalent course at AURAK.
- The language of instruction of the course(s) taken at the host university must be English (except for language courses conducted in other languages) unless otherwise approved by AURAK.
- The student must obtain approval from AURAK prior to registering the summer course(s).
- Some programs may reserve the right not to allow any courses to be taken at another college/university.

Final Semester Grades

Final grades are made available on-line through the Student Portal. Official transcripts and official semester grade reports are obtained through the Department of the Registrar.

Students may access and print an unofficial record of their semester grades and other academic information from the Student Portal on the university website by logging onto their account. Official transcripts are available on request.

Change of Grade

Once a final grade has been recorded, it can be changed only in cases of computational error, recording error, or pursuant of a successful appeal of grade. Additional work of any type submitted to improve a grade after the final grade has been assigned will not be accepted. All changes of final grades must be initiated, approved, and recorded by the second week of classes of the next regular semester (spring for fall grades and fall for spring grades).

Grade Appeal

Although faculty members are generally the best judges of student performance, sometimes a student believes a grade is unfair. This procedure is therefore designed to specify the process through which students can appeal final course grades and the process through which faculty members can initiate a change in a final course grade following the official posting of students' grades in a course.

Where a student believes their final course grades to be in error the student must request a meeting with the member of faculty to ask that they reconsider the grade and ensure that the appeal is submitted within the required timeline. Completed Grade Appeal Forms must be submitted within the following timeframe:

- For Fall Semester courses, completed Grade Appeal Forms must be submitted within the first three weeks of the following Spring Semester.
- For Spring Semester courses, completed Grade Appeal Forms must be submitted within three weeks of the "Grades Available" date listed in the Academic Calendar.

For Summer Session courses, completed Grade Appeal Forms must be submitted within the first three weeks of the following Fall Semester.

The grade appeal will only be considered where:

- There was an error or unfairness in the assessment of the exam grade.
- There was an error or unfairness in the evaluation process for determining the course grade.

The faculty member will meet formally with the student and a record of the meeting will be recorded on the Grade Appeal Review Form. In the meeting with the student, the faculty member must explain how the grade

in question was arrived at. If the faculty member decides, based on the discussion with the student, that a change of grade is warranted, a Change of Grade Form will be completed by the faculty member, explaining the basis for the grade change and this will be forwarded to the Department Chair.

If the faculty member decides a change of grade is not warranted, the faculty member will complete the Grade Appeal Review Form, explaining why the original grade should not be changed and this will be forwarded to the Department Chair. The Change of Grade Form/Grade Appeal Review Form (as applicable) will then be reviewed in succession by the Department Chair, the Dean, and the Associate Provost for Student Affairs and a decision will be made.

The Associate Provost for Academic Affairs will render a decision only after the faculty member has responded to any requests for further consideration made by the Department Chair and the Dean. The decision by the Associate Provost for Academic Affairs will be final and not subject to further appeal. The Registrar will not change a student's final grade in a course for any other reason or through any other procedure.

Where an appeal is successful, the Associate Provost for Academic Affairs will forward the Change of Grade Form to the Registrar's Department so that the student's grade is changed on the Student Information System (SIS). A copy of the Grade Change Form will be filed in the student's record.

Grading System

University course work is measured in terms of quantity and quality. One semester credit hour normally represents fifty minutes per week of lecture throughout a semester. The number of credits is a measure of quantity. The grade is a measure of quality. The university system for graduate grading is as follows:

Grade	GPA Points	Percentage Scores
A	4.0	95-100
A-	3.7	90-94
B+	3.3	86-89
B	3.0	83-85
B-	2.7	80-82
C+	2.3	76-79
C	2.0	70-75
F	0	0-69

Grade Point Average (GPA)

Grade or quality point values are assigned to letter grades as indicated in the grading system table. A quality or grade point score is computed by multiplying the value of a letter grade by the number of credits for the course. For example, a student receiving an A in a three-credit course earns twelve quality points. The GPA is computed by dividing the quality points earned by the number of credits graded A through F (GPA hours).

Additional Grade Notations

Satisfactory/Unsatisfactory (S/U):

An "S" grade reflects satisfactory work and is considered passing in a course; a "U" grade reflects unsatisfactory work and is considered a failure. Courses evaluated with the "S" and "U" assessment have no effect on GPA.

Incomplete (I):

This grade is given to students who are passing a course but are unable to complete all the course work or the final exam for a verified reason beyond their control. The student must then complete all the requirements by the end of the next semester, not including any summer term, and the instructor must turn in the final grade by the end of the that semester's grading deadline. Unless an explicit written extension is filed, the grade of "I" is changed to an "F" if the course requirements have not been fulfilled. Students who have filed their intention to graduate have only six weeks from the date of degree conferral to resolve any incomplete grades. An "I" grade is not calculated in a student's GPA.

In Progress (IP):

The grade of "IP" indicates that the student is making progress in a course, writing a thesis or dissertation, or participating in an internship that extends beyond one semester or summer term. The grade of "IP" has no effect on the student's GPA.

Withdrawal (W):

The grade of "W" on a course indicates that the student has dropped the course. While it has no effect on the GPA, dropped courses are part of attempted course credits that serve as the basis for the student's credit level. A "W" for all courses in a given semester and the transcript notation "withdrawn" indicates that the student withdrew from AURAK.

Repeating a Course

Normally, graduate courses cannot be repeated. With the recommendation of the dean of the student's school and with the approval of the Associate Provost for Academic Affairs, a graduate student may be allowed to repeat any course in which a grade of "C+," "C," or "C-" is received. The original grade and the new grade will appear on the transcript, but only the new grade will be calculated into the GPA.

No course may be taken more than twice.

Academic Progress / GPA Retention Levels

Satisfactory Progress

To make satisfactory progress toward degree completion, the student must attain a CGPA of 3.0 ("B") or better.

Academic Probation and Dismissal

Graduate students receiving a grade of F in any given term will be subject to academic dismissal from the university. If the CGPA of a graduate student falls below 3.0 in any semester and no grade of F is earned, the student will be placed on Probation. Graduate students on probation will be allowed a maximum of 6 semester credit hours in the subsequent semester. Graduate students failing to earn a CGPA of 3.0 or higher in the subsequent semester will be subject to academic dismissal from the university. Probation will be removed at the end of any semester in which the student attains a CGPA of 3.00.

Academic Dismissal

A graduate student on probation who does not achieve good academic standing by the end of the regular semester following the term in which the CGPA fell below 3.00 will be dismissed from the university.

University Withdrawal

Official withdrawal removes students from any academic program and cancels student status at AURAK. In order to return to the university, students need to apply for readmission through the Department of Admissions. Students who wish to withdraw have to complete a "Withdrawal Form" available at the Department of the

Registrar and get clearance from all university departments. If the withdrawal process is completed satisfactorily and all financial obligations to the university are cleared, the effective date of withdrawal is noted on the student's permanent academic record. The effective date is the date used for calculating billing or refunds. No grades other than a "W" for the current semester are recorded.

Leave of Absence

Occasionally, students are forced to leave the university for a semester or two because of circumstances beyond their control. Others find they simply need a break from studying. In such circumstances, taking a leave of absence might be wise. Students who have an approved leave of absence for a semester or a year may register for the semester in which they plan to return without applying for readmission. Unless there are extenuating circumstances such as illness, a leave of absence is not normally given to a student who leaves the university during a term.

Fields of Study and Degree Offerings

The American University of Ras Al Khaimah has three schools that offer both undergraduate and graduate degree programs. Graduate programs are listed below:

- Master of Education: Educational Leadership
- Executive Master of Business Administration
- Master of Business Administration
- Master of Science in Engineering Project Management
- Master of Science in Sustainable and Renewable Energy

Transferring from Non-Degree or Visiting to Degree Status

Students may request a change of status from non-degree to degree status or from visiting to degree status by submitting a complete application through the Department of Admissions. All admissions requirements in place at the time of the change-of-status request must be met. Schools may choose to take into consideration the student's academic performance in the 500-level courses taken with AURAK.

Courses taken while under the non-degree or the visiting status may apply towards the degree program given the approval of the dean of the student's school. Grades earned in courses that are accepted will count in the CGPA. The university rules and regulations governing transfer courses and credits will apply.

The graduation requirements will be determined by the catalog that is effective when the student joins a degree program. Additional leveling courses may apply if appropriate.

Change of Program

Students seeking to change their graduate degree program must complete the Change of Major Form available from the Department of the Registrar. Requests for a change of program should be submitted to the dean of the school of the program of the student's choice by the last day of the 12th week of classes of the fall or spring semester. The dean will forward the approved forms to the Department of Registration. Forms received by the Department of the Registrar by the end of the Add and Drop period will be effective as of the following semester.

To be eligible for a change of program, the graduate student must meet the requirements for admission to the new program. Please refer to the relevant program's catalog section for information on admission requirements.

Student Petitions and Appeals

Students' Responsibility

All official university communications are distributed through the AURAK-issued email address. These are considered official notifications. Students are responsible for checking their AURAK email accounts and for responding to or acting upon messages accordingly.

Students should keep their own records of all transactions with the university (e.g., registration schedules and forms, grade reports, payment records, etc.). It is also advisable to keep copies of all tests, digital files, papers and so forth submitted in fulfillment of course work. Students should keep copies of all course syllabi.

Appeal of Academic-Related Issues

If a student wishes to discuss an issue pertaining to a course, instructor, or other academic-related issues, the student may direct his or her concern to the involved faculty member. If a satisfactory resolution of the situation is not reached, the student may appeal to the Department Chair, then to the Dean of the School, and finally to the VPAASS/Provost (or his/her designee), who has the final authority to decide the merit of the appeal.

If the instructor is no longer associated with the university, the VPAASS/Provost will appoint a faculty surrogate, who will assume the authority of the instructor. If the matter is not resolved at this point, the VPAASS/Provost will hear the case and make a decision that is not subject to further appeal. Academic appeal requests must be submitted no later than the end of the first day of orientation week of the following semester.

Degree Completion

At AURAK, Master's degree programs will typically require at least one year of full-time study, or a minimum of 30 semester credits of course work (or equivalent) beyond the Bachelor's degree. These minimum credits shall not be inclusive of any non-credit bridge courses which may be required. Master's degrees offered by AURAK must include no less than 15 credit hours of courses (excluding thesis/dissertation credits) that are distinctive to that particular Master's program. AURAK also requires that all graduate students hold a CGPA of 3.0 in order to graduate and that Masters students complete a thesis.

AURAK reserves the right to change program or course requirements, grading procedures, or the requirements for acceptance into particular programs, academic standing, or graduation. These changes generally apply to all students enrolled at the time of implementation of the change and thereafter. When changes are made to degree requirements, however, students may elect to complete either the requirements in effect at the time of their admission or the new set of requirements, as detailed in the program requirements section of the catalog. Degree or program requirements do not include modality of program delivery (e.g., online or in-person delivery). With a change in modality of program delivery (e.g., shifting from in-person classes to online delivery of classes), the new modality of program delivery will typically apply only to students admitted following the change in modality.

For all degrees, requirements must be met as stated in the University Catalog. Candidates for graduate degrees who have been continuously enrolled may choose to graduate under the terms of any catalog in effect at or after their admission.

Master's Thesis

In AURAK, a Master's degree will require a substantial thesis of at least six and no more than nine credit hours. A master's thesis documents research conducted by AURAK graduate students undertaking a master's degree program. Preparation of the thesis includes training in research methods, the development of the thesis is

supervised by appropriately qualified and experienced, full-time faculty. The thesis is the culmination of the students' programs of study and is expected to reflect appropriate scholarly depth and rigor.

Students who do not complete their thesis in the final semester of the program of study will be required to maintain continuous enrollment in the thesis in succeeding semesters, with an In-progress (IP) grade assigned in the course designed to be the final thesis course until subsequent thesis coursework leads to the successful completion of the thesis. At that time, all IP grades originally assigned will be converted by the thesis supervisor to standard letter grades.

Where a student wishes to pursue an additional Master's degree from AURAK, the student must meet the admission requirements for the second degree and complete requirements for the additional degree. The requirements of the second degree must include at least 15 credit hours of courses (excluding thesis/dissertation credits) that are distinctive and not taken to meet requirements of the first degree.

It is AURAK policy to maintain master's theses in AURAK's Saqr Library so that they are available to other students and scholars. Students completing their thesis will be required to submit a bound copy of their theses to the Library.

Maximum Graduate Degree Completion Periods

In order to graduate successfully from a graduate degree program after full admission to the program, the following periods for completion will be applied:

- Master's Degree program
 - One year degree program must be completed within four regular semesters.
 - Two year degree program must be completed within six regular semesters.

Academic Standing Requirement

A student must be in good academic standing to be eligible for graduation.

Application for Graduation

Candidates for graduate degrees must file an "Application for Graduation Form" with the Department of the Registrar during the registration period of the last expected term of study. Only after an "Application for Graduation Form" has been filed can the Department of the Registrar begin processing the necessary information for final certification for graduation.

Students who fail to complete all degree requirements by the end of the term for which they apply to graduate need not reapply for graduation. Their previous application will be automatically moved to the following semester.

Conferral of Degrees

Only students who have successfully completed degree requirements and all thesis requirements, including corrections and final submission of the completed thesis to the library, by the end of the term for which they have applied to graduate are certified for conferral of a degree. Degrees are conferred at the end of the semester in which requirements have been met. Conferral of the degree is noted on the academic transcript of the graduate with the date of graduation.

Names on Degrees

The names of AURAK students will be spelled in English exactly as they appear on their passports or identity cards when printed on degrees. If a name on a passport or an identity card does not appear in English, then the spelling of the name will be printed according to the personal preference of the student.

Attestation of Degrees and Transcripts

The Department of the Registrar provides information relevant to the attestation of degrees and transcripts with the UAE Ministry of Education.

ACADEMIC PROGRAMS

Executive Master of Business Administration

Program Description

The Executive Master of Business Administration (EMBA) program is designed for mid-career and senior professionals who want to accelerate their career progression and develop higher level skills for senior leadership roles. It provides students with a comprehensive understanding of business operations and strategy and the ability to analyze and solve complex business problems.

In addition to traditional classroom learning, the program offers a range of experiential learning opportunities, such as case studies, collaborative online international learning, consulting projects and simulations to prepare students to navigate the complexities of the ever-changing business landscape.

The program employs a **blended learning mode of teaching**. This means that some courses are taught on-campus in a traditional in-person mode of instruction. Other courses are taught exclusively online, and a small number of courses are offered in a blended format, meaning that some class sessions are conducted on-campus, while others are conducted online. Overall, 50% of the program is taught on-campus, and 50% is taught online.

The Executive Master of Business Administration is overseen by the School of Business and is designed to satisfy QF Emirates Level 9 requirements.

School Mission Statement

The mission of the School of Business is to work with the AURAK 'family' to prepare tomorrow's business leaders to be thoughtful, creative, collaborative, and responsible with an ability to think critically, communicate their ideas clearly and take initiative in applying what they have learned. Backed by a diverse faculty committed to enrich their teaching and enhance their community impact by engaging in applied, cross-disciplinary, and pedagogical scholarship, the School offers programs based on the North American model and the cultural characteristics of the Gulf region in a student-centered teaching environment.

Program Aims

The overall aims of the program are:

1. To impart contemporary knowledge, and provide an integrated background in advanced concepts, frameworks and methodologies underpinning the high-level management of organizations.
2. To afford candidates the opportunity to concentrate on specific areas and issues within the business field. It also seeks to encourage life-long learning beyond the duration of the program.
3. To enhance professional communications skills and peer-based interaction, through its curriculum and associated group-based coursework.

Program Learning Outcomes

The program learning outcomes for the Executive MBA are as follows:

- PLO1** Evaluate advanced business knowledge for the effective functioning of local and international organizations across business sectors.
- PLO2** Enhance strategic leadership roles by applying program knowledge to solve complex business problems, constructing viable and effective solutions in the process.
- PLO3** Assess effectiveness of strategic leadership in corporate governance, ethical functioning, and socially responsible organizational behavior.
- PLO4** Develop skills of collaborative decision making through teamwork and leadership to communicate complex business requirements to organizational personnel and stakeholders.
- PLO5** Produce critical and entrepreneurial thinking for effective decision-making towards business innovation and sustainability.
- PLO6** Integrate research principles and techniques with prior work experience to execute a major business project.
- PLO7** Combine personal reflection with program learning for professional development to executive level.

EMBA Remedial Courses

If an EMBA applicant holds a baccalaureate degree, or its equivalent, in a non-business field with a minimum CGPA or 3.00, followed by a postgraduate certificate or diploma in a business-related field, he or she can qualify for full admission to the program, assuming all other admission requirements (such as English language proficiency) are met.

However, for those prospective students who have a CGPA of 2.0-2.4 and/or hold a degree from a non-business field, the students will have to complete remedial courses designed to prepare them for graduate-level coursework. The remedial courses consist of three compulsory courses as listed below.

Compulsory Courses	EMFN 500 EMBA Foundation (3 CH) EDUC 500 Foundations of Education (3 CH) ENGR 500 Engineering Foundation for Graduate Students (3 CH)
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The student must achieve a minimum CGPA of 3.0 on a 4.0 scale, or its established equivalent, in these nine credits of remedial courses in order to progress to the graduate program or be subject to dismissal.

Only after passing the three remedial courses within two semesters will students be allowed to register for the EMBA Program. An unsuccessful student who has not completed the courses within two semesters will not be able to continue to the EMBA program and will have to leave the institution.

Where feasible, the foundational business course and the other remedial courses, which do not count toward the required credits for the degree, may be run in a condensed format (e.g., as condensed multi-weekend course modules) provided the learning outcomes remain unchanged and achievable.

AURAK will also accept candidates who have taken and passed the EMBA preparatory courses from other UAE Ministry or Higher Education-accredited universities, provided the courses taken have compatible syllabi and learning outcomes.

Degree Requirements

The Executive Master of Business Administration requires the completion of 36 credits in the following areas:

Area	Credits
I. Core Course Requirements	30
II. Elective Course Requirements	6
Total	36

CORE COURSES

30 Credits

EMBA Core Courses				
Course Code	Course Title	Prerequisite	Teaching Mode	Credits
EMMG 510	Leadership and Managing People	Graduate Standing	Online	3
EMAC 512	Managerial and Financial Accounting	Graduate Standing	On-Campus	3
EMFN 514	Managerial Finance	Graduate Standing	On-Campus	3
EMMG 515	Strategic Management	Graduate Standing	Online	3
EMMK 518	Marketing Management	Graduate Standing	Online	3
EMMI 520	Managing Information Systems Performance	Graduate Standing	On-Campus	3
EMIB 524	International Business	Graduate Standing	Online	3
EMGN 525	Research Methods for Business	Graduate Standing	On-Campus	3
EMGN 555	EMBA Investigative Thesis or Project I (Thesis Research Course)	EMGN 525	Blended	3
EMGN 556	EMBA investigative Thesis or Project II (Thesis Research Course)	EMGN 555	Blended	3

ELECTIVE COURSES

6 Credits

EMBA Elective Course Requirements			
Course Code	Course Title	Prerequisite	Credits
Elective 1	Any course from the EMBA Electives List	-	3
Elective 2	Any course from the EMBA Electives List	-	3

EMBA Elective Courses – Any Two Courses				
Course Code	Course Title	Prerequisite	Teaching Mode	Credits
EMFN 560	International Corporate Finance	EMFN 514	On-Campus	3
EMFN 562	Investment Planning & Management	EMFN 514	On-Campus	3
EMFN 564	Financial Markets	EMFN 514	On-Campus	3
EMHR 590	Applied and Strategic Human Resources Management	EMMG 510	Online	3

EMBA Elective Courses – Any Two Courses				
EMHR 592	Managing Change and Innovation	EMMG 510	Online	3
EMHR 594	Employee Relations and Compensation Management	EMMG 510	Online	3

For Further Detailed information, Please Visit: <https://aurak.ac.ae/graduate-degrees/business/executive-master-of-business-administration>

Executive Master of Business Administration - Two Year Plan

Year 1, First Semester			Year 1, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
EMMG 510	Leadership and Managing People	3	EMGN 525	Research Methods for Business	3
EMFN 514	Managerial Finance	3	EMAC 512	Managerial and Financial Accounting	3
EMMG 515	Strategic Management	3	EMMK 518	Marketing Management	3
Subtotal =		9	Subtotal =		9
Year 2, First Semester			Year 2, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
EMGN 555	EMBA Investigative Thesis or Project I	3	EMGN 556	EMBA Investigative Thesis or Project II	3
EMIB 524	International Business	3	EMMI 520	Managing Information Systems Performance	3
Elective 1	Any course from the EMBA Elective List	3	Elective 2	Any course from the EMBA Elective List	3
Subtotal =		9	Subtotal =		9
Total 36 Semester Credit Hours					

Master of Business Administration

Program Description

AURAK's Master of Business Administration provides students with a well-rounded business education with, for example, courses to manage accounts, finance, information systems, international business, marketing, strategy, to undertake research, and to lead people.

In addition to classroom learning, this program offers a range of experiential learning opportunities, such as case studies, collaborative online international learning, simulations, and consulting projects, preparing students for the challenges of a fast-paced business environment.

Graduates of the Master of Business Administration can look forward to career opportunities as entrepreneurs and leaders in for-profit and not-for-profit organizations.

The Master of Business Administration is overseen by the School of Business and is designed to satisfy QF Emirates Level 9 requirements.

School Mission Statement

The mission of the School of Business is to work with the AURAK 'family' to prepare tomorrow's business leaders to be thoughtful, creative, collaborative and responsible with an ability to think critically, communicate their ideas clearly and take initiative in applying what they have learned. Backed by a diverse faculty committed to enrich their teaching and enhance their community impact by engaging in applied, cross-disciplinary, and pedagogical scholarship, the School offers programs based on the North American model and the cultural characteristics of the Gulf region in a student-centered teaching environment.

Program Aims

The overall aims of the program are:

1. To impart contemporary knowledge, and provide an integrated background in advanced concepts, frameworks and methodologies underpinning the high-level management of organizations.
2. To afford candidates the opportunity to concentrate on specific areas and issues within the business field. It also seeks to encourage life-long learning beyond the duration of the program.
3. To enhance professional communications skills and peer-based interaction, through its curriculum and associated group-based coursework.

Program Learning Outcomes

On successful completion of this program the graduate will be able to:

- PLO 1** Evaluate core business knowledge for the effective functioning of local and international organizations across business sectors.
- PLO 2** Enhance strategic management roles through analysis of the corporate strategy, constructing viable and effective solutions in the process.
- PLO 3** Assess effectiveness of corporate governance, ethical functioning, and socially responsible organizational behavior.
- PLO 4** Develop teamwork, interpersonal, communication and leadership skills for communicating business requirements to organizational personnel and stakeholders.
- PLO 5** Produce critical and entrepreneurial thinking to support decision-making for business innovation and sustainability.
- PLO 6** Integrate research principles and techniques to address business issues.

PLO 7 Combine personal reflection with further learning for professional development of own career.

MBA Remedial Courses

If an MBA applicant holds a baccalaureate degree, or its equivalent, in a non-business field with a minimum CGPA of 3.00, followed by a postgraduate certificate or diploma in a business-related field, he or she can qualify for full admission to the program, assuming all other admission requirements (such as English language proficiency) are met.

However, for those prospective students who have a CGPA of 2.0 – 2.49 and/or hold a degree from a non-business field, the students will have to complete remedial courses designed to prepare them for graduate-level coursework. The remedial courses consist of three compulsory courses detailed in the table below.

Compulsory Courses	MBFN 500 Business Graduate Foundation (3 CH) EDUC 500 Foundations of Education (3 CH) ENGR 500 Engineering Foundation for Graduate Students (3 CH)
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The student must achieve a minimum CGPA of 3.0 on a 4.0 scale, or its established equivalent, in these nine credits of remedial courses in order to progress to the graduate program or be subject to dismissal.

Only after passing the three remedial courses within two semesters will students be allowed to register for the MBA. An unsuccessful student who has not completed the courses within two semesters will not be able to continue to the MBA program and will have to leave the institution.

Where feasible, the foundational business course and the other remedial courses, which do not count toward the required credits for the degree, may be run in a condensed format (e.g. as condensed multi-weekend course modules) provided the learning outcomes remain unchanged and achievable.

AURAK will also accept candidates who have taken and passed the MBA preparatory courses from other UAE Ministry of Higher Education-accredited universities, provided the courses taken have compatible syllabi and learning outcomes.

Program Requirements

The Master of Business Administration requires the completion of 36 credits in the following areas:

MBA DEGREE REQUIREMENTS	
Area	Credit
Core Courses	30
Elective Courses	6
Total	36

Degree Requirements

Core Course Requirements (30 Credits)			
Course Code	Course Title	Prerequisite	Credits
MBMG 510	Leadership and Managing People	Graduate Standing	3
MBAC 512	Managerial and Financial Accounting	Graduate Standing	3

MBFN 514	Managerial Finance	Graduate Standing	3
MBMG 515	Strategic Management	Graduate Standing	3
MBMK 518	Marketing Management	Graduate Standing	3
MBMI 520	Managing Information Systems Performance	Graduate Standing	3
MBIB 524	International Business	Graduate Standing	3
MBGN 525	Research Methods for Business	Graduate Standing	3
MBGN 555	MBA Investigative Thesis or Project I (Thesis Research Course)	MBGN 525	3
MBGN 556	MBA Investigative Thesis or Project II (Thesis Research Course)	MBGN 555	3
MBA Elective Course Requirements (6 Credits)			
Elective 1	Any MBA Elective Course	-	3
Elective 2	Any MBA Elective Course	-	3

Further Detailed information, please visit: <https://aurak.ac.ae/graduate-degrees/business/master-of-business-administration>

MBA Elective Course List – Any 2 Courses			
Course Code	Course Title	Prerequisite	Credits
MBFN 560	International Corporate Finance	MBFN 514	3
MBFN 562	Investment Planning & Management	MBFN 514	3
MBFN 564	Financial Markets	MBFN 514	3
MBHR 590	Applied and Strategic Human Resources Management	MBMG 510	3
MBHR 592	Managing Change and Innovation	MBMG 510	3
MBHR 594	Employee Relations and Compensation Management	MBMG 510	3
MBMK 519	Digital Marketing	MBMG 510	3

Further Detailed Information, please visit: <https://aurak.ac.ae/graduate-degrees/business/master-of-business-administration>

Master of Business Administration - Two Year Plan

Year 1, First Semester			Year 1, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MBMG 510	Leadership and Managing People	3	MBGM 525	Research Methods for Business	3
MBFN 514	Managerial Finance	3	MBAC 512	Managerial and Financial Accounting	3
MBMG 515	Strategic Management	3	MBMK 518	Marketing Management	3
Subtotal =		9	Subtotal =		9
Year 2, First Semester			Year 2, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MBGN 555	MBA Investigative Thesis or Project I	3	MBGN 556	MBA Investigative Thesis or Project II	3
MBIB 524	International Business	3	MBMI 520	Managing Information Systems Performance	3
Elective 1	Any MBA Elective Course	3	Elective 2	Any MBA Elective Course	3
Subtotal =		9	Subtotal =		9
Total 36 Semester Credit Hours					

Master of Education: Educational Leadership

Program Description

The Master of Education Degree is a thorough scholarly experience that provides educators with excellent learning opportunities in leadership theory and practice. Students will acquire practical knowledge in research, data-driven decision-making, and real-world leadership. Upon completing the M.Ed., graduates will have the necessary skills to become positive change agents capable of leading the continuous improvement of schools and student learning.

The program is designed to provide educational professionals with the opportunity to develop the necessary skills to become effective educational leaders, school administrators, cluster managers, educational researchers, employees of governmental agencies, and/or higher education professionals.

The M.Ed program codifies, in an intentional way, the value of learning communities. This is accomplished through the implementation of a Graduate Cohort. In a cohort model, all students take all the same courses at the same times, altogether. Cohort models are common in Graduate education programs across the United States, as this approach yields immensely valuable benefits. Notable, a cohort model allows the program to achieve the following goals: (1) to implement a meaningful progression of knowledge and skills while promoting rigor, and (2) to build an authentic learning community.

The Master of Education consists of a total of thirty-three (33) credit hours over four semesters. All courses are scheduled according to a pre-set timetable, and the language of instruction is English. Courses are offered in a face-to-face format with designated out-of-classroom experiences/activities conducted collaboratively or independently to complement course content and interactions. The program is collaborative with teaching and learning occurring through applied interactive lectures, discussions, case studies, and projects. A student-centered, active learning, and inquiry-based approach is consistently utilized, and the entire program incorporates reflective practice.

The Master of Education: Educational Leadership is designed to satisfy QF Emirates Level 9 requirements. The award is overseen by the School of Arts and Sciences.

Program Mission

AURAK's Master of Education offers a transformational experience to graduate candidates as they actively apply theory, critical reflection and analysis, and best practice. M.Ed. students actively engage in practical diverse discourse on topics related to leadership, change agency, and governance within the educational systems of the UAE and surrounding region. Candidates obtain the requisite knowledge, skills, and tools to meaningfully contribute to positive change in addressing the dynamic challenges that occur when leading educational institutions, communities, faculty, and staff, in both their daily activities and in overall systemic change.

Program Educational Objectives

Graduates of the Master of Education are prepared to actively contribute to K-12 schools, higher education institutions, and governmental entities in leadership roles such as instructional, administration and community building. The program is designed to prepare educational leaders with the requisite knowledge, skills, and tools to meaningfully contribute to positive change in addressing the dynamic challenges that occur when leading educational institutions, communities, faculty, and staff, in both their daily activities and in overall systemic change.

The specific program goals of the Master of Education are to:

1. Prepare scholar-practitioners who engage in lifelong learning to meaningfully develop their careers in education;
2. Produce educators who apply in reflective practice to promote inclusion and diverse discourse grounded in the culture of the region;
3. Prepare scholar-practitioners who engage in data-driven decision making, to effect meaningful, research-based positive change in academic settings; and
4. Develop problem-solvers with advanced knowledge of education and specialized skills in education who contribute to overcoming challenges in education, academic settings, and at the intersection of education and industry.

Master of Education (MEd) Program Learning Outcomes

Graduates of the Master of Education degree are expected to:

1. Assess highly specialized knowledge of leadership theories and/or methodologies as the basis for original thinking, encompassing reflective practice and ethical processes of a detailed body of knowledge of recent developments in the field of educational leadership;
2. Evaluate the specialized and advanced problem-solving skills required in educational leadership, research and analysis, while utilizing specialist communication and technology skills to present, explain and/or critique highly complex matters;
3. Construct a comprehensive, internalized, personal worldview, while accepting responsibility to society, and the sociocultural norms and relationships;
4. Take responsibility for leading strategic performance and development of professional teams, and self, in learning contexts that are complex and may require new strategic approaches within the ethical boundaries of educational leadership;
5. Critically reflect, and actively choose to enhance, leadership skills in the field of Education to contribute to, and implement, ethical leadership standards that advances professional knowledge and practice; and
6. Lead and participate in developing academic settings, and where education meets industry, demonstrating best practice in leadership.

Degree Requirements

The following program requirements must be met for degree completion.

1. Successful completion of 33 credit hours, across 11 courses, identified below
2. Maintenance of a Grade Point Average of 3.00 or higher throughout the program
3. Successful submission of an original research thesis
4. Successful defense of an original research thesis

Assuming students do not interrupt their study, the M.Ed. degree is designed to be completed in 24 months (4 semesters). Students who withdraw or take a leave of absence from the program must meet requirements for returning that are outlined in AURAK's catalog. Students must meet specific standards to progress and do so within the maximum time allowed to complete the program; these standards and time allotment are also detailed in the catalog. If a degree is not completed within a period of six years, all coursework in the degree may be re-evaluated for its current relevance.

The M.Ed. program requires thirty-three credit hours detailed as presented below.

Program Course Requirements for the Master of Education		
Course No.	Title	No. of Credit Hours
EDUC 511	Introduction to Applied Research	3
EDUC 515	Educational Psychology for Inclusive Learning	3
EDUC 525	Theories and Applications of Leadership	3
EDUC 545	Budgeting and Resource Management	3
EDUC 620	Evaluation of Instruction for Continuous Improvement	3
EDUC 640	Inclusive Leading of Diverse Groups	3
EDUC 660	Organizational Change and Continuous Improvement	3
EDUC 675	Educational Policy and Community Engagement	3
EDUC 680	Applied Research I (Thesis Research Course I)	3
EDUC 685	Analytics for Data-Driven Leadership	3
EDUC 695	Applied Research II (Thesis Research Course I)	3
TOTAL =		33

**Master of Education: Educational Leadership
Two-Year Plan**

Year One, First Semester		
Course No	Course Title	Credit Hours
EDUC 511	Introduction to Applied Research	3
EDUC 525	Theories of Leadership and Administration	3
Semester Total =		6
Year One, Second Semester		
EDUC 525	Theories and Applications of Leadership	3
EDUC 545	Budgeting and Resource Management	3
EDUC 680	Analytics for Data-Driven Leadership	3
Semester Total =		9
Year Two, First Semester		
EDUC 620	Evaluation of Instruction for Continuous Improvement	3
EDUC 640	Inclusive Leading of Diverse Groups	3
EDUC 680	Applied Research I (Thesis Course I)	3
Semester Total =		9
Year Two, Second Semester		
EDUC 660	Organizational Change and Continuous Improvement	3
EDUC 675	Educational Policy and Community Engagement	3
EDUC 695	Applied Research II (Thesis Course II)	3
Semester Total =		9
Total Program Credit Hours = 33		

Master of Science in Engineering Project Management

Program Description

The curriculum consists of thirty-six credit hours that provide the students with balanced coverage of project management: seven core courses, three elective courses and a master thesis. The master courses is designed to showcase the graduate student's proficiency in applying the program learning outcomes to a real-world project. This comprehensive course integrates all fundamental aspects of project management, including planning, organizing, securing, managing, leading, and controlling resources, with the ultimate aim of accomplishing specific and attainable goals.

The Master of Science in Engineering Project Management is overseen by the School of Engineering and Computing and is designed to satisfy QF Emirates Level 9 requirements.

Program Mission

The Master of Science in Engineering Project Management (MSEPM) program at AURAK mission is to meet the needs of graduates in disciplines requiring an advanced understanding of the theoretical and practical principles of the project management function.

The MSEPM program requires students to apply principles of engineering and management science to contribute to the growing body of knowledge in project management. The program will prepare project managers from diverse engineering disciplines to efficiently manage complex projects in the UAE and the Gulf Region countries. The program provides students with excellent technical and managerial skills: Change Control, Cost Management, Documentation Management, Information / Communication Management, Procurement, Project Context / Environment, Quality Management, Reporting, Resource Management, Risk Management, Time Management / Scheduling / Planning, Stakeholder / Relationship Management, Team Building / Development / Teamwork, Work Content, and Scope Management to better satisfy customer needs. After completing the requirements for the degree, students will be able to apply engineering and management knowledge to successfully lead and execute engineering projects. Students will be trained to design, model, manage and control complex projects, and prepare to play leadership roles in industry and government.

Program Educational Objectives

The Program Educational Objectives describe the expected accomplishments of graduates during their first few years after graduation. The program objectives have been derived from and support the mission statement of American University of Ras Al Khaimah. On successful completion of the Master of Science in Engineering Project Management program, a graduate should have:

1. Acquired skills in the initiation of projects, methods and techniques to control time cost and quality, resource management and long-term stewardship of assets;
2. Acquired a comprehensive understanding of the whole process of project management;
3. Gained project team leadership skills; and
4. Understood the establishment of project management methodologies.

Program Learning Outcomes

The program learning outcomes of the MSEPM program are aligned with those of the Project Management Institute (PMI), Global Accreditation Center (GAC), and with level nine descriptors of the QF Emirates. Students who successfully complete this program will be able to:

PLO 1: Apply the principles of project management knowledge throughout the project life cycle within a broad range of scope and complexity;

PLO 2: Utilize the full range of project management tools and techniques to satisfy industry and stakeholders' requirements best;

PLO 3: Demonstrate professional responsibilities concerning people, resources, and the environment adherence to the project management code of ethics and professional conduct.

PLO 4: Integrate scope, schedule, budget, resource, quality, procurement, and risk processes in projects to meet stakeholders' requirements.

PLO 5: Direct projects with a clear understanding of project manager responsibilities of professional teams within each domain area of project management to motivate and influence team members and stakeholders.

PLO 6: Create (Design) effective project management communications and problem-solving techniques, project status reporting, conflict management, negotiation, and stakeholder engagement.

PLO 7: Conduct research using highly developed cognitive and creative skills and intellectual independence that significantly contributes to expanding project management knowledge.

PLO 8: Develop a project management framework by tailoring the processes based on project complexity and the competing constraints of scope, schedule, cost, resources, quality, and risk.

Degree Requirements

The requirements of the MSPEM program are thirty-six (36) credit hours as follows: twenty-seven (27) credit hours are compulsory and nine (9) are electives.

Compulsory

27 Credit Hours

Course Code	Course Title	Credits
MEPM 511	Project Management Fundamentals	3
MEPM 512	Engineering Contracts and Procurement	3
MEPM 515	Project Scheduling	3
MEPM 521	Project Cost Accounting and Finance	3
MEPM 532	Engineering Management	3
MEPM 541	Organizational Behavior and Project Team Management	3
MEPM 543	Risk Management for Project Managers	3
MEPM 590	Thesis I (Thesis Research Course)	3
MEPM 591	Thesis II (Thesis Research Course)	3

Electives Selected from the Following List

9 Credit Hours

Course Code	Course Title	Credits
MEPM 513	Project Implementation and Performance	3
MEPM 514	Global Projects Management	3
MEPM 516	Project Quality Management	3
MEPM 522	Financial Analysis and Decision Making	3
MEPM 531	Operation Research	3
MEPM 533	Information Systems for Project Management	3
MEPM 542	Supply Chain Management for Project Managers	3

Master of Science in Engineering Project Management
Two-Year Study Plan

Year 1, First Semester			Year 1, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MEPM 511	Project Management Fundamentals	3	MEPM 512	Engineering Contracts and Procurement	3
MEPM 532	Engineering Management	3	MEPM 515	Project Scheduling	3
MEPM 521	Project Cost Accounting and Finance	3	MEPM 543	Risk Management for Project Managers	3
Subtotal =		9	Subtotal =		9
Year 2, First Semester			Year 2, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MEPM 590	Thesis I	3		MEPM Elective (2)	3
MEPM 541	Organizational Behavior and Project Team Management	3		MEPM Elective (3)	3
	MEPM Elective (1)	3	MEPM 591	Thesis II	3
Subtotal =		9	Subtotal =		9
Total 36 Semester Credit Hours					

Master of Science in Sustainable and Renewable Energy

Program Description

The Master of Science in Sustainable and Renewable Energy program is designed to provide the students with balanced coverage of advanced renewable energy technologies. Students will be exposed to the status of energy resources, their interaction with the environment, different renewable energy sources technologies, different techniques and technologies for energy management and energy conservation along with the economic aspects of renewable energy-based power generation. The curriculum developed for this Master's degree brings together the necessary tools to learn how to identify energy needs and propose the application of techniques for the saving and efficient use of energy as well as integrate the knowledge acquired in the field of sustainability and energy efficiency. The curriculum consists of 36 credit hours composed of six core courses, three elective courses and nine credits for the master thesis. The program can be completed in as little as two years. The program also targets full-time working students.

The Master of Science in Sustainable and Renewable Energy is overseen by the School of Engineering and Computing and is designed to satisfy QF Emirates Level 9 requirements.

Program Mission

The mission of the Master of Science in Sustainable and Renewable Energy program at the American University of Ras Al Khaimah (AURAK) is to provide a high-quality education for United Arab Emirates students in the key aspects of Renewable and Sustainable Energy. Thus, it will enable the students to take responsible, creative, challenging and stimulating posts in research and industry in this exciting field.

The Master of Science in Sustainable and Renewable Energy program is designed to close the strategic gap between the technical aspects of renewable energy and the policy drivers for sustainable development. The program aims also to address the social, economic, and environmental issues involved with renewable energy systems. In this way, students will not only learn about technical and operational details, but also about the policy context, market mechanisms, and business models for renewable energy technologies and installations.

After completing the requirements for the degree, students will be in an excellent position to start or continue a career in renewable and sustainable energy technologies. Students will be trained to design, model, manage and control complex projects and prepared to play leadership roles in industry and government.

Program Learning Objectives (PLOs)

The Program goals describe the expected accomplishments of graduates during their first few years after graduation. The program goals have been derived from and support the mission statement of the American University of Ras Al Khaimah. On successful completion of the Master of Science in Sustainable and Renewable Energy, a graduate should have:

1. Acquired skills and competency in using renewable energy solutions in place of fossil fuel energy sources.
2. Acquired a comprehensive understanding of renewable and sustainable energy systems.

3. Teamwork skills in multidisciplinary projects and the identification of problems and proposal of solutions, including rigorous analysis and design of a process, component, or system.
4. Appreciation of the ethical and societal responsibilities entailed in the environmental engineering profession and the need for continuous education in the field and commitment to life-long learning.

Program Learning Outcomes

The outcomes of the Master of Science in Sustainable and Renewable Energy program have developed the Student Outcomes (SOs) that resulted from an extensive outreach effort and analysis. The outcomes are based on program educational objectives and are in line with level 9 descriptors of the QF Emirates. Students who successfully complete this program will develop:

1. An ability to acquire and apply knowledge in the field of renewable and sustainable energy systems, their components and interactions between the components;
2. An ability to identify and critically evaluate current developments and emerging trends within the renewable energy sector;
3. An ability to contextualize renewable technology projects within a local, national and international framework, and consider social, ethical and environmental impacts and consequences;
4. An ability to analyze, formulate, and handle technical problems from a system perspective, with an overview on their life-cycle, from idea/need to specification, development, maintenance and termination, and the ability to set conditions, decide necessary resource consumption and manage procedures to solve and realize problems;
5. An ability to communicate effectively with peers, other engineering experts and the community using a variety of media and instruments of communication;
6. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts; AND
7. An ability to work effectively and proactively within cross-cultural, multi-disciplinary teams, to accomplish high-level engineering results.

Degree Requirements

The requirements of the Master of Science in Sustainable and Renewable Energy are thirty-six (36) credit hours as follows: eighteen (18) credit hours are compulsory, nine (9) are electives, and nine (9) are for the thesis.

Degree Requirements	Credits
Core Courses	18
Elective Courses	9
Thesis	9
Total	36

Core Requirements**18 Credit Hours**

Code	Course Title	Credit
MEPM 511	Project Management Fundamentals	3
MSRE 511	Energy Conversion	3
MSRE 512	Energy Systems Modeling and Optimization	3
MSRE 513	Energy Economics	3
MSRE 514	Direct Research	3
MSRE 515	Energy Efficiency and Management	3

Electives from the Following List**9 Credit Hours**

Code	Course Title	Credit
MSRE 521	Photovoltaics	3
MSRE 522	Wind Energy Systems	3
MSRE 523	Biofuels	3
MSRE 524	Sustainable Desalination	3
MSRE 525	Concentrated Solar Power	3
MSRE 526	Green Buildings	3

Master Thesis**9 Credit Hours**

Code	Course Title	Credit
MSRE 590	Thesis I (Thesis Research Course)	3
MSRE 591	Thesis II (Thesis Research Course)	6
Program Total 36 Credit Hours		

Master of Science in Sustainable and Renewable Energy Two-Year Study Plan

Year 1, First Semester			Year 1, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MEPM 511	Project Management Fundamentals	3	MSRE 513	Energy Economics	3
MSRE 511	Energy Conversion	3	MSRE 515	Energy Efficiency and Management	3
MSRE 512	Energy Systems Modeling and Optimization	3	MSRE 514	Direct Research	3
Subtotal =		9	Subtotal =		9
Year 2, First Semester			Year 2, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
	MSRE Elective (1)	3		MSRE Elective (3)	3
	MSRE Elective (2)	3	MSRE 591	Thesis II	6
MSRE 590	Thesis I	3			
Subtotal =		9	Subtotal =		9
Total 36 Semester Credit Hours					



FACULTY LIST



FACULTY LIST

Abu Hantash, Tawfiq

Associate Professor, MS, Architecture Studies, Massachusetts Institute of Technology, USA

Abudakka, Mohamed

Senior Instructor, PhD in Materials Science, University of Rajasthan, India

Abu-Raed, Abeer

Assistant Professor, M.Sc., Architectural Engineering, Anhalt University of Applied Sciences, Germany

Adeel, Umar

Instructor, Master of Science, Computer Engineering with Specialization in Digital Design, UET Taxila, Pakistan

Ahmed, Irshad

Professor, PhD, Chemistry, Bhavnagar University, India

Ahmed, Muhammad

Assistant Professor, PhD, Physics from University of Western Ontario, Canada

Al Hazza, Muataz

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Al Mansoori, Khaledah

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Alamir, Islam

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Alamurin, Babatunde

Lab Engineer, BS, Science Laboratory Technology, University of Jos, Nigeria

Alassi, Eman

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Alnahhal, Mohammed

Associate Professor, PhD, Industrial Engineering / Project Management, Duisburg-Essen University, Germany

Alnoman, Ali

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Alsaleh, Nael

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Arman, Abdelfatah

Assistant Professor, PhD, Workforce Education and Development, Pennsylvania State University, USA

Arshi, Tahseen

Associate Professor, PhD, Entrepreneurship and Innovation, University of Bedfordshire, UK

Assaf, Hamed

Professor, PhD, Civil Engineering (Water Resources), University of British Columbia, Canada

Ataby, Ali

Associate Professor, PhD, Signal Processing and Machine Learning, University of Liverpool, UK.

Attia, Hussain
Assistant Professor, PhD, Electronics, University of Malaya, Malaysia

Awad, Mohammed
Professor, PhD, Computer Science, University of Houston, USA

Awadallah, Ibrahim
Instructor, MS, Mathematics and Statistics, Mu'tah University, Jordan

Azzam, Mohammad
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Balhasan, Saad
Associate Professor, PhD, Petroleum Engineering, University of Wyoming, USA

Bhatti, Tariq
Associate Professor, PhD, Business and Industry Operations Management, University of Western Sydney, Australia

Begum, Vazeer Jan
Associate Professor, PhD, Management & Marketing, Mother Teresa Women's University, India

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COURSE DESCRIPTIONS

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Program Title	Program Code
Accounting	ACCT
Architecture	ARCH
Arabic	ARAB
Artificial Intelligence	CSAI
Biology, Bioinformatics, Biotechnology	BIOL
Business, Business Analytics (Undergraduate)	BUAN/BUSN/MGHR/MGMT/ MIST/MKTG/OPMT
Business (MBA/EMBA Graduate)	EMAC/EMFN/EMGN/EMHR/EMIB/ EMMG/EMMI/EMMK/ MBAC/MBFN/MBGN/MBHR/MBIB/MB MG/MBMI/MBMK/ MBOM
Chemical Engineering	CHEN
Chemistry	CHEM
Civil & Infrastructure Engineering	CIEN
Communication	COMM
Computer Engineering	CENG
Computer Sciences	CSCI
Economics	ECON
Education	EDUC
Electrical and Electronics Engineering	EEEN
Electrical Engineering	ELEN
Electronics and Communication Engineering	ECEN
Engineering	ENGR
Engineering Project Management	MEPM
English	ENGL
Environmental Science	ENVS
Finance	FNAN
History	HIST
Hospitality and Tourism	HTMT
Industrial Engineering	IENG
Information Technology	ITEC
Interior Design	INDS
Mathematics	MATH
Mechanical Engineering	MENG
Middle Eastern Studies	MEST
Petroleum Engineering	PENG
Philosophy	PHIL
Physics	PHYS
Political Sciences	POLS
Psychology	PSYC
Sociology	SOCI
Statistics	STAT
Sustainable and Renewable Energy (Graduate)	MSRE
University Studies	UNIV

Course Descriptions

Each course provides a course number, name, semester credit hours, prerequisites and corequisites if any, and a detailed course description. Course titles are followed by values in parentheses (0:0:0) separated by colons. The first number refers to the number of credit hours for the course; the second number refers to the number of lecture hours per week; the third number refers to the number of laboratory hours/studio hours/research or practicum hours per week.

Course Level Designations

AURAK offers degree programs that provide a coherent course of study. The coursework in all undergraduate programs follows a logical pattern from the 100 through the 200, 300, and 400 levels. The distinctions among these levels are generally understood as follows:

000-level course designation

Prerequisite courses serving to prepare students for successful performance in college level courses.

100-level course designation

Courses with no prerequisites, or survey courses, or courses defining basic concepts or presenting the terminology of a discipline.

200-level course designation

Courses of intermediate university-level difficulty; courses with 100-level course(s) as prerequisite(s); or survey courses devoted to specific areas within a discipline.

300-level course designation

Courses of advanced university-level difficulty taken by majors and upper division students who have a clear interest and who are qualified in the subject.

400-level course designation

Advanced courses and/or seminars, tutorials and special courses for majors and upper-division students.

For Masters Programs courses are offered at the 500 and 600 levels:

500-level course designation

Courses that assume bachelor's level familiarity with the subject, typically serve as the core for graduate programs, and are the foundation for more advanced and focused study.

600-level course designation

Courses at a more advanced graduate level through which students attend to focused and specialized topics in the discipline.

The number designations of the course descriptions in this chapter have the following significance:

- The single number (BIOL 100) indicates the course is complete within a single semester, and the semester course may be taken separately with credit toward a degree.
- The double number separated by a comma indicates that the subject matter or content of the course extends through two semesters, but that either semester may be taken by itself.
- Unless otherwise specified, the first semester is not a prerequisite to the second semester.

COURSE DESCRIPTIONS

ACCT 204 3 Semester Credit Hours Prerequisite(s): ECON 103	Principles of Financial Accounting	(3:3:0)
This is an introductory course in financial accounting which incorporates International Financial Reporting Standards (IFRS)/GAAP and emphasizes the use of accounting information in an ethical manner. Topics covered include: Basic accounting concepts and processes with particular emphasis on double entries; the accounting cycle; transaction analysis; accounting for assets, liabilities, owner's equity, expenses and revenues recognition, preparation of financial statements and ethical decision-making.		
ACCT 205 3 Semester Credit Hours Prerequisite(s): ECON 103	Principles of Managerial Accounting	(3:3:0)
This is the first course in cost/managerial accounting. It discusses a range of introductory managerial accounting topics such as cost measurement and terminology, costs systems; analyzing cost-volume-profit relationships, job order and process costing, operating budgets, standard costing, introduction to capital budgeting and ethical managerial accounting-related decision-making.		
ACCT 311 3 Semester Credit Hours Prerequisite(s): ACCT 205	Managerial and Cost Accounting	(3:3:0)
This course extends topics studied in Principles of Managerial Accounting (ACCT 205), as well as covers topics; pricing decisions, and joint product costing. The course examines the application of Managerial Management Accounting concepts and skills in decision making, of contemporary management philosophies and techniques in relation to JIT, quality, transfer pricing, performance measurement and questions the relevance of various aspects of traditional management accounting for internal management decision.		
ACCT 321 3 Semester Credit Hours Prerequisite(s): ACCT 204	Governmental and Not-for-Profit Accounting	(3:3:0)
This course covers the conceptual framework of government accounting, restricted funds theory, instructions and procedures and the accounting treatment for government operations, internal control, control over revenues and expenditures, the government budget principles and stages, the general procedures to control government spending, operations and government receipts, the role of audit bureau of financial control, government accounts and types, the financial statements, the internal control systems on government purchases, and finally tendering and receiving supplies and its registration.		
ACCT 331 3 Semester Credit Hours Prerequisite(s): ACCT 204	Intermediate Accounting I	(3:3:0)
The aim of this course is to develop the learner's understanding of financial accounting that comprises complex topics related to measuring and reporting of cash, inventories, investments, property, receivables, revenue recognition principles, plant, and intangible assets, and equipment. Emphasis is placed on both accounting theory and practice and on the development of professional opinion and significant thoughts.		
ACCT 332 3 Semester Credit Hours Prerequisite(s): ACCT 331	Intermediate Accounting II	(3:3:0)
In this course, students examine the liability and equity sections of the balance sheet. Topics include current and other liabilities; applications to long-term liabilities; dilute securities and earnings per share (EPS); income taxes; pensions and other employee future benefits; leases; accounting changes and error analysis; statement of cash flows; and full disclosure in financial reporting.		
ACCT 352 3 Semester Credit Hours Prerequisite(s): ACCT 331	Taxation and Managerial Decisions	(3:3:0)
The objective of this course is to develop a sound conceptual and technical foundation for the study of federal tax system and thus provide students with the tools necessary to stay up to date with ever-changing tax laws, where these apply. The course will imbue students with sufficient understanding of the tax environment to identify important issues when evaluating business transactions.		
ACCT 361 3 Semester Credit Hours Prerequisite(s): ACCT 331	Accounting Information Systems	(3:3:0)
This course discovers the study and development of accounting information systems. The focus is on business processes covering many industries with an emphasis on accounting data modeling and internal control. Special topics including Extensible Business Reporting Language (XBRL) and commercial systems are incorporated throughout the course.		

ACCT 390 Internship in Accounting**3 Semester Credit Hours****Prerequisite(s): Completion of 90 credit hours**

The Internship is designed for candidates on the undergraduate accounting major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation takes place at the beginning of the internship, and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

ACCT 412 Auditing**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): ACCT 331**

This course will cover: [1] Study and evaluation of internal control; [2] audit planning and assessment of risk; [3] audit evidence and documentation; [4] professional ethics; [5] Application of auditing and [6] audit report.

ACCT 444 Accounting Project Based Internship**3 Semester Credit Hours****Prerequisite(s): ACCT 331**

The accounting project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem- solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the accounting specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.

ACCT 472 International Auditing**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): ACCT 331**

This course discovers a broad overview of the global financial and accounting environment and addresses three areas: financial reporting, financial analysis, and planning and control of multinational enterprises. In addition, the course analyzes how different businesses operating in global environments influence accounting practices.

ACCT 491 Special Topics in Accounting**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): ACCT 332**

This is an advanced subject and seminar course which covers major applications of Classical/current issues in Accounting. The topics to be covered will vary and will be announced by the end of the first week of commencing the course.

ARAB 100 Arabic Language and Culture for Visiting Students**(3:3:0)****3 Semester Credit Hours**

The course provides an introduction to the basics of the Arabic language. It aims to develop students' communicative competence in spoken as well as written Arabic. Intended to resonate with the experiences of non-native learners of Arabic who are visiting the UAE, the course's linguistic content is based within the contemporary culture of the region. The course is intended to stand-alone and not to serve as a prerequisite for any other university courses.

ARAB 101 Arabic Language and Culture for Non- Native Learners I**(3:3:0)****3 Semester Credit Hours**

Arabic for speakers of other languages helps develop students' communicative competence in the skills of speaking, listening, reading, and writing. The course provides for the acquisition of vocabulary, the study of grammar, and structures of spoken and written Arabic. Non-native speakers of Arabic gain insights into target language countries.

ARAB 102 Arabic Language and Culture for Non- Native Learners II**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): ARAB 101**

Beginner Level Arabic II is a direct continuation of Beginner Level Arabic I. This course is designed to enhance the reading, speaking, and listening skills. With this course, students can increase their vocabulary and improve their grammar in Arabic.

ARAB 110 Arabic Language and Culture for Native Arabic Speakers I**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): ENGL 101**

Arabic literature has developed many traditions though originating from a common source. The course is an introduction to representative texts from contemporary Arab writers, and their connections with the traditions of the past.

- ARAB 201 Intermediate Arabic I (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ARAB 102
 The course provides an introduction to academic sources of the language. This program is specially designed to teach Arabic to university level Arabic students.
- ARAB 202 Intermediate Arabic II (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ARAB 201
 Students engage in simple Arabic conversation on a range of everyday subjects, so students properly introduce themselves, and engage in simple conversation on a range of everyday topics. Building upon the basic foundation provided in Level I, topics include the definite article. Proper pronunciation and listening skills continue to be emphasized. In addition to readings and exposure to Arabic culture, students demonstrate further competence with structure and the pattern of words and sentences.
- ARAB 210 Arabic for Media: Native Speakers (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ARAB 110
 This course introduces Arabic language for Media: the language of print, broadcast, and digital media to Mass Communication native-speaking students. It aims to familiarize students with media vocabulary, text structure, and language usage. The course prepares Mass Communication students to apply requisite Arabic language skills and adhere to ethical norms in various mediums used in Mass Communication.
- ARAB 211 Arabic for Media: Non-Native Speakers (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ARAB 101
 This course introduces students to language used in Arabic media. The course covers the diverse media platforms and topics in the Arab world including weather news, deaths, special events, sports, advertisements, and economics. In each of those topics, the students will be equipped with requisite vocabulary and jargon used for discussing these in media.
- ARAB 311 Upper Intermediate Level Arabic Language and Culture for Native Arabic Speakers I (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): Arabic Placement Test
 Upper Intermediate Level Arabic Language and Culture I is the first course in a four-course sequence for the native Arabic speaker. All courses in the sequence are taught in Modern Standard Arabic (MSA). The course is designed to provide native Arabic-speakers with higher level linguistic skills in writing, reading, speaking, and listening. Emphasis is placed on grammar review, vocabulary acquisition, and composition. This course and the subsequent sequence of courses are tailored particularly to students interested in Arabic-English translation in the English Language and Mass Communication Programs.
- ARAB 312 Upper Intermediate Level Arabic Language and Culture for Native Arabic Speakers II (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ARAB 311
 Upper Intermediate Level Arabic Language and Culture II is a direct continuation of ARAB 311. It is designed to provide Arabic-speaking Translation and Mass Communications majors with the linguistic skills (writing, reading, speaking, and listening) that serve as a solid foundation to the journalistic expression in Arabic. Emphasis is placed on grammar review, vocabulary acquisition, and composition.
- ARAB 411 Advanced Level Arabic Language and Culture for Native Arabic Speakers I (3:3:0)**
3 Semester Credit Hours
Prerequisite(s) (s): ARAB 312
 This is a direct continuation of Arabic 312. The course substantially expands students' existing vocabulary and capability of expression, both orally and in writing. Literary texts of increasing sophistication are used in the course.
- ARAB 412 Advanced Level Arabic Language and Culture for Native Arabic Speakers II (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ARAB 411
 Advance -Level Arabic Language and Culture for Native Arabic Speakers II is a direct continuation of ARAB 411 and represents the fourth course in the Arabic Language and Culture for native speakers' sequence. This course provides a survey of themes and genres of Arabic literature from the mid- 19th century to the present. While focus will be on content, students will continue their acquisition of MSA through written and oral assignments designed for their advanced level of competency.
- ARAB 420 Special Topics in Arab Culture and Civilization for Native Arabic Speakers (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ARAB 412
 Supplementing the four-course sequence in Arabic Language and Culture for native speakers, ARAB 420 will be taught in MSA. The course

critically examines issues, values, and institutions of the contemporary Arab world primarily through analysis and discussion of current events. While focus will be on content, students will continue their acquisition of Modern Standard Arabic through written and oral assignments designed for their level of competency.

ARCH 121 Basic Design I

(4:0:8)

4 Semester Credit Hours

An introduction to the basic design components: subject, form, content, with a focus on their dynamic interrelation. Through observation, analysis, discussion of traditional and contemporary approaches in fine arts, architecture and interior design, students learn to apply the elements and principles of visual organization to create projects using various methods.

ARCH 122 Basic Design II

(4:0:8)

4 Semester Credit Hours

Prerequisite(s): ARCH 121

The course introduces basics of design process and theory as it relates to the 3-dimensional world. Students are involved in hand-on exploration of simple structures and spaces using diverse materials and textures. Space function, volume, mass are discovered through designing a small-scale residential space(s), where design elements/principles are applied.

ARCH 221 Architectural Design Studio 1

(4:0:8)

4 Semester Credit Hours

Prerequisite(s): ARCH 122

Introduction to the application of design principles in architecture and to the technical conventions and expressions. Focus on elementary architectural design as it relates to function, structure, site, land plot and the environment.

ARCH 222 Architectural Design Studio 2

(4:0:8)

4 Semester Credit Hours

Prerequisite(s): ARCH 221

Design of medium scale buildings with an integrated program and moderate site conditions. Emphasis on human centered design process, impact of human and climatic factors on design solutions. Exploration of sustainability and particularly natural ventilation, daylighting, building materials in regards to the design intent and environment.

ARCH 223 Computer Architectural Drawing

(3:2:3)

3 Semester Credit Hours

Prerequisite(s): INDS 122

Concepts, features and applications related to Building Information Modeling (BIM) are introduced. Students explore Autodesk® Revit Architecture structure, features, modeling and editing techniques, sheet creation and organization. Emphasis on modelling BIM models and rendering 2D and 3D interiors and exteriors (including site), furnishings, accessories and details for architectural projects.

ARCH 231 Building Construction 1

(3:2:3)

3 Semester Credit Hours

Prerequisite(s): INDS 122

This course aims to introduce building materials, their properties, the impact of building materials and construction on environment and building codes. It focuses on sustainable methods of building technology as it relates to the site and construction and explores building systems, structure's types and elements (including foundations).

ARCH 232 Building Construction 2

(3:2:3)

3 Semester Credit Hours

Prerequisite(s): ARCH 231

This course explores materials and buildings technologies employed in building envelope (including windows, doors, other openings) and interior construction (including fixtures, fitments, built-ins) and systems, thermal and acoustic insulation, waterproofing and damp proofing. Accents on selection of materials and building technologies with minimal negative impact on environment.

ARCH 241 History and Theory of Architecture 1

(3:3:0)

3 Semester Credit Hours

Introduction to history of art and architecture, and their development through the ancient civilizations of Egypt, Mesopotamia, Greece, and Rome. Analysis of historical monuments and buildings and their architectural characteristics. Emphasis on the Architectural principles, theories, and impact on the development of buildings throughout ancient times. Examining ancient historical patterns.

ARCH 242 History and Theory of Architecture 2

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): ARCH 241

Introduction to Early Christianity Architecture during the Middle Ages in Europe. The course begins with the origins of Christian architecture of the Early Christianity, Byzantine Architecture, Romanesque Architecture Gothic, Gothic Style in France and England, Gothic Style in Italy. The course concludes with the Architecture of the Baroque and Renaissance age.

- ARCH 321 Architectural Design Studio 3 (4:0:8)**
4 Semester Credit Hours
Prerequisite(s): ARCH 222
 Design of a building or a group of buildings with moderate complexity through different spatial, formal, and functional strategies. Consideration of psychological, environmental, and technical requirements influencing architecture and design. Experiment with green buildings design and explore passive design solutions.
- ARCH 322 Architectural Design Studio 4 (4:0:8)**
4 Semester Credit Hours
Prerequisite(s): ARCH 321
 Create sustainable multifunctional buildings or a group of buildings design, within an urban context, that responds to the nature of the site and intricate topography. Solve design problems related to building structure, building materials and technologies as influenced by the building type, function, urban context and environmental protection.
- ARCH 331 Building Construction 3 (3:2:3)**
3 Semester Credit Hours
Prerequisite(s): ARCH 232
 This course explores advanced building technologies and complex building structural configurations for long span buildings such as structural frames, space structure, suspended roof structure and air-supported structures. Offsite constructions, prefabricated and modular housing, envelopes to above listed types of building structures in relation to impact on environment are discussed.
- ARCH 336 Construction Drawings (3:2:3)**
3 Semester Credit Hours
Prerequisite(s): ARCH 223, ARCH 331
 This course will lead the students to explore the principles of BIM used to visualize, document and communicate professional architectural working drawings respecting national, regional and international building codes and regulations.
- ARCH 341 History and Theory of Architecture 3 (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ARCH 242
 The study of the development of Islamic art and architecture and its characteristics since the dawn of Islam and the impact of social, cultural, environmental and functional factors on architecture in the Moslem world. Analysis of selected examples of historical monuments from Islamic countries and interpretations of contemporary Islamic Architecture.
- ARCH 342 History and Theory of Contemporary Architecture (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ARCH 341
 The formation of modern theories and trends in contemporary art and architecture since the industrial revolution until today. Analysis of works by the pioneers of modern movements in architecture. Regional and vernacular architecture with special regards to environmental, social and technological issues. Selected examples of characteristic contemporary architecture in the UAE.
- ARCH 372 Sustainable Architecture (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ENVS 102
 The course addresses the relationship of the built environment to natural environment through a whole system approach, with a focus on sustainable design of buildings. The emphasis is on local green design approaches, energy efficiency, renewable energy, and the appropriate use and conservation of resources, including materials, water, and land.
- ARCH 390 Internship I**
3 Semester Credit Hours
Prerequisite(s): 90 credits and a CGPA of 2.0 or higher
 This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.
- ARCH 391 Internship II**
3 Semester Credit Hours
Prerequisite(s): Completion of 130 credits and a CGPA of 2.0 or higher
 This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

- ARCH 421 Architectural Design Studio 5 (4:0:8)**
4 Semester Credit Hours
Prerequisite(s): ARCH 322
 Introduce students to the integrated design process with multiple stakeholders and multidisciplinary approaches (structural, mechanical, environmental) through buildings with shared functions. Via independent study and creative solutions students explore approaches and processes of integrated design process and produce environmentally comprehensive and human-centered design solutions.
- ARCH 422 Architectural Design Studio 6 (4:0:8)**
4 Semester Credit Hours
Prerequisite(s): ARCH 421
 Design of a complex site and buildings through an integrated process taking in consideration the boundaries and exploring their mutual reciprocities. Emphasis on professional design thinking approaches that reflect student's personal values and ethical attitude that address issues of sustainability and technology in response to the project's functional requirements and regional context.
- ARCH 449 Technology of the Built Environment (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ARCH 223
 This is an advanced level course that will allow students to discover the fundamentals of Parametric modeling for design applications in Architecture. Students will investigate rule-based and parametric design concepts and techniques in the context of a generative modeling environment: Dynamo for Revit.
- ARCH 454 Urban Planning (3:1:6)**
3 Semester Credit Hours
Prerequisite(s): ARCH 322
 Introduction to the environmental, social and economic aspects necessary in the formation of an urban planning process; the characteristics of the urban fabric and the development of cities and their urban elements into integrated, harmonious and functional unity; social services and infrastructure and their necessary provision according to norms and standards.
- ARCH 456 Sustainable Housing (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ENV5 102
 The course begins with a section on society and housing, exploring how sustainable housing practices build community. The second part of this course focuses on the relationship between society and the environment. We explore questions about how community growth has influenced the environment and how natural events affects our communities.
- ARCH 457 Urban Design (3:1:4)**
3 Semester Credit Hours
Corequisite(s): ARCH 463
 This course focuses on the definition, development and designing urban space concepts that both challenges and demonstrates students' knowledge and skills in urban design. Through advanced investigation of the history, theory and practice of urban design approaches, students learn the principles of place making, connections between people and places within the spatial layers of a city.
- ARCH 459 Conservation of Historic Environment (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): ARCH 341
 This course aims to provide students with knowledge about development of historic conservation theory and practice. Students will learn about the principal schools of thoughts, defining practices and new approaches to definition and assessment of cultural heritage. Students will explore a range of subjects and issues that affect contemporary heritage conservation practice, including its historical and philosophical underpinnings and its relation to sustainable environment.
- ARCH 463 Landscape Architecture (3:1:6)**
3 Semester Credit Hours
Prerequisite(s): ARCH 322
 The natural environmental equilibrium affecting different site scales. History of gardens, evolution through civilizations of the Mediterranean region & plant species in UAE. Landscape as gardens, open spaces & gathering areas. Introduction to environment-friendly concepts, energy conservation implementation on building sites. The influence of local climates to landscape designs.
- ARCH 466 Building Utilities II: Illumination, Acoustics, and Electrical Building Services (3:2:3)**
3 Semester Credit Hours
Prerequisite(s): MENG 468
 This course focuses on sustainable-green architecture design supported by the computer simulation of building energy use, natural ventilation, and daylighting. The simulation process utilizes research and simulation software to produce design evidence of the sustainability criteria. The course also may prepare undergraduate students for their further graduate studies in sustainability.

ARCH 473	Environmental Control	(3:3:0)
3 Semester Credit Hours		
The course focuses on sustainable-green architecture design supported by the computer simulation of building energy use, natural ventilation, and daylighting. The simulation process utilizes research and simulation software to produce design evidence of the sustainability criteria. This course also may prepare undergraduate students for their further graduate studies in sustainability.		
ARCH 485	Professional Practice	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): ARCH 422		
An overview of the architectural practice complexities in historic and contemporary context; the roles and responsibilities of the architect with emphasis on the characteristics of best practices; the architects in the office from starting a practice, marketing and strategic planning of practice, methods of project delivery to design contracts and agreements and AIA documents as well as ethics and existence in the society.		
ARCH 487	Project Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): ARCH 336		
The course covers key components of project management, including project integration, project scope management, project time and cost management, construction project, contract documents and types of specification depending on project nature, quantity surveying and quality management, human resource considerations, communications, risk management, and procurement management.		
ARCH 521	Architectural Design Studio 7	(4:0:8)
4 Semester Credit Hours		
Prerequisite(s): ARCH 422		
Contextual design of a group of buildings through complex design approaches and integrated urban design framework that might urban infill, upgrading existing built environment, or heritage buildings conservation via reuse, etc. Focus of demographics of the site, socio-economic, environmental conditions for users' wellness and life quality improvement.		
ARCH 591	Graduation Project Thesis Research	(2:1:3)
2 Semester Credit Hours		
Prerequisite(s): ARCH 422		
The course is oriented towards systematic analysis of the graduation project on selected topic, its functional and spatial relationships, location and setting, physical and human environments, use of case studies, and provision of synthesis and alternative concepts as design solutions. The outcome is presented in both forms, written as a report and visual as posters, to a jury of professionals. Writing Intensive Course		
ARCH 592	Graduation Project Design	(5:0:10)
5 Semester Credit Hours		
Prerequisite(s): ARCH 521 and ARCH 591		
Under the supervision of a faculty advisor(s), students independently develop design solutions on investigated architectural topic of personal interest selected for ARCH 591. Through final project jury presentation and portfolio, students demonstrate their ability to professionally develop design solutions and solve problems related to spatial, functional, aesthetic, social, environmental, technical and other aspects of the built environment.		
BIOL 095	Introductory Biology	(3:3:0)
3 Semester Credit Hours		
This course is offered for new science students without high school science backgrounds. The course provides an introduction to basic concepts in biology, cell energetics, genetics, diseases and biotechnology. This course is not for degree credit.		
BIOL 100	Humankind in a Biological World	(3:3:0)
3 Semester Credit Hours		
In this course, students will learn how human beings interact with, affect, and are affected by other living organisms. Ways in which human activities have had an impact on other life on earth, humankind and disease, and the development of scientific thought how we could solve this issue are examined.		
BIOL 112	University Biology I	(3:3:0)
3 Semester Credit Hours		
Corequisite(s): BIOL 113 University Biology I Laboratory		
Introduction to cell chemistry, metabolism, and genetics.		
BIOL 113	University Biology I Lab.	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): BIOL 112		
Introduction to Cell Structure & Function and Basics on plant biology.		

BIOL 114	University Biology II	(3:3:0)
3 Semester Credit Hours		
Corequisite(s): BIOL 115		
This course is intended for science majors and pre-professionals in life sciences and provides an introduction to Genetics, Microbiology and animal form and function.		
BIOL 115	University Biology II Lab.	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): BIOL 114		
This course encompasses the basic techniques in Biology for science majors and pre-professionals in life sciences that accompany BIOL 114.		
BIOL 230	General Microbiology	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 114		
Corequisite(s): BIOL 231		
This course covers the classification of microorganisms, microbial metabolism, microbial growth, microbial genetics, structure of eubacteria, archaea, bacteria, fungi, and viruses, specific defenses, and diseases.		
BIOL 231	General Microbiology Lab	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): BIOL 230		
This course focuses on laboratory techniques involved in culturing, staining, and identifying microorganisms.		
BIOL 250	Biochemistry I	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 111 or CHEM 120		
Corequisite(s): BIOL 251		
Structure and chemical behavior of biochemical compounds, levels of protein structure, steady state enzyme kinetics and activities, protein purification, protein functions. Structure and functions of Carbohydrates, Nucleotides and Nucleic Acids, DNA-Based Information Technologies. Lipids and Biological Membranes and Transport.		
BIOL 251	Biochemistry I Lab	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): BIOL 250		
This experimental course will introduce students with hand practice of some major biochemistry techniques. This course has two parts, part A is wet lab where students learn and practice separation, identification, and quantization techniques that exploit properties of biological molecules. To analyze results, data and write reports to develop analytical reasoning and problem-solving skills. The second part B is dry/computer lab where students will use Geneious to study the 3D structures of biological macromolecules from PDB files (Protein Data Bank).		
BIOL 270	General Genetics	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 112 or BIOL 114		
Corequisite(s): BIOL 271		
This course focuses on the basic concepts of heredity and modern developments in Genetics.		
BIOL 271	General Genetics Lab.	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): BIOL 270		
This course covers basic laboratory techniques in Genetics that accompany the course BIOL 270.		
BIOL 322	Microbial Genetics	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 270		
This course covers the basic concepts of microbial genetics including gene structure, gene expression and its control, and evolution of bacterial genetic diversity. The emphasis will be on topics such as mechanisms of genetic exchange, recombination, plasmids, analysis of bacteria and phage genomes, mutagenesis, and DNA repair. The course will also discuss the common methods using bacteria in gene cloning and identification and isolation of mutants		

BIOL 330	Applied and Industrial Microbiology	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 230		
Corequisite(s): BIOL 331		
This course covers the biology of microorganisms of ecological and industrial significance and includes topics such as food production, spoilage and preservation, fermentation technology, waste disposal, water purification, biodeterioration, and decomposition.		
BIOL 331	Techniques in Applied and Industrial Microbiology	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): BIOL 330		
This course includes laboratory exercises that illustrate applied methodologies in microbiology, including isolation of commercially useful strains, and discusses the production and purification of industrial products.		
BIOL 350	Biochemistry II	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 250		
This course covers the bioenergetics, Glycolysis, Gluconeogenesis, the Citric Acid Cycle and the oxidative phosphorylation and photophosphorylation. Lipid, amino acids and nucleotides Biosynthesis. DNA and RNA metabolism and gene expression regulation.		
BIOL 356	Virology	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 230		
This course provides an overview of the characteristics of major families of viruses; the intrinsic properties of viruses that cause disease, their development, and life cycles; interaction with host cells; genetics; and tumor-inducing properties and epidemiology.		
BIOL 357	Artificial Intelligence in Biotechnology	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): STAT 100 and Completion of 60 credits		
This course provides an overview of Artificial Intelligence (AI) in Biotechnology. It examines various AI-based techniques impacting the Biotechnology field. It provides guidelines to a potent AI decision framework to help understand the considerations associated with implementing AI in Biotechnology.		
BIOL 380	Biotechnology and Genetic Engineering	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 270		
Corequisite(s): BIOL 381		
This course focuses on the theory and applications of Biotechnology to real-world problems, emphasizing significance and societal applications.		
BIOL 381	Biotechnology Lab Methods and Techniques	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): BIOL 380		
This course focuses on the fundamental practical techniques used in Biotechnology research and industry.		
BIOL 390	Internship	
3 Semester Credit Hours		
Prerequisites(s): Prior completion of 90 credits and CGPA of 2.0 or higher		
This is a supervised training experience for biotechnology-related duties for a duration of 240 hours at an approved biotechnology or biotechnology-related organization under the guidance of a designated site supervisor in coordination with a faculty supervisor.		
BIOL 411	Drug Discovery, Design and Development	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 350		
This course will explore the process of drug development, from target identification to final drug registration. It will present drug development as a process involving target selection, lead discovery using computer-based methods and combinatorial chemistry/high-throughput screening. Safety evaluation, bioavailability, clinical trials, and the essentials of patent law will also be discussed. Along the way, you will learn about molecular recognition, computer aided drug design, and toxicology as applied to the development of new medicines.		
BIOL 420	Molecular Biology	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 270		

This course focuses on the basic concepts of structure and function of genetic material at the molecular level. Topics such as DNA replication, chromosome structure, gene expression and mutation, DNA repair, and transposable elements are covered in molecular detail.

BIOL 421 Molecular Biology and Recombinant DNA Lab. (1:0:3)

1 Semester Credit Hour

Corequisite(s): BIOL 420

This course focuses on the techniques in modern molecular biology techniques such as restriction mapping, isolation of DNA and RNA from eukaryotic cells, Southern hybridization, reverse transcription, and polymerase chain reaction (PCR and RT-PCR).

BIOL 423 Advanced Molecular Biology and Forensic Sciences (3:3:0)

3 Semester Credit Hours

Prerequisite(s): BIOL 270

This is for forensic DNA practitioners, those interested in a career in forensic DNA and all students interested in broadening their understanding of the genetic principles that underlie the forensic application of DNA typing methods.

BIOL 424 DNA Fingerprinting and Serology Lab (1:0:3)

1 Semester Credit Hour

Prerequisite(s): BIOL 270

Corequisite(s): BIOL 423

Laboratory course to accompany BIOL 423 (Advanced Molecular Biology and Forensic Sciences). Focusing on the basic techniques used in forensic DNA laboratories, this course will introduce the students to the most recent DNA fingerprinting techniques and serological analysis.

BIOL 425 Advanced Genetics (3:3:0)

3 Semester Credit Hours

Prerequisite(s): BIOL 270

This course discusses the principles of genetics and the role of genetic variations on the expressed phenotype in unicellular and multicellular organisms. The topics include genome structure and function, gene mutations, recombination, transposable elements, experimental genetics analysis, population genetics and diseases.

BIOL 430 Cell Culture Theory and Technology (3:3:0)

3 Semester Credit Hours

Prerequisite(s): BIOL 270

This course aims at providing a strong foundation in the structure, growth, and function of plant and animal cells as well as the technology involved in cell and tissue culture, cell preservation, protoplast culture and fusion, cell cloning and fusion, monoclonal antibody production, breeding and genetic engineering. The applications of stem cells to therapy will also be emphasized in this course.

BIOL 431 Cell Culture Techniques Lab. (1:0:3)

1 Semester Credit Hour

Corequisite(s): BIOL 430

This course deals with the principles and practical applications of animal tissue culture. Molecular techniques or manipulation and maintenance of cells in culture using animal systems.

BIOL 434 Cell Communication and Signal Transduction (3:3:0)

3 Semester Credit Hours

Prerequisite(s): BIOL 270

This course focuses on how cells communicate with each other and their microenvironment, the signaling events that result from these interactions, and their translation into cellular responses.

BIOL 435 Stem Cell Biology and Regenerative Medicine (3:3:0)

3 Semester Credit Hours

Prerequisite(s): BIOL 270

This course provides a broad overview of the biological principles influencing stem cell populations and emphasizes the functional role of stem cells in regulating normal development and contributing to disease, as well as examines the therapeutic potential of stem cells in regenerative medicine.

BIOL 436 Immunology (3:3:0)

3 Semester Credit Hours

Prerequisite(s): BIOL 270

This course focuses on the structure and function of immunoglobulins, the role of cell-mediated immunity, the protective role of the immune system, and disease and injury related to malfunctions of the immune system.

BIOL 440	Medical Microbiology	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 230		
This course focuses on mechanisms of microbial pathogenesis, the host response, and the scientific approaches that are used to investigate these processes. The course provides a strong foundation in the methods to prevent, diagnose and treat infectious diseases. It also discusses laboratory tests used for the diagnosis of infectious diseases.		
BIOL 441	Medical Microbiology Lab.	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): BIOL 440		
The course covers a collection of techniques used for culture, isolation, and identification of common microorganisms. The topics aim to provide students with practical training on standardized methods for microbial identification including staining and microscopy, antibiotic susceptibility, and biochemical tests. In addition, the course provides in depth emphasis on some methods of disinfection, sterilization, and antimicrobial therapy.		
BIOL 442	Molecular Diagnostics	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 380		
This course covers techniques used to analyze biological markers in the genome and proteome to diagnose human disease. The course examines modern tools employed to identify variations in the genetic material or proteins associated with specific health conditions. Additionally, the course emphasizes specimen handling and quality assurance in the molecular lab.		
BIOL 443	Molecular Diagnostics Lab	(1:0:3)
1 Semester Credit Hour		
Prerequisite(s): BIOL 380 and BIOL 381		
Corequisite(s): BIOL 442		
This course covers the basic concepts and techniques that are used in the diagnosis of disease. The students will be trained to have hands-on experience on several assays such as nucleic acid isolation, PCR, gel electrophoresis, restriction fragment length polymorphisms analysis, and ELISA. In addition, the course will prepare students to evaluate the controls for validating results, standardize tests, and discuss quality control for diagnostic laboratory.		
BIOL 450	Bioethics	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): 90 credits		
This course examines selected ethical issues arising from the application of modern biotechnology to microorganisms, plants, animals, and humans. It also discusses the ethics and misconduct in research, authorship, and peer-review. (Writing Intensive Course)		
BIOL 462	Crime Scene Investigation	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 270		
This course will introduce students to crime scene investigation procedures and techniques. With a focus on crime scene processing, this course discusses the most recent topics on database technologies and DNA analyses.		
BIOL 464	Criminal Justice	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Completion of 90 credit hours		
This course will introduce students to the real world of criminal justice with a focus on topics related to understanding crimes, roles of the police, and the court system.		
BIOL 473	Bioinformatics and Computational Biology	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 270		
This course is designed to provide instruction in the scientific concepts and computer skills currently used in searching biological databases, comparing sequences, looking at protein and DNA characteristics and answering biological and biomedical questions using databases.		
BIOL 480	Food Biotechnology	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 270		
This course introduces students to the impact of Biotechnology on food production, the genetic tools applied in Food Biotechnology, the improvement of microbes used in food production by modern biotechnological approaches, and also discusses related ethical, legal, and regulatory issues.		

- BIOL 481 Bioprocessing Technology in Pharmaceuticals Industry (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): BIOL 380
This course provides an overview of the main elements, theory and application of biotechnology procedures, and bioprocess design settings and operations related to the development of biopharmaceutical products including good manufacturing practices and bioprocessing equipment standards. The course lays emphasis on fermentation, harvest, primary and final purification, media and buffer preparation, equipment cleaning and sterilization, and critical process utilities.
- BIOL 490 Research Methods in Biology (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): Completion of 90 credits
This course provides in-depth knowledge of research design and methodology and trains the student to evaluate scientific literature and write a research proposal. The emphasis is on interpreting and evaluating recent studies from scientific literature and communicating biological findings in a written and oral format to a range of audiences
- BIOL 491 Senior Seminar (2:2:0)**
2 Semester Credit Hours
Prerequisite(s): 90 credits
This is a capstone course and involves student participation in seminars on various Biotechnology topics.
- BIOL 493 Research Methodology in the Biological Sciences (2:2:0)**
2 Semester Credit Hours
Prerequisite(s): Completion of 90 credits
This course provides in-depth knowledge of research design and methodology and to train the student in writing a research proposal and critically reviewing scientific literature. The emphasis will be on interpreting and evaluating recent studies from the scientific literature and communicating biological findings in a written and oral format to a range of audiences. The course focus is to train students to critically review literature, identify a research question, lay out a detailed experimental plan including the statistical analysis, and discuss anticipated results.
- BIOL 494 Senior Project in Cell and Molecular Biotechnology (4:0:12)**
4 Semester Credit Hours
Prerequisite(s): Completion of 90 credits and BIOL 490
Under the guidance of Biotechnology faculty mentors, students will conduct a research project in the field of Cell and Molecular Biotechnology and present their research findings in both oral and written formats.
- BIOL 495 Senior Project in Medical Biotechnology (4:0:12)**
4 Semester Credit Hours
Prerequisite(s): Completion of 90 credits and BIOL 490
Under the guidance of Biotechnology faculty mentors, students will conduct a research project in the field of Medical Biotechnology and present their research findings in both oral and written formats.
- BIOL 496 Senior Project in Forensic Biology (4:0:12)**
4 Semester Credit Hours
Prerequisite(s): Completion of 90 credits and BIOL 490
Under the guidance of Biotechnology faculty mentors, students will conduct a research project in the field of Forensic Biology and present their research findings in both oral and written formats.
- BUAN 301 Analytics for Decision Making (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): OPMT 311
This course is an introductory survey course in business analytics and is geared towards providing a broad coverage of descriptive, predictive, and prescriptive analytics techniques and applications within a business context. The course aims to introduce students to foundational concepts, methodologies, and tools required to understand the emerging role of business analytics in organizations, with a focus on decision making for efficiency, profitability, and sustainability. Through hands-on practice tutorials and assignments, students will also be introduced to computer-based modeling approaches to facilitate formulation, resolution, and interpretation of the results, and the process of transforming data into actions.
- BUAN 311 Business Data Mining (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): BUAN 301
Data Mining (DM) is the non-trivial process of extracting implicit, novel, and useful information from large volume of data. It has emerged as a unique combination of several fields of science and technology including statistics, database systems, computer programming, machine learning, and artificial intelligence. This course is designed to provide undergraduate students with an understanding of data mining

techniques, including data pre-processing, classification, clustering and association rules, as well as their applications in business intelligence, customer relationship management, marketing, credit scoring, churn and survival analysis. The students will also be introduced to Big Data and Big Data analytics. Students gain hands-on experience with the Rapid Miner software application. They will perform pre-processing and apply data mining techniques to analyze the selected data to discover useful and actionable patterns and information.

BUAN 312 Visual Analytics

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): BUAN 301

This course provides an in-depth view of working with data to extract and present valuable information. Students will learn to collect, clean, manipulate, analyze, and visualize data from various sources correctly and efficiently. Through hands-on application, students will gain an understanding of what problems can occur when dealing with a variety of data and what solutions exist. This course introduces data visualization as an analytical tool, a medium of communication, and the basis for interactive information dashboards. Students will learn best practices in data visualization, sharpen analytical skills, and learn how to design dashboards for use by stakeholders. Computing is a major component of this course, and students will learn a number of in-demand technical skills.

BUAN 390 Internship in Business Analytics

3 Semester Credit Hours

Prerequisite(s): Completion of 90 credit hours

The Internship is designed for candidates on the undergraduate business analytics major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

BUAN 401 Business Data Engineering

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): MIST 301, BUAN 311

This course introduces students to the foundations of database management systems (DBMS), data warehousing fundamentals, practices, and technologies, and the challenges faced by businesses in managing large amounts of data and making meaningful use of this data for informed decision making. The course focuses on the basics of data pipelines, data pipeline flows and associated business use cases, and how organizations derive value from data and data engineering. Throughout the course, students will be introduced to various computing platforms for developing, deploying, configuring a wide range of data science applications for different business use-cases.

BUAN 411 Business Intelligence & Big Data Analytics

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): MIST 301, BUAN 311

As an intermediate-level course in business analytics, this course covers foundational theories and practices concerned with business intelligence and big data analytics. The course focuses on business applications of big data and provides an introduction to supporting technologies for big data analytics including MapReduce, Hadoop, Spark, and their ecosystems. The course also provides an introduction to the data analytics lifecycle to address business challenges that leverage big data. Examples of various types of BI use-cases for business decision-making will be discussed to allow students to identify and apply appropriate tools and methodologies to real-world business challenges and to facilitate data-driven decision-making.

BUAN 444 Business Analytics Project Based Internship

3 Semester Credit Hours

Prerequisite(s): BUAN 301

The business analytics project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem-solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the business analytics specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.

BUSN 301 Business Communications

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): ENGL 101

This course introduces fundamentals of business models and writing as a learning tool. The interrelationships between accounting, finance, information systems, marketing, and operations are the subject of several “learning by writing” deliverables. The course teaches business students effective methods for communicating business-related issues to clients, management, and fellow employees. **(Writing Intensive Course)**

BUSN 304 Business Law

(3:3:0)

3 Semester Credit Hours**Prerequisite(s): Completion of 58 Credit Hours**

This course provides exposure to general legal environment of business, emphasizing government regulatory process, business ethics, and social responsibility. Regulatory topics include torts and crimes, product liability, intellectual property and cyber law, contracts, and issues related to employment and competition.

BUSN 306 International Business**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): ECON 104**

This course offers a multi-disciplinary approach to global economic issues from the viewpoint of managing international business. It introduces salient aspects of managing in a globalized environment, including consideration of theoretical concepts. It provides a practical treatment of political-economic aspects of international trade. The course discusses foreign direct investment, global monetary systems, and strategy formulation for international business.

BUSN 307 Research Methodology**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): Completion of 58 Credit Hours**

This course provides essential underpinning support for research elements of higher-level business courses. It allows candidates to make informed decisions and appropriate choices pertaining to research methodology. A range of business research tools, approaches and analytical techniques are discussed, and guidance is given on optimal structuring of research work in the business field, and the generation of research proposals.

BUSN 312 Public Policy and Business Ethics**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): ECON 103**

This course introduces the policy and ethical dimensions of business as they relate to the various stakeholders inside and outside the organization. It includes topics such as actors, institutions and ideas in the policy process, business ethical theory, ethical decision making, ethical leadership, ethical and policy dilemmas, corporate social responsibility, and environmental sustainability.

BUSN 390 Internship**3 Semester Credit Hours****Prerequisite(s): Completion of 90 credit hours**

The Internship is designed for candidates on the undergraduate finance, marketing, human resource management and accounting programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship and will serve to introduce the student to real-world business practices, culture, and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

BUSN 391 International Internship**3 Semester Credit Hours****Prerequisite(s): Completion of 90 credit hours**

This course is designed for candidates on the Bachelor of Science in Business Administration to provide a field experience in which students are able to apply knowledge and skills gained through course work, and to demonstrate their capacity to function successfully in a professional setting. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues in international settings. Through this internship, the students will learn, and apply real-world management skills on an international level, and throughout the process, the student will complete the internship report to fulfil the course requirements. The vast majority of the work and value of this course, however, will come in the form of the international, intercultural, and interpersonal experiences gained throughout the period in the international workplace.

BUSN 498 Strategic Management**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): Completion of 90 credit hours**

This course is designed to help students apply functional knowledge gained from earlier courses and integrate these with more practical information and real-life experiences. The course takes the general management point of view, emphasizing the creation, implementation, and evaluation of strategy in organizations. Students adopt the viewpoint of upper management, strategize, and make essential "Big Picture" decisions, using standard strategic analysis frameworks. Students develop expertise in the analysis of complex business situations and in clearly presenting strategic findings both orally and in writing.

CENG 315 Microprocessors**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CSCI 112 and EEEN 331**

Microprocessors as components in a computer system; programmer's view of a microprocessor's architecture; microprocessor instruction set; assembly language programming; interrupts; input and output; interfacing a microprocessor to memory and I/O devices from the programmer's view. At the end of the course, the students should be able to program a modern microprocessor in assembly or C language and perform hardware I/O interfacing.

(1:0:3)

1 Semester Credit Hour

Prerequisite(s): CSCI 112

Corequisite(s): CENG 315

This course introduces microprocessors in a laboratory environment. Topics include: assembly language programming; hardware interface and design with common microprocessor peripherals, uVision Simulation, Software Setup, GPIO: LEDs and Joystick, LCD Display, Keypad, LED in Assembly, Stepper Motor in C and Assembly, System timer, PWM, ADC/Infrared Sensor and DAC / Digital Music.

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EEEN 331

Corequisite(s): CENG 336

This course covers details of microprocessor design including the instruction set architecture, memory design, and data path and control design. The course also emphasizes memory performance related concepts such as associativity and multi-level caching. Additional topics include virtual memory and performance speed-up techniques using pipelining, multithreading, and multiprocessing.

(1:0:3)

1 Semester Credit Hour

Prerequisite(s): EEEN 331

Corequisite(s): CENG 335

This course covers modern computer system architecture and computer design principles. A Hardware Description Language is used to design basic components of a microprocessor datapath and control. Additional topics covered include Adders, MUX, Counters, ALU, registers/shift registers, RAM, pipelining, and cache memory.

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSCI 462

This course discusses socket programming, client-server applications, peer-to-peer networks and applications, Web servers, datacenters and load balancing, CDNs, IPV6, NAT, anycasting, DHCP, multimedia networking, network operation and management and basic Internet measurements.

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSCI 215

This course examines in detail the software development process. Topics include concepts such as software processes, software lifecycle models, requirement engineering, system modeling, architectural design, software design implementation, software testing, software evolution, and software reuse.

(3:3:0)

3 Semester Credit Hours

Prerequisite(s):CSCI 232 or CENG 315

Corequisite(s): CENG 432

This course introduces the design and analysis of embedded computer systems that interact with the physical world. The course provides an integrated approach to the development of low-power systems with hardware, software, sensors, actuators, processors and networks. Topics include hardware and software architectures, assembly and C programming, real-time design, interrupts, multitasking, embedded software tools, and embedded systems performance. Students will work on a comprehensive project to design, implement, and evaluate a prototype embedded system.

(1:0:3)

1 Semester Credit Hour

Corequisite(s): CENG 431

In this lab course, students will design microcontroller-based embedded systems for various applications. Topics covered include interfacing to sensors and actuators from both hardware and software perspectives, assembly and C programming for embedded systems I/O, real-time design, and interrupts. Additionally, students will implement and evaluate a complete embedded system.

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CENG 335

This course provides students with a deep understanding of parallel computing architectures including multicore CPUs and GPUs. The course emphasizes trade-offs involved in design of modern parallel computing systems, and techniques necessary to effectively utilize these machines. Topics covered include data-level parallelism, thread-level parallelism, GPU architectures, cache coherence, memory consistency, interconnection networks, reconfigurable computing and domain-specific parallel programming systems.

CENG 461 **Network Security**

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): MATH 225 and CSCI 462 or EEEN 462

This course examines information security services and mechanisms in a network context. Topics include Cryptographic tools, User Authentication, Authorization and Access Control, Database Security, Malicious Software, Denial of Service Attacks, Firewall and Intrusion Prevention, Software Security, Operating System, Internet Security Protocols and Standards. Stack and Buffer Overflow, Security Auditing, Forensics, Ethical and Legal aspects, and Wireless Network Security.

CENG 492 **Senior Design Project I**

(2:0:6)

2 Semester Credit Hours

Prerequisite(s): Completion of 90 credit hours

Corequisite(s): ENGR 450

This is the first course of a capstone project that requires students to develop, design, and implement a solution to an engineering problem under the supervision of a faculty advisor. Students are required to consider the ethical, social, and economic implications of their project. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resource, procurement, and communication, with consideration of ethical and professional conduct.

CENG 493 Senior Design Project II

(4:0:12)

4 Semester Credit Hours

Prerequisite(s): CENG 492

Implementation of the project for which preliminary work was done in CENG 492. Project includes designing and constructing software and/or hardware, conducting experiments or studies, and testing and validating a complete system. At the end of the term, each team presents to a committee information related to its project in both written and oral formats.

CHEM 095 Introductory Chemistry

(3:3:0)

3 Semester Credit Hours

This course is an introduction to the basic chemistry notions such as atomic structure, molecules and mole concept, periodic classification of elements, naming of compounds, chemical formulas and equations and its balancing, the basic understanding of simple compounds of carbon and its applications, the influence of chemistry in the environment, and its scope for application in other fields of study. This course is not for degree credit. Placement in this course is based on the student's score on the Chemistry Placement Test.

CHEM 100 Chemistry in Everyday Life

(3:3:0)

3 Semester Credit Hours

This course focuses on chemistry and its impact in everyday life. Basic chemical principles, the impact, and practical everyday applications are examined. Environmental concerns including energy, resource consumption and pollution as well as the crucial role involved in making informed opinions and better consumer choices in our society is addressed.

CHEM 101 Chemistry in Everyday Life Lab

(1:0:3)

1 Semester Credit Hour

A chemistry appreciation course for non-science majors. Covers basic chemistry concepts with technical/fundamental aspects. Emphasizes the practical chemical world of human beings and the chemical nature of everyday products. Hands-on laboratory sessions provide the opportunity to put the scientific method into action allowing further exploration and reinforcement of chemical principles.

CHEM 111 Principles of General Chemistry

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): None

Corequisite(s): CHEM 112

This course introduces students to the fundamental principles and laws of chemistry. This course covers coordination compounds, acid-base equilibrium, chemical equilibrium, solubility and complex-ion equilibria, thermochemistry, gas laws, and solutions. Introduction to organic chemistry, electrochemistry and nuclear chemistry.

CHEM 112 Principles of General Chemistry Lab

(1:0:3)

1 Semester Credit Hour

Corequisite(s): CHEM 111

This course introduces chemistry lab techniques and provides hands-on experience in conducting chemical reactions that accompany the course CHEM111. It provides experience in basic techniques for the synthesis of coordination complexes and physical chemistry-based experiments.

CHEM 120	Analytical Chemistry	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): None		
Corequisite(s): CHEM 121		
This course emphasizes quantitative chemical analysis and underlying principles. Specific analytical techniques or concepts covered are: basic understanding of analytical chemistry including experimental error, statistical analysis of data, quality assurance, calibration methods, introduction of titrations, and polyprotic acid base equilibria. This course will introduce spectroscopic, separation, gravimetric, and combustion methods for quantitative chemical analysis.		
CHEM 121	Analytical Chemistry Lab	(1:0:3)
1 Semester Credit Hours		
Corequisite(s): CHEM 120		
Laboratory techniques and experiments arranged to accompany course Analytical Chemistry (CHEM 120). The aim of this course is to teach students the basic principles of analytical chemistry. The course covers the preparation of reagent, statistical evaluation of replicated data measurements and studies the validity of results. In addition, the analysis of unknowns using different analytical qualitative and quantitative methods is also enclosed.		
CHEM 211	General Chemistry I	(3:3:0)
Prerequisite: None		
3 Semester Credit Hours		
This course forms the foundation of chemical concepts: basic facts and principles of chemistry, including atoms, molecules, ions, chemical reactions, gas theory, thermochemistry, electrochemistry, chemical kinetics and equilibrium, molecular geometry, and states of matter.		
CHEM 212	General Chemistry I Lab	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): CHEM 211		
This is a Laboratory class that is a companion class to CHEM 211 with laboratory techniques and reactions arranged to reinforce concepts. This laboratory course provides experience in physical-chemical measurement, synthesis-based experiments, qualitative, and quantitative analysis.		
CHEM 213	General Chemistry II	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 211		
Corequisite(s): CHEM 214		
This course covers the foundations of chemical concepts including: solutions, chemical reactions, acid base equilibrium, acids and bases, quantum theory of atoms, solubility and complex-ion equilibrium, chemistry of main group elements, polymer chemistry, and introduction of organic and nuclear chemistry, transition elements and coordination complexes.		
CHEM 214	General Chemistry II Lab	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): CHEM 213		
This course introduces Laboratory techniques and reactions arranged to accompany CHEM-213. This laboratory course provides experience in physical /chemical measurement, synthesis-based experiments, qualitative, and quantitative analysis.		
CHEM 215	Organic Chemistry I	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 111 or CHEM 120 or CHEM 211		
Corequisite(s): CHEM 216		
This course provides a foundation in Organic Chemistry and includes the theoretical and synthetic aspects of the chemistry of carbon compounds.		
CHEM 216	General Organic Chemistry Lab I	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): CHEM 215		
This course provides hands-on laboratory experience in techniques and chemical reactions in organic chemistry that accompany CHEM 215. The course includes basic techniques for the preparation, isolation, purification, inter-conversion, and spectroscopic study of common classes of organic compounds.		

CHEM 217	Organic Chemistry II	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 215		
Corequisite(s): CHEM 218		
This course focuses on aromatic organic chemistry, stereochemistry, organic chemistry reaction mechanisms, bio-molecules and the methods used for the synthesis of organic compounds.		
CHEM 218	General Organic Chemistry Lab II	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): CHEM 217		
This course provides practical experience in common techniques for synthesis, separation, purification, and identification of organic molecules such as NMR, Mass Spectrometry, and Fourier Transform Infrared spectroscopy, as well as the qualitative analysis of functional groups.		
CHEM 315	Physical Chemistry	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 215		
Corequisite(s): CHEM 316		
This course will introduce the concepts of physical chemistry, properties of gases, first and second law of thermodynamics, physical transformation, phase diagram, chemical equilibrium, atomic structure and spectra, molecular structure, spectroscopy, chemical reactions, statistical thermodynamics, kinetics and dynamics of chemical reactions.		
CHEM 316	Physical Chemistry Lab.	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): CHEM 315		
Introductions to the techniques and reasoning of experimental chemistry. Lab techniques and reactions arranged to accompany Physical Chemistry.		
CHEM 411	Medicinal Chemistry	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 217		
This course examines principles of structural, physical, and physical-organic chemistry, including mechanistic considerations involved in synthetic organic chemistry, bioorganic chemistry, and the design of chemotherapeutic agents. This course includes mechanisms of action and factors that influence drug action within specific drug classes of pharmaco-dynamic and chemotherapeutic nature, drug structures, chiral drugs, and structure-activity relationships of drugs.		
CHEM 420	Forensic Chemistry	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 120		
Corequisite(s): CHEM 421		
Foundation of forensic chemistry: applications of the principles, methods, and instrumentation of chemistry to forensic problems. Covers topics most commonly applicable to the functions of a crime laboratory professional.		
CHEM 421	Forensic Chemistry Lab	(1:0:3)
1 Semester Credit Hour		
Prerequisite(s): CHEM 121		
Corequisite(s): CHEM 420		
This course introduces laboratory techniques to the forensic analysis of trace evidences according to established forensic procedures. The focus of the course will be on identifying and understanding the nature of the samples, sample preparation methods, chemical and instrumental methods for analysis.		
CHEN 202	Principles of Chemical and Petroleum Engineering	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 211		
This course introduces the students to chemical and petroleum engineering profession. The course is designed to equip students with the fundamental of chemical and petroleum engineering such as introduction to the chemical and petroleum industry, crude oil formation, exploration and drilling, introduction to chemical engineering calculations, problem solving techniques, material and energy balance for reactive and nonreactive systems.		
CHEN 301	Computational Tools in Chemical and Petroleum Engineering	(1:0:3)
1 Semester Credit Hour		
Prerequisite(s): CHEN 202		

The course covers the use of various software in solving basic chemical and petroleum engineering problems related to thermodynamics, fluid mechanics, reaction engineering and separation problems. In addition to utilize computational tools for simulation of oilfield economics.

CHEN 312 Chemical Engineering Thermodynamics (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CHEM 315

This course aims to introduce the principles of Chemical Engineering Thermodynamics and illustrate their application. The content comprises the fundamentals of thermodynamics, such as thermodynamic properties (energy, entropy, enthalpy, heat capacity, etc.), the first and second law of thermodynamics (energy and entropy balance), heat of reactions, thermodynamics of ideal and real gases and liquids, vapor-liquid equilibrium and thermodynamics of chemical reaction processes. In addition, to the use of software tools to solve different thermodynamic problems.

CHEN 351 Chemical Reaction Engineering (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CHEN 202 and CHEM 315

This course provides a detailed analysis to the principles of chemical kinetics, and reactor analysis and design. The course covers kinetics of homogeneous and heterogeneous reactions, design of isothermal reactors such as Batch, Continuous Stirred Tank Reactor (CSTR) and Plug Flow Reactor (PFR). Other topics include data collection and handling, non- isothermal reactor design and multiple reactions. The last part of the course considers homogeneous and heterogeneous catalytic reactions. Further, the course introduces the link between the theoretical part and the laboratory scale chemical reactors through physical tour of the Chemical Engineering Laboratory facility.

CHEN 352 Chemical Reaction Lab (1:0:3)

1 Semester Credit Hour

Corequisite(s): CHEN 351

This course covers experiments that deal with chemical reactions over different set of reactors including batch, tubular and continuous stirred tank reactors. Performing theoretical calculations and experimentation on real and virtual reactor systems for homogeneous phase as well as catalytic reactions.

CHEN 370 Heat and Mass Transport (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CIEN 251

Corequisite(s): CHEN 312

This course introduces the physical origins and the governing laws for heat and mass transfer. The principal topics covered include identification of the driving forces for heat and mass diffusion, development of transport models from first principles, steady state and transient solutions, fundamentals and engineering treatment of convection heat and mass transfer, heat transfer with phase change, radiation heat transfer in addition to interfacial mass transfer and continuous stage-wise contact operations. Further, the course introduces the link between the theoretical and experimental design of heat and mass transfer equipment.

CHEN 372 Thermal and Fluid Sciences Lab (1:0:3)

1 Semester Credit Hour

Corequisite(s): CHEN 370

This course is composed of a set of selected experiments which demonstrate and apply the concepts of fluid, heat and mass transfer. The course aims to develop student's basic knowledge in: volumetric properties of pure fluids, principles and concepts related to heat transfer, principles of steam and power systems, principles of fluid flow (laminar flow and turbulent flow) and principles of heat exchangers. The lab also includes an open-ended design of experiment.

CHEN 403 Fundamentals of Biochemical Engineering (3:3:0)

3 Semester Credit Hours

Prerequisite(s): BIOL 112 and CHEN 351

Corequisite(s): CHEN 370

This course aims to apply the principles of Chemical Engineering in bioprocess, in which biocatalysts, i.e. cells and enzymes, play key roles in producing new bioproducts and chemicals. The purpose of this course is to introduce the fundamental principles of biochemical engineering and present a wide spectrum of potential technological applications. This course includes introduction to biochemical and microbiological applications to commercial and engineering processes, including industrial fermentation, ultrafiltration and resulting waste treatment. In addition to introduction to enzyme kinetics, cell growth, energetics and mass transfer.

CHEN 404 Safety and Environmental Impact (3:3:0)

3 Semester Credit Hours

Prerequisite(s): Completion of 90 credit hours

The course introduces students to sources and type of material and energy hazards in chemical and petroleum industries. Topics include fire and explosion risks, loss prevention strategies, safety program implementation, and adherence to safety regulations. Additionally, it

explores the significance of engineering ethics in process safety, models for the release of materials, energy, and pollutants, as well as industrial hygiene and risk analysis.

CHEN 452 Introduction to Hydrogen Technologies (3:3:0)

3 Semester Credit Hours

Pre-requisite(s): CHEN 351 and CHEN 370

This course covers the fundamentals of hydrogen technologies including: methods of hydrogen production (with emphasis on “green” and ‘blue’ hydrogen production), storage, transportation and utilization for energy applications. The course provides a broad knowledge of hydrogen as an energy carrier, the way it will play an important role in various sectors towards decarbonization, current limitations and future scenarios.

CHEN 462 Petroleum Refining Engineering (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CHEN 351 and CHEN 370

This course aims at introducing different techniques for petroleum refining. Topics include refinery feed stocks and products, field processes, crude distillation, coking and thermal processes, catalytic reforming and cracking, hydro- processing, and solvent treating processes. Solve petroleum-refining problems including material balances and process economics. Conduct lab experiments to characterize crude oils and final petroleum products. Students will do a case study of a typical refinery.

CHEN 463 Natural Gas Engineering (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CHEN 370

This course introduces the natural gas properties including real gas mixtures behavior in addition to flow of gas in reservoir, wellbore and surface pipelines. The course also deals with natural gas water systems, natural gas condensate systems, hydrate formation and inhibition, separation processes, field treatment of natural gas, absorption and adsorption processes, natural gas dehydration, sweetening and sulfur recovery, designing and sizing of the main equipment and performing nodal analysis for gas production system.

CHEN 464 Industrial Catalysis (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CHEN 351

This course will provide a comprehensive overview of homogenous and heterogeneous catalytic science and technology for different chemical engineering industry. Course modules will cover the preparation and characterization of catalyst systems, kinetics and mechanism of catalyzed reactions including adsorption, desorption and surface reactions. Emphasis will be placed on understanding the design, synthesis, characterization and industrial application of various catalytic systems.

CHEN 471 Unit Operation Lab (1:0:3)

1 Semester Credit Hour

Corequisite(s): CHEN 472

This course aims to deepen the students’ knowledge of the unit operations with a focus on distillation, absorption, liquid-liquid extraction, membrane and drying processes. This course requires students to integrate and apply knowledge from previous chemical engineering courses to design experiments, collect data, analyse results and make recommendations. Students will also work on developing oral and written technical communication skills. **(Writing Intensive Course)**

CHEN 472 Separation Processes (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CHEN 370

This course aims to introduce the fundamental principles of Separation Processes through examining both equilibrium-controlled separation processes as well as separation processes that involve both mass transport and equilibrium considerations. In order to probe the key concepts in depth, the course will focus primarily on binary and multi-component distillation, leaching, absorption, membranes and liquid-liquid extraction. Furthermore, Computer instruction will be employed throughout the course to illustrate important characteristics of these separation systems.

CHEN 473 Water Desalination (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CHEN 370

This course covers water resources, chemistry of saline water, modeling and analysis of single effect desalination combined with mechanical vapor compression and thermal vapor compression, modeling of multiple effect desalination (MED), modeling of single and multistage desalination processes, reverse osmosis. Also, cogeneration, scale formation, pre and post treatment operations will be discussed.

CHEN 474 Industrial and Wastewater Treatment (3:3:0)

3 Semester Credit Hours

Prerequisite(s): Completion of 90 credit hours

This course covers characteristics of industrial wastewater; physical, chemical and biological unit operations/processes in industrial wastewater treatment; non-conventional (advanced) wastewater treatment technologies; case study of industrial wastewater treatment processes.

CHEN 476 Fundamentals of Nanotechnology (3:3:0)

3 Semester Credit Hours

Pre-requisite(s): CHEN 351 and CHEN 370

The general goal of the course is to provide an introduction to and an overview of nanotechnology (NT). The course handles the basics of nanomaterials and nanotechnology and its applications. It will provide an insight into the chemical materials and fabrication lines nowadays used in nanotechnology. The course will also demonstrate how the applications of NT will influence science of tomorrow and will change many sides of our life as well as discussing the impact of NT on our society.

CHEN 481 Process and Plant Design (3:3:0)

3 Semester Credit Hours

Prerequisite(s): MENG 241 and MENG 323

Corequisite(s): CHEN 472

Application of chemical engineering principles to the development of process flow diagrams including flow and process conditions. Estimation of capital and manufacturing cost and application of heuristic methods to the design of process equipment. Design of heat and separation sequences of chemical processes. Utilization of process simulators to develop process flow diagram and size equipment.

CHEN 482 Chemical Process Dynamics and Control (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CHEN 301 and CHEN 351

This course aims to introduce process dynamics and principles of control for chemical processes. Topics covered include block diagrams, Laplace transforms, development of dynamic equations for elements of control loops, transient response of feed-back systems, stability characteristics of dynamic systems, gain and phase margins. In addition to development of process characterization using mathematical models, controller design, and implementation.

CHEN 483 Advance Simulation Lab (1:0:3)

1 Semester Credit Hours

Prerequisite(s): CHEN 312

Corequisite(s): CHEN 472

This course focus on the use of process simulators such as HYSYS, ASPEN PLUS and COMSOL in designing and simulation of chemical processes and its applications. In addition to data generation, analysis, thermodynamic modeling, and process economic analysis using simulation software.

CHEN 484 Process Integration and Optimization (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CHEN 481

This course aims to introduce the concepts, tools and applications of heat and mass integration using pinch technology and application of linear and nonlinear programming for process integration and optimization. In addition to application of heat and mass integration in process design and retrofit. Graphical and mathematical procedures are emphasized for analyzing process performance. Case studies and examples will be used to illustrate how process integration helps in optimizing actual industrial processes.

CHEN 485 Carbon Capture, Utilization and Storage (3:3:0)

3 Semester Credit Hours

Pre-requisite(s): Completion of 90 credit hours

The course aims to provide students with a comprehensive overview of Carbon Capture, Utilization and Storage (CCUS) technologies. The course will cover methods and technologies related to removing CO₂ from the flue gas and the atmosphere, followed by recycling the CO₂ for utilization and determining safe and permanent storage options. In addition, the full spectrum of economic, environmental, regulatory, and their implications for regional and global carbon management strategies of the future will be discussed.

CHEN 486 Data Science in Chemical Engineering (3:3:0)

3 Semester Credit Hours

Prerequisite(s): ENGR 200

Corequisite(s): Completion of 90 credit hours

This course is an introduction to data science and machine learning with specific focus on chemical engineering applications. It will cover data access and management, databases and data warehousing, statistical methods including classification and clustering, time series, various regression methods and multivariate statistics and data visualization. In addition to the use of artificial intelligence and machine learning to solve chemical engineering problems.

CHEN 491 Senior Design Project I (3:0:9)

3 Semester Credit Hours

Prerequisite(s): Completion of 90 credit hours

The Senior Design Project entails team work involving development of a two-semester long workflow that includes preparing and presenting a proposal for a major design work. The design work may involve physic-chemical process design and testing, or simulation workflow design related to one or more chemical engineering activities, such as, reaction, separation, mass or heat transfer and dynamic and control– with due consideration of economic, environmental and societal impacts. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resource, procurement, communication, with consideration of ethical and professional conduct.

**CHEN 492 Senior Design Project II
3 Semester Credit Hours****(3:0:9)****Prerequisite(s): CHEN 491**

The Senior Design Project entails team work involving development of a two-semester long workflow that includes designing, implementing and evaluate the propose solution to a certain engineering problem. The design work may involve process design and testing, or simulation workflow design related to one or more chemical engineering activities, such as, reaction, separation, mass or heat transfer and dynamic and control – with due consideration of economic, environmental and societal impacts. The SDP-II entails finalizing the tasks planed in SDP-I. Students must present their work in a detailed technical report and defend their project in oral presentation.

**CHEN 493 Special Topics in Chemical Engineering
3 Semester Credit Hours****(3:3:0)****Prerequisite(s): Department Consent**

A specific topic in Chemical Engineering that is not covered in other program courses is presented in a course format.

**CIEN 201 Computer Aided Drawings
3 Semester Credit Hours****(3:1:4)****Prerequisite(s): None**

Introduction to computer graphics. Includes the following topics: geometric construction; line convention; orthographic projections, isometric projections; oblique projections; Descriptive projections; dimensioning, and sectional views.

Computer-aided design and problem-solving techniques. Uses computer software AutoCAD in data analysis, data display and visualization techniques. Prepare drawings in civil and infrastructure engineering.

**CIEN 211 Statics
3 Semester Credit Hours****(3:3:0)****Prerequisite(s): MATH 113 and PHYS 110**

Vectors, force systems (2D and 3D), equilibrium of particles and rigid bodies (2D and 3D), structures (trusses, cables, frames and machines), distributed forces (centroids and centers of mass), internal forces (shearing force and bending moment diagrams), friction, and moment of inertia.

**CIEN 212 Mechanics of Materials
3 Semester Credit Hours****(3:3:0)****Prerequisite(s): CIEN 211**

Types of loads, axial stress and strain of determinate and indeterminate system, normal and bending moment diagrams, torsion of determinate and indeterminate system, bending of beams, combined stresses, shearing stress and strain, Mohr's circle of stress and strain, thin walled pressure vessels.

**CIEN 216 Structures for Architecture 1
3 Semester Credit Hours****(3:3:0)****Prerequisite(s): MATH 113 and PHYS 110**

Equilibrium of a rigid body, force systems, geometric properties of various shapes, internal forces in beams, trusses, and frames, analysis of three-hinged and tied arches, construction of shear force and bending moment diagrams for beams and frames, mechanical properties of materials, axial, shear, and bending stresses in members, stability, determinacy and indeterminacy of beams and trusses.

**CIEN 241 Infrastructure Management
3 Semester Credit Hours****(3:3:0)**

An introduction to infrastructure management systems, including management process, data collection technologies, interdependence, benchmarking and best practices. Other related issues, such as, resilience security of infrastructure systems are addressed.

**CIEN 251 Fluid Mechanics
3 Semester Credit Hours****(3:3:0)****Prerequisite(s): MATH 213 & CIEN 211 (Civil Engineering) and MATH 114 and PHYS 110 (Chemical, Mechanical, and Petroleum Engineering)**

Properties of fluids, flow regimes, pressure and force calculations under hydrostatic conditions, manometers, buoyancy and stability of floating and submerged bodies, elementary fluid dynamics, conservation equations: mass, energy and momentum, continuity and Bernoulli equations. Boundary layer theory, fluid flow in conduits and determination of head loss in pipes.

CIEN 261	Surveying	(1:0:3)
1 Semester Credit Hour		
Prerequisite(s): CIEN 201 (for Civil and Infrastructure Engineering); INDS 122 (for Architecture Students)		
Principles of surveying by conducting tests on distance measurements, levels and theodolites, directions and angular measurements, topographic surveys, areas and volumes; traverse surveys; setting out horizontal and vertical curves, training on Total Station.		
CIEN 265	Geographic Information System (GIS)	(2:1:3)
2 Semester Credit Hours		
Prerequisite(s): CSCI 112 or CSCI 114		
This course exposes students to the principles of GIS (hardware, software, people, data, and methods) and its environmental and infrastructure management applications. Subjects also include the acquisition and compilation of data from maps, field surveys, air photographs and satellite images. The course includes hands-on GIS state-of-the-art software.		
CIEN 301	Numerical Analysis	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MATH 214 and (CSCI 112 or CSCI 114)		
Matrix Algebra: Matrix operations, inverse, determinant, Eigen vectors, Eigen Values and solutions of systems of linear equations. Application of computers to solve engineering problems using various numerical methods, mathematical modeling and error analysis, solution of algebraic and differential equations, numerical differentiation and integration and curve fittings. Presents application examples from civil and infrastructure engineering, industrial engineering and mechanical engineering.		
CIEN 311	Structural Analysis	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CIEN 212		
Structural forms, reactions, determinate structures, degree of determinacy, shear and moment diagrams for beams and frames, influence lines for beams, deflections (principle of moment area theorems, virtual work and conjugate-beam methods), Analysis of indeterminate structures by moment force method, slope-deflection method and distribution method.		
CIEN 316	Structures for Architecture 2	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CIEN 216		
Introduction to concrete, structural steel, and masonry as construction materials. Design of rectangular reinforced concrete beams reinforced concrete one-way slabs, and simple design of reinforced concrete short columns according to ACI 318 design code. Design of simple masonry walls. Design of structural steel beams and structural steel columns according to LRFD design method.		
CIEN 321	Reinforced Concrete Design	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CIEN 311 and CIEN 331		
Properties of concrete and steel, cracked and uncracked section analysis, strength design, stress block, design for bending and shear, singly, doubly reinforced sections, rectangular sections, and T-sections, load cases and moment envelopes, bond requirements, development length and bar cutoffs, one-way solid and one-way ribbed slabs, design of short columns.		
CIEN 331	Construction Materials	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CIEN 212 and CHEM 211		
Cement (types, manufacture, properties and hydration), aggregates, fresh concrete, hardened concrete (strength, strength development, shrinkage, creep), concrete in severe environment (hot and cold), durability, mix design by ACI and DoE methods, use of masonry, fiber reinforcement and metal form decking and structural steel joists.		
CIEN 332	Construction Materials Lab.	(1:0:3)
1 Semester Credit Hour		
Prerequisite(s): CIEN 212 and CHEM 211		
Corequisite(s): CIEN 331		
Introduction to testing & specifications, concrete and mortar tests, aggregate testing, fresh and hardened concrete testing, non-destructive tests, design & testing of concrete mixes, brick testing, tests on steel reinforcing bars.		
CIEN 333	Geotechnical Engineering	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): CIEN 212 and CHEM 211		
Index and classification of soils, water flow in soils (one- and two-dimensional water flow), soil stresses, soil compaction, distribution of stresses in soil due to external loads, consolidation and consolidation settlement, shear strength of soils, slope stability.		
CIEN 334	Civil Engineering Testing and Materials	(1:0:3)

1 Semester Credit Hour**Prerequisite(s): CIEN 331, CIEN 333****Corequisite(s): CIEN 361**

Tests on soils: specific gravity, grain size distribution, consistency limits, coefficient of permeability (constant and falling head), consolidation test, direct shear and tri-axial and CBR. Tests on fluids: Center of pressure, orifice and jet flow, Pressure variation in flowing fluid, momentum principle, flow through pipelines and closed conduits. Tests on asphalt properties, and marshal mix design. **(Writing Intensive Course)**

CIEN 351 Environmental Engineering**(2:2:0)****2 Semester Credit Hours****Prerequisite(s): None**

Introduces principles of environmental engineering management and design pertaining to water supply and treatment, wastewater treatment, solid waste management, air pollution control, noise pollution measurement and control, and environmental impact assessment. Includes case studies from UAE.

CIEN 361 Highway Engineering and Design**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CIEN 331**

Pavement types; Materials used in flexible pavement layers (soil, aggregate, and asphalt); Calculation of Equivalent Standard Axle Loads (ESAL); flexible pavement thickness design, Hot Mix Asphalt (HMA) design; Highway construction operations; highway performance.

CIEN 362 Transportation Engineering**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CIEN 261**

Role of transportation engineering; Transportation system issues and challenges; Modes of transportation, Main components of highway, mode of transportation (driver, pedestrian, traffic, road); Geometric design of highways and highway facilities; Highway functional classification and special facilities; Intersection design and control. Introduction to rail, air, and water transportation.

CIEN 421 Structural Steel Design**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CIEN 311**

Introduction to steel structures and practical design methods, steel sections, load factors and load combinations, design of various steel elements using LRFD-method, design of tension and compression members, beam design: Compact section criterion, lateral-torsional buckling, lateral supports, and various design aspects of beams. Beam-column elements, design of column base plates, design of simple bolted (or welded) steel connections.

CIEN 422 Advanced Reinforced Concrete Design**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CIEN 321**

Structural layout, estimation of dead and live loads, serviceability, deflections and crack control, design for torsion, design of frames, moment redistribution, slender columns, approximate methods for two-way slabs, detailing of reinforcement.

CIEN 424 Bridge Design**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CIEN 321**

Covers design of highway bridges; history, classification and aesthetics of bridge structures; design philosophy; loading, girder distribution factors; and load combinations; design of concrete deck slab; design of reinforced concrete T-beam and box girders bridges; and design of piers, bearings and abutments. Introduces pre-stressed concrete bridges.

CIEN 431 Foundation Engineering**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CIEN 333**

Site investigation, bearing capacity of shallow foundation, distribution of stresses in soils, settlement of shallow foundation, factors to be considered in foundation design, introduction to deep foundation, lateral earth pressure and retaining walls.

CIEN 440 Infrastructure Financing**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CIEN 241**

Fundamentals of infrastructure project financing, Public Private Partnerships (PPP), Build Own Transfer (BOT), variation models, financing structures, framework, bonds, life cycle analysis, sponsors and risk management.

CIEN 441 Construction Management**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CIEN 331**

Basic concepts of construction project management, Construction planning, project time Management, project cost management, project quality management, value engineering and project life cycle, construction process optimization, construction contracts, contracting methods, project specifications, bidding, procurement methods and contractor applications for payment procedures.

CIEN 442 Construction Planning and Scheduling

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CIEN 441

Principles of planning, monitoring, and controlling construction projects. Developing schedules using bar charts, precedence diagrams, program evaluation and review techniques (PERT), and linear schedules. Resource histograms and s-curves. Resource allocation and resource leveling. Schedule constraints. Earned value concept. Includes MS project and Primavera Project Planner software.

CIEN 444 Construction Cost Analysis and Estimating

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CIEN 441

Perceptions of construction cost, engineering economic analysis, risk and uncertainty, range estimating, cost fundamentals, types of cost estimating, estimating construction materials cost, estimating construction labor cost, direct versus indirect costs, estimating construction equipment cost, cost of concrete structures, estimating project cost, time/cost trade-off analysis and contractor general requirements.

CIEN 451 Infrastructure Systems

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CIEN 351 and CIEN 362

Role of planning, system demands and networking in infrastructure systems, energy systems, water and waste water infrastructure, transportation systems, waste disposal and resources conservation, smart growth and effects of infrastructure on the environment, models of creating sustainable future development, planning, design and architecture in sustainable communities, and integrated infrastructure system models.

CIEN 456 Sustainable Urban Building Sites

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CIEN 351

Concept of sustainability, how sustainability applies to infrastructure projects and programs, measuring sustainability, identification and design of sustainable technologies associated with water and energy management for infrastructure projects, green buildings and sustainable housing, sustainable transportation, energy, use of materials and waste management and water use. Includes case studies in sustainability.

CIEN 457 Water Resources Sustainability

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CIEN 351

This course provides a comprehensive overview of the challenges and solutions related to the sustainable management of water resources. Students will gain a deep understanding of water supply systems, integrated water resources management, management of water resources in arid regions, and sustainable use of water in urban, agricultural, and industrial sectors. Several case studies will be presented and discussed.

CIEN 462 Advanced Pavement Design

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CIEN 361

Pavement types, Pavement materials; subgrade stabilization methods; Principles of mix design using SUPERPAVE; Analysis of stresses in flexible and rigid pavement, Design methods of highway flexible and rigid pavements; Overlay design, Computer applications.

CIEN 464 Airport Planning and Design

(3:3:0)

3 Semester Credit Hours

Corequisite(s): CIEN 362

Examines airport master planning, forecasting air travel demand and sustainable design of airports, including lighting, terminal facilities, noise-level control, aircraft control, airspace utilization and automobile parking.

CIEN 491 Senior Design Project (1)

(2:0:6)

2 Semester Credit Hours

Prerequisite(s): Completion of 90 credit hours

This is the first course of a capstone project that requires students to develop, design and implement a solution to an engineering problem under the supervision of a faculty advisor. Students are required to consider ethical, social and economic implications of their project. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resources, procurement, communication with consideration of ethical and professional conduct.

CIE 492 Senior Design Project (2)	(4:0:12)
4 Semester Credit Hours	
Prerequisite(s): CIE 491	
Continuation of the senior design project I. The project is a multidisciplinary interaction for infrastructure design and management that includes system analysis and inculcates sustainable engineering principles. Includes use of engineering software's especially project management software such as MS project, Primavera Project Planner and CYCLONE, writing a technical report and developing project drawings, specifications and details.	
CIE 493 Special Topics in Civil Engineering	(3:3:0)
3 Semester Credit Hours	
Prerequisite(s): Department Consent	
Special up-to-date topic in the civil and infrastructure engineering.	
COMM 101 Interpersonal Communication and Group Interaction	(3:3:0)
3 Semester Credit Hours	
The course presents the principles to develop appropriate and effective communication strategies in one-to-one and small group communication settings. It emphasizes analyzing and assessing communication skills to create and sustain effective communication in personal and professional relationships.	
COMM 102 Reading Image and Film	(3:3:0)
3 Semester Credit Hours	
Prerequisite(s): ENGL 101	
This course allows learners to study aesthetic effects and strategies of persuasion in contemporary visual images. Learning engagement includes images in electronic media, film, photographs, cartoons, advertisements, and public performances and events.	
COMM 111 Introduction to Mass Communication	(3:3:0)
3 Semester Credit Hours	
This course surveys the history and characteristics of mass communication as a field and set of intertwining industries and professions. Learners explore the role of mass media in modern society by considering the impact of technology, culture, government, and economics. Trends are considered in historical context.	
COMM 112 Introduction to Public Relations	(3:3:0)
3 Semester Credit Hours	
This course provides learners with basic knowledge about the theories, concepts, and best practices in communication within the public relations field.	
COMM 113 Introduction to Digital Media	(3:3:0)
3 Semester Credit Hours	
Introduction to Digital Media, allows students to learn and understand the basics of Digital Media. The course content discusses various technologies that will aid students to learn the language of visual imagery. The applications that will be focused on industry standard applications for many graphic design positions. In this course, students will build a blog as a means of communicating and presenting their work as producing print and digital layouts to a wide audience.	
COMM 212 Media Writing	(3:3:0)
3 Semester Credit Hours	
Prerequisite(s): COMM 111 or COMM 112 or COMM 113	
This course introduces learners to the basic norms, values, standards and practices for writing for the mass media.	
COMM 213 Public Relations Writing	(3:3:0)
3 Semester Credit Hours	
Prerequisite(s): COMM 111 or COMM 112 or COMM 113	
Learners improve their media-related writing skills with a focus on standard, professional presentation of information and messages in the public relations setting.	
COMM 214 New Media and Digital Culture	(3:3:0)
3 Semester Credit Hours	
Prerequisite(s): COMM 111 or COMM 112 or COMM 113	
This course examines the cultural impact of new digital technologies, such as the Internet and new telephone and audiovisual media. Students survey the origins of digital communication and the Internet; and they are introduced to contemporary scholarship on digital technologies, the Internet, and the institutions that control these technologies.	

COMM 215	Feature Writing	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 111 or COMM 112 or COMM 113		
This course is designed to give students experience in recognizing and producing high-quality feature articles. Course materials and lectures will cover the basics of writing the newspaper and magazine feature story. Students will be exposed to, and write in, a diverse variety of approaches and techniques.		
COMM 221	Communication Analysis and Criticism	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 111 or COMM 112 or COMM 113		
This course investigates media through a variety of theories and methods. The focus will be on some of the dominant critical perspectives that have contributed to our understanding of media and its roles in society. The course provides the basic vocabulary and concepts used to analyze different types of media.		
COMM 222	Intercultural Mass Communication	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 111 or COMM 112 or COMM 113		
This course explores the communication strategies and techniques within a specific cultural milieu and how those strategies and techniques differ among various cultural milieu, and learners practice in a variety of communication modes how best to accommodate their rhetorical strategies in communicating to both intracultural and intercultural audiences.		
COMM 223	Globalization and Media Cultures	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 111 or COMM 112 or COMM 113		
Learners in this course apply communication strategies to and investigate real-world case study challenges with a global worldview of various issues of global media cultures. Students learn how to develop a <i>Weltanschauung</i> from which to best communicate in modern technologies with world audience.		
COMM 224	Visual Storytelling	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 111 or COMM 112 or COMM 113		
This course combines storytelling arts with social, mobile, and digital media technologies. In this course, students apply digital storytelling theory and techniques to write, produce and publish digital stories. They integrate images, text, video and audio to create digital stories, and acquire competency in the use of digital media applications.		
COMM 225	Visual Communication and Digital Photography	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): ENGL 101 or COMM 111		
This course explores photography, photographic composition and aesthetics as a form of visual communication. Through a blend of theoretical and practical approaches, students will develop an aesthetic and creative understanding of photography as well as gain knowledge of technical photographic terms and digital photography skills.		
COMM 226	Graphic Communication Design and Print Production	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): ENGL 101 or COMM 111		
This course is designed to expose students to study and practice graphic communication design and print production fundamentals. In this course, students apply layout theory and varying techniques to write, produce, and publish news stories in the production process. They integrate images, and text, to create a page makeup, with the use of different digital media applications.		
COMM 227	Film and Video Editing	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): ENGL 101 or COMM 111		
This course is designed to provide students with the essential skills and knowledge required for video and film editing. Students will learn the principles and techniques of video editing, including the use of industry-standard software tools and technology.		
COMM 311	Media Law and Ethics	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 111 or COMM 112 or COMM 113		
Learners engage the philosophical underpinnings of ethics and the core principles of journalism and mass communication to develop an understanding and appreciation of the field's normative ethical values. Students will learn how to apply an ethical decision-making framework to a variety of challenges.		

COMM 321	Theories of Media Communication	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 212 or COMM 222 or COMM 223		
Learners identify the major concepts, issues, and theories of media communication, and learners identify and use communication theories in a variety of best practices to demonstrate effective use of the theories learned in the course.		
COMM 322	Digital Resources and Content	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 111 or COMM 112 or COMM 113		
This course is designed to present fundamentals of open /data collecting, gathering, cleaning, analyzing, and create data visualizations. Students will learn how database journalism turns to data journalism; and how data became a strong role in transferring more efficient growing online resources, tools, and techniques.		
COMM 323	News Reporting	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 212 or COMM 215		
This course will engage learners in the values, norms and professional practices newsgathering and writing. Emphasis is on traditional reporting methods, including interviewing and observation, and on the ethical, reliable presentation of news in print/digital formats.		
COMM 334	Digital Journalism	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 212 or COMM 213 or COMM 222		
This course introduces students to the multimedia production skills and techniques associated with broadcast media. Learners will integrate text, audio, photos, and video to produce broadcast journalism content that is attractive, accessible, easy to navigate, and appropriate for the platform and the audience.		
COMM 337	Public Relations Cases	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 212 or COMM 213 or COMM 222		
This course gives learners the context, knowledge, and skills to examine and critically analyze real-world public relations problems and cases. Students will study and apply communication and public relations theories to case studies and problems. Learners examine the four steps included in the design of public relations programs that include research, planning, implementation, and evaluation.		
COMM 344	Public Relations & Event Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 212 or COMM 213 or COMM 222		
This course introduces students to the public relations process, focusing on the strategic planning of PR programs, public relations tactics, media relations management, crisis management, and event management. Students will learn how to develop goals, design strategies, and select tactics that can better serve an organization's goals.		
COMM 390	Internship in Mass Communication	
3 Semester Credit Hours		
Prerequisite(s): Prior completion of 90 credits		
The Internship for Mass Communication is eight weeks (240 hours), and provides exposure to practices, challenges, and opportunities in real-world Media settings. Students apply real-world work strategies throughout the internship, and engage in self-reflection, and incorporate employer feedback to enhance their professional skills in creating an initial record of employment.		
COMM 391	Communication Research Methodology	(2:2:0)
2 Semester Credit Hours		
Prerequisite(s): COMM 212 or COMM 222 or COMM 223		
This course provides an introduction to research methods and the philosophical underpinnings of research inquiry in the field of communication. It includes the topic adherence, overviews of quantitative, qualitative, and mixed-method research methodologies, a range of alternative research methods, including observation, archival research, questionnaire surveys, case studies, and experimentation, research design, data collection, and data analysis, the ethical implications of research with human and non-human subjects, and appropriate connections between research questions and methodologies.		
COMM 421	Social Media: Audiences and Messages	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): COMM 311 or COMM 321 or COMM 334		
Learners explore social media as a unique platform for interaction with multiple audiences and learn how to plan strategies to engage with key stakeholders and disseminate persuasive, effective messages.		

- COMM 422 Streaming Media, Podcasting and Radio Production (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 334
 This advanced course covers the technical aspects of live video streaming, podcasting and radio production including live production techniques, audio recording, mixing, and editing techniques.
- COMM 423 Interactive Multimedia (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 334
 The course will examine the history, aesthetics, and cultural implications of interactivity through media. Students will gain experience and critical analysis skills by engaging with interactive multimedia works. Students will learn design strategies and development techniques for interactive media and will have the opportunity to explore the latest innovations in this area.
- COMM 424 Capstone: Multimedia Storytelling (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 321 or COMM 323 or COMM 334 or COMM 391
 In this course, learners apply skills in newsgathering, writing, and multimedia to produce an individual and a group project of publication quality that demonstrates their proficiency in multimedia storytelling.
- COMM 425 Media Management and Entrepreneurship (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 212 or COMM 223
 This course introduces students to key concepts, models, and thinkers in media management and examines the practice of and emerging models in entrepreneurial journalism. The course aims to equip learners with analytical tools to understand the current state of media and develop new models for the future.
- COMM 427 Capstone: Media Production in Public Relations (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 321 or COMM 323 or COMM 334 or COMM 391
 The course provides learners with practical instruction and experience in a variety of media modes of communication. Students focus on producing clear, persuasive messages in attractive formats.
- COMM 450 Selected Topics in Communication (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 212 or COMM 222 or COMM 223
 This course examines current and emerging issues in communication theory, research, and practice, focusing on specific media and public relations topics, trends, and challenges. Students will explore various topics, ideas, and issues that are not typically covered in standard courses. The course will provide an opportunity for students to engage in critical analysis and discussion, as well as research and writing, on a range of contemporary communication issues (**Writing Intensive Course**).
- COMM 492 Mass Media and Public Opinion (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 311 or COMM 321 or COMM 323 or COMM 391
 This course is designed to prepare students to be informed and critical consumers of polls and media coverage of them, to introduce students to basic theories and findings regarding the influence of mass media on public opinion, to provide students with firsthand experience in conducting and writing about public opinion research.
- CSAI 350 Introduction to Artificial Intelligence (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): CSCI 215
 This course provides an introduction to the different sub-areas of Artificial Intelligence (AI). In addition, students learn basic concepts, methods, and algorithms of AI and how they can be used to solve practical AI problems. The topics include classical and adversarial search & heuristic, knowledge representation, probabilistic reasoning, convex optimization methods, Bayesian methods, reinforcement learning, and supervised and unsupervised learning techniques. Particular focus will be placed on real-world applications of the material.
- CSAI 351 Data Science (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): CSAI 350 and MATH 203
 This course provides an introduction to data science and highlights its importance in real world context. Topics include data science concepts, project lifecycle, tools & programming environment, fundamentals of Python programming, numerical processing, data visualization, exploratory data analysis, data preprocessing, parameter optimization, model performance evaluation, and applications of machine learning algorithms in Python (i.e., Naïve Bayes, k-Nearest Neighbors, Linear/Multiple/Logistic Regressions, Decision Trees, and Clustering Applications), natural language processing, and real-world data science case studies.

CSAI 390 Internship I**3 Semester Credit Hours****Prerequisite(s): Completion of 60 credits and a CGPA of 2.0 or higher**

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

CSAI 391 Internship II**3 Semester Credit Hours****Prerequisite(s): Completion of 90 credits and a CGPA of 2.0 or higher**

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

CSAI 450 Machine Learning**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CSAI 350 and ENGR 200****Corequisite(s): CSAI 451**

This course introduces fundamental concepts of machine learning, and provides students with knowledge and understanding of the methods, mathematics, and algorithms used in machine learning. Topics include statistical learning concepts, linear & quadratic discriminant analysis, resampling methods, model selection and regularization, regression & smoothing splines, generalized additive models, regression trees, bagging and boosting, support vector machines, principal components analysis, k-means clustering, hierarchical clustering, and neural networks.

CSAI 451 Machine Learning Lab**(1:0:3)****1 Semester Credit Hour****Corequisite(s): CSAI 450**

This course, which is conducted within a laboratory environment, aims to familiarize students with several techniques used in machine learning. The topics covered include Linear Regression, Classification, Resampling, Linear Model Selection, Tree-Based Methods, Support Vector Machines, and Neural Networks. **(Writing Intensive Course)**

CSAI 452 Natural Language Processing**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CSAI 351**

This course introduces the fundamental concepts and techniques of natural language processing (NLP). Topics include text corpora and conditional frequency distributions, lexical resources and WordNet, raw text processing and regular expressions, text normalization and lemmatization, structured natural language processing (NLP) programs, part-of-speech tagging, automatic tagging, n-gram, & transformation-based tagging, document and sequence classification, maximum entropy classifiers and modeling linguistic patterns, information extraction, linguistic structure, named entity recognition, & relation extraction, grammatical structure & context free grammar, context free grammar parsers & dependency grammar, and feature based grammars.

CSAI 453 Data Visualization**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CSAI 350**

Data visualization is an essential skill required in today's data-driven world. This course presents principles and techniques to design and create data visualization based on gathered data and the goals of the task at hand. Topics include the value of visualization, data, tasks, validation, marks and channels, design guidelines, tables, networks and trees, spatial, temporal and textual data, interaction and navigation, and data reduction.

CSAI 480 Big Data**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CSCI 326**

This course provides an in-depth coverage of various topics in big data from data generation, storage, management, transfer, to analytics. Topics include Hadoop ecosystem, Hadoop architecture, YARN & HDFS, Sqoop, Apache Hive & Impala, Apache Flume, Apache Spark, resilient distributed datasets (RDDs) in Spark, map reduce in Spark, Spark applications, Spark parallel processing, SparkSQL, workflow management with Apache Oozie, machine learning in Spark, and big data applications.

CSAI 481 Computer Vision**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CSAI 450**

This course provides introduction to fundamental topics in computer vision and the application of statistical estimation techniques in this field. It is intended to give students a good basis for work in this important field. Topics include image representation, image processing, image analysis, image segmentation, object tracking, 3D shape reconstruction, feature detection and tracking, object detection, and convolutional neural networks.

CSAI 482 Data Mining

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSAI 350

Data Mining studies algorithms and computational paradigms that allow computers to find patterns and regularities in datasets, then perform prediction/forecasting and generally improve their performance through interaction with data. The course introduces the fundamental concepts of data mining techniques. Topics include data preparation, data classification, performance measures, association rule mining, clustering, outlier detection, collaborative filtering and data mining case studies.

CSAI 483 Information Retrieval

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSCI 326

The course covers basic and advanced techniques necessary to define, design, and implement Information Retrieval (IR) systems, including the following topics: basic IR models, basic tokenizing, indexing, and implementation of vector-space retrieval, experimental evaluation of IR, query operations and languages, text representation, web search, text categorization, language-model based retrieval, recommender systems, information extraction, and integration.

CSAI 484 Internet of Things System

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): (CSCI 232 or CENG 315) and CSCI 462

The Internet of Things is an emerging technology in which interconnected devices are embedded within everyday objects enabling them to send and receive data. This course introduces core concepts and networking protocols for IoT applications. Topics covered include: IoT architectures, smart objects, sensing and actuation, displaying information, WPANs, WBANs, IoT clouds, security and privacy for IoT systems, IoT robustness and reliability, and IoT applications.

CSAI 485 Introduction to Deep Learning

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSAI 450

The course provides an introduction to neural networks and deep learning. The topics to be covered include basic conceptual understanding of neural networks, deep learning programming environments, shallow neural networks, radial basis function networks, recurrent neural networks, convolutional neural networks, and deep reinforcement learning.

CSAI 486 Special Topics in Artificial Intelligence

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSAI 350

This course gives instructors the opportunity to cover the latest developments and contemporary issues in Artificial Intelligence. Instructors will provide a detailed course outline at the beginning of the semester.

CSAI 487 Introduction to Robotics

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): (MATH 213 or MATH 203) & (CSCI 112 or CSCI 114)

This course introduces fundamentals of robotic systems to the students. Topics covered include robotic system architecture, configuration space, and Robot Operating System (ROS). Moreover, forward and reverse kinematics, robot control systems, and robotic sensors are covered in this course. Students will also explore robot navigation, obstacle avoidance, and Simultaneous Localization and Mapping (SLAM), enabling them to develop and assess robotic systems that can autonomously navigate and interact with their environments. The last part of the course focuses on motion planning to enable students to devise, implement, and optimize algorithms for planned robotic motions.

CSAI 490 Professional Software Practice

(2:2:0)

2 Semester Credit Hours

Prerequisite(s): Completion of 90 credit hours

Corequisite(s): CSCI 492

The course develops student understanding about various non-technical issues related to the field of computing. The course covers the field from economic, legal, ethical, and professional perspectives. Covered topics include code of conduct, ethics, intellectual property, cyber security, privacy, risk management, and teamwork.

CSCI 104 Introduction to Computing

(3:3:0)

3 Semester Credit Hours

This course serves as an introduction to the field of computer science and the computer's various layers. The course provides exposure to the following layers: information, hardware, programming, operating systems, applications, and communications. Additional topics include ethics, security, privacy, the impact of computing, and widely used software applications.

CSCI 112 Introduction to Computer Programming (3:3:0)

3 Semester Credit Hours

Corequisite(s): CSCI 113

This course introduces students to computers and programming languages and more specifically the C++ language. Additionally, students learn to use computer programming as a problem-solving tool. The topics covered include basic operations, data types, input/output, selection statements, control structures, arrays, functions, and strings, knowledge representation, neural networks and natural language, data summarization and visualization.

CSCI 113 Introduction to Computer Programming Lab (1:0:3)

1 Semester Credit Hour

Corequisite(s): CSCI 112

This course introduces the use of computer programming as a problem-solving tool in laboratory environment. Topics in procedural programming include, simple data types, input/output selection statements, control loops, testing, debugging, and programming environments.

CSCI 114 Applied Computational Thinking (3:3:0)

3 Semester Credit Hours

Prerequisite(s): None

This introductory course equips students with essential programming skills while emphasizing Python as a versatile problem-solving tool. Topics include core operations, data types, input/output techniques, conditional statements, control structures, lists, functions, and string manipulation, GUI programming and Python libraries usage.

CSCI 115 Applied Computational Tools (1:0:3)

1 Semester Credit Hour

Prerequisite(s): None

In this course, students will acquire the skills to manage complex equations and both linear and nonlinear systems with practical engineering applications. The course emphasizes the utilization of computational tools for effective problem-solving and optimization. Through an introductory exploration of MATLAB and VBA in Excel, the course focuses on fostering competencies in data visualization, mathematical problem-solving, and the customization of methods.

CSCI 211 Object Oriented Programming (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSCI 112 or CSCI 114

This course is an introduction to object-oriented programming principles and techniques using Java. Topics include Java elementary programming, and Java object-oriented features such as methods, objects, classes, access modifiers, constructors, immutable objects & classes, abstraction, encapsulation, inheritance, polymorphism, dynamic binding, object castings, abstract and interface classes, and exception handling.

CSCI 215 Data Structures and Algorithms (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSCI 211 and MATH 225

This course introduces data structures and various fundamental computer science algorithms. The course covers abstract data-type concepts, stacks, queues, lists, and trees. Several sorting and searching algorithms are covered. Additional topics include an introduction to graphs and their implementation and running time and time complexity measurement.

CSCI 232 Computer Organization (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSCI 112

This course provides a programmer's view of the execution of programs in computer systems. Topics covered include instruction sets, machine-level code, assembly language, performance evaluation and optimization, memory organization and management, address translation, and virtual memory.

CSCI 312 Operating System Fundamentals (3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSCI 215

This course covers the principles, components, and design of modern operating systems, focusing on the UNIX platform. Topics include system structure, process concept, multithreaded programming, process scheduling, synchronization, atomic transaction, deadlocks, memory management, and file system.

CSCI 315 Design and Analysis of Algorithms (3:3:0)

3 Semester Credit Hours**Prerequisite(s): CSCI 215**

This course introduces the design and analysis principles for various algorithms. The topics covered include searching algorithms, dynamic programming, greedy algorithms, Huffman coding, graph traversing algorithms, shortest path algorithms, linear programming, and NP-completeness.

CSCI 326 Database Systems**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CSCI 211**

This course is an introductory course on database management systems. The goal of the course is to present a comprehensive introduction to the use of data management systems. Some of the topics covered are the following: The Entity-Relationship Model, the Relational Data Model, the SQL language, the database design, and the database integrity and security.

CSCI 372 Compiler Design**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CSCI 232**

The course introduces the theory and practice of compilation. Topics include compiler architecture, components, phases, software tools, lexical roles and specifications, regular expressions, syntax roles and specifications, context-free grammars, top-down parsing, bottom-up parsing, LR parsers & parse trees, syntax directed translation, syntax tree, abstract syntax tree, and finite automata.

CSCI 388 Programming Languages**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CSCI 215**

This course is an introduction to basic concepts in the design of programming languages. The course focuses on programming languages within the functional and logic programming paradigms such as Scheme and Prolog. Topics include history of programming languages, language design criteria, functional programming, syntax, logic programming, semantics, and object-oriented principles.

CSCI 390 Internship I**3 Semester Credit Hours****Prerequisite(s): Completion of 60 credits and a CGPA of 2.0 or higher**

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

CSCI 391 Internship II**3 Semester Credit Hours****Prerequisite(s): Completion of 90 credits and a CGPA of 2.0 or higher**

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

CSCI 411 Computer Graphics**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CSCI 215****Corequisite(s): CSCI 412**

This course introduces interactive computer graphics and drawing algorithms. The topics covered include graphics output primitives and their implementations, two-dimensional and three-dimensional geometric transformations and viewing, hierarchical modeling, computer animation, spline representations, visible-surface detection methods, illumination models, and surface-rendering methods, and texturing and surface-detail methods.

CSCI 412 Computer Graphics Lab**(1:0:3)****1 Semester Credit Hour****Corequisite(s): CSCI 411**

This course introduces computer graphics and drawing algorithms in a laboratory environment. The topics covered include graphics output primitives and their implementations, two-dimensional and three-dimensional geometric transformations and viewing, hierarchical modeling, computer animation, spline representations, visible-surface detection methods, illumination models, and surface-rendering methods, and texturing and surface-detail methods.

CSCI 415 Introduction to Parallel Programming**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CSCI 215**

This course is an introduction to parallel programming principles and techniques. Topics include parallel computing memory architecture, memory organization, parallel programming models, parallel program design, performance evaluation, thread-based parallelism, process-based parallelism, message passing, asynchronous programming, and heterogeneous programming.

CSCI 416 Human Computer Interaction

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSCI 215

This course provides an introduction to and overview of the field of human-computer interaction (HCI). The topics include usability principles, predictive evaluation, design management processes, graphic design, understanding users' requirements gathering, task analysis, handling errors & help, prototyping & UI software, interaction styles, user models, evaluation, and universal design.

CSCI 450 **Information Security and Privacy**

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSCI 215 or Instructor permission

This course is a survey of information security considerations as they apply to information systems analysis, design, and operations. Topics include information security vulnerabilities, threats, and risk management. Furthermore, the course introduces several cryptographic algorithms in addition to the privacy and secrecy of statistical databases and e-government applications.

CSCI 462 Data Communications and Computer Networks

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSCI 112

This course introduces computer networks. Topics include layering approach, functions of different layers, Internet applications (HTTP, DNS), reliable and unreliable transport (TCP and UDP), routing and IP addressing, data link layer services and protocols, and Ethernet.

CSCI 463 Data Communications and Computer Networks Lab

(1:0:3)

1 Semester Credit Hour

Corequisite(s): CSCI 462

This course provides students with hands on training on design, troubleshooting, modeling and evaluating of computer networks. Topics include network addressing, Address Resolution Protocol (ARP), basic troubleshooting tools, IP routing, and route discovery. Additionally, student will perform network modeling, simulation, and analysis using Packet tracer and WireShark analyzer. **(Writing Intensive Course)**

CSCI 492 Senior Design Project I

(2:0:6)

2 Semester Credit Hours

Prerequisite(s): Completion of 90 credit hours

This is the first course of a capstone project that requires students to develop, design, and implement a solution to a computing problem under the supervision of a faculty advisor. Students are required to consider the ethical, social, and economic implications of their project. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resource, procurement, and communication, with consideration of ethical and professional conduct.

CSCI 493 Senior Design Project II

(4:0:12)

4 Semester Credit Hours

Prerequisite(s): CSCI 492

Implementation of the project for which preliminary work was done in CSCI 492. Project includes designing and constructing software and/or hardware, conducting experiments or studies, and testing and validating a complete system. At the end of the term, each team presents to a committee information related to its project in both written and oral formats.

CSCI 499 Special Topics in Computing

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): CSCI 215

This course gives instructors the opportunity to cover the latest developments and contemporary issues in computing. Instructors will provide a detailed course outline at the beginning of the semester.

ECON 103 Principles of Microeconomics

(3:3:0)

3 Semester Credit Hours

This course introduces microeconomics in the context of current problems. This course explores how market mechanism allocates scarce resources among competing uses; uses supply, demand, production, and distribution theory to analyze problems.

ECON 104 Macroeconomic Policy

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): ECON 103

This course introduces students to the applications of macroeconomics in the context of current problems to business decision making and economic policy. Students will be introduced to the economic way of thinking and how key concepts; theories and methods of modern economic analysis can be applied to everyday economic issues and problems. Key areas covered include the national economy; national

income accounting; business cycles; inflation; economic growth and stability; unemployment; and monetary and fiscal policy. Particular emphasis is placed on reviewing contemporary economic issues and how economics permeates almost every aspect of business, highlighting economics as the fundamental discipline underpinning the study of most business specializations.

EDUC 500 Foundations of Education (3:3:0)
3 Semester Credit Hours

This course examines historical, philosophical, social, and psychological foundations of education. Students will explore the nature of teaching and learning as well as UAE education policy.

EDUC 511 Introduction to Applied Research (3:3:0)
3 Semester Credit Hours

This course provides students and introduction to quantitative, qualitative, and mixed-methods research design in the educational sciences. This is a practical course designed to guide students through the development of their applied research proposals into a gap in practice.

EDUC 515 Educational Psychology for Inclusive Learning (3:3:0)
3 Semester Credit Hours

This course provides an overview of child and adolescent development, teaching and learning theories, strategies, and approaches for inclusive education. Students will utilize covered concepts as the framing for the development of a personal philosophy of teaching and learning.

EDUC 525 Theories and Applications of Leadership (3:3:0)
3 Semester Credit Hours

This course provides students with conceptual and practical knowledge in leadership theories, with an emphasis on application in real world education settings. Students will acquire skills grounded in various leadership theories and use this conceptual knowledge to create an individual, metacognitive Leadership Profile.

EDUC 545 Budgeting and Resource Management (3:3:0)
3 Semester Credit Hours

This course provides students with a broad overview of school finance, budgeting, and the ways in which resources are allocated. Students develop an understanding of the economic, social, and legal principles that inform how schools fund day-to-day operations.

EDUC 620 Evaluation of Instruction for Continuous Improvement (3:3:0)
3 Semester Credit Hours

This course will provide students with the fundamental concepts of instructional evaluation, observation, peer coaching and mentoring. The observation findings will be used to by students to design, create, present, and evaluate staff development.

EDUC 640 Inclusive Leading of Diverse Groups (3:3:0)
3 Semester Credit Hours

This leadership course is designed to prepare students to lead diverse groups and/or teams of individuals in schools and where education meets industry. Throughout the duration of this course, students will engage in meaningful dialogue centered on these authentic, real-world scenarios.

EDUC 660 Organizational Change and Continuous Improvement (3:3:0)
3 Semester Credit Hours

This course provides students with the knowledge and skills to carry out continuous school improvement initiatives which includes determining needs, planning for improvement, understanding challenges and roadblocks to initiatives, and implementing and evaluating results toward established goals.

EDUC 675 Educational Policy and Community Engagement (3:3:0)
3 Semester Credit Hours

In this course students explore the governance of schools and the building of constructive relationships between schools and communities from socio-economical and sociological perspectives grounded in educational research.

EDUC 680 Applied Research I (3:1:6)
3 Semester Credit Hours

Prerequisite(s): EDUC 511

This field-based course provides students with the tools needed to undertake data collection and analysis in real-world settings. Throughout the duration of this course, students will undertake data collection and analysis for their IRB-approved research project.
Thesis Research Course I

EDUC 685 Analytics for Data Driven Leadership (3:3:0)
3 Semester Credit Hours

This course focuses on the role of research in the work of school leaders; ways leaders use research to identify opportunities for improvement and change; relationship between theory, research, and practice; role of research and benchmarking in school improvement; and strategies for change in education.

EDUC 695	Applied Research II	(3:1:6)
3 Semester Credit Hours		
Prerequisite(s): EDUC 680		
This research capstone prepares students in the Master of Education program to finalize their original research for submission to an academic journal. This course is designed to enable students with requisite skills in disseminating research. Thesis Research Course II		
EEEN 220	Signals and Systems	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MATH 113		
Corequisite(s): EEEN 221		
Introduction to methods of representing continuous-time signals and systems, and interaction between signals and systems. Analysis of signals and systems through differential equations and transform methods; Laplace and Fourier transforms; frequency response of systems; stability in time and frequency domains. Presents application examples from communications, circuits, control, and signal processing.		
EEEN 221	Signals and Systems Lab	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): EEEN 220		
Laboratory course to accompany EEEN 220. In this course, the student will acquire hands-on experience with programming in MATLAB. Topics include representation of different signals, system linearity and time invariance, analysis of a first - order system, implementing matched filter for Barker codes, response of second - order systems and damping ratio, synthesis periodic signals		
EEEN 280	Electric Circuit Analysis I	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): PHYS 220		
Basic circuit concepts and DC analysis, circuit analysis techniques, circuit theories, fundamental operation of operational amplifiers and their applications, transient and steady state analysis of RL, RC, and RLC circuits and basic AC analysis.		
EEEN 281	Electric Circuit Analysis Lab	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): EEEN 280 or MENG 231		
Laboratory course to accompany EEEN 280. In this course, students will experimentally verify circuit analysis concepts under DC and AC excitations and transient response. They will use different measurement instruments and build DC and AC electric circuits.		
EEEN 282	Electric Circuit Analysis II	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): EEEN 280		
Review of AC sinusoidal circuit analysis with active and reactive power. Covers magnetically coupled inductors and ideal transformers, three phase circuits, Laplace transform, application of Laplace transform in circuit analysis, passive and active filter analysis and design, two port networks.		
EEEN 283	Electric Circuit Analysis II Lab	(1:0:3)
1 Semester Credit Hour		
Prerequisite(s): EEEN 281		
Corequisite(s): EEEN 282		
Laboratory course to accompany EEEN 282. In this course, students will experimentally verify circuit analysis concepts under steady state AC excitation. Students will use different measurement instruments and build AC electric circuits.		
EEEN 305	Electromagnetic Theory	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): PHYS 220 and MATH 214		
Course uses vector algebra and vector calculus. Covers topics related to electrostatic and magnetostatic fields, electric and magnetic properties of media, electric boundary value problems, Maxwell's equations, electromagnetic waves and plane wave propagation, Poynting theorem and transmission line theory.		
EEEN 331	Digital System Design	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): PHYS 220		
Corequisite(s): EEEN 332		
This course covers principles of digital logic and digital system design. Topics include number systems; Boolean algebra; analysis, design, and minimization of combinational logic circuits; analysis and design of synchronous and asynchronous finite state machines; and an introduction to VHDL and behavioral modeling of combinational and sequential circuits.		

EEEN 332	Digital Systems Design Lab	(1:0:3)
1 Semester Credit Hour		
Prerequisite(s):		
Corequisite(s): EEEN 331		
Laboratory course to accompany EEEN 331. In this course, the student will acquire hands-on experience with basic logic components, combinational and sequential logic circuits, and the use of VHDL.		
EEEN 333	Linear Electronics I	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): EEEN 280		
Principles of operation and application of electron devices and linear circuits. Topics include semiconductor properties, diodes, bipolar and field effect transistors, biasing, amplifiers, frequency response, operational amplifiers, and analog design.		
EEEN 334	Linear Electronics I Lab	(1:0:3)
1 Semester Credit Hour		
Prerequisite(s): EEEN 281		
Corequisite(s): EEEN 333		
Laboratory course to accompany EEEN 333. In this course, the student will acquire hands-on experience with basic electronic components and circuits. Topics covered include: Semiconductor diodes, rectification, Zener diodes, BJT and FET transistors and Amplifiers.		
EEEN 350	Electric Machines	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): EEEN 282		
Corequisite(s): EEEN 351		
The general theory of electro-mechanical motion devices relating to electric variables and electromagnetic forces. Basic concepts and operational behavior of DC motors, induction and brushless DC Motors, and stepper motors used in control applications.		
EEEN 351	Electric Machines Lab	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): EEEN 350		
Laboratory course to accompany EEEN 350. In this course, students will acquire hands-on experience with the characteristics of dc motors and dc generators (separate, series, shunt and compound). They will learn to find the parameters of transformers and evaluate their performance characteristics. The starting, speed control and performance of 3-phase induction motors are also studied. (Writing Intensive Course)		
EEEN 360	Random Signal and Noise	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): ENGR 200 and EEEN 220		
Introduction to set theory, probability, random variables, and random processes. Modelling various types of noise encountered in communication systems as random processes. Analyzing the system response in the presence of random noise processes and calculating the power spectral density.		
EEEN 412	Power Systems	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): EEEN 350		
Electric Power Systems, Elements of a Power Systems; The analysis of power systems starting with the calculation of line resistance, line inductance, and line capacitance of power transmission lines; Analysis of power systems in terms of current, voltage, and active/reactive power; Per-Unit Quantities; Load Flow Study; Economic Dispatch; Symmetrical Components; Fault Study; System Protection; System transient and Stability issues.		
EEEN 413	Microcontrollers and Embedded Systems	(3:3:0)
3 Semester Credits Hours		
Prerequisite(s): CSCI 112 and EEEN 331		
Corequisite(s): EEEN 414		
This course provides an in-depth exploration of microcontroller architecture, programming, and their applications in embedded systems. Topics include microcontroller architecture, programming in C and assembly, interfacing, real-time operating systems (RTOS), and the development of embedded systems applications.		
EEEN 414	Microcontrollers and Embedded Systems Lab	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): EEEN 413		

This laboratory course provides practical experience in microcontroller programming, hardware interfacing, and embedded system design, complementing the theoretical course ELEN 413. Students will work directly with microcontrollers, development boards, sensors, and actuators to design and implement embedded systems.

EEEN 421 Power System Protection

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EEEN 412

Power systems protection schemes for transmission and distribution networks, connection and standards of current and voltage instrument transformers for protection and metering applications, Protective relays, Protection of generators; Differential protection - Problems with differential protection - Biased differential protection - Biased differential protection of generator - Over current and earth fault protection, P; Buchholz relay - Biased differential protection of transformers - Harmonic restraint – Harmonic blocking - Other transformer protections, Protection of transformers, Protection of transmission lines.

EEEN 422 High Voltage Engineering

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EEEN 282

The components of power system and their characteristics. Fundamental electric field calculations (Laplacian fields) in insulation systems of simple geometries, introduction to gas discharge physics, Townsends theory of electric breakdown in air and Paschens law and its implications on gas insulation strength. Experimental techniques applied in high voltage engineering.

EEEN 423 Electrical Energy Systems & Fault Analysis

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EEEN 412

Energy and power; forms of energy, energy conversion from energy sources including wind , solar, tidal, bio-fuel, wave, hydro, nuclear and fossil fuel. Structure of a modern power system: operating charts, voltage control, and matrix representation of transmission lines. Two port network representation of transmission lines, per unit system, fault analysis: symmetrical components, transformers: construction, operation, connections, and relevant calculations. Load flow analysis: network matrix representation, Gauss-Seidel and Newton-Raphson solution techniques. AC/DC conversion: converter types, dc transmission, advantages compared to AC transmission. Over-voltages: switching and fault over-voltages, Bewley Lattice diagrams, switchgear principles, current chopping, insulation coordination. Modal component theory: wave propagation.

EEEN 425 Smart Power Grid Systems Theory & Implementation

(3:3:0)

3 Semester Credit Hours

Pre0requisite(s): EEEN 412

This course explores a set of emerging concepts, technologies, applications and business models, and the related trade-off decisions involved in transforming the traditional centralized power grid into a climate and renewable energy-friendly “Smart Grid.” A cross-disciplinary approach intended to deepen individual areas of expertise in the context of multidisciplinary teamwork. Basic Smart Grid literacy, applications of this knowledge base to specific “real world” case studies.

EEEN 426 Renewable Energy Systems

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): Completion of 90 credit hours

The course aims to introduce a general engineering/science audience to the basic concepts of renewable energy. In the interest of time some mathematical criteria will be covered, e.g. Betz limit for wind, limit of efficiency of WEC point absorber. Each lecture contains several examples from real world applications and in-progress industrial developments.

EEEN 431 Digital Circuit Design

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EEEN 331 and EEEN 333

Analysis and design of discrete and integrated switching circuits. Topics include transient characteristics of diodes, bipolar, and field-effect transistors; MOS and bipolar inverters; no regenerative and regenerative circuits; TTL, ECL, IIL, NMOS, and CMOS technologies; semiconductor memories; VLSI design principles; and SPICE circuit analysis.

EEEN 432 Nanotechnology Fundamentals & Applications

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): Completion of 90 credit hours

Students will be provided an introduction to the basic principles of the subject of nanotechnology with minimum mathematics involved. The applications of nanotechnology to various fields such as environment, health, alternative energy and electronic/optics will be addressed and explained. Students will be required to do research projects and presentations in class.

EEEN 433 Linear Electronics II

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EEEN 333

Differential amplifiers, feedback circuits, power amplifiers, feedback amplifier frequency response, analog integrated circuits, operational amplifier systems, oscillators, wide band and microwave amplifiers, and computer-aided design.

EEEN 434 Linear Electronics II Lab

(1:0:3)

1 Semester Credit Hour

Prerequisite(s): EEEN 334

Corequisite(s): EEEN 433

Laboratory course to accompany EEN 433. In this course, the student will acquire hands-on experience with Electronic Amplifiers, active filters and oscillators. Topics covered include: Cascade amplifiers, differential amplifier, active filters, oscillators, and feedback amplifier concepts. **(Writing Intensive Course)**

EEEN 435 Introduction to Optical Electronics

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EEEN 305

Corequisite(s): EEEN 333

Introduces optoelectronic devices for generation, detection, and modulation of light. Topics include electro-optic modulators, gas, solid state and semiconductor lasers, photodetectors, and detector arrays.

EEEN 437 Power Electronics

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EEEN 333

Course examines the application of electronics to energy conversion and control. The subject covers modern power semiconductor devices e.g., diodes, thyristors, MOSFETS, and other insulated gate devices; Static and switching characteristics, gate drive and protection techniques; Various DC-DC, AC-DC and DC-AC converter circuit topologies, their characteristics and control techniques; Analysis of input and output waveforms of these circuits; and their applications. Utility interference and Harmonic issues for power electronics circuits.

EEEN 451 Control Theory

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EEEN 220 or MATH 214

Introduction to feedback control systems; Block diagram and signal flow Graph representation; Mathematical modeling of physical systems; Stability of linear control systems; Time-domain and frequency-domain analysis tools and performance assessment; Lead and lag compensator design; Multi input multi output systems; Routh, Nyquist; Bode and root locus diagrams; Introduction to state variable techniques; state transmission matrix and state variable feedback.

EEEN 452 Control Lab

(1:0:3)

1 Semester Credit Hour

Corequisite(s): EEEN 451

Laboratory course to accompany EEN 451. In this course, the student will acquire hands-on experience with feedback control circuits/systems. Topics covered include; Introduction in control systems, DC motor speed controller, open loop/closed, Proportional Integral Derivative PID controller and position control.

EEEN 455 Digital Image Processing

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EEEN 220

An exploration of digital image processing fundamentals, including image enhancement, restoration, segmentation, and transformation techniques. Emphasis on practical applications and the use of image processing software tools (MATLAB/Python).

EEEN 460 Communication Systems

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EEEN 220 and ENGR 200

Introduction to analog and digital communications. Topics include review of important concepts from signals and systems theory and probability theory; Gaussian processes and power spectral density; digital transmission through additive white Gaussian channels; sampling and pulse code modulation; analog signal transmission and reception using amplitude, frequency and phase modulation; and effects of noise on analog communication systems.

EEEN 461 Communication Systems Lab

(1:0:3)

1 Semester Credit Hour

Prerequisite(s): EEEN 460

Corequisite(s): EEEN 464.

Laboratory course to follow EEEN 460 and accompany EEEN 464. In this course, the student will acquire hands-on experience with fundamental blocks of Analog and Digital communication systems. Topics covered include: Amplitude and Angle Modulation and demodulation, sampling and reconstruction, PCM Encoding & PCM Decoding and digital modulation and demodulation.

EEEN 462	Data and Computer Communications	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): EEEN 220 and ENGR 200		
Introduction to modern data communications and computer networks. Topics include point -to-point communication links and transmission of digital information, modems, and codecs; packet switching, multiplexing, and concentrator design; multi-access and broadcasting; local area and wide area networks; ISDN; architectures and protocols for computer networks; OSI reference model and seven layers; physical interfaces and protocols; and data link control layer and network layer.		
EEEN 464	Digital Communication Systems	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): EEEN 460		
Corequisite(s): EEEN 461		
Introduces digital transmission systems. Topics include quantization, digital coding of analog waveforms, PCM, DPCM, DM, base band transmission, digital modulation schemes, ASK, FSK, PSK, MSK, QAM, pulse shaping, inter symbol interference, partial response, voice band and wideband modems, digital cable systems, regenerative repeaters, clock recovery and jitter, multi path fading, digital radio design, optimal receiver design, MAP receiver, and probability of error.		
EEEN 466	Digital Signal Processing	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): EEEN 220 and ENGR 200		
This course provides a thorough treatment of digital signal processing including the fundamental theorems and properties of discrete-time linear systems, filtering, sampling, and discrete-time Fourier Analysis.		
EEEN 467	Mobile and Wireless Communications	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): EEEN 460		
Cellular systems design fundamentals, fading and multipath channels, Modulation techniques for mobile radio systems, Diversity and combining techniques for mobile radio systems, multiple access techniques for mobile systems, Mobile systems and standards.		
EEEN 472	Antenna Theory and Design	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): EEEN 305		
Course provides the fundamental knowledge in the theory and design of antennas. The theory of electromagnetic radiation is introduced and the fundamental antenna properties and parameters are explained. Standard antenna characterization parameters such as impedance, far-field radiation pattern, gain, directivity, bandwidth, beam width, polarization, efficiency, antenna temperatures are studied. The electromagnetic theory behind antenna operation and an overview of different antenna systems such as monopoles, dipoles, wire antennas and loop antennas etc... are discussed. The principles of analysis and design of antenna arrays are discussed.		
EEEN 473	Radio Frequency and Microwave Engineering	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): EEEN 305		
This course covers a broad range of topics in the field of radio frequency (RF) and microwave engineering. This includes transmission lines, waveguides, impedance matching, microwave resonators, RF filters, RF amplifiers, and passive RF and microwave devices (mixers, diplexers, etc.). Furthermore, RF/microwave communications link design will be provided.		
EEEN 474	Advanced Information theory and coding	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): EEEN 460		
Advanced topics in information theory and coding. The course is divided into two main parts, namely, Source coding and data compression, and channel coding and error detection/correction codes. The first part covers, entropy, amount of information source coding techniques, Shannon Fano, Huffman, and Lempel-Ziv codes. The second part covers binary symmetric channels, Z-channels, and E-channels, channel capacity, mutual information, linear block codes and convolutional codes, Viterbi decoders and cyclic redundancy check codes.		
EEEN 481	Concepts of Multimedia Processing and Transmission	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): EEEN 220 or CSCI 462		
The course introduces the fundamentals of signal processing and communications for multimedia applications. It covers various topics relating to audio, image and video processing, storage and transmission. It discusses the human visual and hearing systems and relates them to image and sound digitization processes. The course also covers various lossless and lossy methods for audio, image and video compression. In addition, it gives the student hands on experience on applying the presented processing techniques using suitable software packages.		
EEEN 487	Robotics Engineering	(3:3:0)

3 Semester Credit Hours**Prerequisite(s): (MATH 203 or MATH 213) and EEEN 331**

This course introduces the fundamental principles of robotics, focusing on the design, operation, and application of robotic systems. Topics include robot kinematics, dynamics, control systems, sensors and actuators, and robot programming. The course emphasizes the integration of electrical and electronic components in robotic systems.

EEEN 492 Senior Design Project I**(2:0:6)****2 Semester Credit Hours****Prerequisite(s): Completion of 90 credit hours**

This is the first course of a capstone project that requires students to develop, design, and implement a solution to an engineering problem under the supervision of a faculty advisor. Students are required to consider the ethical, social and economic implications of their project. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resource, procurement, and communication, with consideration of ethical and professional conduct.

EEEN 493 Senior Design Project II**(4:0:12)****4 Semester Credit Hours****Prerequisite(s): EEEN 492**

Implementation of project from EEEN 492. Project includes designing and constructing hardware, writing required software, conducting experiments or studies, and testing complete system. Requires oral and written reports during project and at completion.

EEEN 499 Special Topics in EEEN**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): Completion of 90 credit hours**

Advanced and emerging topics in electronics and communication engineering. Topics are announced through the Schedule of Classes.

EMAC 512 Managerial and Financial Accounting**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): Graduate Standing**

The course deals with managerial and financial accounting techniques and practices. The course examines the impact of cost and cost allocation on business performance analysis, using a variety of costing schemes. It presents accounting as an information system construct, and looks at methods for recording, presenting and analyzing accounting information. The course also looks at budgeting, ratio analysis and other fiscal management measures. The course discusses the impact, on Management and Financial Accounting, of contemporary management philosophies and techniques including JIT practices, quality assurance and performance measurement.

EMFN 514 Managerial Finance**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): Graduate Standing**

The course provides detailed critical treatment of the theory and practice of financial management within organizations. Topics covered include financial modeling, breakeven analysis, investment decision-making (NPV, IRR, etc.), capital budgeting and structure, working capital management, analysis of financial statements, and risk analysis. There is additional treatment of managerial economics.

EMFN 560 International Corporate Finance**(3:3:0)****3 Semester Credit Hours****Prerequisites: EMFN 514**

This course focuses on corporate finance at a global environment context. It explores advanced concepts and practices applicable to multinational company's financing and investment decision- makings. Topics covered include financial analysis for investment decisions in an international setting, international money operations and capital markets, management of foreign exchange risk, direct foreign investment, funding of international projects, political risk analysis, currency derivatives and swap markets..

EMFN 562 Investment Planning and Management**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): EMFN 514**

The course deals with the advanced analysis of equity securities and investment portfolios, predicated on relevant market hypothesis and capital market theory. The course looks at investment risk-return trade-off, asset pricing models, and stock price behavior. The course adds emphasis on stocks, bonds, and financial futures and options.

EMFN 564 Financial Markets and Institutions**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): EMFN 514**

The course deals with financial markets, primarily global equity markets. It looks at markets for handling government debt instruments, and exchange-traded and over-the-counter financial derivative instruments such as futures, options, swaps, and asset-backed securities. There is detailed treatment of derivatives theories, derivatives exchanges, and valuation of derivatives, including standard and other non-standard options on a variety of underlying assets, in relation to relevant financial markets. The course also discusses emerging financial markets, and the effective management of risks emanating from these markets.

EMGN 525 Research Methods for Business

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): Graduate Standing

This course provides the necessary underpinning support for the EMBA dissertation and general business domain research. It allows candidates to make informed decisions and appropriate choices pertaining to research methodology. A range of business research tools, approaches and analytical techniques are discussed, and guidance is given on optimal structuring of business research documents and EMBA dissertations.

EMGN 555 EMBA investigative Thesis or Project I

(3:0:9)

3 Semester Credit Hours

Prerequisite(s): EMGN 525

EMGN 555 is the first part of the EMBA investigative Thesis or Project course, serving to integrate the various topics studied in the School of Business EMBA program. Students undertake a work-based study that focuses on a chosen business /management topic. The outcome for EMGN 555 is to submit a research introduction, literature review, and research methodology that will be used for the completion of this course. Satisfactory performance of the students in this course will move the student into EMBA 556.

EMGN 556 EMBA Investigative Thesis or Project II

(3:0:9)

3 Semester Credit Hours

Prerequisite(s): EMGN 555

EMGN 556 is the second part of the six-credit hour research course, serving to integrate the various topics studied in the School of Business EMBA program. Students undertake a work-based study that focuses on a chosen business/management topic. The outcome is a thesis, an extended piece of work necessitating investigation of secondary data and empirical fieldwork, through collection of primary data, pertaining to an organization and/ or a business domain problem. Analysis of collected data is expected to generate critical commentary and recommendations.

EMHR 590 Applied and Strategic Human Resource Management

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EMMG 510

The course deals with key areas of applied and strategic human resource management, essential for providing supervisors and human resource specialists with a thorough understanding of the strategic role of human resource management and the techniques available to management for making effective use of the human resources of an organization. The material is delivered from the perspective of strategic and applied HRM practice in both the UAE/GCC and internationally.

EMHR 592 Managing Change and Innovation

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EMMG 510

This course discusses issues related to the management of change in organizations. The different types of change typically encountered in organizations are characterized using well- defined analysis frameworks. The effective management of corporate innovation is also considered, looking in detail at the ideas generation, development and diffusion phases.

EMHR 594 **Employee Relations and Compensation Management**

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): EMMG 510

This course deals with legal and regulatory issues pertaining to employee relations from several perspectives. The compensation management aspect focuses on the theories underlying direct compensation and reward systems in organizations, and the practical administrative practices used to implement such systems, with particular reference to the UAE/GCC sub-region. Compensation management practices, including the analysis and evaluation of jobs, criteria and procedures for determining wage levels, individual wage determination, forms of payment, and incentive systems will be covered.

EMIB 524 International Business

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): Graduate Standing

The course looks in-depth at salient aspects of managing in a globalized environment, including consideration of theoretical concepts. It offers a practical treatment of political-economic aspects of international trade. The course discusses foreign direct investment, global monetary systems, and strategy formulation for international business practice. The course additionally places stress on international business case analysis, within the UAE/GCC context.

EMMG 510	Leadership and Managing People	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Graduate Standing		
This course provides an in-depth study of concepts relating to leadership and people management within organizations. The work integrates theory, research and applications, with an emphasis on context-driven learning. Students apply principles of leadership and people management to their own occupational situations and will investigate topical case studies. The material is delivered from the perspective of people-management practice in both the UAE/GCC and internationally.		
EMMG 515	Strategic Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Graduate Standing		
The course deals with the formulation, implementation and evaluation of strategies designed to give organizations a competitive edge. Corporate, business and operational – level strategy types are treated, and the use of various strategy analysis frameworks and models are discussed. Additional emphasis is placed on strategy formulation in the context of business practice in the UAE/GCC.		
EMMI 520	Managing Information Systems Performance	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Graduate Standing		
The course studies techniques and practices for analyzing business information systems performance with emphasis on support for business processes. The course gives a strategic and integrated view of the exploitation of information and communication technology, with particular emphasis on e-business for improving business performance and business efficiency. The course discusses IT-driven business strategy, and reviews IT-related factors that are necessary for enhancing the performance of organizations in the current competitive global business environment.		
EMMK 518	Marketing Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Graduate Standing		
This course emphasizes managerial aspects of marketing, including developing marketing strategies and plans, and integrating specific elements of the marketing process. The course additionally places stress on case analysis, as a means for learning the material and for students to develop problem-solving, group work, judgement and decision-making skills.		
ENGL 091	ELP 1: Intensive Listening and Speaking	(6:6:0)
6 Semester Credit Hours		
Prerequisite(s): None		
This course is designed to equip students in the English Language Program (ELP) with the essential listening and speaking skills necessary to communicate effectively in English at the A1-A2 level. The course will provide students with a solid foundation in understanding spoken English, responding appropriately, and engaging in basic conversations.		
ENGL 092	ELP 2: Intensive Reading	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): None		
This course is designed to equip students in the English Language Program (ELP) with the essential reading skills necessary to understand and interpret basic written English texts at the A1-A2 level. Students will develop their ability to read and comprehend simple texts, basic instructions, and everyday materials as well as increase their vocabulary acquisition.		
ENGL 093	ELP 3: Intensive Writing	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): None		
This course is designed to equip learners in the English Language Program (ELP) with the essential writing skills necessary to express themselves effectively in written English at the A1-A2 level. The course will provide learners with a solid foundation in sentence structure, grammar, vocabulary, and basic writing mechanics. Through a variety of activities and exercises, students will develop their ability to write simple sentences and paragraphs on familiar topics.		
ENGL 094	ELP 4: Communication Skills Integration	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): None		
This course is designed for students in the English Language Program (ELP) at the A1-A2 level who need to build a strong foundation in the English language. It focuses on developing and practicing essential communication skills in listening, speaking, reading, and writing, with an emphasis on everyday vocabulary and basic grammar structures. Through interactive lessons, practical exercises, and real-world scenarios, students will learn to engage in simple conversations, comprehend short texts and audio, and express themselves in both spoken and written English.		

ENGL 095	Academic Bridge	(3:3:1)
3 Semester Credit Hours and 1 Lab Hour		
Prerequisite(s): None		
ENGL 095 Academic Bridge uses an integrated skills approach to improve students' English competency in reading, writing, speaking, and listening to prepare them for ENGL 099 Foundations of College English. This course is not for degree credit. Placement in this course is based on the student's score on the English placement test.		
ENGL 099	Foundations of College English	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): None		
This course focuses on developing students' ability to use academic English in college settings. It refines students' writing and reading skills; improves their ability to understand spoken language and deliver oral presentations in a college setting; and develops their ability to express themselves at an effective operational level of proficiency. This course is not for degree credit. Placement in this course is based on the student's score on the English placement test.		
ENGL 101	Composition	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Requisite score on the EmSAT, IELTS, TOEFL, OOPT, or completion of required English foundation course(s)		
The course introduces students to the reading, writing, and critical thinking skills central to academic discourse. Students will engage critically with contemporary issues through college-level readings. They will practice analytical reading skills, presenting the argument of others, and critically responding to texts through intense practice in the writing process. (<i>Writing Intensive Course</i>)		
ENGL 102	Public Speaking	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): ENGL 101		
This course is designed to provide an overview of principles to develop effective presentations for public and professional settings while integrating appropriate technologies. Course content and assignments emphasize audience analysis, effective and coherent composition development, research strategies and skills, and presentation delivery methods to strengthen confidence and credibility.		
ENGL 200	Advanced Composition	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Completion of a minimum of 45 credit hours and ENGL 101		
In this advanced writing course, students learn to communicate persuasively about and respond to literature to enhance their skills in academic writing and communication. Building on the writing skills acquired in ENGL 101, students extensively study and practice argumentative techniques, considering differing audiences and purposes when responding to literature. (<i>Writing Intensive Course</i>)		
ENGR 107	Introduction to Engineering	(2:1:3)
2 Semester Credit Hours		
Prerequisite(s): Math Placement Test score qualifying student for MATH 113		
This course introduces functions of the engineer, professionalism, engineering ethics, problem-solving techniques, presentation of technical information, estimation, analysis, and design. The course will include a workshop where students develop basic skills in hand filing, turning, welding, carpentry, sheet metal fabrication, metal forming, and basic electric circuit connections.		
ENGR 200	Engineering Statistics	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Math 114		
The course introduces principles of statistics and probability for undergraduate students in Engineering. The course covers the basic concepts of probability, discrete and continuous random variables, probability distributions, expected values, joint probability distributions, and independence. The course also covers statistical methods and topics including data summary and description techniques, sampling distributions, hypothesis testing, and regression analysis.		
ENGR 210	Sustainability in Engineering	(2:2:0)
2 Semester Credit Hours		
This course introduces engineering students to the contemporary issue of sustainability and how it should guide engineering practice and design. Topics covered include sustainability concepts, SDGs, climate change, environmental issues, life cycle assessment, sustainable engineering systems (Infrastructure, buildings, clean energy and smart cities, and sustainable business management.		
ENGR 390	Internship I	
3 Semester Credit Hours		
Prerequisite(s): Completion of 60 credits and a CGPA of 2.0 or higher		
This is one of two supervised field experiences of professional-level duties where each is for 240 to 320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the		

end of the internship.

ENGR 391 Internship II

3 Semester Credit Hours

Prerequisite(s): Completion of 90 credits and a CGPA of 2.0 or higher

This is one of two supervised field experience of professional-level duties where each is for 240 to 320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designate site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

ENGR 399 Undergraduate Research Project

(3:0:9)

3 Semester Credit Hours

Prerequisite(s): A minimum of 90 C.H., 3.0 CGPA or Department Chair approval

Student in this course will be engaged in a creative research project under the guidance of the faculty member from school of engineering. The student will prepare a paper suitable for publications in peer review journal, conference or scientific gathering. Fixed credit hours; 3 credits are assigned.

ENGR 410 Climate Change

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): ENGR 210 or ENVS 102

This course offers a comprehensive overview of climate change, exploring its causes, and its environmental and socio-economic impacts. Students will learn about the greenhouse effect and its main drivers, climatic changes in the near and distant past, and the role of climate models in assessing future climatic changes. They will also learn about adaptation and mitigation strategies, and international climate agreements to address climate change. They will actively engage in this course by researching and presenting a topic on climate change.

ENGR 420 Life Cycle Assessment

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): None

This course aims to provide students with a comprehensive understanding of the principles, methodologies, and applications of Life Cycle Assessment (LCA) in assessing the environmental impacts of products, processes, and systems. Students will acquire the knowledge and skills necessary to conduct LCAs and make informed decisions aimed at reducing environmental impacts in various industries. Additionally, students will gain hands-on experience by using LCA software.

ENGR 450 Engineering Seminar

(1:1:0)

1 Semester Credit Hour

Prerequisite(s): Completion of 90 credit hours

Engineering ethics, professionalism, the role of engineers in society, current topics, and employment opportunities.

ENGR 500 Engineering Foundation for Graduate Students

(3:3:0)

3 Semester Credit Hours

This course is designed to provide a foundation in engineering principles, including strategic and systems thinking, statistical analysis of scientific data and sustainability. It is required for students who do not meet the CGPA for enrollment in graduate programs. The course aims to provide students with the skills and education required to pursue a Master's program.

ENVS 101 Energy and Environmental Science

(3:3:0)

3 Semester Credit Hours

The course is an inter-disciplinary study of environmental disruption and management, natural environmental systems, and the human impact on them. Other topics include energy procurement and use, waste management, water resources and water pollution, acid rain, global warming and ozone depletion.

ENVS 102 Sustainability and Human-Environment Relations

(3:3:0)

3 Semester Credit Hours

This course is designed to present the student with an overview of environmental issues. Humans are closely related to their environment and depend upon it to supply all the necessary resources to maintain life and to provide inputs for industrial activities.

FNAN 300 Principle of Business Finance

(3:3:0)

3 Semester Credit Hours

Prerequisite ACCT 204

This course teaches students the fundamental principles of finance. It covers the financial decisions that financial managers should make to maximize shareholders' wealth or a firm's value. These financial decisions include investment, financing, dividend, and working capital decisions. Students are taught how to apply financial theories to real life financial situations, financial ratio analysis, time value of money, stock valuation, bond valuation, capital budgeting and risk and return. It also discusses the difference between profit maximization and shareholders' wealth maximization. The course entails calculations and arriving at conclusions that help financial managers make financial

decisions and measure financial performance.

FNAN 303 Financial Analysis, Forecasting, and Valuation (3:3:0)

3 Semester Credit Hours

Prerequisites: ACCT 204

This course teaches students a range of techniques for examining, analyzing, and interpreting financial statements, and applying the information for both forecasting and making operational or a strategic financial decision that maximize a firm's value. The course aims at providing comprehensive understanding of the role of financial statements play and their impact on financial decision-making. Topics covered in this course include financial statement analysis, analysis of risk and return, development of financial models, financial forecasting, and firm valuation.

FNAN 304 Principles of Risk Management and Insurance (3:3:0)

3 Semester Credit Hours

Prerequisites: FNAN 303

This course introduces students to the techniques of risk management and insurance. The course also discusses the theory and application of risk and insurance to real world financial decisions. Different disciplines define risk differently. However, in finance, we define risk as the possibility or chances of not achieving the firm's set objective of maximizing shareholder wealth or firm value. Students will study how financial managers identify, measure, control, and monitor risk in order to eliminate, reduce, transfer or avoid it in their operational and strategic financial decision-making.

FNAN 321 Financial Institutions (3:3:0)

3 Semester Credit Hours

Prerequisites: FNAN 300

The course examines the functions of financial institutions with particular references to industry structure and regulatory environment. It analyses decision variables that management should concentrate on to achieve objectives. Topics include types of financial institutions, the structure and functioning of financial institutions, regulation of financial institutions, measuring and managing the risk financial institutions are exposed to, and liquidity and liability management.

FNAN 331 Fundamentals of Sustainable Finance (3:3:0)

3 Semester Credit Hours

Prerequisites: FNAN 300

The course offers business students with a comprehensive exploration of finance principles within the context of sustainability. It focuses on how financial decisions can effectively contribute to environmental and social objectives. Using theoretical frameworks, case studies, and practical applications, students will learn essential concepts such as integrating environmental, social, and governance (ESG) factors in financial decision-making, evaluating sustainable finance products, interpreting ESG metrics disclosed by companies, assessing risks associated with sustainability issues like climate change, and understanding the importance of stakeholders and regulatory frameworks in advancing sustainable finance initiatives.

FNAN 390 Internship in Finance

3 Semester Credit Hours

Prerequisites: Completion of 90 credit hours

The Internship is designed for candidates on the undergraduate finance major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

FNAN 401 Advanced Financial Management (3:3:0)

3 Semester Credit Hours

Prerequisites: FNAN 303

The course analyzes financial decision-making in firms. It emphasizes the conceptual structure of problems and the use of advanced analytic techniques. The topics covered include investment decisions, financing decisions, dividend decisions, working capital management, current asset management, capital structure, and corporate planning models.

FNAN 411 Investment Analysis and Portfolio Management (3:3:0)

3 Semester Credit Hours

Prerequisites: STAT 100, FNAN 303

This course focuses on analysis and valuation of investment alternatives and strategies. It will mainly focus on classifying investment alternatives and examining how investors allocate their financial assets and manage their portfolio. The course analyzes modern techniques of portfolio management including evaluating standards for selecting individual securities to include or remove from portfolio. The course also discovers risk-return analysis for portfolio, and portfolio performance measures. The course is practically oriented, but also aims to equip students with conceptual and theoretical background of investment analysis and portfolio management.

- FNAN 412 Futures and Options Markets (3:3:0)**
3 Semester Credit Hours
Prerequisite OPMT 311, FNAN 303, FNAN 411
 This course demonstrates the theoretical and practical aspects of derivative securities and their applications in financial risk management. Over the past few decades, the use of derivatives to manage financial risk has been increasing tremendously. The main goal of this course is to enhance students' knowledge and skills on the use of derivative securities and application of derivative trading strategies in financial risk management. The course emphasizes on the understanding of the features and functions of different derivative securities, and the operation of the derivative market. Topics covered include an overview of the derivative securities, the mechanics of forwards, futures, swaps and options, hedging, option trading strategies, valuation of derivative securities, option Greeks and Volatility smiles.
- FNAN 421 Money and Capital Markets (3:3:0)**
3 Semester Credit Hours
Prerequisite(s): FNAN 303
 This course illustrates the conceptual and real-world applications of the functions and operations of the money and capital markets. The course starts with an introduction of the financial system and proceeds with a detailed analysis of the functions of the central bank, demonstrates the interest rates behaviors and term structure, emphasizes the practical applications related to the function, operation and performance of money and capital markets and the instruments traded within. The course also develops an analytical framework to understand the factors that affect the performance of the financial market with a special emphasis on financial regulations and financial crisis. It also discusses the importance of the mortgage and foreign exchange markets in the financial system. These issues are presented within a unified framework intended to encourage critical thinking and problem-solving skills of students.
- FNAN 431 International Financial Management (3:3:0)**
3 Semester Credit Hours
Prerequisite FNAN 303
 The course focuses on the management of contemporary financial operations of international corporations. The topics include foreign exchange risk, political risk, returns and risks of international projects, international money and capital markets, financial accounting, capital structure, and cost of capital.
- FNAN 444 Finance Project-Based Internship**
3 Semester Credit Hours
Prerequisite(s): FNAN 303
 The finance project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem-solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the finance specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.
- HIST 100 Contemporary Middle Eastern History (3:3:0)**
3 Semester Credit Hours
 This course examines the major issues affecting the Middle East in the 21st century while reviewing the origins and development of the modern Middle East. Students will develop an understanding of the social, economic, and political foundations that set the stage for the region this century.
- HIST 101 Ancient History of the Arabian Peninsula (3:3:0)**
3 Semester Credit Hours
 This course concentrates on the geographical background of the Arabian-peninsula including location, descriptions of its provinces, routes, flora and fauna. It also focuses on the commercial importance and its political situation during the period from third Millennium B.C. to the rise of Islam or to the seventh century A.D.
- HIST 111 Introduction to World History (3:3:0)**
3 Semester Credit Hours
 Analytical approach to an overview of world history that surveys the major features of principal existing civilizations of the world, as originally formed and as altered by key global processes, including forces of modernity.
- HIST 281 Survey of Middle Eastern Civilization I (3:3:0)**
3 Semester Credit Hours
 Survey of Middle Eastern history from rise of Islam to present, emphasizing processes that led to emergence of economic, cultural, social, and political institutions that characterize region today. HIST 210 surveys the period from rise of Islam in 570 to medieval period (ca. 1258).

HIST 282	Survey of Middle Eastern Civilization II	(3:3:0)
3 Semester Credit Hours		
Second half of the survey of Middle Eastern history. HIST 282 surveys the period from about 1258 to the present.		
HTMT 311	Survey of the Hospitality & Tourism Industry	(3:3:0)
3 Semester Credit Hours		
A survey of the history, trends, organizational structure, and economic impact of the hospitality and tourism industry on UAE economy. Some study of the problems originating in the operation and management of various segments of the hospitality industry in UAE will be introduced.		
HTMT 312	Sustainable Tourism	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): HTMT 311		
The course will introduce the issues associated with the sustainable management of destinations and tourist attractions. Students will: learn the variety of ways that sustainability can manifest in the hospitality and tourism industry in UAE; understand the roles of international stakeholders involved in sustainable tourism; analyze case studies in key areas related to social, economic, and environmental impacts and consequences; and learn broad perspectives on how tourism relates to ecosystems, poverty, health, education, and social equity.		
HTMT 321	Hospitality Operations I - F&B Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): HTMT 311		
This course involves the concepts and applications of management in food and beverage operations in various types of hospitality firms across UAE. Topics such as menu development, beverage management, catering, service, sanitation, foodservice design, and cost controls will be presented. Management approaches will be developed to provide quality products and services.		
HTMT 322	Hotel Operations	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): HTMT 311		
This course will offer a thorough overview of the management of various lodging properties, taking a generalist view of lodging management, moving from department to department in a typical hotel. While this course will be general in nature, we will take a much more strategic perspective, identifying and considering issues of concern to general managers of all types of lodging properties, with a particular focus on profit maximization (yield/revenue management) and distribution channel management. As such, we will consider the broader political, economic, social, and technological environments and trends and their impact on lodging operations. For example, the recent global economic downturn has had a devastating effect on the lodging industry. Thus, the recession will form a basic issue that we will deal with throughout the semester.		
HTMT 390	Internship in Hospitality and Tourism Management	
3 Semester Credit Hours		
Prerequisite(s): Completion of 90 credit hours		
The Internship is designed for candidates on the undergraduate hospitality and tourism management major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.		
HTMT 411	Destination Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): HTMT 311		
The course will introduce the various issues associated with the management of a convention visitor's bureau (CVB). The course will analyze the mission, structure, and business activities of organizations that develop and promote a complex tourism destination in UAE. This includes the management functions of membership services, visitor services, financial and marketing concerns, research activities, and the strategic planning and evaluation of these efforts. Topics covered will include developing the following consumer segments: meetings/conventions, pleasure/tours, and festivals/special events.		
HTMT 412	Meetings and Conventions	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Prior Completion of 84 Credit Hours		
A course dealing with the many issues impacting the management of large convention and exposition centers. The course is taught from an organizational marketing base. Topics include meeting site selection, program planning and budgeting, legal issues and insurance problems, housing, food and beverage arrangements, transportation, exposition management, and audio-visual services."		
HTMT 444	Hospitality & Tourism Management Project Based Internship	

3 Semester Credit Hours**Prerequisite(s): HTMT 311**

The hospitality and tourism management project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem-solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the hospitality and tourism management specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.

INDS 111 Color Theory and Rendering**(3:2:2)****3 Semester Credit Hours****Prerequisite(s): None**

The aim of this course is to introduce color theory as a component of the design process that influences human perception of the interior space, behavior, performance and wellbeing and therefore influences design solutions. It offers skills in color selection and rendering for diverse spaces function and design concept communication.

INDS 112 Design Communication 1: Sketching and Drawing for Interiors**(3:2:2)****3 Semester Credit Hours**

This course introduces students to the principles and techniques of interior design's form and space representation and communication. The course makes the students familiar with manual drawing tools and modes of documenting to develop students' sketching, technical drawing and ideation skills.

INDS 121 Interior Design Process, Human Factors and Ergonomics**(3:3:0)****3 Semester Credit Hours**

The aim of this course is to provide knowledge in design process steps, components, human factors, anthropometrics and ergonomics and how they influence design solutions and contribute to the human wellbeing. It offers skills in gathering, analyzing human-centered evidence, application of ergonomics and other human factors in inclusive, universal design.

INDS 122 Design Communication 2: Digital Media for Interiors**(3:2:3)****3 Semester Credit Hours****Prerequisite(s): INDS 112**

This course introduces students to the most common 2D and 3D software used in the field of interior design to create and develop graphical and technical presentations. Emphasis will be on giving the students tools to digitize a sketches, drawings and concepts to examine design idea in terms of accuracy and workability of schematic design.

INDS 211 Interior Design Studio 1**(4:0:8)****4 Semester Credit Hours****Prerequisite(s): ARCH 122****Corequisite(s): INDS 111 and ARCH 223**

This course introduces a small-scale residential environment interior design project. A scenario-based design brief focuses on relationships between the users' needs and space's features leads students through the early phases of the design process (pre-design, schematic design, design development) with an aim of meeting space functional requirements and user(s) needs.

INDS 212 History of Art and Interior Design 1**(3:3:0)****3 Semester Credit Hours**

The course aims to provide knowledge in history as it relates to decorative arts, architecture, interior design, furniture, material culture and understanding of how the design of the built environment changes along with historical periods that are influenced by the social, political, physical changes in design of the built environment.

INDS 213 Interior Construction Methods, Materials and Finishes 1**(3:2:3)****3 Semester Credit Hours****Prerequisite(s): INDS 121****Corequisite(s): INDS 112**

The course aims to provide students with understanding of building material, finishes, and furnishings, their fabrication process, installation methods and maintenance requirements and skills in selecting materials, finishes and furnishings to support the design intent, project criteria and human and environmental wellbeing. The courses focuses on classic materials.

INDS 221 Interior Design Studio 2**(4:0:8)****4 Semester Credit Hours****Prerequisite(s): INDS 211****Corequisite(s): INDS 223**

This studio introduces medium size interior design of office spaces. The studio encourages the students to investigate social interaction in multiusers environments and to use elements that reflect the nature and identity of work activities through the phases of data collection and analyzes, space programming, materials, furnishings and finishes selection.

INDS 222 History of Art and Interior Design 2 (3:3:0)

3 Semester Credit Hours

Prerequisite(s): INDS 212

The course aims to provide knowledge in history as it relates to decorative arts, architecture, interior design, furniture, material culture and understanding of how the design of the built environment changes along with historical periods that are influenced by the social, political, physical changes in design of the built environment.

INDS 223 Interior Construction Methods, Materials and Finishes 2 (3:2:3)

3 Semester Credit Hours

Prerequisite(s): INDS 213

The course aims to provide students with understanding of building material, finishes, and furnishings, their fabrication process, installation methods and maintenance requirements and skills in selecting materials, finishes and furnishings to support the design intent, project criteria and human and environmental wellbeing. The course focuses on modern materials.

INDS 224 Sustainable Approaches in Interior Design (3:3:0)

3 Semester Credit Hours

Prerequisite(s): ENVS 102

The course aims to explore the complex context of sustainability, its application in interior design, sustainability approaches in interior design and how sustainable approaches in interior design contribute to environmental and human wellbeing. The course provide practice in systems thinking for understanding how social, economic, cultural sustainability influences the design.

INDS 311 Interior Design Studio 3 (4:0:8)

4 Semester Credit Hours

Prerequisite(s): INDS 221

This design studio increases the complexity of the interior design projects by introducing design solutions for commercial spaces in urban, suburban, and shopping mall settings. Students explore space utilization, fixtures, and display as well as the role interior designer plays in supporting the business and branding of the commercial space.

INDS 312 Construction Drawings and Interior Detailing (3:2:3)

3 Semester Credit Hours

Prerequisite(s): INDS 223

This course will lead the students to continue exploring more advanced software application and the principles of Building Information Modelling used to visualize, document and communicate interior design working drawings.

INDS 313 Environmental Systems 1: Indoor Comfort and Human Wellbeing (3:3:0)

3 Semester Credit Hours

Prerequisite(s): INDS 224

Corequisite(s): INDS 312

The course aims to explore environmental control systems, types, size, their integration into interior design project, and awareness of these systems in relation to indoor thermal comfort, indoor air quality, environmental and human wellbeing. The course focuses on water, waste, HVAC, electrical, fire safety, conveyance, and communication and security systems.

INDS 321 Interior Design Studio 4 (4:0:8)

4 Semester Credit Hours

Prerequisite(s): INDS 311

Corequisite(s): INDS 322 and INDS 323

This design studio deals with contemporary complex interior design project within the hospitality industry and will enable the students to explore the combination of wellness, aesthetics, functionally and operational aspects in designing city restaurants, cafes, and hotels.

INDS 322 Furniture Design (3:2:3)

3 Semester Credit Hours

Prerequisite(s): INDS 121 and INDS 312

The aim of this course is to provide knowledge and skills in materials, construction methods and details for furniture design and making. The course analyzes materials for furniture design in relation to their impact on environment, healthy and safe living and efficient work within the interior space.

INDS 323 Environmental Systems 2: Design with light and sound (3:2:3)

3 Semester Credit Hours

Prerequisite(s): INDS 313

The course aim to provide knowledge in space's lighting and acoustical design for space quality enhancement in relation to safety, comfort, performance, and wellbeing. The course suggests skills in light and sound integration into design of residences, stages, exhibitions, theaters, halls, etc. with the application of high-tech materials and technologies.

INDS 351 Parametric Design**(3:3:0)****3 Semester Credit Hours****Prerequisite(s) INDS 312 or ARCH 336**

This course is an advanced level course that will allow students to discover the fundamentals of Parametric modeling for design applications in Interior Design and Architecture. Students will investigate rule-based and parametric design concepts and techniques in the context of a generative modeling environment: Dynamo for Revit.

INDS 352 Sketching and Painting**(3:2:2)****3 Semester Credit hours**

This course aims to improve students' skills in sketching and painting and focuses on two-dimensional media. Students continue to explore and experiment various techniques in art making. Two-dimensional media art works such as still life, compositions, collage are practiced with the application of various two-dimensional media.

INDS 353 Landscape and Garden Design**(3:3:0)****3 Semester Credit Hours**

This course aims to introduce principles of landscape design throughout all its phases. Students study types of landscape, materials and its relationship with the context, environmental and human wellbeing and apply various representation techniques in landscape design projects.

INDS 390 Internship I**3 Semester Credit Hours****Prerequisite(s): Completion of 60 credits and a CGPA of 2.0 or higher**

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

INDS 391 Internship II**3 Semester Credit Hours****Prerequisite(s): Completion of 97 credits and a CGPA of 2.0 or higher**

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

INDS 411 Interior Design Studio 5**(4:0:8)****4 Semester Credit Hours****Prerequisite(s): INDS 321**

This course extends the scope of interior design within a medium scale project of a healthcare environment. Students explore design approaches related to this complex and sensitive environment with respect to safety, efficiency, codes and regulations and that will satisfy all end users (including patients, staff, and visitors)' needs and wellness.

INDS 412 Graduation Final Thesis Research**(2:1:3)****2 Semester Credit Hours****Prerequisite(s): INDS 321****Corequisite(s): INDS 411**

The course aims to prepare students for Graduation Project Design that follows next semester. It offers students opportunity to consider their evolving interest, targeted career plan and select a project theme, gather, analyze, systemize relevant to the theme data, and develop a design brief, project statement, design concept and program. **(Writing Intensive Course)**

INDS 414 Professional Practice 1: Codes and Regulations**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): INDS 323**

The aim of this course is to equip students with knowledge of regional and international laws, codes, standards and guidelines that influence the encounter of the inhabitants with interior spaces and skills in application of laws, codes, standards and guidelines concerning built environment, sustainability, wellness, inclusive and universal design.

INDS 421	Graduation Project Design	(5:0:10)
5 Semester Credit Hours		
Prerequisite(s): INDS 412		
Corequisite(s): INDS 422		
The graduation project is a capstone self-directed course under the supervision of one the program's senior faculty member where the project identified during the Graduation Final Thesis Research course is implemented. Students apply gained knowledge and skills in developing a holistic design approach to a design project.		
INDS 422	Professional Practice II: Ethics and Project Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): INDS 414		
The aim of the course is to provide awareness of the context of interior design practice, the influence of local and global context on design practices, understanding of the interior design profession principles, processes, responsibilities and its value to the society, interior design management, business formation, practices, and ethics.		
INDS 451 Portfolio Design		(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): None		
This course aims to assist and mentor students in assembling a comprehensive portfolio of their academic work, enhancing students' graphic techniques, and provides skills in organizing the layouts of their projects in both book and digital formats.		
INDS 452	Photography	(3:3:0)
3 Semester Credit Hours		
This course aims to provide knowledge in types of cameras and skills in camera operating, taking pictures in different contexts, enhancing the pictures, processing and organizing pictures and application of photographs in interior design presentations.		
INDS 453	Advanced BIM: Scheduling and Coordination	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): INDS 312 or ARCH 336		
The course aims to introduce technologically based collaboration of interior design with multiple disciplines and provides knowledge and training on collaboration tools and methods using Building Information Modelling applications. It focuses on developing, reading, exploring, and coordinating working drawings and executive documents to avoid conflicts and ensure construction documents consistency.		
INDS 454	Research Methods for Interior Design	(3:3:0)
3 Semester Credit Hours		
This course aims to introduce research processes in design related tasks within the framework of Evidence-Based Design Approach. It offers a review of design research methods such as literature review, sampling procedures, questionnaires interviews, and observations.		
ITEC 103	Fundamentals of Information Technology	(3:3:0)
3 Semester Credit Hours		
The course focuses on the nature and uses of computers with an introduction to word processing, spreadsheets, presentation software and effective presentation information, in addition to related lab projects. Additional topics include computer systems organizations, communications and networking, legal and ethical issues, computer security, the Internet, and trending technologies.		
MATH 091	Pre-algebra	(3:3:0)
3 Semester Credit Hours		
This course prepares students to take MATH 101 (Numbers and Data interpretation). It enables them to gain a command in operations with integers, fractions, decimals and percent, geometric figures and their measures, and pre-algebra topics including properties of rational numbers, operations of rational numbers, simplification of polynomials and equation-solving techniques. This course is not for degree credit.		
MATH 093	Intermediate Algebra	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): None		
This course covers basic algebraic skills including factoring, solving basic linear and quadratic equations, the rules of exponents and radicals, and basic concepts of functions with specific examples, such as polynomial and rational functions. The course serves as a prerequisite to MATH 108. This course is not for degree credit.		
MATH 095	Pre-calculus	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): None		
This course is a prerequisite to MATH 113. It reviews mathematical skills essential to studying calculus. Topics include inequalities, absolute values, graphs, functions, exponential and logarithmic functions, and trigonometry. This course is not for degree credit.		

MATH 101	Numbers and Data Interpretation	(3:3:0)
Prerequisite(s): MATH 093 or Appropriate Score on the Placement Test		
3 Semester Credit Hours		
This course meets the quantitative reasoning requirement, one of the Foundation requirements of the University General Education program. Topics include Logic Mathematics, Sets and Counting, probability and descriptive statistics. This course is broad in scope, emphasizing applications.		
MATH 102	Mathematics in Civilization	(3:3:0)
3 Semester Credit Hours		
Throughout the history of mankind, the mathematical spirit has been a driving force in the development of the civilized world. This course creates an understanding of the impact of mathematical ways of thinking on the arts and sciences, on the development of technologies, and on the study of history. The goal is to illuminate the role of mathematics in the development of Eastern and Western civilizations and, thereby, to shed on human civilization from a mathematical point of view.		
MATH 108	Calculus with Business Applications	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MATH 093 or Appropriate Score on Math Placement Test		
The course covers standard topics such as functions, limits, derivative, and integral calculus as well as applications of differentiation and integration. Simple introductory examples and applications are drawn from, but do not require, advanced knowledge of business applications.		
MATH 111	Calculus with Life Sciences Applications	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): None		
The course covers standard topics such as functions, limits, derivative, and integral calculus as well as applications of differentiation and integration. Simple introductory examples and applications are drawn from, but do not require advanced knowledge of the life sciences.		
MATH 113	Calculus I	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): None		
This course covers the concepts of limits and continuity, derivatives of functions, extreme values, curve sketching, L'Hopital's rule, definite and indefinite integrals and integration by substitution.		
MATH 114	Calculus II	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MATH 113		
This course covers techniques and applications of integration, transcendental functions, infinite sequences and series and parametric equations.		
MATH 203	Linear Algebra	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MATH 113		
This course covers systems of linear equations, linear independence, linear transformations, inverse of a matrix, determinants, vector spaces, eigenvalues, eigenvectors, and diagonalization.		
MATH 213	Calculus III	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MATH 114		
This course covers partial differentiation, multiple integrals, line and surface integrals, and three-dimensional analytic geometry.		
MATH 214	Elementary Differential Equations	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MATH 114		
This course covers first-order ODEs, higher-order ODEs, Laplace transforms, linear systems, nonlinear systems, numerical approximations, and modeling.		
MATH 225	Discrete Mathematics	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MATH 113 and (CSCI 112 or CSCI 114)		
This course covers elementary discrete mathematics. Topics include propositional logic, tautology and equivalence rules, logical inference rules, logical consistency, predicate logic and quantifiers, graphs, planar graphs, binary and decision trees, and complexity of algorithms.		

MBAC 512	Managerial and Financial Accounting	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Graduate Standing		
The course deals with managerial and financial accounting techniques and practices. The course examines the impact of cost and cost allocation on business performance analysis, using a variety of costing schemes. It presents accounting as an information system construct, and looks at methods for recording, presenting and analyzing accounting information. The course also looks at budgeting, ratio analysis and other fiscal management measures. The course discusses the impact, on Management and Financial Accounting, of contemporary management philosophies and techniques including JIT practices, quality assurance and performance measurement.		
MBFN 500	Business Graduate Foundation	(3:3:0)
3 Semester Credit Hours		
The course is designed to provide a foundation in business for Graduate students who graduated from a non-business discipline. This course offers a comprehensive introduction to aspects of business. This course aims to equip students with business education and skills to successfully complete the business graduate program. This course does not count toward degree credit.		
MBFN 514	Managerial Finance	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Graduate Standing		
This course focuses on corporate finance at a global environment context. It explores advanced concepts and practices applicable to multinational company's financing and investment decision- makings. Topics covered include financial analysis for investment decisions in an international setting, international money operations and capital markets, management of foreign exchange risk, direct foreign investment, funding of international projects, political risk analysis, currency derivatives and swap markets.		
MBFN 560	International Corporate Finance	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MBFN 514		
This course focuses on corporate finance at a global environment context. It explores advanced concepts and practices applicable to multinational investing, financing and dividend decision making that maximize firm value. Topics covered include financial analysis for investment decisions in an international setting, international money operations and capital markets, management of foreign exchange risk, direct foreign investment, funding of international projects, political risk analysis, currency derivatives and swap markets.		
MBFN 562	Investment Planning and Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MBFN 514		
The course deals with the advanced analysis of equity securities and investment portfolios, predicated on relevant market hypothesis and capital market theory. The course looks at investment risk-return trade-off, asset pricing models, and stock price behavior. The course adds emphasis on stocks, bonds, and financial futures and options.		
MBFN 564	Financial Markets	(3:3:0)
3 Semester Credit Hours		
Prerequisite MBFN 514		
The course deals with financial markets, primarily global equity markets. It looks at markets for handling government debt instruments, and exchange-traded and over-the-counter financial derivative instruments such as futures, options, swaps, and asset-backed securities. There is detailed treatment of derivatives theories, derivatives exchanges, and valuation of derivatives, including standard and other non-standard options on a variety of underlying assets, in relation to relevant financial markets. The course also discusses emerging financial markets, and the effective management of risks emanating from these markets.		
MBGN 525	Research Methods for Business	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Graduate Standing		
This course provides the necessary underpinning support for the MBA dissertation and general business domain research. It allows candidates to make informed decisions and appropriate choices pertaining to research methodology. A range of business research tools, approaches and analytical techniques are discussed, and guidance is given on optimal structuring of business research documents and MBA dissertations.		
MBGN 555	MBA Investigative Thesis or Project I	(3:0:9)
3 Semester Credit Hours		
Prerequisite(s): MBGN 525		
MBGN 555 is the first part of the MBA Investigative Thesis or Project course, serving to integrate the various topics studied in the School of Business MBA program. Students undertake a work-based study that focuses on a chosen business/management topic. The outcome for MBGN 555 is to submit a research introduction, literature review and research methodology that will be used, for the completion of this course. Satisfactory performance of the students in this course will move the student into MBGN 556.		

MBGN 556	MBA Investigative Thesis or Project II	(3:0:9)
3 Semester Credit Hours		
Prerequisite MBGN 555		
MBGN 556 is the second part of the six-credit hour research course, serving to integrate the various topics studied in the School of Business MBA program. Students undertake a work-based study that focuses on a chosen business/ management topic. The outcome is a thesis, an extended piece of work necessitating investigation of secondary data and empirical fieldwork, through collection of primary data, pertaining to an organization and/or a business domain problem. Analysis of collected data is expected to generate critical commentary and recommendations.		
MBHR 590	Applied and Strategic Human Resource Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MBMG 510		
The course deals with key areas of applied and strategic human resource management, essential for providing supervisors and human resource specialists with a thorough understanding of the strategic role of human resource management and the techniques available to management for making effective use of the human resources of an organization. The material is delivered from the perspective of strategic and applied HRM practice in both the UAE/GCC and internationally.		
MBHR 592	Managing Change and Innovation	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MBMG 510		
This course discusses issues related to the management of change in organizations. The different types of change typically encountered in organizations are characterized using well- defined analysis frameworks. The effective management of corporate innovation is also considered, looking in details at the ideas generation, development and diffusion phases.		
MBHR 594	Employee Relations and Compensation Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MBMG 510		
This course deals with legal and regulatory issues pertaining to employee relations from several perspectives. The compensation management aspect focuses on the theories underlying direct compensation and reward systems in organizations, and the practical administrative practices used to implement such systems, with particular reference to the UAE/GCC sub-region. Compensation management practices, including the analysis and evaluation of jobs, criteria and procedures for determining wage levels, individual wage determination, forms of payment, and incentive systems will be covered.		
MBIB 524	International Business	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Graduate Standing		
The course looks in-depth at salient aspects of managing in a globalized environment, including consideration of theoretical concepts. It offers a practical treatment of political-economic aspects of international trade. The course discusses foreign direct investment, global monetary systems, and strategy formulation for international business practice. The course additionally places stress on international business case analysis, within the UAE/GCC context.		
MBMG 510	Leadership and Managing People	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Graduate Standing		
This course provides an in-depth study of concepts relating to leadership and people management within organizations. The work integrates theory, research, and applications, with an emphasis on context-driven learning. Students apply principles of leadership and people management to their own occupational situations and will investigate topical case studies. The material is delivered from the perspective of people-management practice in both the UAE/GCC and internationally.		
MBMG 515	Strategic Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Graduate Standing		
The course deals with the formulation, implementation and evaluation of strategies designed to give organizations a competitive edge. Corporate, business and operational – level strategy types are treated, and the use of various strategy analysis frameworks and models are discussed. Additional emphasis is placed on strategy formulation in the context of business practice in the UAE/GCC.		
MBMI 520	Managing Information Systems Performance	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Graduate Standing		
The course studies techniques and practices for analyzing business information systems performance with emphasis on support for business processes. The course gives a strategic and integrated view of the exploitation of information and communication technology, with particular emphasis on e-business for improving business performance and business efficiency. The course discusses IT-driven business strategy, and reviews IT-related factors that are necessary for enhancing the performance of organizations in the current competitive global		

MBMK 518	Marketing Management	(3:3:0)
3 Semester Credit Hours		

This course looks in-depth at techniques for crafting effective marketing strategies, for a variety of market environments. Customer-focused marketing strategy design implementation and management are discussed using contemporary frameworks within a market-driven setting. The course emphasizes managerial aspects of marketing, including integrating specific elements of the marketing process. The course additionally places stress on case analysis, within the UAE/GCC context.

This course equips students with skills and knowledge needed to excel in today's digital landscape. It covers Digital Marketing Analytics, Digital Retail, Content Marketing, Customer Experience Design, Social Media and Viral Marketing, Digital Marketing Metrics, and Neuromarketing. The course offers a blend of theory and practical application, preparing students to implement effective marketing strategies in their future careers.

The course introduces the fundamental rules of an international language that enables ideas to be expressed and communicate in an easy and clear way through visual illustration. Includes the following topics: geometric construction; line convention; orthographic projections, Position student feet on the first step of the engineering design based on the visual representation. The course covers the following topics: isometric projections; oblique projections; Perspective projections; dimensioning, and sectional views. Computer software design, prepare parts drawing and assembly it using the software CREO.

This course will consider the fundamental science of classical thermodynamics and its practical applications. Problem solving will be emphasized, including problem formulation, analytic, and computational solutions. Topics include the first law of thermodynamics, work, heat, properties of substances and state equations, the second law of thermodynamics and applications to engineering systems.

This course is a continuation of MENG 211 - Thermodynamics I. It provides more depth in the study of cycles, with applications to gas power and refrigeration cycles, and vapor and combined power cycles; mixtures of gases and vapors, psychrometrics, chemical reactions, and energy analysis.

Kinematics of a Particle, Kinetics of a Particle: Force and Acceleration, Kinetics of a Particle: Work and Energy, Kinetics of a Particle: Impulse and Momentum, Planar Kinematics of a Rigid Body, Planar Kinetics of a Rigid Bodies.

This course introduces the theory behind engineering measurements. Basic definitions, error analysis, characteristics of mechanical systems, system response and signal analysis. Students will apply the knowledge learned in this course in experiments preformed in the BSME program's required laboratory courses.

This course describes the material science and why should an engineer know about it. It covers: Classification of engineering materials, bonding forces and atomic structure and structure of crystalline materials, imperfections and defects, diffusions in solids. Phase diagrams and phase transformation diagrams and heat treatment also covered. Properties of materials such as mechanical, thermal, corrosive and electrical properties of materials included. Moreover, the mechanical failure of engineering materials and the application and processing of metal alloys will be discussed.

Corequisite(s): MENG 241

This laboratory course provides an introduction to material science that covers the main material experimentations. General introduction and safety procedures are introduced. Physical and mechanical properties will be inspected. This include microstructure, hardness, creep, impact, tensile, compression and torsion test. Further the microstructure/processing properties relationships are also investigated. Steel heat treatments is also included.

MENG 311 Internal Combustion Engines (3:3:0)

3 Semester Credit Hours

Prerequisite(s): MENG 212

Engine classifications and terminology. Engine operating characteristics and performance parameters. Air standard engine cycles including: Otto, Diesel, Dual and two-stroke cycles. Common fuels used in IC engines, combustion reactions and the associated thermochemical calculations. Engine emissions and their control technologies and strategies. Air and fuel induction methods and technologies, the physics of the combustion phenomena. Friction losses, lubricants and lubrication systems.

MENG 312 Manufacturing Processes (3:3:0)

3 Semester Credit Hours

Pre-requisite(s): MENG 241

This course introduces the principles and importance of manufacturing processes with some practical applications. It describes and differentiate between manufacturing processes groups. That's is forming and shaping processes of metals including casting, forging, rolling and drawing and for polymers including injection molding and extrusion. Machining processes such as turning, milling and drilling and their associated machining processes also identified as well as introducing the advanced machining processes. Joining processes including fusion welding processes, solid-state processes as well as brazing, soldering, adhesive bonding, and mechanical-fastening processes also explained. Surface modifications and surface treatments and coatings processes also included. The product design and process selection and considerations in a competitive environment is highlighted and identified.

MENG 313 Manufacturing Processes Lab (1:0:3)

1 Semester Credit Hour

Corequisite(s): MENG 312

This laboratory course provides an introduction to manufacturing processes experimentation. Experiments include Oxy, Arc, Spot welding as well as mechanical fastening by riveting, screwing and assembling, metal fabrication and sheet metal, machining processes of milling, turning and CNC. Advance manufacturing technologies of LASER and FDM are introduced. (Writing Intensive Course).

MENG 321 Vibration and Control (3:3:0)

3 Semester Credit Hours

Prerequisite(s): MATH 214, MENG 221

Introduction to Vibration, Oscillatory Motion, Free Vibration, Forced Vibration, Rotating Unbalance, Multiple-Degree-of-Freedom Systems, Introduction to control, block diagrams, modeling of systems, state space representation, Laplace transform, solution of linear systems, stability, input/output description, PID controllers, transfer function methods, tracking.

MENG 323 Engineering Economy (3:3:0)

3 Semester Credit Hours

Pre-requisite(s): MATH 113

Principles of economic analysis and methods in engineering including: time value of money, discounted cash flow techniques equivalence, economic measures of worth, single and multiple alternatives evaluation and selection, replacement decisions, cost estimation, equipment depreciation, the use of Minimum Attractive Rate of Return MARR and Benefit/cost analysis.

MENG 342 Fluid Mechanics Lab (1:0:3)

1 Semester Credit Hour

Prerequisite(s): MENG 231

Corequisite(s): CIEN 251

This course is composed of a set of selected experiments about general fluid mechanics. The experiments will be either performed in groups by the students or demonstrated by the instructor. Individual class work will be strongly encouraged as well as teamwork. The lab also includes an open-ended design of experiment.

MENG 361 Heat Transfer (3:3:0)

3 Semester Credit Hours

Prerequisite(s): MENG 212 and MATH 214

Introduction to heat transfer mechanisms, heat conduction equation, steady heat conduction including the thermal resistance networks, transient heat conduction, lumped systems, fundamental of convection and thermal boundary layers, external and internal forced convection, natural convection, boiling and condensation, thermal radiation, and heat exchangers.

MENG 362 Thermal Sciences Lab (1:0:3)

1 Semester Credit Hour

Prerequisite(s): MENG 211 and MENG 231

Corequisite(s): MENG 361

This course is composed of a set of selected experiments which demonstrate and apply the concepts of thermodynamics and heat transfer. The experiments will be either performed in groups by the students or demonstrated by the instructor. Individual class work will be strongly encouraged as well as teamwork. The lab also includes an open-ended design of experiment.

MENG 421 Theory of Machines**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): MENG 221**

This course focuses on the kinematic and kinetic analysis of mechanisms. It introduces the fundamental concepts, definitions and terminologies in mechanisms, basic mechanisms and applications, linkages and mobility, dynamic analysis of cams, gears and gear trains, velocity and acceleration analysis in mechanisms, and static and inertia force analysis of machinery.

MENG 422 Building Utilities II: Illumination, Acoustics, and Electrical Building Services**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): PHYS 110, MENG 468**

This course explores building electrical system and controls, lighting and acoustic design and assist integrating them into architectural design of the building. Evaluates systems types, components, and installation and maintenance procedures as it relates to building regulations and systems impact human's indoor comfort and on environment.

MENG 441 Turbo Machinery**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CIEN 251**

Turbomachinery classifications and terminology. Implementation of dimensional analysis for predicting performance of turbomachines and designing engineering systems. Understand the fundamentals of energy transfer between rotating rotors and fluid flow. Demonstrate the ability to construct velocity diagrams for various turbomachines (axial-flow compressors and turbines, radial-flow compressors and turbines, pumps, fans, blowers, hydraulic turbines) and their relation to design. Perform elementary analysis for determining input/output work of various turbo devices. Design and selection of turbomachines for various engineering applications.

MENG 451 Mechanical Design I**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): CIEN 212, MENG 241**

Introduction to Mechanical Engineering Design, Materials Properties, Load and Stress Analysis, Deflection and Stiffness, Failure Prevention, Fatigue Failure, Design of Mechanical Elements, Screws Fasteners and Nonpermanent Joints, Mechanical Springs.

MENG 452 Mechanical Design II**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): MENG 451**

This course is a continuation to the machine design I course. Students will be introduced to the analysis and design concepts of various types of machine elements that include: bearings (journal and anti-friction); spur, helical and bevel gears; flexible drives and flywheels; clutches and brakes.

MENG 453 Computer Aided Design**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): MENG 201**

This is an upper-year mechanical engineering course. It introduces students to the analytical basis to CAD software and the three main ways to represent an entity, namely wireframe, surface and solid modeling. The course aims at introducing the concept and importance of CAD as part of the design process. Also it focuses on mathematical representation and manipulation of geometry. The course introduces students to Computer-Aided Mechanical Design (CAMD) tools and their applications to mechanical systems design.

MENG 455 Finite Elements in Machine Design**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): MENG 451**

The objective of this course is to learn how to design and analyze structural components of machine system, especially using the finite element method. The course exposes students to analytical and numerical methods for computing stresses and strains in structures, use of finite element software for static structural analysis and the application of design and failure criteria to ensure that mechanical components can carry the design load without failure. Another important area of the course is to make the students recognize the importance of self-education and life learning.

MENG 461 HVAC and Refrigeration Systems**(3:3:0)****3 Semester Credit Hours****Prerequisite(s): MENG 361 and MENG 212**

Review of psychrometry. Air conditioning processes. Thermal comfort, inside and outside design conditions. Ventilation and infiltration. Heating and cooling load calculations. Solar radiation. Water heating systems layout and design. Air systems design. Under floor heating.

Review of vapor compression and absorption cycles; compressors, condensers, evaporators, expansion devices; refrigerants (including new ones); cooling towers; components of an absorption cycles, controls.

MENG 462 Design of Thermal Systems

(3:3:0)

Semester Credit Hours

Prerequisite(s): MENG 361 and CIEN 251

Application of principles of fluid mechanics heat transfer and thermodynamics in the component design of thermal systems. Examples are drawn from power generations, environmental control, and industrial processes such as design and sizing of piping systems, piping networks, pumps sizing and selection, and heat exchangers selection and performance evaluation. Students work individually and on group to conduct assignments and projects for integration of these components in the design of thermal systems.

MENG 463 Energy Conversion and Management

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): MENG 361

An introduction to the basic technical and economic criteria for the design of efficient energy conversion systems, including traditional as well as alternative power systems. To discuss strategies for increased energy efficiency and more environmentally sound operation. To assess design alternatives and selection criteria based on long-term economic viability and overall energy management strategies.

MENG 468 **Building Utilities I: HVAC and Mechanical Building Services**

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): PHYS 110, ARCH 336

This course explores building systems such as heating, ventilating and air-conditioning, water supply, drainage, fire safety, vertical transportation, building control systems, and assist integrating them into architectural design of building. Evaluates systems types, components, and installation and maintenance procedures as it relates to building regulations, systems impact on indoor comfort and environment.

MENG 471 Sustainable Product Design

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): None

This course is designed to equip engineering students with a comprehensive knowledge of the principles, methods, and tools needed to design environmentally responsible and sustainable products. It covers various aspects of sustainability, including ecological, economic, circular economy, and social considerations. This course aims to equip students with the tools to design products that minimize their ecological footprint while meeting user needs and market demands.

MENG 491 **Senior Design Project I**

(2:0:6)

2 Semester Credit Hours

Prerequisite(s): Completion of 90 credit hours

This is the first course of a capstone project that requires students to develop, design, and implement a solution to an engineering problem under the supervision of a faculty advisor. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resource, procurement, and communication, with consideration of ethical, economical, social and professional conduct. In this course, students will learn several personal and professional skills such as teamwork, time management, leadership and oral and written presentations.

MENG 492 **Senior Design Project II**

(4:0:12)

4 Semester Credit Hours

Prerequisite(s): MENG 491

The Mechanical Engineering Senior Design Projects Program coordinates the completion of the second half of the capstone design sequence required of all Mechanical Engineering seniors. Students apply engineering design methodology, using both analysis and synthesis, to solve open-ended problems. The range of design problems considered spans other engineering fields as well as non-engineering disciplines. At the end of the term, each student design team is expected to present information related to their project in both written and oral formats.

MENG 493 **Special Topics in Mechanical Engineering**

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): Dept. Approval

Special up-to-date topic in one of the mechanical engineering streams of applied mechanics or thermal sciences.

MEPM 511 Project Management Fundamentals

(3:3:0)

3 Semester Credit Hours

The fundamentals of project management including: overview and concepts of project management (principles, body of knowledge, strategies); planning successful projects (defining, specifying, delivery options, scheduling, budgeting); implementing (organizing the team, work assignments, team building, effective leadership); executing (performance measurement, maintaining the schedule, adjustments/mid-course corrections, record keeping, status reporting, communications, managing conflict, time management); and

closeout (performance measurement, maintaining the schedule, adjustments/mid-course corrections, record keeping, status reporting, communications, managing conflict, time management).

MEPM 512	Engineering Contracts and Procurement	(3:3:0)
3 Semester Credit Hours		
This course presents fundamental concepts and techniques for project procurement. Students are introduced to the PMBOK Guide four-step procurement process and expected to develop an in-depth understanding of project contracting, negotiation, and procurement execution.		
MEPM 513	Project Implementation and Performance	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MEPM 511		
This course examines various topics related to project initiation and performance, Key performance indicators and criteria for measuring project success. The course includes using project performance software. Topics include: team building and management, organization structure, performance and success measures and earned-value technique. Other topics include: work breakdown structure, stakeholder management and project communication management.		
MEPM 514	Global Projects Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MEPM 511		
This course provides an overview of the global project management process with specific emphasis on cross-cultural considerations, environmental factors, challenges, benefits & risks of global projects and leadership in global projects. The course also includes best practices that apply to global project management.		
MEPM 515	Project Scheduling	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MEPM 511		
The course will prepare students to master project scheduling. Project scheduling methods are covered including: network construction, forward pass and backward pass calculations, activity slakes or floats calculations, critical path method (CPM), critical chain scheduling, Program Evaluation Review Technique (PERT), resource loading and resource leveling, Project crashing, and scheduling risk analysis. Students will have the opportunity to learn and use one of the project management software's for project scheduling.		
MEPM 516	Project Quality Management	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): MEPM 511		
This course covers the fundamentals of quality control and management, quality principles, quality control techniques, quality control tools (Check Sheets, Histograms, Pareto Diagram, Cause and Effect Diagram, Scatter Diagram, Flow Process Charts and Control Charts), Control Charts for Variables and for Attributes, Lot-by-Lot acceptance sampling, acceptance sampling system, quantitative techniques, quality decision-making techniques. Examples and case studies on project quality management are presented and discussed from a wide variety of engineering discipline.		
MEPM 521	Project Cost Accounting and Finance	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): None		
This course covers the fundamentals of project cost accounting and finance. It reviews the fundamentals of accounting; examines cost accounting principles, applications, and the impact on profitability. It also examines the principles of project costing and covers the elements and introduces a framework for using an effective project cost system. Moreover, it introduces a framework for how projects are financed.		
MEPM 522	Financial Analysis and Decision Making	(3:3:0)
3 Semester Credit Hours		
This course covers some of the theory and practice of decision-making, as well as basic procedures for the application and interpretation of financial statement analysis. Topics covered in this context will include the concept of time value of money, project analysis and evaluation, cost of capital, stock valuation and capital budgeting.		
MEPM 531	Operation Research	(3:3:0)
3 Semester Credit Hours		
This course introduces the modeling techniques, theory and computation of linear programming. Topics include: The structure of linear programming, basic feasible solutions, simplex method, sensitivity analysis and linear programming duality. Linear programming modeling techniques for different applications: blending problem, inventory problem, minimum cost network flow problem, transportation problem, capital budgeting and fixed charge problems. Basic integer programming, dynamic programming and queening theory methodologies and techniques will be introduced.		
MEPM 532	Engineering Management	(3:3:0)
3 Semester Credit Hours		

The fundamentals of engineering management including: overview and concepts of engineering management (Introduction to engineering management, Management cycle includes (Planning, organizing, Leading and controlling. Different aspects of engineering managements will be introduced; like managing research and development, managing engineering design and managing production activities and operations. Market management and service activities for engineers will be clarified as well. Engineers as managers and Engineers code of ethics concepts and elements will be clarified. Finally, the globalization concept and the new challenges for engineering managers in the future will be discussed.

MEPM 533 Information Systems for Project Management

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): MEPM 511

This course will assist the student in understanding the challenges, opportunities and risks involved in information technology management. It introduces the management information systems foundations; current trends; MIS technology fundamentals; its applications to business functions and management practice and in an ethical manner. The impact of information technology on project management will be stressed since it affects the planning process. This course will also cover various system applications for specific project and business functions and their importance to today's manager.

MEPM 541 Organization Behavior and Project Team Management

(3:3:0)

3 Semester Credit Hours

This course deals with human behavior as individual and as groups in organizations. Individual-level characteristics such as personality, attitudes and values, perceptions and judgment, motivations, career development and ethics are emphasized. Topics include group formation, development, structure, leadership, diversity, and dynamics, as well as the processes of communication, decision making, power, and conflict. Class sessions and assignments are intended to help participants acquire skills and analytic concepts to improve organizational relationships and project team management effectiveness.

MEPM 542 Supply Chain Management for Project Managers

(3:3:0)

3 Semester Credit Hours

This course focuses on management and improvement of supply chain processes and performance. It will be valuable for project managers who would like to pursue a career in consulting or take a position in operations, marketing or finance functions in a manufacturing or distribution firm. We explore important supply chain metrics, primary tradeoffs in making supply chain decisions, and basic tools for effective and efficient supply chain management, production planning and inventory control, order fulfillment and supply chain coordination.

MEPM 543 Risk Management for Project Managers

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): MEPM 511

This course addresses fundamental issues, principles, and theory of project risk management and planning. It covers the quantitative and qualitative approaches to identifying, analyzing, assessing, and managing risks inherent to engineering projects. Other topics include risk response strategies and planning.

MEPM 590 Thesis I

(3:0:9)

3 Semester Credit Hours

Prerequisite(s): Department Approval

A research-based thesis course that offers students the opportunity to work on a comprehensive, individual project. The thesis work must make a significant contribution to knowledge in the field of infrastructure engineering. The project will be of suitable complexity for results to be published for an expert audience. **Thesis Research Course**

MEPM 591 Thesis II

(3:0:9)

3 Semester Credit Hours

Prerequisite(s): MEPM 590

Continuation of the Master Thesis I. A research-based thesis course that offers students the opportunity to work on a comprehensive, individual project. The thesis work must make a significant contribution to knowledge in the field of engineering project management. The project will be of suitable complexity for results to be published for an expert audience. **Thesis Research Course**

MEST 100 Introduction to Islam in World Culture

(3:3:0)

3 Semester Credit Hours

This course is the introductory course on Islamic Studies that is required for all majors. The course provides an introduction to the basic sources and historical contexts for the origins of Islam. (**Writing Intensive Course**)

MGHR 301 Human Resource Management

(3:3:0)

3 Semester Credit Hours

Prerequisites: MGMT 301

This course is a broad survey of key areas of human resources management, designed to provide line students wishing to become future supervisors and/or staff human resource specialists with a thorough understanding of the role of human resource management and the

techniques available to management in making effective use of the human resources of an organization. The material is delivered from the perspective of HRM practice in both the UAE/GCC and internationally.

MGHR 302 Compensation Management

(3:3:0)

3 Semester Credit Hours

Prerequisites: MGHR 301

The Compensation Management course focuses on the theories underlying direct compensation and reward systems in organizations, and the practical administrative practices used to implement such systems, with particular reference to the UAE/GCC sub-region. Compensation management practices, including the analysis and evaluation of jobs, criteria and procedures for determining wage levels, individual wage determination, forms of pay, and incentive systems will be covered.

MGHR 304 Human Resource Recruitment and Selection

(3:3:0)

3 Semester Credit Hours

Prerequisites: MGHR 301

The Human Resource recruitment and selection course focuses on employee staffing, training and development processes in organizations. Topics covered include human resource planning, job analysis for selection, multiple recruitment and selection methods are investigated, and the course looks in detail at how these HRM elements are typically implemented in the UAE/GCC region. Through a blend of theory and application, the course introduces students to a wide range of issues, principles, practices and trends in recruitment and selection.

MGHR 390 Internship in Human Resource Management

3 Semester Credit Hours

Prerequisites: Completion of 90 credit hours

The Internship is designed for candidates on the undergraduate human resource management major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship, and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

MGHR 403 Cross Cultural and International Human Resource Management

(3:3:0)

3 Semester Credit Hours

Prerequisites: MGHR 301

The course introduces students to the field of international human resource management (I-HRM). Topics covered include: HR planning for international operations, the impact of national cultures on I-HRM, including standardized cultural models; Divergent and Convergent International-HRM practices; Integration versus Differentiation; International people-resourcing; HR cultural adaptation and adoption issues; Organizational Culture and I-HRM practice; comparative I-HRM studies featuring country comparisons. A number of contemporary I-HRM models are also discussed and utilized, with particular reference to I-HRM within and out of the UAE/GCC.

MGHR 431 Employee Relations

(3:3:0)

3 Semester Credit Hours

Prerequisites: MGHR 301

The course is aimed at students who are interested in pursuing a career in human resource management. The course focuses on legal and regulatory issues that affect employee relations in the GCC and UAE. Students will compare and contrast policies and procedures related to employee benefits and possible labor violations. They will also examine the core issues and applications of UAE/GCC labor regulations in managing employee regulations. At the end of this course, students will have developed an understanding of individual and collective labor relations in the UAE/GCC region and the appropriate methods of dealing with employee disputes.

MGHR 444 HRM Project Based Internship

3 Semester Credit Hours

Prerequisite(s): MGHR 301

The HRM project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem-solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the HRM specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.

MGHR 461 Diversity in Organizations

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): MGMT 301

The course enables the student to understand the role of diversity in human affairs particularly in the workplace. The course discusses and analyzes the role of diversity in the workplace, along with the procedures and practices of diversity management. Students will be able to apply these concepts to discussions of specific dimensions of diversity, including gender, race-ethnicity, sexual orientation,

religion and disability. It also offers students the opportunity to develop their critical thinking on topics such as identity, relationships across difference and bias, and equality of opportunity in organizations around the world and how they relate to organizational issues of equality of opportunity, inclusion, and effectiveness.

MGHR 462 Strategic Human Resource Management

(3:3:0)

3 Semester Credit Hours

Prerequisite MGHR 301

This course will explore the contribution strategic Human Resource Management (HRM) makes towards the development and support of sustainable organizational strategies. On successful completion of this unit students will have the confidence to contribute to strategic decision-making in an HR context. This will be based on strong theoretical and applied foundations which will add value to an organization's HR function and role.

MGHR 463 Negotiations in Organizations

(3:3:0)

3 Semester Credit Hours

Prerequisite MGHR 301

This course focuses on developing the conflict management and negotiating skills of the learner. By the end of this course, the learner will be able to evaluate negotiation situations, strategize and plan upcoming negotiations, and engage in more effective negotiations. The course content will contextualize conflict management and negotiation to the local UAE/GCC environment, looking at cross-cultural influences. Consequently, to assist the learner in developing effective negotiation and conflict management skills, this course emphasizes experiential learning through student participation in a variety of exercises and role-plays.

MGHR 464 Training and Development

(3:3:0)

3 Semester Credit Hours

Prerequisite MGHR 301

The purpose of this course is to provide students a firm understanding of human resource training and development systems in today's business environment. A constant theme setting the foundation for this course will be on the various kinds of changes facing organizations, with a focus of UAE/GCC organizations, and how these changes relate to human resource training and development. Topics will cover assessing training needs, developing and delivering training, evaluating training outcomes, career planning and career management. Aspiring HR professionals will gain essential knowledge to effectively manage employee training and development systems in a variety of companies. The course looks in detail as to how these HRM elements are typically implemented in the UAE/GCC region.

MGMT 301 Principles of Management

(3:3:0)

3 Semester Credit Hours

Prerequisites: Sophomore Standing

This course presents essential management theories and concepts. It examines the nature of managerial work under a range of business models and under rapidly changing business conditions. Managerial functions and activities such as planning, organizing, leading, and controlling are examined in depth, and in the context of current organizational practices and scenarios.

MGMT 401 Organizational Behavior

(3:3:0)

3 Semester Credit Hours

Prerequisite Completion of 90 credit hours

This course applies essential organizational behavior and leadership theories and concepts to organizational practices. It examines the complexities of human behavior in a range of organizational work settings under rapidly changing business environment. Individual personality, group/team dynamics, motivation, performance, communication, and leadership concepts are examined in depth, and in the context of current organizational practices.

MIST 301 Introduction to Business Information Systems

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): Sophomore Standing

This course introduces fundamentals of computer hardware, software, networking, Internet and its technology components. The discussions are centered on the role of technology in contemporary business, and include basic relational storage concepts, with hands-on experience in building business database applications and web sites.

MKTG 301 Principles of Marketing

(3:3:0)

3 Semester Credit Hours

Prerequisites: ECON 103

This course examines marketing principles, concepts, strategies, tactics, and analytical tools used by profit and nonprofit organizations to market ideas, products, or services to selected target groups. The course emphasizes how to promote, distribute, and price firm's offerings in dynamic economic, social, political, and international environments.

MKTG 311 Sales Management

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): MKTG 301

The course familiarizes students with marketing –sales interfaces including sales force role and capabilities, personal selling strategies, organizational relationships, and responsibilities of sales managers including training, motivating, and evaluating sales force.

MKTG 312 Consumer Behavior

(3:3:0)

3 Semester Credit Hours

Prerequisites: MKTG 301

The Consumer Behavior course studies the purchasing behavior of consumers and the exchange process involved in acquiring, consuming and disposing of goods, services, experiences and ideas. In this course students learn to deal with an unstructured situation by the usage of a buyer decision – making model and buyer behavior concepts enabling them to identify important considerations and their possible resolutions.

MKTG 313 **Integrated Marketing Communications**

(3:3:0)

3 Semester Credit Hours

Prerequisites: MKTG 301

This course involves an in-depth study on the application of advertising and other forms of marketing communication with an emphasis on their role in marketing planning. The course includes a study of the identification of relevant data to analyze the marketing situation, the development of a product position, marketing and marketing communications objectives, creative strategy, media planning, and evaluation. This is an applied course that utilizes real-world examples, course materials, and project assignments whenever possible in order to allow students to gain practical experience related to the course topics.

MKTG 315 Social Media and Digital Marketing

(3:3:0)

3 Semester Credit Hours

Prerequisites: MKTG 301

This course provides a comprehensive overview of the ever-changing landscape of social media and digital marketing. It aims to equip students with a strong grasp of the strategies, tools, and analytical techniques employed in the field. By integrating technology, innovation, and data-driven decision-making, the course seeks to enable students to create effective digital campaigns, evaluate their performance, and interact with audiences in a meaningful manner. Students will delve into content creation, curation, social media advertising, search engine optimization, email marketing, and influencer marketing. By combining theoretical knowledge with hands-on practice, students will examine how these tools influence consumer behavior, brand communication, and overall marketing approaches.

MKTG 316 Luxury Brand Management

(3:3:0)

3 Semester Credit Hours

Prerequisites: MKTG 301

This course provides an in-depth exploration of the complexities and subtleties of luxury brand management. It sheds light on the strategies and strategic challenges crucial for navigating the affluent luxury goods market. Students will develop a comprehensive understanding of the principles that govern the marketing and management of prestigious brands, with a focus on creating a luxury brand identity that resonates with consumer perceptions and stands out in a competitive landscape. The course will examine the strategic opportunities and challenges of luxury goods while also refining practical marketing skills tailored to the high-end market. Additionally, students will learn through an analysis of current luxury market trends, consumer behavior, and the essential ethical and sustainable practices that are increasingly important in responsible luxury brand management.

MKTG 332 Retailing and E-Commerce

(3:3:0)

3 Semester Credit Hours

Prerequisite MKTG 301

Examination of retailing as a specialized economic and social institution within the distribution process and as it relates to society, fashion, and overall marketing activities. The planning and implementing of store and non-store (catalog, Internet) retail marketing strategies are addressed. Critical decision alternatives, variables, forces, and processes are considered from a managerial perspective. This is an applied course that utilizes real-world examples, course materials, and project assignments whenever possible in order to allow students to gain practical experience related to the course topics.

MKTG 351 Marketing Research Techniques & Applications

(3:3:0)

3 Semester Credit Hours

Prerequisites: MKTG 301 & OPMT 311

The course introduces the student to tools used in marketing research which are both quantitative and qualitative. Quantitative tools used are questionnaires, experiments, and conjoint analysis and qualitative tools used are focus groups and interviews. In the process, both primary and secondary research may be used. The course emphasizes problem formulation skills and takes a managerial perspective on research methodology that focuses on the kinds of decisions that each method can support. This is an applied course that utilizes real-world examples, course materials, and project assignments whenever possible in order to allow students to gain practical experience related to the course topics.

MKTG 390 Internship in Marketing

3 Semester Credit Hours

Prerequisites: Completion of 90 credit hours

The Internship is designed for candidates on the undergraduate marketing major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship, and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

MKTG 407 International Marketing

(3:3:0)

3 Semester Credit Hours

Prerequisite MKTG 301

The course is multidisciplinary approach to international marketing from viewpoint of business management, examines major marketing issues affecting companies operating in a global environment. Students understanding of economics, political and cultural differences among nations as they affect marketing opportunities and operations and develop skills to identify international marketing opportunities.

MKTG 412 Marketing of Services

(3:3:0)

3 Semester Credit Hours

Prerequisites: MKTG 301

This course introduces the student to various aspects of Services Marketing. Specifically, the course examines the unique characteristics of services marketing, management and evaluation of service quality, techniques for service recovery and improvement, and services marketing in global environments. Particular emphasis is placed on analyzing the service encounter and the perception of service quality by service recipients and the service providers. This is an applied course that utilizes real-world examples, course materials, and project assignments whenever possible in order to allow students to gain practical experience related to the course topics.

MKTG 444 Marketing Project Based Internship

3 Semester Credit Hours

Prerequisites: MKTG 301

The marketing project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem- solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the marketing specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.

MKTG 471 Marketing Strategy

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): MKTG 301

The course emphasizes managerial aspects of marketing, including developing marketing strategies and plans, and integrating specific elements of the marketing process. The course additionally places stress on case analysis, as a means for learning the material and for students to develop problem-solving, group work, judgement and decision-making skills. This is an applied course that utilizes real-world examples, course materials, and project assignments whenever possible in order to allow students to gain practical experience related to the course topics.

MSRE 511 Energy Conversion

(3:3:0)

3 Semester Credit Hours

Forms of energy. Energy needs and available sources of energy. Conversion of thermal energy into mechanical energy, including power, and heat engine cycles, internal and external combustion systems and turbines. Basic technical and economic criteria for the design of efficient energy conversion systems, including traditional as well as alternative power systems.

MSRE 512 Energy Systems Modeling and Optimization

(3:3:0)

3 Semester Credit Hours

Review of renewable energy resources; solar energy and photovoltaic, wind energy, wave energy, biomass energy conversion, fuel cells and batteries. Hydroelectric power and geothermal energy. Modeling methodology including system conceptualization. Model construction and validation (computational accuracy). Model evaluation and calibration. Simulation of energy and environmental systems. Optimization techniques; Classical direct search-for-optimum methods, Golden Mean, Conjugate Gradients, Modified Newton Method. Methods for constrained optimization such as Lagrange Multipliers, Search methods, Linear and Dynamic Programming. Use of software packages.

MSRE 513 Energy Economics

(3:3:0)

3 Semester Credit Hours

Comprehensive understanding of energy markets. Technological, cost, and environmental fundamentals of energy sources and environmental systems. Economic principles underlying the supply and demand for energy in a modern economy, through considerations of topics such as energy demand at the individual and economy-wide level, the supply of renewable and non-renewable energy resources,

MSRE 514 Direct Research

3 Semester Credit Hours

(3:0:9)

Under the guidance of an engineering faculty member, the Directed Research Project provides the student with a meaningful research experience. It requires that the student conduct a research topic or issue of significance to the field of renewable and sustainable energy.

Techniques and approaches adapted to improve the efficiency of energy generation, utilization, conversion, transport, storage and management. Energy audits. Energy conservation opportunities for efficiency improvements in different sectors: industrial, commercial, transportation and domestic. Economic evaluation of energy conservation opportunities using engineering economic formulas, simple pay-back analysis, and life-cycle cost models. Short- and long-term planning. Restructuring and Privatization. Models of electricity industry, contract issues, markets and transmission pricing. Deregulation of the energy market around the world. Electrical demand control and power factor correction. Load forecast and generation side management.

Introduction to solar radiation and some important related concepts like blackbody, solar spectrum, irradiance, irradiation, air mass and peak sun hour. Semiconductor and P-N junctions. PV cells properties, design, interconnection and module fabrication. Modules and arrays, stand-alone PV schemes with battery energy storage and grid-connected PV schemes.

Basic characteristics of wind, site characterization, statistical methods of wind analysis, wind resources assessment, fundamental principles of wind energy utilization, aerodynamics, mechanical and electrical design aspects. Wind machine technologies and wind turbines performance analysis. Wind farm planning and design. Wind power integration into the power systems, and the environmental impact of wind power utilization.

Fundamentals and applications of biofuels and bioenergy produced from biomass including processes of production, availability and attribution of biofuel/bioenergy production. Types of biomass raw material and methods of selection. First, second and third generation of biofuels. Methods of biomass conversion to fuel including thermochemical conversion, biochemical conversion and catalytic conversion. Production of hydrogen, biodiesel and microbial fuel cells. Environmental impacts of biofuel production. Economics and life-cycle analysis of biofuel. Value-added processing of biofuel residues.

Basic science and technology of water desalination to ensure sustainable water supply. Water production via desalination within the water-energy-cost nexus, evaluation of renewable energy- powered desalination processes, power-desalination cogeneration analysis, evaluation and applications of novel desalination systems, such as thermal desalination, membrane distillation and forward osmosis. Recent technological improvements for enhanced desalination processes, and fouling issues in current technologies. Assessing economic feasibility and the environmental impact of new desalination processes.

Introduction to solar energy, solar radiation; review of the basics of thermodynamics and heat transfer, power plant technologies; types of CSP systems including CSP parabolic trough systems, CSP dish technology, CSP Fresnel technology and solar tower; heat storage systems; hybridization; secondary use of CSP systems; operation and maintenance of CSP systems; power quality control and grid integration; CSP plant project planning: economic, social and environmental considerations and site assessment.

Green building initiatives, their origin, characteristics of a green building, certification of green buildings rating systems, criteria for rating. Policies and drivers that are leading to the more widespread uptake of green building technologies; green building codes, policies and planning from the past, present and future from around the globe. Integrated design: urban micro-climate design, passive and active architectural interventions.

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3 Semester Credit Hours**Prerequisite(s):** Department Approval

A research-based thesis course that offers students the opportunity to work on a comprehensive, individual project. Topic to be agreed in consultation with a supervisor. The project will be of suitable complexity for results to be published for an expert audience. This is the first part of the master thesis course. **Thesis Research Course**

MSRE 591 Thesis II**(6:0:18)****6 Semester Credit Hours****Prerequisite(s):** MSRE 590

Continuation of the Master Thesis I. A research-based thesis course that offers students the opportunity to work on a comprehensive, individual project. Topic to be agreed in consultation with a supervisor. The project will be of suitable complexity for results to be published for an expert audience. **Thesis Research Course**

OPMT 301 Operations Management**(3:3:0)****3 Semester Credit Hours****Prerequisite(s):** Sophomore Standing

The course examines principal aspects of organizational operations in various settings. It emphasizes the application of planning and decision-making activities associated with managing operations, with particular focus on production and service operations. It uses analytical models to describe key operations planning and control activities.

OPMT 311 Methods and Models of Management Science**(3:3:0)****3 Semester Credit Hours****Prerequisite(s):** STAT 100

This is a course in applied operations research for business management. Business situations are represented by analytical mathematical models solved by the effective application of the methods mathematical programming and probabilistic process analysis. This course introduces operation research and management sciences techniques for supporting business management decisions. Specific mathematical programming and probabilistic topics include linear programming, transportation models, network flow models, decision analysis, inventory models, queuing models, and Monte Carlo simulation.

OPMT 405 Supply Chain Management**(3:3:0)****3 Semester Credit Hours****Prerequisite(s):** OPMT 311

A comprehensive study of the concepts, processes, and strategies used in the development and management of global supply chains. Supply-chain management (SCM) is a systems approach to managing the entire flow of information, materials, and services from raw material suppliers through factories and warehouses to the final end-customer. SCM represents a philosophy of doing business that stresses processes and integration. Specific topics include global supply chain management, procurement, electronic commerce, information technologies, and logistics activities. This course will be taught through the use of textbook materials, outside readings, and case analysis.

PENG 322 Drilling Fluids Laboratory**(1:0:3)****1 Semester Credit Hour****Corequisite(s):** PENG 361

This course deals with lab measurements of cement and mud properties. Mud preparation, mud rheology, filtration, wall building and resistivity, mud weight control, drilling fluid contamination test, oil well cementing experiment. (Writing Intensive Course)

PENG 351 Reservoir Rock and Fluid Properties**(3:3:0)****3 Semester Credit Hours****Prerequisite(s):** CHEM 215

This course covers the origin of the earth and its shells, composition of the earth's crust and oceans, and their geological characteristics. Furthermore, it covers the fundamental properties of reservoir rocks, hydrocarbon phase behavior and the fundamental properties of reservoir fluids. The course also deals with the measurement of fundamental properties of reservoir rocks and fluids.

PENG 361 Drilling Engineering I**(2:2:0)****2 Semester Credit Hours****Prerequisite(s):** PENG 231**Corequisite(s):** CIEN 251

This course introduces basic drilling techniques and drilling fluid properties. Topics include rock characteristics, drilling fluids, mud weight calculations, components of rotary drilling rig, drilling hydraulics, drilling bits, factors affecting rate of penetration, cementing operations.

PENG 362 Drilling Engineering I**(3:3:0)****3 Semester Credit Hours****Prerequisite(s):** CIEN 251

The course aims to provide students with a fundamental understanding of petroleum well drilling procedures, its mechanics, and design methodology. The course covers topics include rock characteristics, drilling fluids and its properties, components of rotary drilling rig,

drilling hydraulics, pore pressure, fracture pressure, casing design and directional and horizontal drilling. Lab measurement of mud and cement properties is an integral part of the course.

PENG 371 Petroleum Reservoir Engineering (3:3:0)

3 Semester Credit Hours

Prerequisite(s): PENG 351

This course deals with material balance (MB) techniques to estimate reserves. Topics covered in this course include generalized MB equations, fluid drive mechanisms, PVT data selection, water influx, flow through porous media, displacement of oil, fractional flow, and oil recovery by the internal drive as well as decline curves analysis.

PENG 381 Well Logging (3:3:0)

3 Semester Credit Hours

Prerequisite(s): PENG 351

Logging is an essential operation to determine the key formation parameters, such as porosity, permeability, reservoir thickness, water saturation. All types of well logging techniques such as Logging while Drilling, Cased Hole Logging, and Production Logging are included. Logs are also run to gain information about well casing and cementing.

PENG 382 Petroleum Production Engineering (Well Performance) (4:4:0)

4 Semester Credit Hours

Prerequisite(s): CIEN 251

This course covers basic well performance calculations necessary for the design and analysis of naturally flowing and artificially lifted wells. Topics include Inflow Performance Relationship (IPR), Tubing Performance Relationship (TPR), Flow line Performance Relationship (FPR), Choke Performance Relationship (CPR), Gas-Lift, Electric Submersible Pumps (ESP), and production forecasting.

PENG 384 Petroleum Production Engineering (3:3:0)

4 Semester Credit Hours

Prerequisite(s): CIEN 251

This course covers basic well performance calculations necessary for the design and analysis of naturally flowing and artificially lifted wells, and the handling of the produced oil, water and gas at the surface. Topics include Inflow Performance Relationship (IPR), Tubing Performance Relationship (TPR), Flowline Performance Relationship (FPR), Choke Performance Relationship (CPR), Gas-Lift, and Electric Submersible Pumps (ESP). It also covers phase behavior of water/hydrocarbon systems, surface production facilities, oil, water and gas separations, flash calculations, separator sizing and design.

PENG 403 Petroleum Property Evaluation (3:3:0)

3 Semester Credit Hours

Prerequisite(s): PENG 371 and MENG 323

Applications of reservoir engineering techniques, reserve calculations, decline curve analysis, rate of return calculations to project design and evaluation.

PENG 404 Petroleum Economics (3:3:0)

3 Semester Credit Hours

Pre-requisite(s): PENG 371 and PENG 384

This course provides engineers with skills to assess the costs and benefits of engineering investments. Applications of reservoir engineering techniques, reserve calculations, decline curve analysis, rate of return calculations to project design and evaluation.

PENG 422 Fluid Flow in Porous Media Lab (1:0:3)

1 Semester Credit Hour

Corequisite(s): PENG 485

This course deals with the design aspects of oil displacement by another fluid in rock samples. It builds on the experiences of students obtained in lab measurements of individual reservoir rock and fluid properties in PENG 321 to create an integrated lab measurement of all properties needed to analyze oil displacement by a displacing fluid. The displacing fluid can be chosen to study the relative permeability and displacement efficiency of water flooding, gas flooding, or any enhanced oil recovery fluids (acidic water, microbial water, polymer solution, or steam) using cores, fractured cores (sand packs and glass beads may be considered as alternatives) in one-dimensional geometry or packed layers in two-dimensional geometry.

PENG 464 Data Science in Petroleum Engineering (3:3:0)

3 Semester Credit Hours

Pre-requisite(s): ENGR 200

This course will cover the fundamentals of data science with applications in the oil and gas industry. Covering topics related to descriptive data analytics, predictive data analytics, prescriptive data analytics, distribution, probability, decision tree analysis, parametric vs. non-parametric, confidence intervals, hypothesis testing, F-Test and ANOVA, correlations, linear and multiple linear regression, nonlinear linear regressions, and state-of-the-art machine learning and AI algorithms.

PENG 472	Applied Reservoir Geology	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): PENG 101		
Corequisite(s): PENG 381		
Oil distribution in the world and in the UAE: geology of reservoirs, which includes the formation of reservoir rocks, cap rocks, source rocks and the environments of depositions: petrophysical parameters of parameters of reservoir fluids: oil field waters, crude oil and natural gas; reservoir conditions: pressure, temperature and their effects on oil maturation, migration and accumulation: oil generation. Oil migration: types of oil traps: methods of exploration.		
PENG 473	Reservoir Simulation	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): PENG 371		
This course covers fundamental concepts of reservoir simulation to model single-phase flow in petroleum reservoirs. Topics include reservoir engineering concepts, mathematical concepts, derivation of reservoir flow equations, finite difference approximations, and their solutions, and applications to predict reservoir performance.		
PENG 474	Applied Reservoir Simulation	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): PENG 371		
Corequisite(s): PENG 473		
This course covers advanced topics in reservoir simulation. These include reservoir fluid flow equations in multiphase, multidimensional flow, up-scaling of rock properties, pseudo functions, vertical equilibrium, analysis of data for consistency, history matching, and applications to field cases.		
PENG 482	Natural Gas Engineering	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): PENG 371 and PENG 382		
This course introduces the properties of natural gas, covers flow of gas in reservoir, wellbore and surface pipelines. Based on such knowledge, students are able to run nodal analysis for gas production system design. This course also covers gas compression and gas operation issues.		
PENG 483	Well Testing	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): PENG 371 and MATH 214		
This course covers reservoir characterization by pressure test analysis. Topics include fluid flow equations in porous media under transient and pseudo-steady state flow conditions, pressure buildup and pressure drawdown tests, average reservoir pressure, type curve matching, well testing of heterogeneous reservoirs, pressure derivatives analysis technique, multiple well testing, and test design and instrumentation.		
PENG 484	Stimulation and Intervention in Petroleum Production Operations	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): PENG 382		
Topics include well completions, perforations, wellbore damage sources and detection, hydraulic fracturing, fracturing fluids, acid/rock interactions, and acid treatment of oil wells, design and evaluation of treatments, evaluation by nodal system analysis.		
PENG 485	Water Flooding	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): PENG 371		
This is a senior level course on reservoir engineering series courses. The course covers the reservoir engineering aspects of water flooding. Topics include introduction to reservoir forces and concepts of surface and interfacial tension, wettability, capillary pressure, relative permeability; trapping and mobilization of residual oil; fluid distribution and frontal displacement theory, concept of mobility ratio, flood patterns and areal sweep efficiency considerations, pattern injection rates and pressures; characterization of reservoir heterogeneity; vertical and volumetric sweep efficiency, waterflood performance prediction models; waterflood pilot test; designing, monitoring and evaluating a waterflooding operation; water for water flooding, its sources and treatment.		
PENG 486	Enhanced Oil Recovery	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): PENG 351		
This course covers chemical and thermal method of EOR. Specific topics include interfacial tension, entrapment and mobilization of oil in porous media, residual oil, miscibility, adsorption at solid/liquid interfaces, surfactants and micro-emulsions, miscible gas flooding, polymer flooding, thermal methods, and the effect of reservoir heterogeneity.		

PENG 491	Senior Design Project I	(3:0:9)
3 Semester Credit Hours		
Prerequisite(s): Completion of 90 credit hours		
The Senior Design Project entails team work involving development of a two-semester long workflow that includes preparing and presenting a proposal for a major design work. Throughout the course students are required to develop, design and implement a solution to one of the petroleum engineering problem under the supervision of a faculty advisor. The course also introduces project management topics including project life cycle, integration, scope, time, cost, risk, quality, resource, procurement, and communication, with consideration of ethical, economic, social and professional conduct.		
PENG 492	Senior Design Project II	(3:0:9)
3 Semester Credit Hours		
Prerequisite(s): PENG 491		
Continuation of phase (1).		
PENG 493	Special Topics in Petroleum Engineering	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): Department Consent		
A specific topic in Petroleum Engineering that is not covered in other program courses is presented in a course format.		
PHIL 100	Critical Thinking and Reasoning	(3:3:0)
3 Semester Credit Hours		
This introduction to basic principles of reasoning and critical thinking enhances the learner's abilities to evaluate various forms of reasoning in everyday life and in academic disciplines. The course explores such topics as inductive and deductive reasoning, the nature and function of definitions, fallacy types, statistic use and misuse, and the rudiments of logic. (<i>Writing Intensive Course</i>)		
PHIL 101	Ethics in Today's World	(3:3:0)
3 Semester Credit Hours		
This course addresses theoretical foundations of ethical thought, and background to traditions and movements in the development of ethical theory and methods of reasoning. Students apply ethical principles and perspectives to analyze, compare and critically evaluate relevant personal, social and professional problems and engage in ethical reasoning and decision-making processes.		
PHIL 102	World Philosophies	(3:3:0)
3 Semester Credit Hours		
The course surveys major philosophers from the most important world philosophical traditions. Some of the topics addressed include the internal world of personal identity, the nature of knowledge, the concept of happiness, the nature of reality and the external world, the relation of language to the world, meaning, and truth.		
PHYS 095	Introductory Physics	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s):		
This course is intended for students who need a general introduction to the fundamental principles of Physics and laws of nature before taking University-level Physics courses required in their major. Topics covered include on Dimensional Motion, Fluids, Heat and Thermodynamics. This course is not for degree credit.		
PHYS 110	University Physics I	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): None		
Corequisite(s): PHYS 111		
This is a calculus-based physics course covering the fundamental principles of mechanics. It concentrates on the conservation of energy, the particle motion, the collisions, the rotation of solid bodies, simple machines and on the fluid mechanics. The focus lies on the resolution of one and two-dimensional mechanical problems.		
PHYS 111	University Physics I Lab	(1:0:3)
1 Semester Credit Hour		
Corequisite(s): PHYS 110		
This course is intended to be taken with Physics 110. It primarily includes experiments on classical mechanics. Particular emphasis is placed on laboratory technique, data collection and analysis and on reporting.		
PHYS 220	University Physics II	(3:3:0)
3 Semester Credit Hours		
Prerequisite(s): PHYS 110		

This second calculus-based physics course includes a detailed study of the fundamental principles of classical electricity and magnetism, as well as an introduction to electromagnetic waves. The course's focus targets the resolution of dc- and alternating circuits.

PHYS 221 University Physics II Lab

(1:0:3)

1 Semester Credit Hour

Corequisite(s): PHYS 220

This course is intended to accompany Physics 220. It includes experiments on electricity, magnetism and RLC circuits. Particular emphasis is placed on three aspects of experimentation: laboratory technique, data analysis (including the treatment of statistical and systematic errors) and written communication of experimental procedures and results.

PHYS 222 University Physics III

(3:3:0)

3 Semester Credit Hours

Prerequisite(s): PHYS 220

Corequisite(s): PHYS 223

This course covers important theoretical bases and experiments that support the development of modern physics with applications in various technological and scientific fields. Topics include physical optics, thermodynamics, special relativity, quantum physics, atomic and molecular physics and nuclear physics. The course's focus is on applications that have led to key discoveries.

PHYS 223 University Physics III Lab

(1:0:3)

1 Semester Credit Hour

Prerequisite(s): PHYS 221

Corequisite(s): PHYS 222.

This course is intended to be taken with Physics 222. It includes experiments on optics, thermodynamics and atomic physics. Particular emphasis is placed on three aspects of experimentation: laboratory technique (including both the execution and the documentation of an experiment); data analysis (including the treatment of statistical and systematic errors, as well as computer-aided analysis of experimental data); and written communication of experimental procedures and results.

POLI 100 Contemporary Global Issues

(3:3:0)

3 Semester Credit Hours

This course gives students a broad, interdisciplinary, comprehensive, and balanced, introduction to key debates, theories, major trends, and topics, relating to Globalization and global developments.

POLI 101 **Politics of Scarcity**

(3:3:0)

3 Semester Credit Hours

The problems of scarcity and security are as much political as they are economic or technological. This course identifies the political aspects of global economic exchange and distribution, flows of labor and capital, and international cooperation, global security and conflict.

POLI 102 State and Society in the UAE

(3:3:0)

3 Semester Credit Hours

The course traces the history of the UAE, the establishment of the federation and the development of the UAE as a nation with significant global impact. It covers contemporary life, the economy, society, population, political system, social customs and traditions, and current changes.

PSYC 100 Introduction to Psychology

(3:3:0)

3 Semester Credit Hours

This course provides an overview of major areas in the field of psychology including: the history of psychology, psychology research methods, organization of human brain and biological basis of behavior, sensation, perception, learning, language, intelligence, emotion, motivation, developmental psychology, personality theories /assessment, stress, abnormal behavior and therapies, social psychology.

PSYC 102 Self and Society

(3:3:0)

3 Semester Credit Hours

Study of how society shapes us as individuals and how we as individuals shape society. Explores how we experience life as individuals and as members of cultural systems that shape our personalities, behavior, and perceptions of the world.

SOCI 100 **Introduction to Sociology**

(3:3:0)

3 Semester Credit Hours

Introduction to basic sociological concepts. Examines aspects of human behavior in cultural framework including: individual and group interaction, social mobility and stratification, status and class, race and gender relations, urbanism, crime and criminology, and social change and reform.

SOCI 101	Contemporary Social Issues	(3:3:0)
3 Semester Credit Hours		
The course is an introduction to basic sociological concepts and examines aspects of human behavior in a cultural framework including: individual and group interaction, social mobility and stratification, status and class, race and gender relations, urbanism, crime and criminology, and social change and reform.		
STAT 100	Introductory Probability and Statistics	(3:3:0)
3 Semester Credit Hours		
This course introduces students to the basics of probability theory and statistical inference with examples and applications in sciences. At the end of this course, students will acquire the necessary quantitative competency in the program.		
STAT 210	Probability and Statistics for Life Sciences	(3:3:0)
3 Semester Credit Hours - Prerequisite(s): STAT 100		
This course involves the use of probability and descriptive and inferential statistical techniques in interpreting biological data.		
STAT 346	Probability for Engineers	(3:3:0)
3 Semester Credit Hours - Prerequisite(s): MATH 114		
The course introduces principles of statistics and probability for undergraduate students in Engineering. The course covers the basic concepts of probability, discrete and continuous random variables, probability distributions, expected values, joint probability distributions, and independence. The course also covers statistical methods and topics including data summary and description techniques, sampling distributions, hypothesis testing, and regression analysis.		
UAES 200	Survey of United Arab Emirates Studies	(3:3:0)
3 Semester Credit Hours		
This course presents the UAE from multiple perspectives in an attempt to expose the students to the distinct qualities of the UAE. The purpose of this course is to give students a broad, interdisciplinary, and comprehensive introduction to key features and issues in the UAE's historical, cultural, political and economic landscapes. (Writing Intensive Course)		
UNIV 100	University First-Year Transition	(1:1:0)
1 Semester Credit Hour		
This course is designed to assist students with the transition from secondary school to university life. Students in this course transition to university life by focusing on academic adjustment, by developing decision-making skills, and by learning about services and opportunities for involvement.		
UNIV 200	Innovation, Entrepreneurship, and Sustainability	(3:3:0)
3 Semester Credit Hours		
The course aims at equipping next generation of leaders with an innovative and entrepreneurial mindset and related core skills. The course explains key qualities of entrepreneurs and analyses how entrepreneurs make utilize innovation in creating new ventures. Design-thinking and business model innovation are key frameworks that underpin the course. (Writing Intensive Course)		
UPAS 100	University Preparation for Academic Success	(1:1:0)
1 Semester Credit Hour		
The focus of this course is on the development and application of college-level study skills, personal success strategies, and the use of campus resources that enhance individual student achievement. Designed for matriculated students who are at risk of being placed on Academic Probation 1 due to having a CGPA of less than 2.0. Graded as Satisfactory/Unsatisfactory.		



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